SESHADRIRAO GUDLAVALLERU ENGINEERING COLLEGE

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Academics Strengthening & Advancement (AS&A)

R–23

III B.Tech. I Semester

CONPUTER NETWORKS

Learning Material

UNIT I
INTRODUCTION

Course Objectives

- To familiarize with the basics of data communication, various types of computer networks and topologies.

Course Outcomes

Students can able to

- Defining, using and implementing Computer Networks and the basic components of a Network system.
- defining the concept of local area networks, their topologies, protocols and applications
- Identify and understand various techniques and modes of transmission
- Differentiating the various types of network models and applying them to meet the changing and challenging networking needs of organizations.
- analyze the requirements for a given organizational structure and select the most appropriate networking architecture and technologies
- Determines the functionality of layered network architecture

Syllabus

1.1 Data Communications
- 1.1.1components
- 1.1.2Data flow

1.2 Physical Structures
- 1.2.1 Networks
- 1.2.2 Type of Connection
- 1.2.3 Physical Topology

1.3 Categories of Network
- 1.3.1 Local Area Network
- 1.3.2 Wide Area Network
- 1.3.3Metropolitan Area Network
- 1.3.4 Interconnection of networks: Internetwork

1.4 Reference Models
- 1.4.1 The OSI reference model
- 1.4.2TCP/IP reference model
- 1.4.3A comparison of the OSI and TCP/IP reference models

1.5 Physical Layer
- 1.5.1 Introduction to Guided Media
- 1.5.2Twisted Pair Cable
- 1.5.3Coaxial cable
- 1.5.4 Fiber optic Cable
- 1.5.5 Introduction about unguided Media

What is Computer Network?

A computer network is a system of interconnected devices (such as computers, servers, routers, and switches) that communicate with each other to share resources, exchange data, and enable collaboration. These connections can be established using wired (e.g., Ethernet cables) or wireless (e.g., Wi-Fi) communication technologies. It provides two main functionalities.

Key Features of Computer Networks

- Data Sharing: Enables users to share files, applications, and information.
- Resource Sharing: Allows multiple devices to access shared resources like printers and storage.
- Communication: Supports email, messaging, video conferencing, and other forms of digital communication.
- Scalability: Can grow from small local networks to large global systems like the internet.

1.1 Data Communications

When we communicate, we are sharing information. This sharing can be local or remote. Between individuals, local communication usually occurs face to face, while remote communication takes place over distance. The term telecommunication, telephony, telegraphy, and television, means communication at a distance (tele is Greek for "far").

- The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data.
- Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable. For data communications to occur, the communicating devices must be part of a communication system made up of a combination
- of hardware (physical equipment) and software (programs).
- The effectiveness of a data communications system depends on four fundamental characteristics: delivery, accuracy, timeliness, and jitter

1.1.1 Components

A data communications system has five components

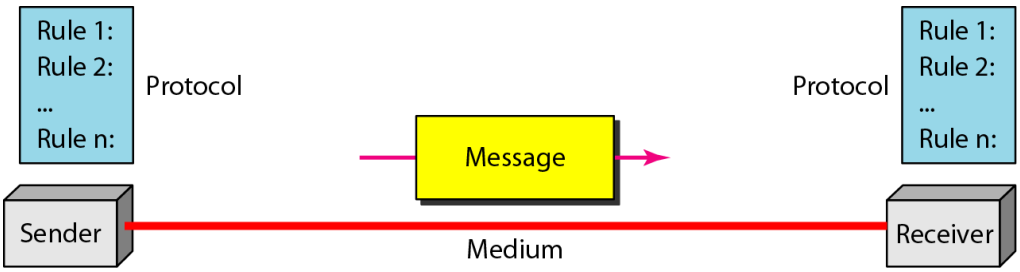


Fig 1.1: Relationship between packets and frames

1. **Message**: The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.

2. **Sender:** The sender is the device that sends the data message. It can be a computer,
workstation, telephone handset, video camera, and so on.
3. **Receiver:** The receiver is the device that receives the message. It can be a computer,
workstation, telephone handset, television, and so on.
4. **Transmission medium**: The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.
5. **Protocol**: A protocol is a set of rules that govern data communications. It represents
an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

1.1.2 Data flow:

Communication between two devices can be simplex, half-duplex, or full-duplex.

Simplex

- In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive (Figure a). Keyboards and traditional monitors are examples of simplex devices.
- The keyboard can only introduce input; the monitor can only accept output. The simplex mode can use the entire capacity of the channel to send data in one direction.

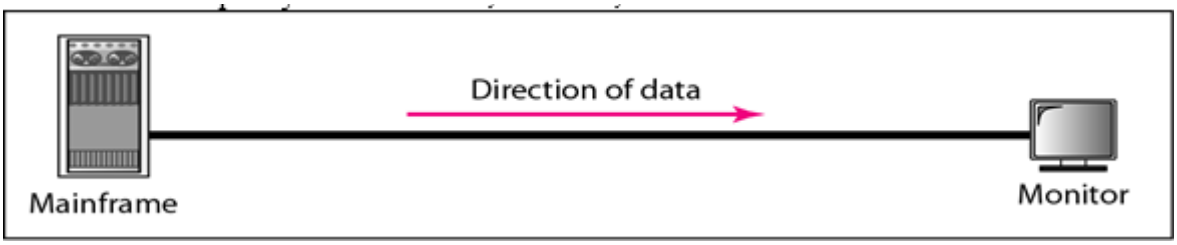


Fig 1.2: simplex Dataflow

Half-Duplex

- In half-duplex mode, each station can both transmit and receive, but not at the same time.
- When one device is sending, the other can only receive, and vice versa (Figure b). The half-duplex mode is like a one-lane road with traffic allowed in both directions.
- When cars are traveling in one direction, cars going the other way must wait. In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time. Walkie-talkies and CB (citizens band) radios are both half-duplex systems.
- The half-duplex mode is used in cases where there is no need for communication in both directions at the same time; the entire capacity of the channel can be utilized for each direction.

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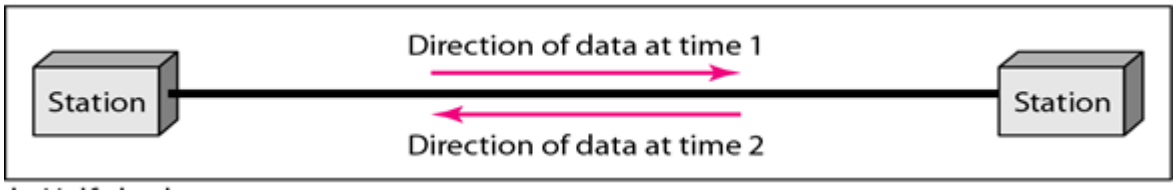


Fig 1.3: Half Duplex Dataflow

Full-Duplex

- In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously.
- The full-duplex mode is like a two-way street with traffic flowing in both directions at the same time.
- In full-duplex mode, signals going in one direction share the capacity of the link with signals going in the other direction This sharing can occur in two ways: Either the link must contain two physically separate transmission paths, one for sending and the other for receiving; or the capacity of the channel is divided between signals traveling in both directions.
- One common example of full-duplex communication is the telephone network. When two people are communicating by a telephone line, both can talk and listen at the same time.
- The full-duplex mode is used when communication in both directions is required all the time. The capacity of the channel, however, must be divided between the two directions.

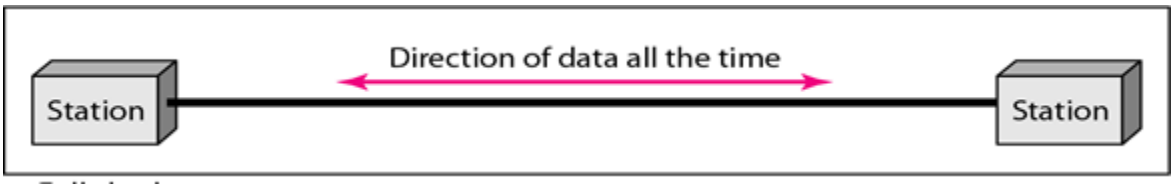


Fig 1.4: Full Duplex Dataflow

1.2 PhysicalStructures

Networks:

- A network is a set of devices (often referred to as nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

1.2.1 Network Criteria

- A network must be able to meet a certain number of criteria. The most important of these are performance, reliability, and security.

Performance

- Performance can be measured in many ways, including transit time and response time. Transit time is the amount of time required for a message to travel from one device to another. Response time is the elapsed time between an inquiry and a response. The performance of a network depends on a number of factors, including the number of users,
- the type of transmission medium, the capabilities of the connected hardware, and the efficiency of the software. Performance is often evaluated by two networking metrics: throughput and delay.

Reliability

- In addition to accuracy of delivery, network reliability is measured by the frequency of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.

Security

Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data losses.

1.2.2 Type of Connection

A network is two or more devices connected through links. A link is a communications pathway that transfers data from one device to another. For visualization purposes, it is simplest to imagine any link as a line drawn between two points. For communication to occur, two devices must be connected in some way to the same link at the same time. There are two possible types of connections: point-to-point and multipoint.

Point-to-Point

A point-to-point connection provides a dedicated link between two devices. The entire capacity of the link is reserved for transmission between those two devices. Most point-to-point connections use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links, are also possible.

When you change television channels by infrared remote control, you are establishing a point-to-point connection between the remote control and the television's control system.

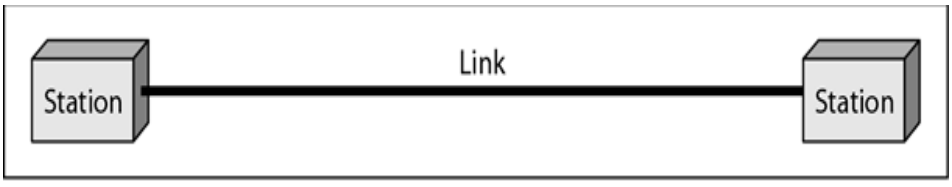


Fig 1.5: Point to point connection

Multipoint:

A multipoint (also called multidrop) connection is one in which more than two specific devices share a single link (see figure b). In a multipoint environment, the capacity of the channel is shared, either spatially or temporally.

- If several devices can use the link simultaneously, it is a spatially shared connection. If users must take turns, it is a timeshared connection.

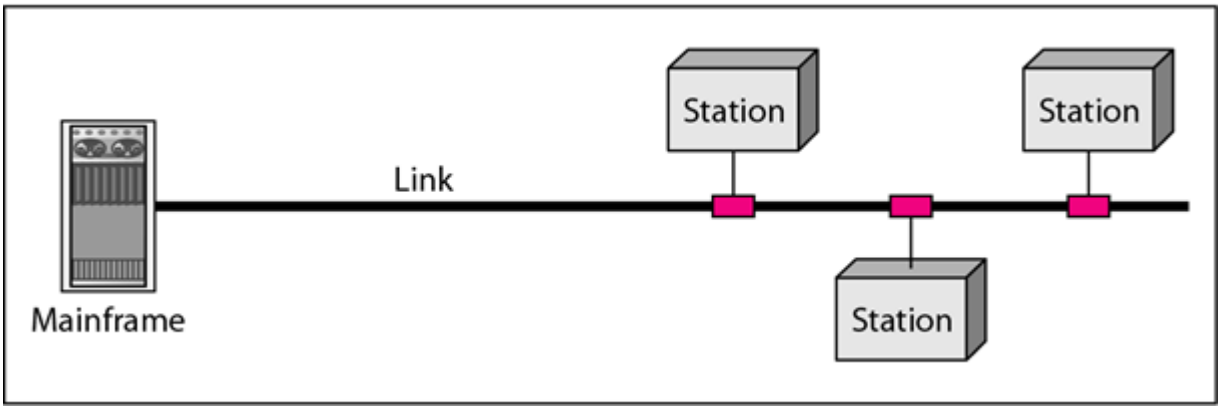


Fig 1.5: Point to Multipoint connection

1.2.3 Physical Topology

- The term physical topology refers to the way in which a network is laid out physically. Two or more devices connect to a link; two or more links form a topology. The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another.
- There are four basic topologies possible: mesh, star, bus, and ring

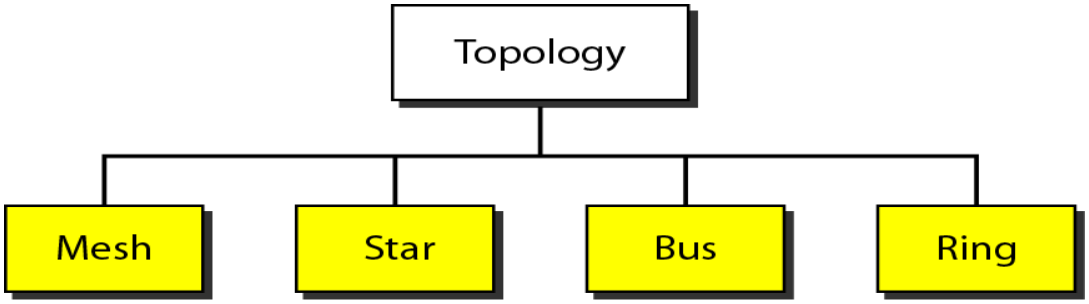


Fig 1.6 : Classification of Topologies

Mesh Topology

- In a mesh topology, every device has a dedicated point-to-point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects.
- To find the number of physical links in a fully connected mesh network with n nodes, we first consider that each node must be connected to every other node.

Node 1 must be connected to n - I nodes, node 2 must be connected to n – 1 nodes, and finally node n must be connected to n - 1 nodes. We need n(n - 1) physical links.

- However, if each physical link allows communication in both directions (duplex mode), we can divide the number of links by 2. In other words, we can say that in a mesh topology, we need n(n -1) /2 duplex-mode links.
- To accommodate that many links, every device on the network must have n – 1 input/output (VO) ports (see below figure1.7) to be connected to the other n - 1 stations.

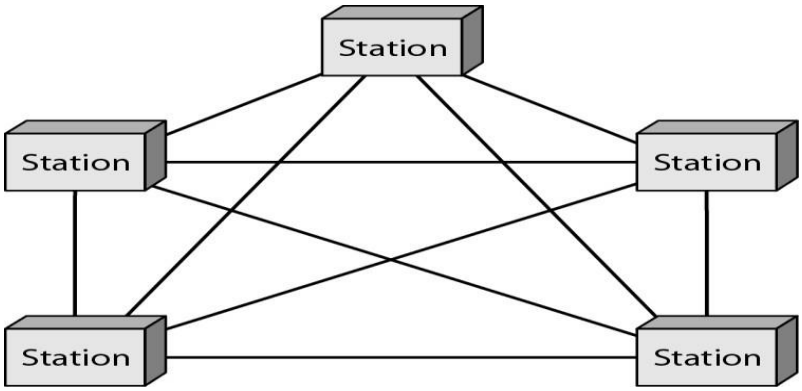


Fig 1.7 : Mesh Topology

Advantages:

- The use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.
- A mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system.
- privacy or security. When every message travels along a dedicated line, only the intended recipient sees it.

Disadvantages:

- The amount of cabling because every device must be connected to every other device, installation and reconnection are difficult.
- the sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate.
- Finally, the hardware required to connect each link (I/O ports and cable) can be prohibitively expensive.

Star Topology

- In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub. The devices are not directly linked to one another. Unlike a mesh topology, a star topology does not allow direct traffic between devices.

- The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device (see figure 1.8).

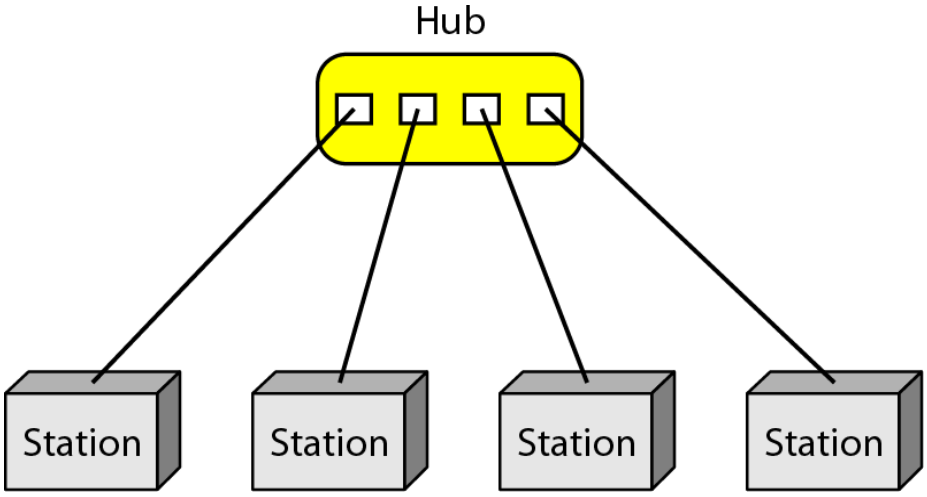


Fig 1.8 : Star Topology

- A star topology is less expensive than a mesh topology. In a star, each device needs only one link and one I/O port to connect it to any number of others. This factor also makes it easy to install and reconfigure.
- Other advantages include robustness. If one link fails, only that link is affected. All other links remain active.
- One big disadvantage of a star topology is the dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.
- Although a star requires far less cable than a mesh, each node must be linked to a central hub. For this reason, often more cabling is required in a star than in some other topologies (such as ring or bus).

Bus Topology

- The preceding examples all describe point-to-point connections. A bus topology, on the other hand, is multipoint. One long cable acts as a backbone to link all the devices in a network (see figure1.9).

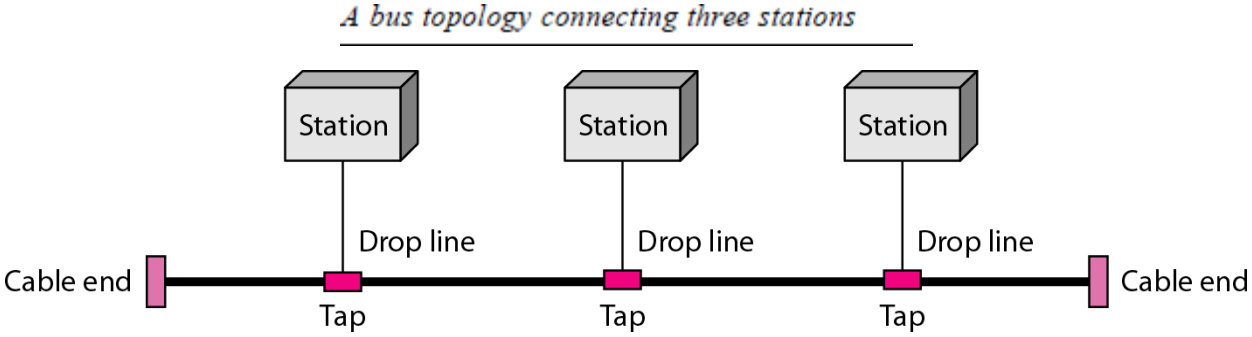


Fig 1.9 : Bus Topology

- Nodes are connected to the bus cable by drop lines and taps.
- A drop line is a connection running between the device and the main cable.

- A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.
- As a signal travels along the backbone, some of its energy is transformed into heat.
- Therefore, it becomes weaker and weaker as it travels farther and farther.
- For this reason, there is a limit on the number of taps a bus can support and on the distance between those taps.

Advantages of a bus topology

- Include ease of installation
- Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths.

Ring Topology:

- In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- Each device in the ring incorporates a repeater.
- When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along (see figure1.10).

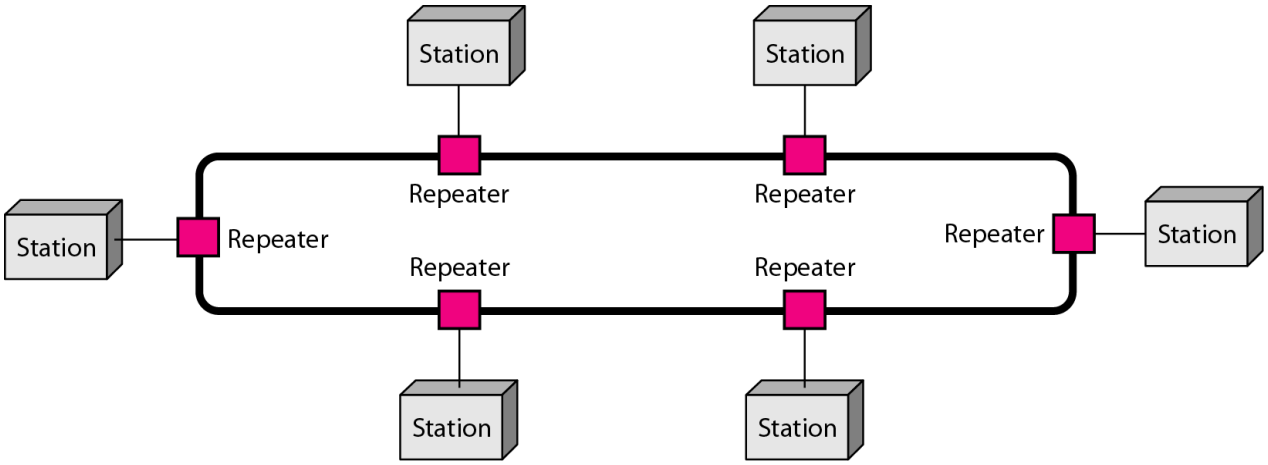


Fig 1.10 : Ring Topology

- A ring is relatively easy to install and reconfigure. Each device is linked to only itsimmediate neighbors (either physically or logically).
- To add or delete a device requireschanging only two connections. The only constraints are media and traffic considerations(maximum ring length and number of devices.

Hybrid Topology:

- A hybrid topology in computer networking is a network structure that combines two or more different network topologies—such as star, bus, ring, or mesh—to leverage the advantages of each while minimizing their weaknesses.

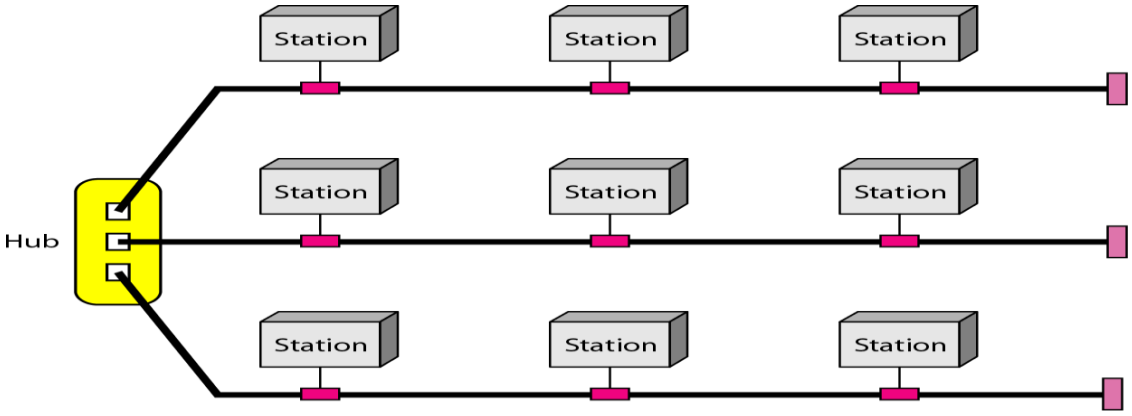


Fig 1.11 : Hybrid Topology

1.3 Categories of Network:

Types of Networks based on size:

- The types of networks are classified based upon the size, the area it covers and its physical architecture.
- The three primary network categories are LAN, WAN and MAN.
- Each network differs in their characteristics such as distance, transmission speed, cables and cost.

1.3.1 LAN (Local Area Network)

- Group of interconnected computers within a small area. (room, building, campus) Two or more pc's can from a LAN to share files, folders, printers, applications and other devices as shown in figure 1.12.
- Coaxial or CAT 5 cables are normally used for connections.
- Due to short distances, errors and noise are minimum.
- Data transfer rate is 10 to 100 mbps.

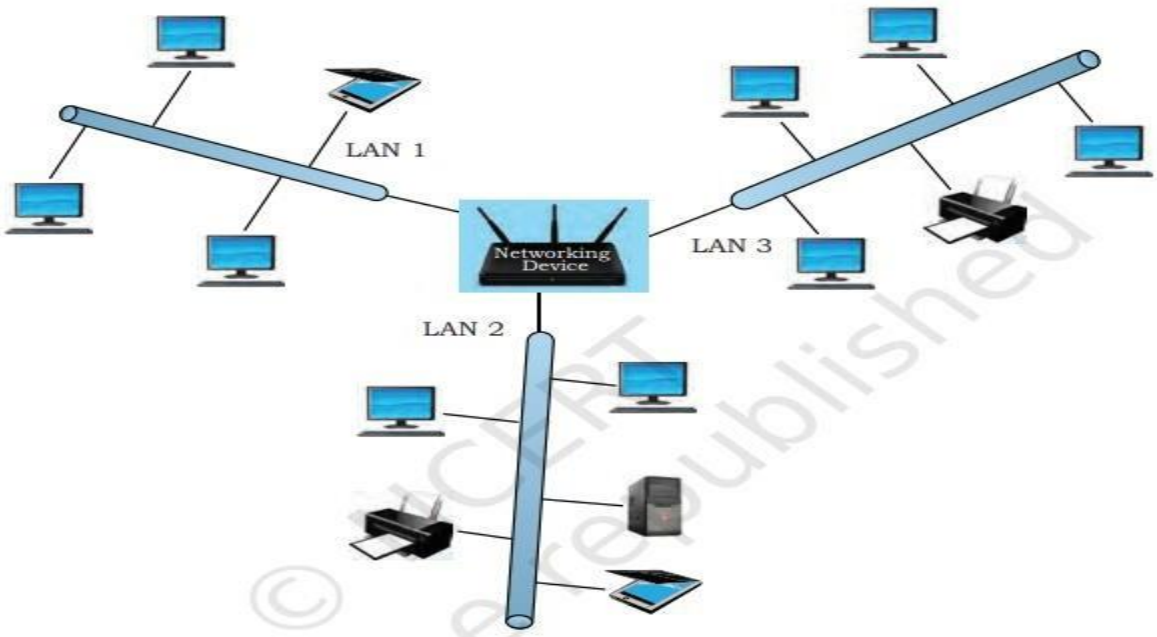
Example: A computer lab in a school.



Fig 1.12 :LocalAreaNetwork

1.3.2 Metropolitan Area Network (MAN)

- Metropolitan Area Network (MAN) is an extended form of LAN which covers a larger geographical area like a city or a town.
- Data transfer rate in MAN also ranges in Mbps,but it is considerably less as compared to LAN.
- Cable TV network or cable based broadband internet services are examples of MAN.
- This kind of network can be extended up to 30-40 km.
- Sometimes, many LANs are connected together to form MAN, as shown in Figure 1.13.



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Fig 1.13 :Metropolitan Area Network

1.3.3 Wide Area Network (WAN)

- Wide Area Network connects computers and other LANs and MANs, which are spread across different geographical locations of a country or in different countries or continents.
- A WAN could be formed by connecting a LAN to other LANs (Figure 1.13) via wired/wireless media.
- Large business, educational and government organizations connect their different branches in different locations across the world through WAN.
- The Internet is the largest WAN that connects billions of computers, smartphones and millions of LANs from different continents.

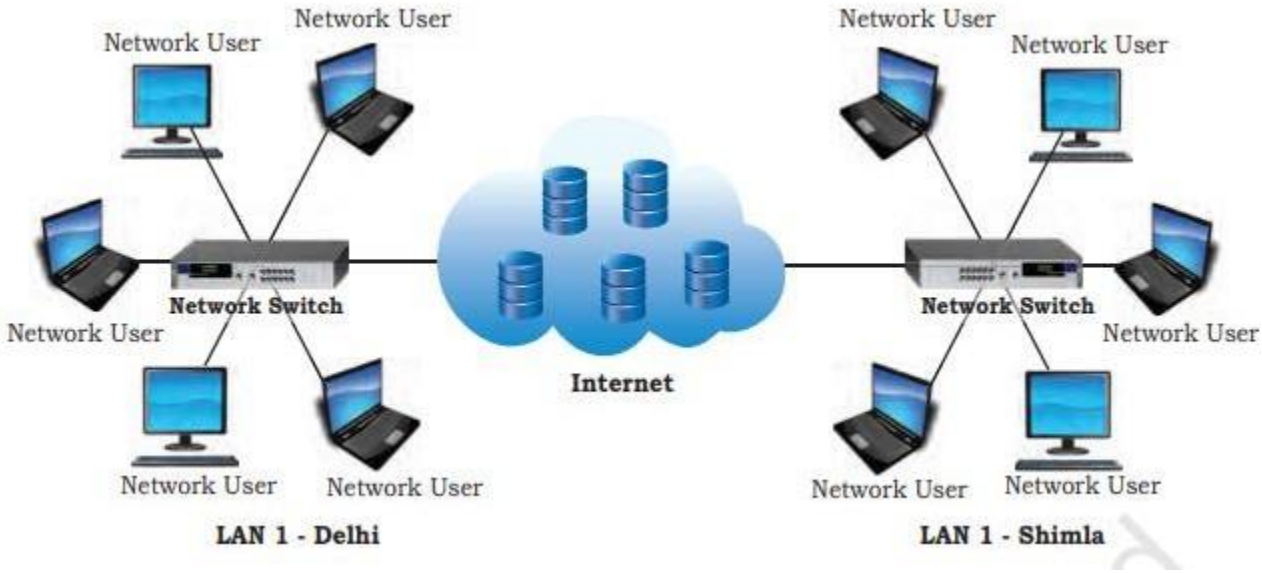


Fig 1.14: Wide Area Network

1.3.4 Internet:

- The Internet is a global network of interconnected computer networks that allows devices worldwide to communicate and exchange information.
- Essentially, the internet is the world's largest WAN, connecting numerous LANs and MANs.
- It utilizes the TCP/IP protocol suite to enable communication between networks and devices.
- It supports various applications and services, such as the World Wide Web, email, and social media platforms
- Today, it is very rare to see a LAN, a MAN, or a LAN in isolation; they are connected to one another. When two or more networks are connected, they become an internetwork, or internet.
- As an example, assume that an organization has two offices, one on the east coast and the other on the west coast. The established office on the west coast has a bus topology LAN; the newly opened office on the east coast has a star topology LAN. The president of the company lives somewhere in the middle and needs to have control over the company from her home. To create a backbone WAN for connecting these three entities (two LANs and the president's computer), a switched WAN (operated by a service provider such as a telecom company) has been leased. To connect the LANs to this switched WAN, however, three point-to-point WANs are required. These point-to-point WANs can be a high-speed DSL line offered by a telephone company or a cable modern line offered by a cable TV provider as shown in below figure 1.14.

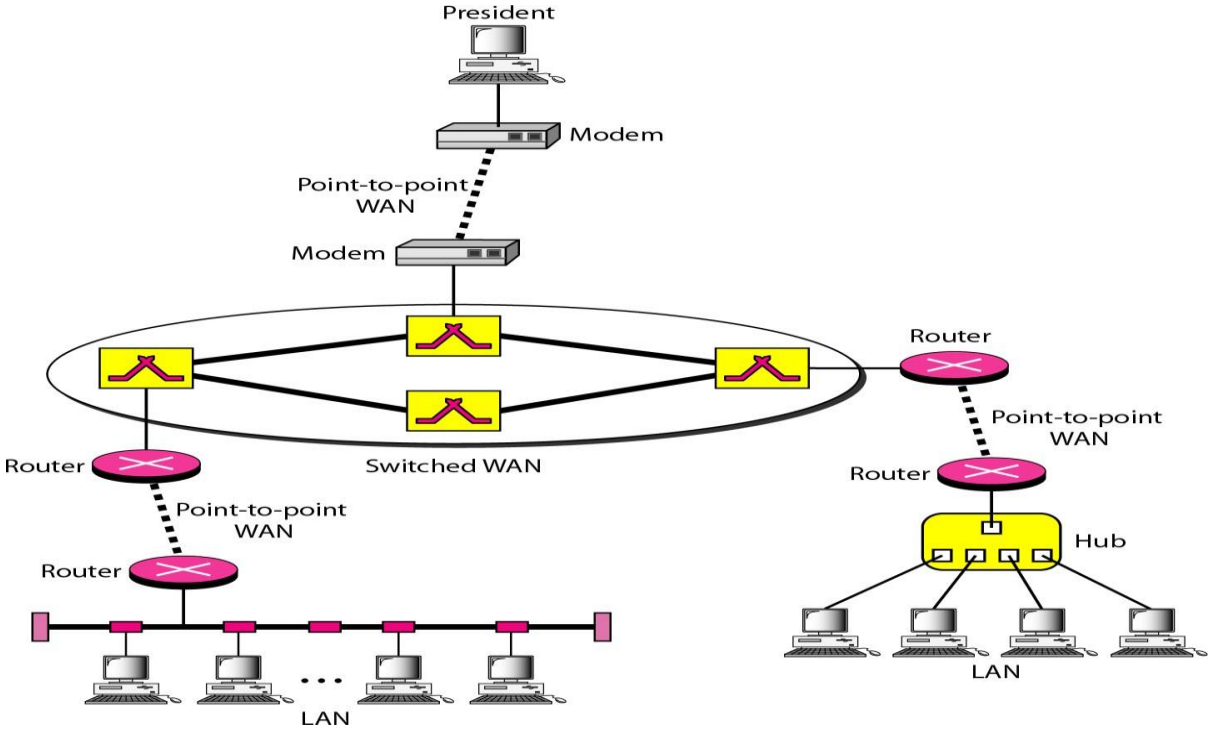


Fig 1.14 : Internet

1.4 Reference Models:

Network Reference Models: A computer network connects two or more devices together to share information and services. Multiple networks connected together form an internetwork.

- Internetworking present challenges - interoperating between products from different manufacturers requires consistent standards. Network reference models were developed to address these challenges. A network reference model serves as a blueprint, detailing how communication between network devices should occur.
- The two most recognized network reference models are:
    - The Open Systems Interconnection (OSI) model
    - TCP/IP model
- Without the framework that network models provide, all network hardware and software would have been proprietary. Organizations would have been locked

into a single vendor's equipment, and global networks like the Internet would have been impractical, if not impossible.

- Network models are organized into layers, with each layer representing a specific networking function. These functions are controlled by protocols, which are rules that govern end-to-end communication between devices.
- Protocols on one layer will interact with protocols on the layer above and below it, forming a protocol suite or stack. The TCP/IP suite is the most prevalent protocol suite, and is the foundation of the Internet.

1.4.1 OSI Reference Model:

OSI model is a reference model describing how information from software application in a computer is transmitted to a software application in another computer.

OSI Model in Computer Network

- The International Organization for Standardization (ISO) developed the OSI Model in the year 1984.
- Open System Interconnection (OSI) is a reference model that explains how information is transmitted between software applications on different computers. It defines the steps and processes involved in this data transmission.
- The structure of this model is designed with a 7-layer architecture, where each layer serves a distinct purpose. The seamless collaboration among these layers facilitates the transmission of data between individuals worldwide. In this model, the entire task is broken down into seven smaller and more manageable tasks.
- Each layer is then assigned a specific task to handle.

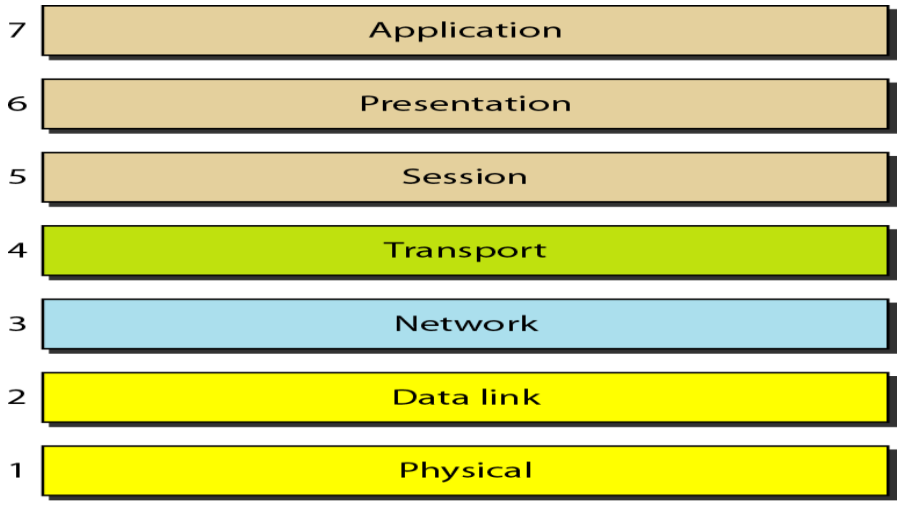


Fig 1.15 : OSI 7 Layers

OSI Model - The Upper Layers

- The top three layers of the OSI model are often referred to as the upper layers:
  - Layer-7 - Application layer
  - Layer-**6 -** Presentation layer
  - Layer-5 - Session layer

- Protocols that operate at these layers manage application-level functions, and are generally implemented in software.
- The function of the upper layers of the OSI model can be difficult to visualize. Upper layer protocols do not always fit perfectly within a layer, and often function across multiple layers.

OSI Model - The Application Layer:

- The Application layer (Layer-7) provides the interface between the user application and the network. A web browser and an email client are examples of user applications.
- The user application itself does not reside at the Application layer - the protocol does.
- The user interacts with the application, which in turn interacts with the application protocol.

  Examples of Application layer protocols include:
  - FTP, via an FTP client
  - HTTP, via a web browser
  - POP3 and SMTP, via an email client
  - Telnet
- The Application layer provides a variety of functions:
  - Identifies communication partners
  - Determines resource availability
  - Synchronizes communication
- The Application layer interacts with the Presentation layer below it. As it is the top-most layer, it does not interact with any layers above it.

OSI Model - The Presentation Layer:

- The Presentation layer (Layer-6) controls the formatting and syntax of user data for the application layer.
- This ensures that data from the sending application can be understood by the receiving application.
- **Standards have** been developed for the formatting of data types, such as text, images, audio, and video.

- **Examples of Presentation layer formats include:**
  - Text - RTF, ASCII, EBCDIC
  - Images - GIF, JPG, TIF
  - Audio - MIDI, MP3, WAV
  - Movies - MPEG, AVI.

- If two devices do not support the same format or syntax, the Presentation layer can provide conversion or translation services to facilitate communication.
- Additionally, the Presentation layer can perform encryption and compression of data, as required. However, these functions can also be performed at lower layers as well.

OSI Model - The Session Layer:

The Session layer (Layer-5) is responsible for establishing, maintaining, and ultimately terminating sessions between devices.

- If a session is broken, this layer can attempt to recover the session. Sessions communication falls under one of three categories:
  - Full-Duplex – simultaneous two-way communication
  - Half-Duplex – two-way communication, but not simultaneous
- Simplex – one-way communication Many modern protocol suites, such as TCP/IP, do not implement Session layer protocols.
- Connection management is often controlled by lower layers, such as the Transport layer.
- The lack of true Session layer protocols can present challenges for high availability and failover.
- Reliance on lower-layer protocols for session management offers less flexibility than a strict adherence to the OSI model.

OSI Model - The Lower Layers

- The bottom four layers of the OSI model are often referred to as the lower layers:
  - Layer-4 – Transport layer
  - Layer-3 – Network layer
  - Layer-2 – Data-Link layer
  - Layer-1 – Physical layer

- Protocols that operate at these layers control the end-to-end transport of data between devices, and are implemented in both software and hardware.

OSI Model - The Transport Layer :

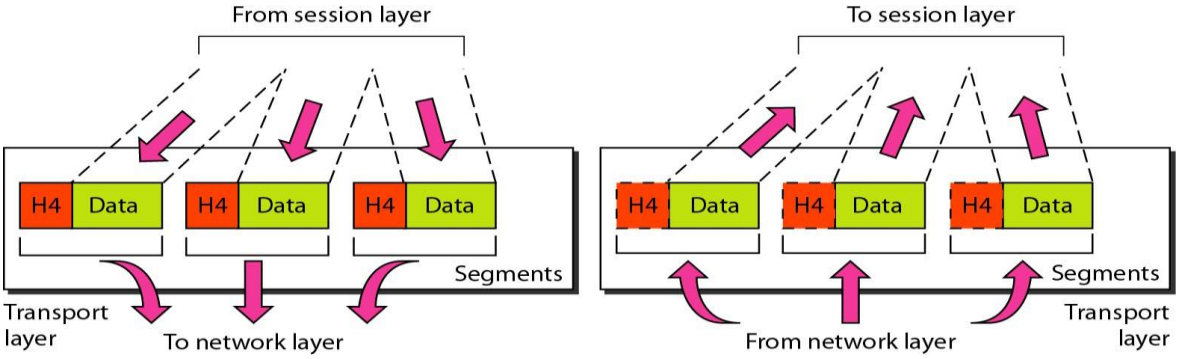
- The Transport layer (Layer-4) does not actually send data, despite its name.
- Instead, this layer is responsible for the reliable transfer of data, by ensuring that data arrives at its destination error-free and in order.
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Fig 1.16 :Transport Layer

Transport layer communication falls under two categories:

- Connection-oriented – requires that a connection with specific agreed-upon parameters be established before data is sent.
- Connectionless – requires no connection before data is sent.

Connection-oriented protocols provide several important services:

- **Service-point addressing:** Computers often run several programs at the same time. For this reason, source-to-destination delivery means delivery not only from one computer to the next but also from a specific process (running program) on one computer to a specific process (running program) on the other. The transport layer header must therefore include a type of address called a service-point address (or port address). The network layer gets each packet to the correct computer; the transport layer gets the entire message to the correct process on that computer.
- **Segmentation and reassembly**: A message is divided into transmittable segments, with each segment containing a sequence number. These numbers enable the transport layer to reassemble the message correctly upon arriving at the destination and to identify and replace packets that were lost in transmission.
- **Connection control**: The transport layer can be either connectionless or connection oriented. A connectionless transport layer treats each segment as an independent packet and delivers it to the transport layer at the destination machine. A connection oriented transport layer makes a connection with the transport layer at the destination machine first before delivering the packets. After all the data are transferred, the connection is terminated.

- **Flow control**: Like the data link layer, the transport layer is responsible for flow control. However, flow control at this layer is performed end to end rather than across a single link.
- **Error control:** Like the data link layer, the transport layer is responsible for error control. However, error control at this layer is performed process-to process rather than across a single link. The sending transport layer makes sure that the entire message arrives at the receiving transport layer without error (damage, loss, or duplication). Error correction is usually achieved through retransmission.

  The TCP/IP protocol suite incorporates two Transport layer protocols:
  - Transmission Control Protocol (TCP) – connection-oriented
  - User Datagram Protocol (UDP) – connectionless

OSI Model - The Network Layer

The Network layer (Layer-3) controls internetwork communication, and has two key responsibilities:

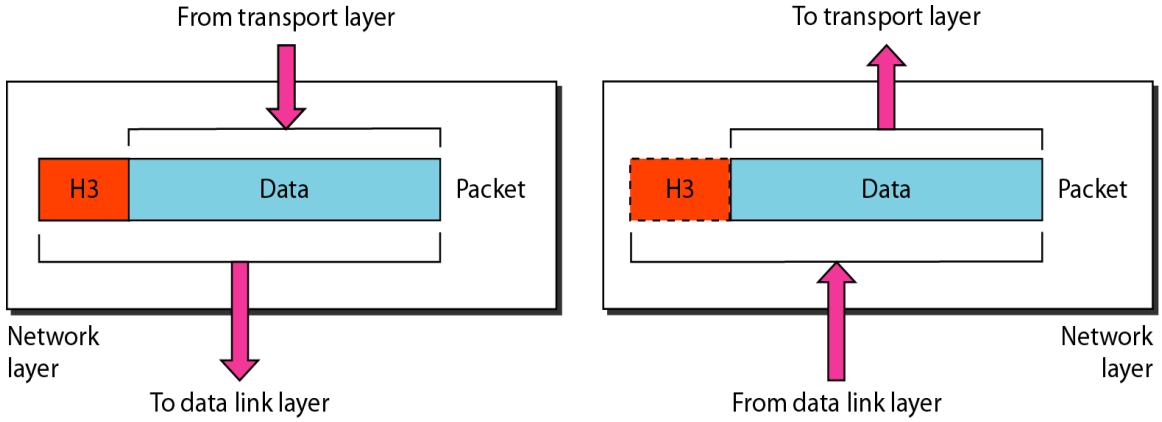


Fig 1.17 :Network Layer

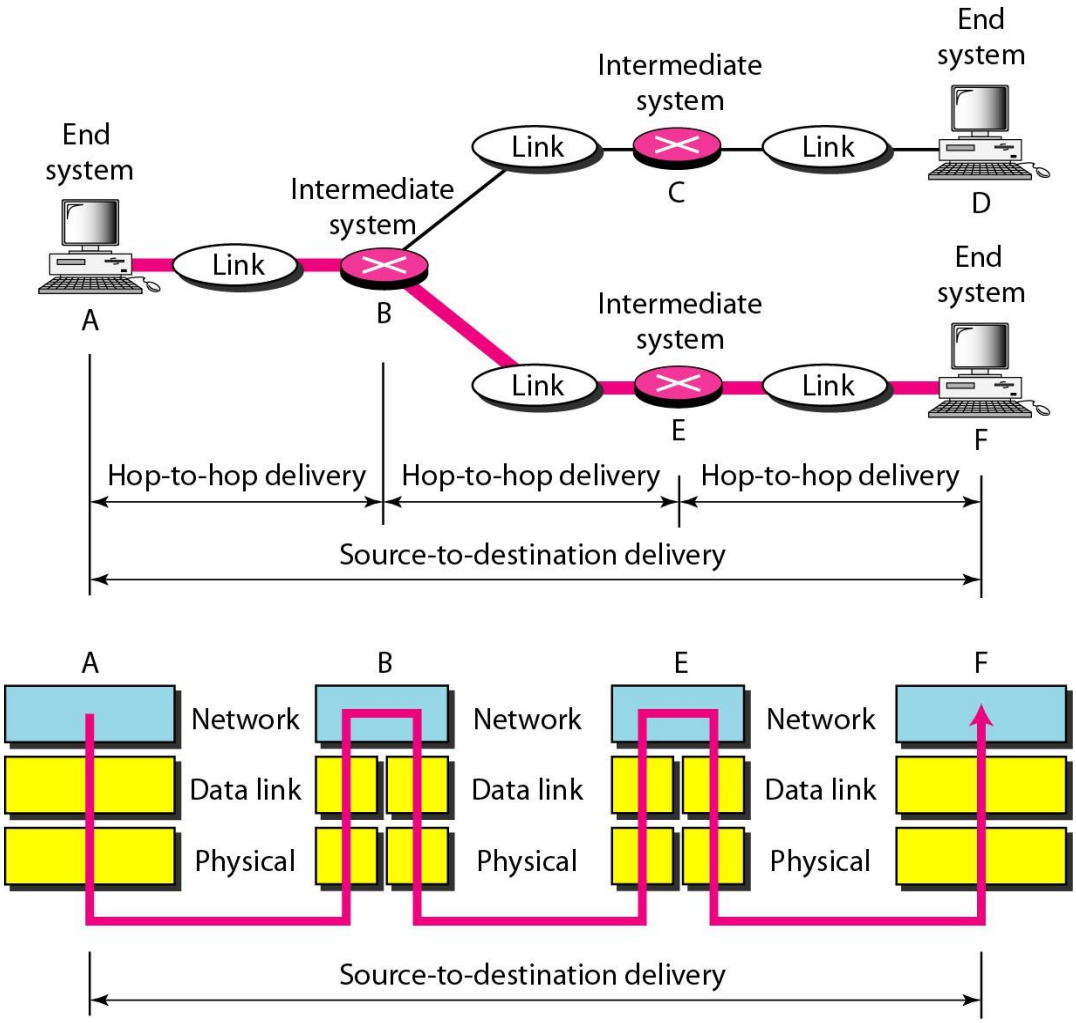
Other responsibilities of the network layer include the following:

Logical addressing -provides a unique address that identifies both the host, and the network that host exists on.

Routing – determines the best path to a particular destination network, and then routes data accordingly.

Two of the most common Network layer protocols are:

- Internet Protocol (IP)
- Novell's Internetwork Packet Exchange (IPX).



OSI Model - The Data-Link Layer

While the Network layer is concerned with transporting data between networks, the Data-Link layer (Layer-2) is responsible for transporting data within a network

The Data-Link layer consists of two sublayers:

- Logical Link Control (LLC) sub layer
- Media Access Control (MAC) sub layer

- The LLC sublayer serves as the intermediary between the physical link and all higher layer protocols.
- It ensures that protocols like IP can function regardless of what type of physical technology is being used.

- Additionally, the LLC sublayer can perform flow-control and errorchecking, though such functions are often provided by Transport layer protocols, such as TCP.
- The MAC sublayer controls access to the physical medium, serving as mediator if multiple devices are competing for the same physical link.
- Datalink layer technologies have various methods of accomplishing this - Ethernet uses Carrier Sense Multiple Access with Collision Detection (CSMA/CD), and Token Ring utilizes a token.

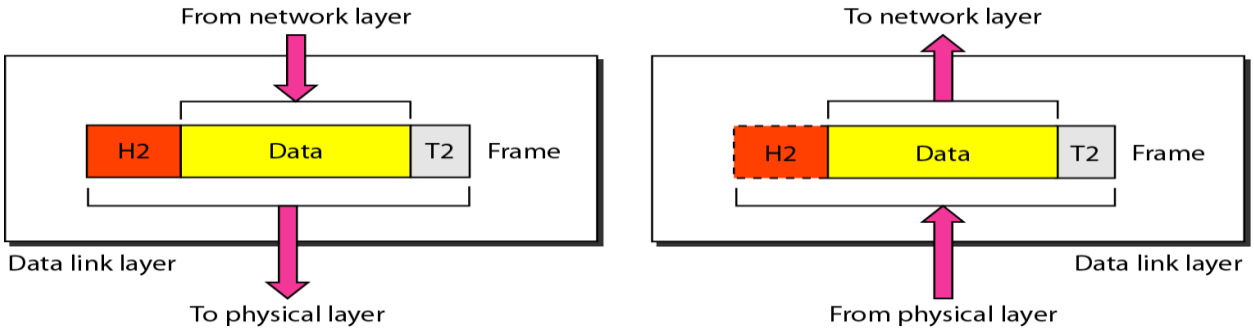
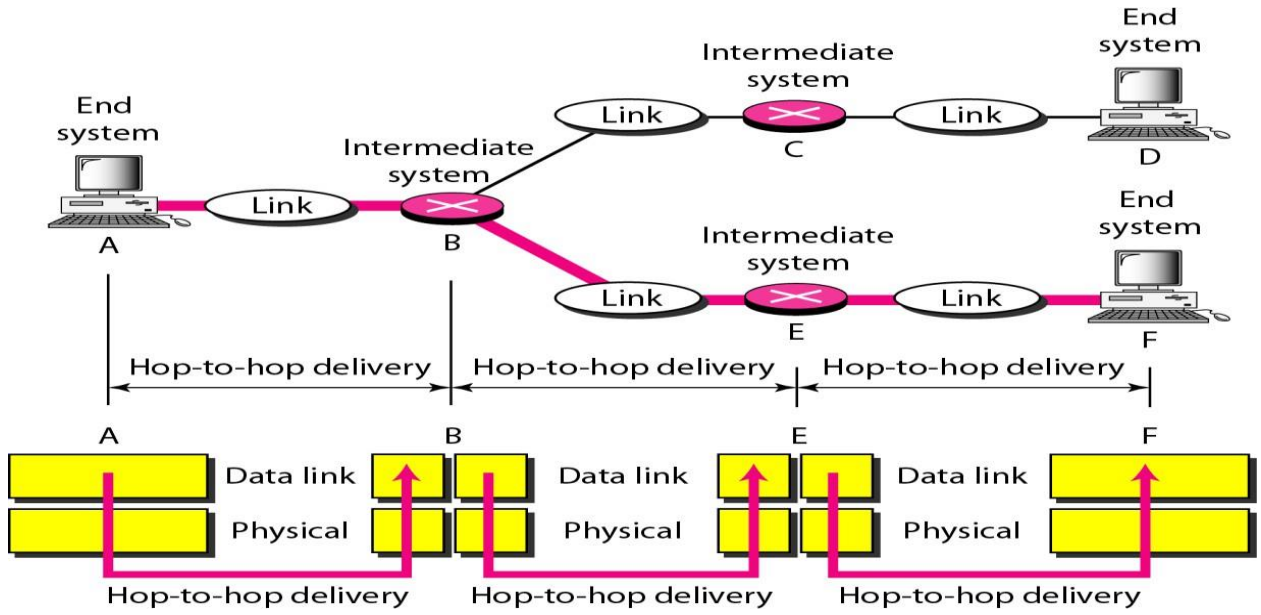


Fig 1.18 :Data Link Layer

Other responsibilities of the data link layer include the following:

- **Framing:** The data link layer divides the stream of bits received from the network layer into manageable data units called frames.
- **Physical addressing:** If frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the sender and/or receiver of the frame.
- **Flow control:** If the rate at which the data are absorbed by the receiver is less than the rate at which data are produced in the sender, the data link layer imposes a flow control mechanism to avoid overwhelming the receiver.
- **Error control:** The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames. It also uses a mechanism to recognize duplicate frames. Error control is normally achieved through a trailer added to the end of the frame.
- **Access control:**When two or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any given time.



OSI Model - The Physical Layer

- The Physical layer (Layer-1) controls the signaling and transferring of raw bits onto the physical medium.
- The Physical layer is closely related to the Data-link layer, as many technologies (such as Ethernet) contain both data link and physical functions.
- The physical layer coordinates the functions required to carry a bit stream over a physical medium. It deals with the mechanical and electrical specifications of the interface and transmission medium.
- It also defines the procedures and functions that physical devices and interfaces have to perform for transmission to Occur. Below figure shows the position of the physical layer with respect to the transmission medium and the data link layer.

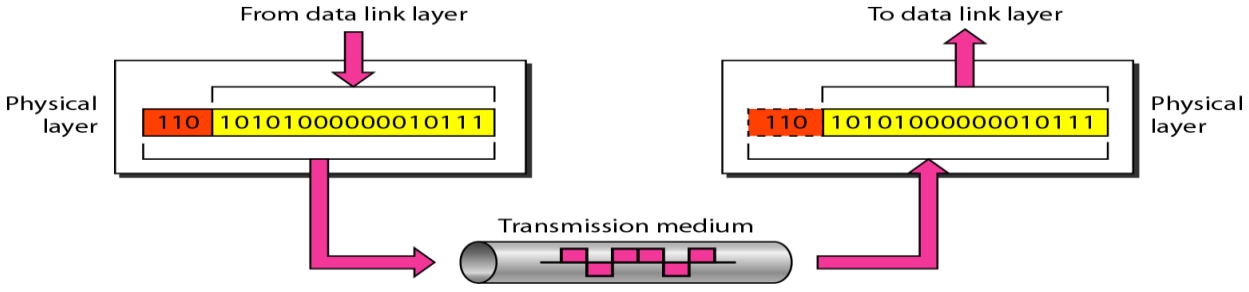


Fig 1.19 :Physical Layer

The physical layer is also concerned with the following:

- **Physical characteristics of interfaces and medium**: The physical layer defines the characteristics of the interface between the devices and the transmission medium. It also defines the type of transmission medium.

- **Representation of bits**: The physical layer data consists of a stream of bits (sequence of 0's or 1's) with no interpretation. To be transmitted, bits must be encoded into signals--electrical or optical. The physical layer defines the type of encoding (how 0's and 1's are changed to signals).
- **Data rate**: The transmission rate-the number of bits sent each second-is also defined by the physical layer. In other words, the physical layer defines the duration of a bit, which is how long it lasts.
- **Synchronization of bits:** The sender and receiver not only must use the same bit rate but also must be synchronized at the bit level. In other words, the sender and the receiver clocks must be synchronized.
- **Line configuration:** The physical layer is concerned with the connection of devices to the media. In a point-to-point configuration, two devices are connected through a dedicated link. In a multipoint configuration, a link is shared among several devices.
- **Physical topology:** The physical topology defines how devices are connected to make a network. Devices can be connected by using a mesh topology (every device is connected to every other device), a star topology (devices are connected through a central device), a ring topology (each device is connected to the next, forming a ring), a bus topology (every device is on a common link), or a hybrid topology (this is a combination of two or more topologies).
- **Transmission mode:** The physical layer also defines the direction of transmission between two devices: simplex, half-duplex, or full-duplex. In simplex mode, only one device can send; the other can only receive. The simplex mode is a one-way communication. In the half-duplex mode, two devices can send and receive, but not at the same time. In a full-duplex (or simply duplex) mode, two devices can send and receive at the same time.

Encapsulation and Layered Communication

As data is passed from the user application down the virtual layers of the OSI model, each layer adds a header (and sometimes a trailer) containing protocol information specific to that layer. These headers are called Protocol Data Units (PDUs), and the process of adding these headers is called encapsulation.

For example, a Transport layer protocol such as TCP will add a header containing flow control, port numbers, and sequencing.

The Network layer header contains logical addressing information, and the Data-link header contains physical addressing and other hardware specific information.

The PDU of each layer is identified with a different term:

Layer	*PDU Name*
Application	-
Presentation	-
Session	-
Transport	**Segments**
Network	**Packets**
Data-Link	**Frames**
Physical	**Bits**

- Each layer communicates with the corresponding layer on the receiving device.
- For example, on the sending device, source and destination hardware addressing is placed in a Data-link header.
- On the receiving device, that Data-link header is processed and stripped away (decapsulated) before being sent up to the Network and other upper layers.
- Network devices are commonly identified by the OSI layer they operate at; or, more specifically, what header or PDU the device processes.
- For example, switches are generally identified as Layer-2 devices, as switches process information stored in the Data-Link header of a frame, such as Ethernet MAC addresses.
- Similarly, routers are identified as Layer3 devices, as routers process logical addressing information in the Network header of a packet, such as IP addresses.

Encapsulation Illustrated

The following illustrates how basic encapsulation occurs with the TCP/IP stack, which typically performs encapsulation only at the lower layers:

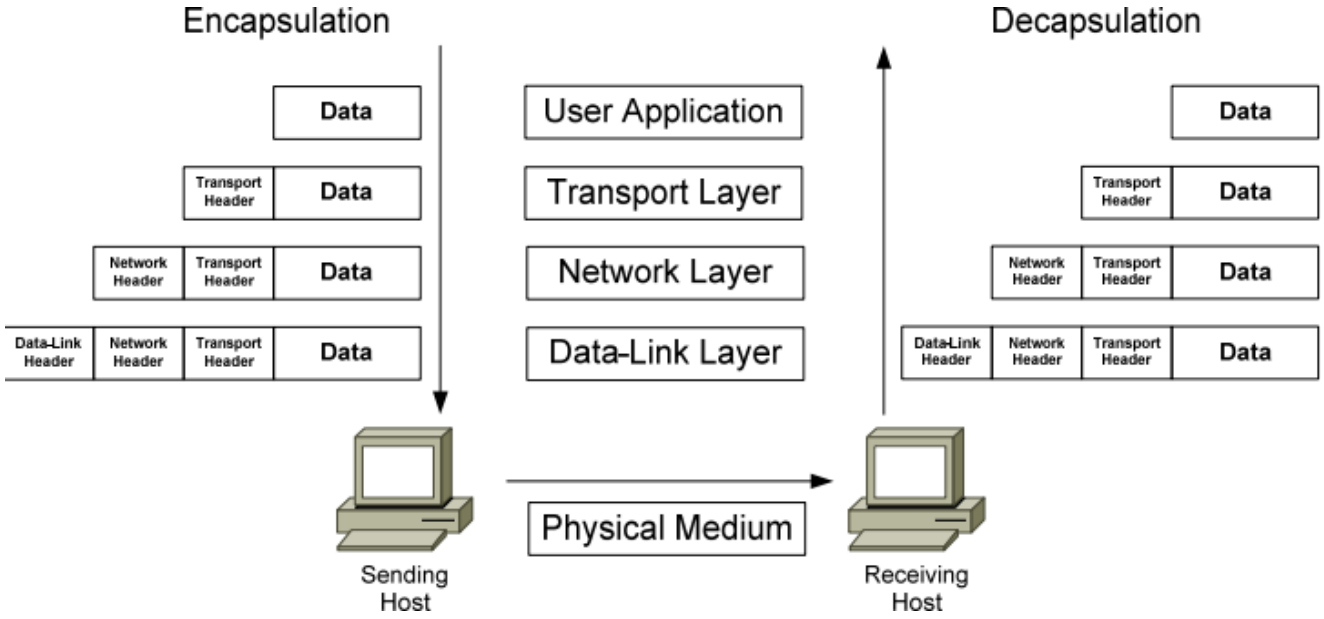


Fig:1.20 Encapsulation and Decapsulation

During encapsulation on the sending host:

- Data from the user application is handed off to the Transport layer.
- The Transport layer adds a header containing protocol-specific information, and then hands the segment to the Network layer.
- The Network layer adds a header containing source and destination logical addressing, and then hands the packet to the Data-Link layer.
- The Data-Link layer adds a header containing source and destination physical addressing and other hardware-specific information.
- The Data-Link frame is then handed off to the Physical layer to be transmitted on the network medium as bits.

During decapsulation on the receiving host, the reverse occurs:

- The frame is received from the physical medium.
- The Data-Link layer processes its header, strips it off, and then hands it off to the Network layer.
- The Network layer processes its header, strips it off, and then hands it off to the Transport layer.
- The Transport layer processes its header, strips it off, and then hands the data to the user application.

Layered Architecture:

The OSI model is composed of seven ordered layers: physical (layer 1), data link (layer 2),network (layer 3), transport (layer 4), session (layer 5), presentation (layer 6), and application (layer 7). Below figure shows the layers involved when a message is sent from device A to device B. As the message travels from A to B, it may pass through many intermediate nodes. These intermediate nodes usually involve only the first three layers of the OSI model.

- In developing the model, the designers distilled the process of transmitting data to its most fundamental elements. They identified which networking functions had related uses and collected those functions into discrete groups that became the layers. Each layer defines a family of functions distinct from those of the other layers.
- Within a single machine, each layer calls upon the services of the layer just below it. Layer 3, for example, uses the services provided by layer 2 and provides services for layer 4. Between machines, layer x on one machine communicates with layer x on another machine. This communication is governed by an agreed-upon series of rules and conventions called protocols. The processes on each machine that communicate at a given layer are called peer-to-peer processes. Communication between machines is therefore a peer-to-peer process using the protocols appropriate to a given layer.

Peer-to-Peer Processes

- At the physical layer, communication is direct: In below figure , device A sends a stream of bits to device B (through intermediate nodes). At the higher layers, however, communication must move down through the layers on device A, over to device B, and then

The interaction between layers in the OSI model

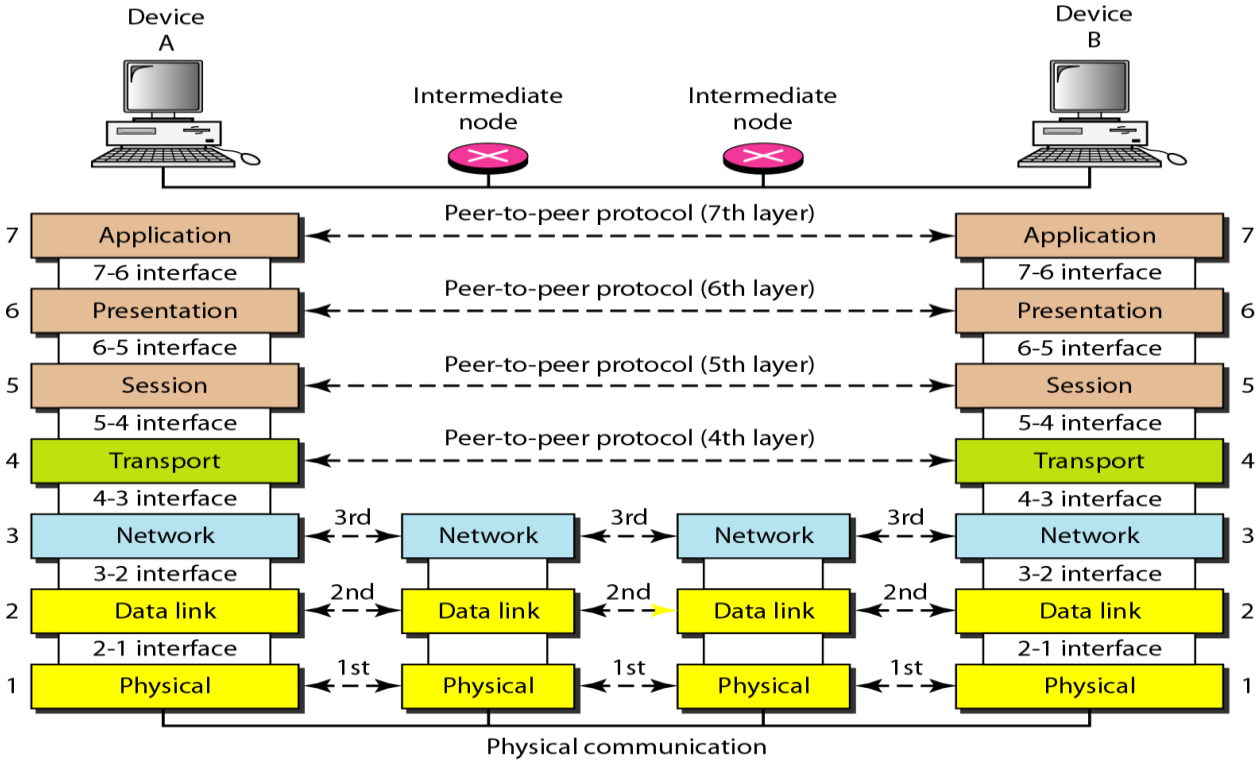


Fig:1.21 Layered Architecture

- back up through the layers. Each layer in the sending device adds its own information to the message it receives from the layer just above it and passes the whole package to the layer just below it.
- At layer 1 the entire package is converted to a form that can be transmitted to the receiving device. At the receiving machine, the message is unwrapped layer by layer, with each process receiving and removing the data meant for it. For example, layer 2 removes the data meant for it, then passes the rest to layer 3. Layer 3 then removes the data meant for it and passes the rest to layer 4, and so on.

Interfaces Between Layers

- The passing of the data and network information down through the layers of the sending device and back up through the layers of the receiving device is made possible by an interface between each pair of adjacent layers. Each interface defines the information and services a layer must provide for the layer above it. Well-defined interfaces and layer functions provide modularity to a network.

Organization of the Layers

- The seven layers can be thought of as belonging to three subgroups.
- Layers 1, 2, and 3-physical, data link, and network-are the network support layers; they deal with the physical aspects of moving data from one device to another (such as electrical specifications, physical

connections, physical addressing, and transport timing and reliability).

- Layers 5, 6, and 7-session, presentation, and application-can be thought of as the user support layers; they allow interoperability among unrelated software systems.
- Layer 4, the transport layer, links the two subgroups and ensures that what the lower layers have transmitted is in a form that the upper layers can use.
- The upper OSI layers are almost always implemented in software;
- Lower layers are a combination of hardware and software, except for the physical layer, which is mostly hardware.

In below figure1.22 , which gives an overall view of the OSI layers, D7 means the data unit at layer 7, D6 means the data unit at layer 6, and so on. The process starts at layer 7 (the application layer), then moves from layer to layer in descending, sequential order. At each layer, a header, or possibly a trailer, can be added to the data unit. Commonly, the trailer is added only at layer 2. When the formatted data unit passes through the physical layer (layer 1), it is changed into an electromagnetic signal and transported along a physical link.

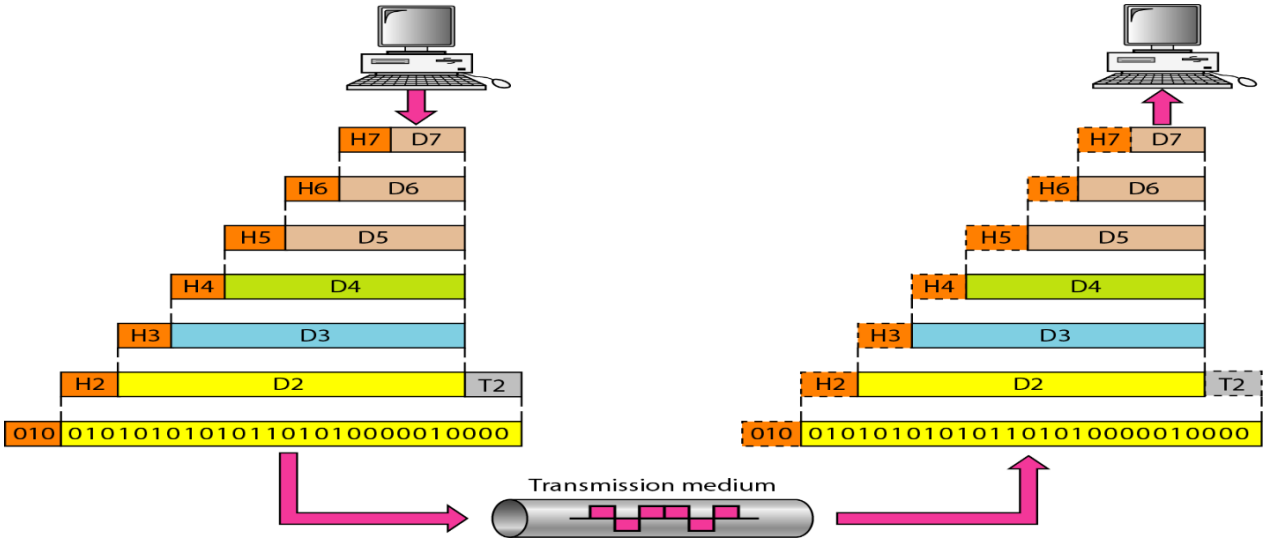


Fig:1.22 Organization of the Layers

Upon reaching its destination, the signal passes into layer 1 and is transformed back into digital form. The data units then move back up through the OSI layers. As each block of data reaches the next higher layer, the headers and trailers attached to it at the corresponding sending layer are removed, and actions appropriate to that layer are taken. By the time it reaches layer 7, the message is again in a form appropriate to the application and is made available to the recipient

1.4.2 TCP/IP PROTOCOL SUITE

- The TCPIIP protocol suite was developed prior to the OSI model.
- Therefore, the layers in the TCP/IP protocol suite do not exactly match those in the OSI model.
- The original TCP/IP protocol suite was defined as having four layers: host-to-

network, internet, transport, and application.

- However, when TCP/IP is compared to OSI, we can say that the host-to-network layer is equivalent to the combination of the physical and data link layers.
- The internet layer is equivalent to the network layer, and the application layer is roughly doing the job of the session, presentation, and application layers with the transport layer in TCPIIP taking care of part of the duties of the session layer.

Fig: TCP/IP and OSI model

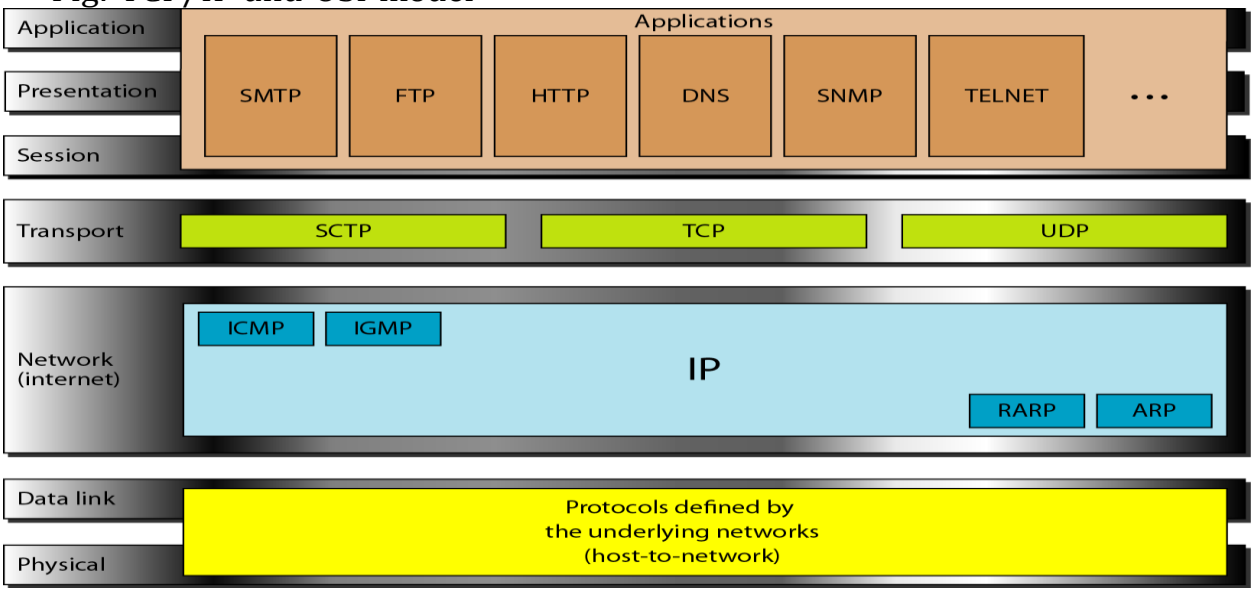


Fig:1.23 TCP/IP Protocol Suite

- TCP/IP is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality; however, the modules are not necessarily interdependent. Whereas the OSI model specifies which functions belong to each of its layers, the layers of the TCP/IP protocol suite contain relatively independent protocols that can be mixed and matched depending on the needs of the system.
- The term hierarchical means that each upper-level protocol is supported by one or more lower-level protocols.
- At the transport layer, TCP/IP defines three protocols: Transmission Control Protocol (TCP), User Datagram Protocol (UDP), and Stream Control Transmission Protocol (SCTP). At the network layer, the main protocol defined by TCP/IP is the Internetworking Protocol (IP); there are also some other protocols that support data movement in this layer.

Physical and Data Link Layers

At the physical and data link layers, TCPIIP does not define any specific protocol. It supports all the standard and proprietary protocols. A network in a TCPIIP internetwork can be a local-area network or a wide-area network.

Network Layer

At the network layer (or, more accurately, the internetwork layer), TCP/IP supports the Internetworking Protocol. IP, in turn, uses four supporting protocols: ARP, RARP, ICMP, and IGMP.

Internetworking Protocol (IP)

The Internetworking Protocol (IP) is the transmission mechanism used by the TCP/IP protocols. It is an unreliable and connectionless protocol-a best-effort delivery service. The term best effort means that IP provides no error checking or tracking. IP assumes the unreliability of the underlying layers and does its best to get a transmission through to its destination, but with no guarantees.

- IP transports data in packets called datagrams, each of which is transported separately.
- Datagrams can travel along different routes and can arrive out of sequence or be duplicated. IP does not keep track of the routes and has no facility for reordering datagrams once they arrive at their destination.
- The limited functionality of IP should not be considered a weakness, however. IP provides bare-bones transmission functions that free the user to add only those facilities necessary for a given application and thereby allows for maximum efficiency

Address Resolution Protocol

- The Address Resolution Protocol (ARP) is used to associate a logical address with a physical address. On a typical physical network, such as a LAN, each device on a link is identified by a physical or station address, usually imprinted on the network interface card (NIC).
- ARP is used to find the physical address of the node when its Internet address is known.

Reverse Address Resolution Protocol

- The Reverse Address Resolution Protocol (RARP) allows a host to discover its Internet address when it knows only its physical address.
- It is used when a computer is connected to a network for the first time or when a diskless computer is booted.

Internet Control Message Protocol

- The Internet Control Message Protocol (ICMP) is a mechanism used by hosts and gateways to send notification of datagram problems back to the sender.
- ICMP sends query and error reporting messages.

Internet Group Message Protocol

The Internet Group Message Protocol (IGMP) is used to facilitate the simultaneous transmission of a message to a group of recipients.

Transport Layer

Traditionally the transport layer was represented in TCP/IP by two protocols: TCP and UDP. IP is a host-to-host protocol, meaning that it can deliver a packet from one physical device to another. UDP and TCP are transport level protocols responsible for delivery of a message from a process (running program) to another process.

User Datagram Protocol

- The User Datagram Protocol (UDP) is the simpler of the two standard TCPIIP transportprotocols. It is a process-to-process protocol that adds only port addresses, checksum error control, and length information to the data from the upper layer.

Transmission Control Protocol

- The Transmission Control Protocol (TCP) provides full transport-layer services to applications. TCP is a reliable stream transport protocol. The term stream, in this context,means connection-oriented: A connection must be established between both ends of a transmission before either can transmit data.
- At the sending end of each transmission, TCP divides a stream of data into smaller units called segments. Each segment includes a sequence number for reordering after receipt, together with an acknowledgment number for the segments received.
- Segments are carried across the internet inside of IP datagrams. At the receiving end, TCP collects each datagram as it comes in and reorders the transmission based on sequence numbers.

Stream Control Transmission Protocol

The Stream Control Transmission Protocol (SCTP) provides support for newer applications such as voice over the Internet. It is a transport layer protocol that combines the best features of UDP and TCP.

Application Layer

The application layer in TCPIIP is equivalent to the combined session, presentation, and application layers in the OSI model. Many protocols are defined at this layer.

1.4.3 A comparison of the OSI and TCP/IP reference models

Aspect	OSI Model	TCP/IP Model
Full Form	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Number of Layers	7 layers	4 layers
Layers	Application, Presentation, Session, Transport, Network, Data Link, Physical	Application, Transport, Internet, Network Access (Link)
Layer Functions	Each layer has a specific function; Presentation & Session are separate	Presentation & Session functions are included in Application layer
Development	Developed by ISO in 1984	Developed by DARPA in the 1970s

Aspect	OSI Model	TCP/IP Model
Approach	Theoretical model to standardize networking	Practical model used for the Internet
Protocol Dependency	Protocol independent	Protocol dependent (based on TCP/IP suite)
Communication Flow	Defines how applications can communicate over a network	Specifies how data should be packaged, addressed, transmitted, and routed
Error Handling	Handled at Data Link, Transport layers	Handled at Network Access, Transport layers
Usage	Used mainly for teaching and reference	Used for actual network communication over the Internet
Examplesof Protocols	No specific protocols (conceptual)	HTTP, FTP, SMTP, TCP, UDP, IP, Ethernet

1.5 Physical media
Transmission Media

A transmission medium can be anything that can carry signals or data between the source (transmitter) and destination (receiver). For example, as we switch on a ceiling fan or a light bulb, the electric wire is the medium that carries electric current from switch to the fan or bulb. Two men are talking as shown in Figure 1.24. Here the medium is air.

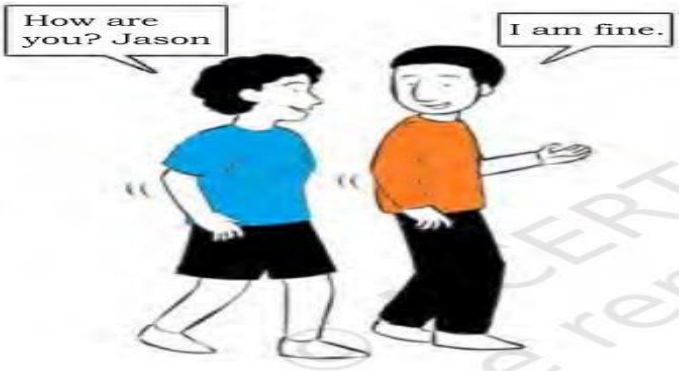


Figure:1.24 Two men are talking

- In data communication, transmission media are the links that carry messages between two or more communicating devices.
- Transmission can be classified as guided or unguided.
- Figure 11.7 shows the classification of communication media.
- In guided transmission, there is a physical link made of wire/cable through which data in terms of signals are propagated between the

nodes. These are usually metallic cable, fiber-optic cable, etc. They are also known as wired media.

- In unguided transmission, data travels in air in terms of electromagnetic waves using an antenna. They are also known as wireless media.

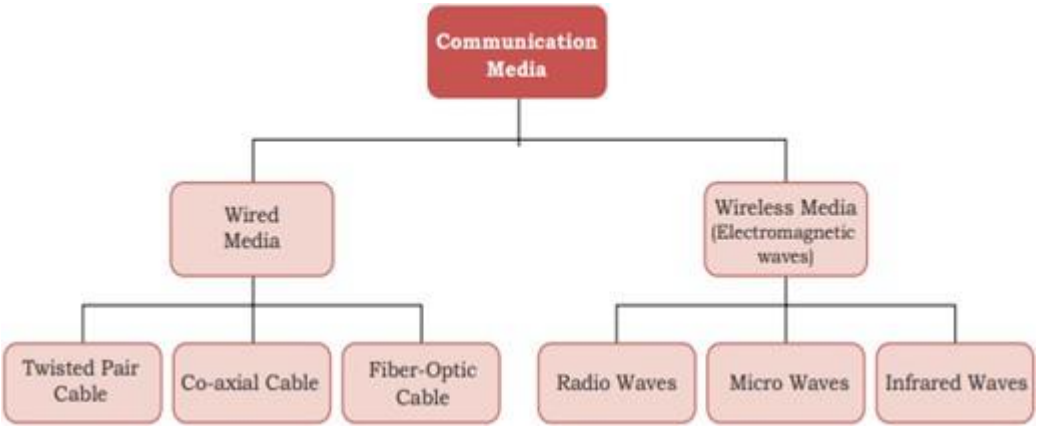


Figure:1.25 Classification of Communication Medium

1.5.1 Wired Transmission Media

- Any physical link that can carry data in the form of signals belongs to the category of wired transmission media. Three commonly used guided/wired media for data transmission are, twisted pair, coaxial cable, and fiber optic cable. Twisted-pair and coaxial cable carry the electric signals whereas the optical fiber cable carries the light signals.
- **1.5.2Twisted Pair Cable**

A twisted-pair consists of two copper wires twisted like a DNA helical structure. Both the copper wires are insulated with plastic covers. Usually, a number of such pairs are combined together and covered with a protective outer wrapping, as shown in Figure 1.26

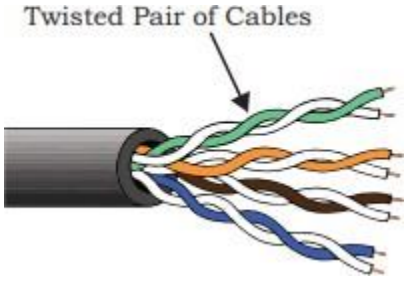


Fig1.26 Twisted Pair cables

Each of the twisted pairs act as a single communication link. The use of twisted configuration minimizes the effect of electrical interference from similar pairs close by. Twisted pairs are less expensive and most commonly used in telephone lines and LANs. These cables are of two types: Unshielded twisted-pair (UTP) and Shielded twisted-pair (STP), as shown in Figure 1.26

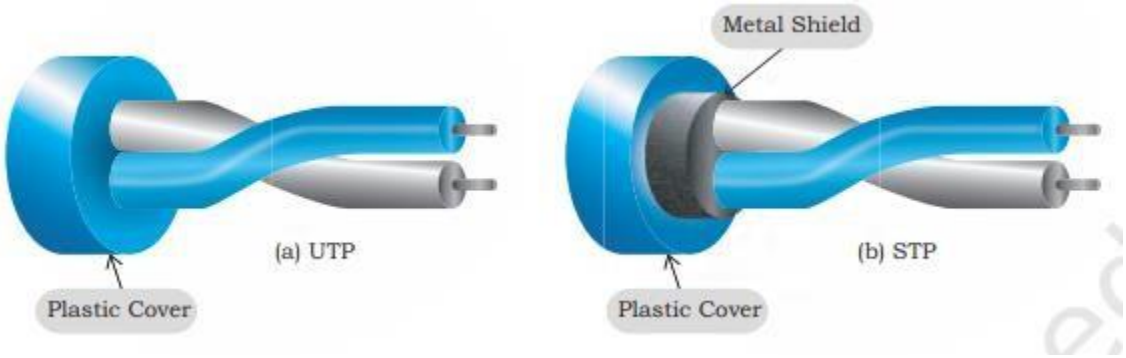


Fig:1.26 UTP cable and STP cable

1.5.3 Coaxial cable

- Coaxial cable is another type of data transmission medium.
- It is better shielded and has more bandwidth than a twisted pair.
- As shown in Figure 1.27, it has a copper wire at the core of the cable which is surrounded with insulating material.
- The insulator is further surrounded with an outer conductor (usually a copper mesh). This outer conductor is wrapped in a plastic cover.
- The key to success of coaxial cable is its shielded design that allows the cable's copper core to transmit data quickly, without interference of environmental factors.
- These types of cables are used to carry signals of higher frequencies to a longer distance.

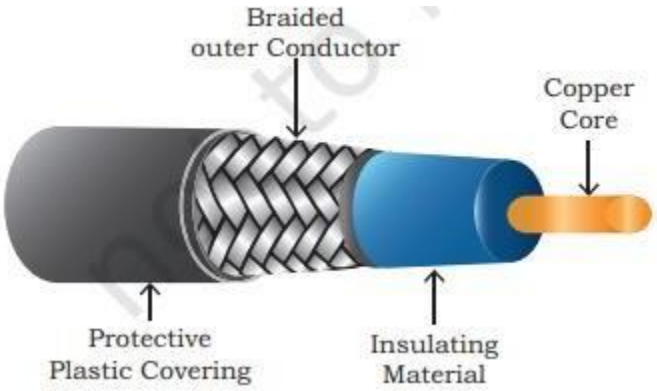


Fig:1.27 A coaxial Cable

1.5.4 Optical Fibre:

- The optical fiber cable carries data as light, which travels inside a thin fiber of glass (Figure 1.28).
- Optic fiber uses refraction to direct the light through the media.
- A thin transparent strand of glass at the centre is covered with a layer of less dense glass called cladding.
- This whole arrangement is covered with an outer jacket made of PVC or Teflon.
- Such types of cables are usually used in backbone networks.
- These cables are of light weight and have higher bandwidth which means higher data transfer rate.
- Signals can travel longer distances and electromagnetic noise cannot affect the cable. However, optic fibers are expensive and unidirectional.
- Two cables are required for full duplex communication.

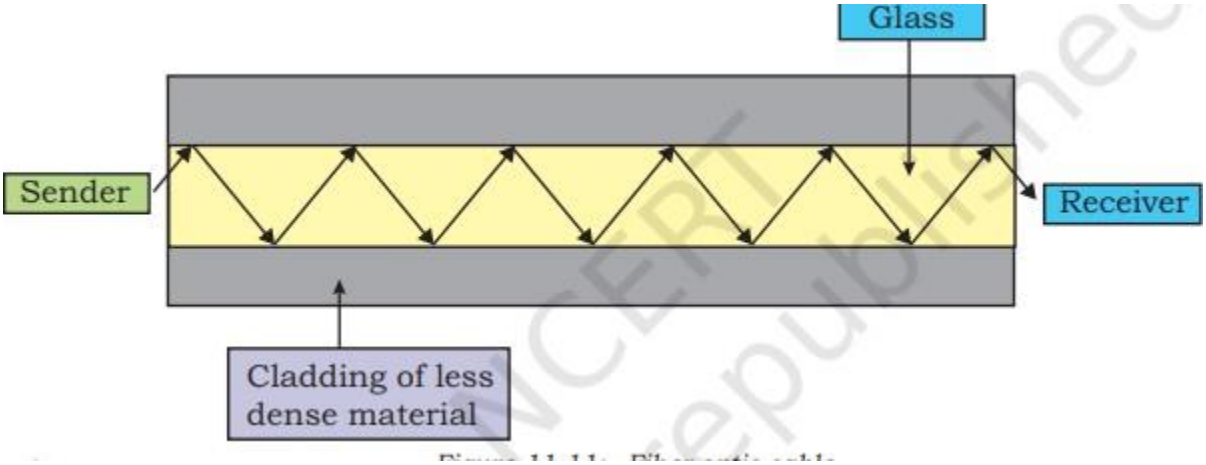


Fig:1.28 Wave propagation in optical fiber

Propagation Modes:

Current technology supports two modes (multimode and single mode) for propagating light along optical channels, each requiring fiber with different physical characteristics.

Multimode can be implemented in two forms: step-index or graded-index.

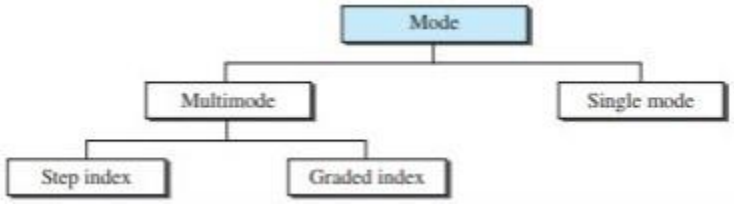


Fig 1.29 Propagation Modes

Multimode:

Multimode is so named because multiple beams from a light source move through the core in different paths. How these beams move within the cable depends on the structure of the core.

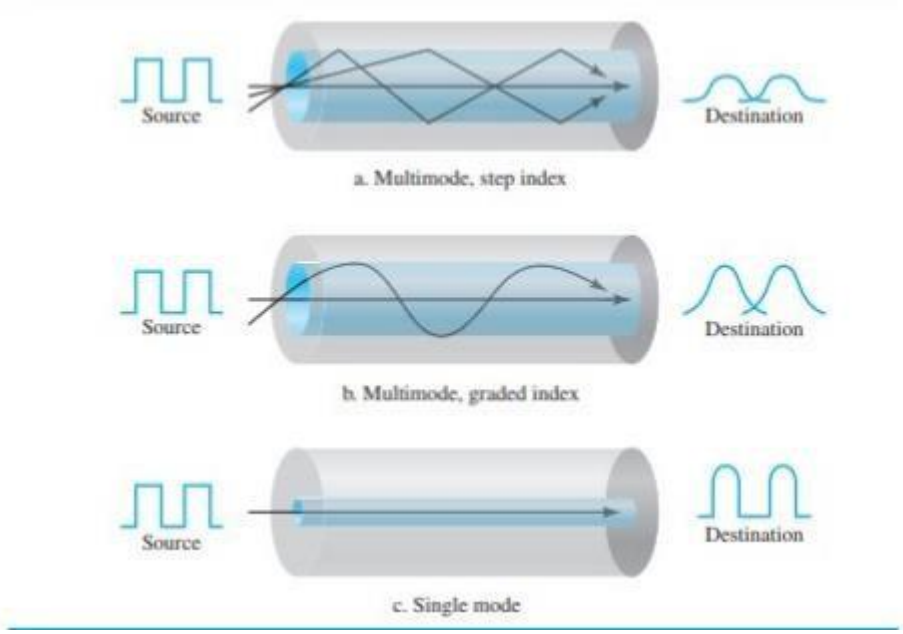


Fig 1.30 Modes

- In multimode step-index fiber, the density of the core remains constant from the center to the edges. A beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding
- A second type of fiber, called multimode graded-index fiber, decreases this distortion of the signal through the cable. The word index here refers to the index of refraction.
- Single-mode uses step-index fiber and a highly focused source of light that limits beams to a small range of angles, all close to the horizontal.

1.5.5 Wireless Transmission Media:

- In wireless communication technology, information In wireless communication technology, information travels in the form of electromagnetic signals through air.
- Electromagnetic spectrum of frequency ranging from 3 KHz to 900 THz is available for wireless communication (Figure 1.31).
- Wireless technologies allow communication between two or more devices in short to long distance without requiring any physical media.
- There are many types of wireless communication technologies such as Bluetooth, WiFi, WiMax etc.

- The electromagnetic spectrum range (3KHz to 900THz) can be divided into 4 categories (Figure 1.31) - Radio waves, Microwaves, Infrared waves and Visible or Light waves, according to their frequency ranges.
- Someof the properties of each wave are listed in Table 11.1. of these, three are useful for wireless communication.

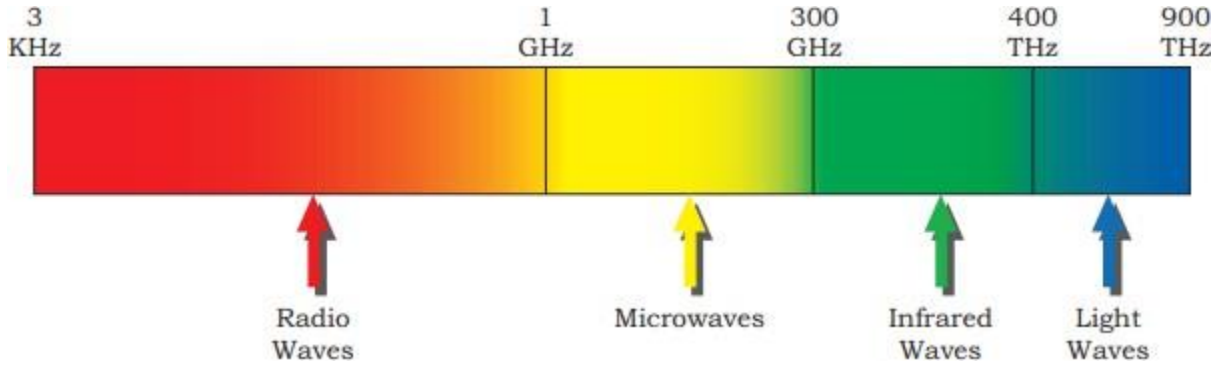


Fig:1.31 Electromagnetic Wave spectrum

Types of Propagation:

Unguided signals can travel from the source to destination in several ways: ground propagation, sky propagation, and line-of-sight propagation, as shown in Figure 1.32



Fig:1.32 Types of propagation

Ground Propagation:

- In ground propagation, radio waves travel through the lowest portion of the atmosphere, hugging the earth.
- These low frequency signals emanate in all directions from the transmitting antenna and follow the curvature of the planet.
- Distance depends on the amount of power in the signal: The greater the power, the greater the distance.

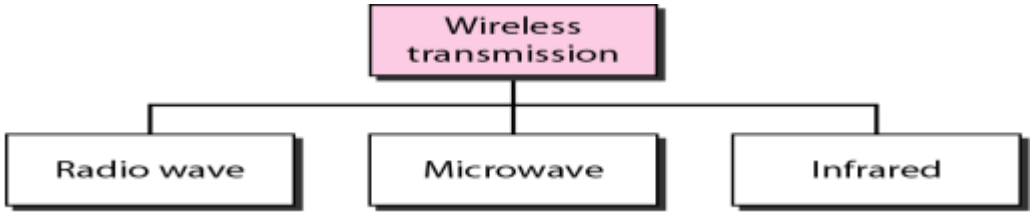
Sky propagation:

- In sky propagation, higher-frequency radio waves radiate upward into the ionosphere where they are reflected back to earth.
- This type of transmission allows for greater distances with lower output power.

Line of sight:

- In line-or-sight propagation, very high-frequency signals are transmitted in straight lines directly from antenna to antenna.

- Antennas must be directional, facing each other, and either tall enough or close enough together not to be affected by the curvature of the earth.
  - Line-of-sight propagation is tricky because radio transmissions cannot be completely focused



Radio Waves

- Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called radio waves.
- Radio waves are omni directional.
- When an antenna transmits radio waves, they are propagated in all directions.
- This means that the sending and receiving antennas do not have to be aligned.
- A sending antenna sends waves that can be received by any receiving antenna.
- The omni directional property has a disadvantage, too.
- The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signals using the same frequency or band.
- Omni directional Antenna Radio waves use omnidirectional antennas that send out signals in all directions.
- Based on the wavelength, strength, and the purpose of transmission, we can have several types of antennas. Figure 1.34 shows an omnidirectional antenna.



Fig:1.34 Omni Directional Antenna

Applications:

The Omni directional characteristics of radio waves make them useful for multicasting, in which there is one sender but many receivers. AM and FM radio, television, maritime radio, cordless phones, and paging are examples of multicasting

Microwaves:

- Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves.
- Microwaves are unidirectional. The sending and receiving antennas need to be aligned.
- The unidirectional property has an obvious advantage.
- A pair of antennas can be aligned without interfering with another pair of aligned antennas Unidirectional Antenna Microwaves need unidirectional antennas that send out signals in one direction.
- Two types of antennas are used for microwave communications: the parabolic dish and the horn

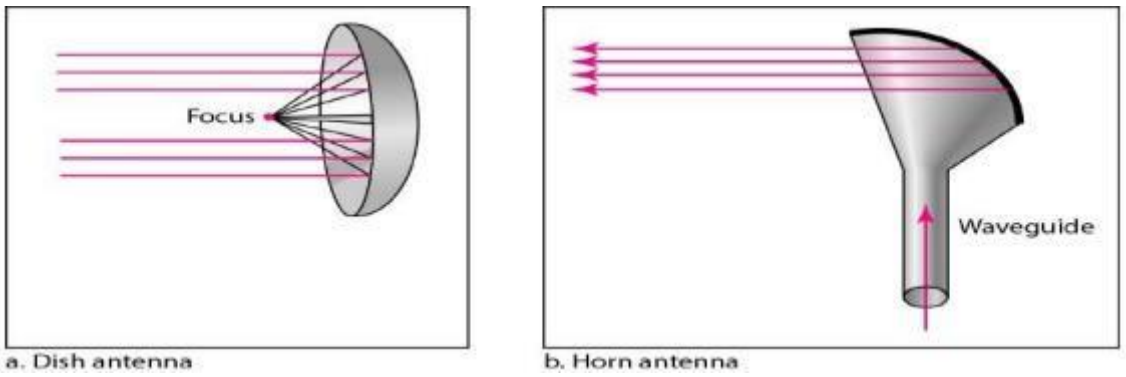


Fig:1.35 Dish and Horn Antenna

Applications: Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs

Infrared

- Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication. Infrared waves, having high frequencies, cannot penetrate walls.
- This advantageous characteristic prevents interference between one system and another; a shortrange communication system in one room cannot be affected by another system in the next room.
- When we use our infrared remote control, we do not interfere with the use of the remote by our neighbors. Infrared signals useless for long-range communication. In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication.

Applications:

Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.

Assignment-Cum-Tutorial Questions

1.1 Data Communications

1.1.1 Components

I) Objective Questions

1. Which of the following is an example of a sender?

a) Monitor
b) Keyboard
c) Printer
d) All of these

2. The message in a communication system refers to:

a) Rules for transmission
b) Information to be communicated
c) Medium of transmission
d) None of these

3. The term "protocol" includes which of the following aspects?

a) Syntax
b) Semantics
c) Timing
d) All of these

3. In a data communication system, the role of the receiver is to:

a) Generate signals
b) Accept and interpret the message
c) Provide routing information
d) Control flow of bits

4. Which component primarily determines the speed of data transfer?

a) Protocol
b) Sender
c) Medium
d) Receiver

II) Short Answer Questions

1. List the five basic components of a data communication system. *(Remember)*
2. Define message in data communication. *(Remember)*
3. What is the role of a transmitter in communication? *(Understand)*

4. Give an example of a receiver in computer networks. *(Understand)*

III) Descriptive Questions

1. Explain the components of a data communication system with a neat diagram. (BL1)
2. Discuss the importance of protocol in communication systems.(BL3)

1.1.2 Data Flow

4. Example of simplex communication:

a) Keyboard to CPU

b) Phone conversation

c) Walkie-talkie

d) Video conferencing

5. Match the example to the mode:
   - Radio broadcasting → Simplex
   - Walkie-talkie → Half-duplex
   - Telephone → Full-duplex

6. Which mode requires two channels for simultaneous two-way communication? [CO1/BL1]

a) Simplex

b) Half-duplex

c) Full-duplex

d) Multiplex

7. Which mode is most bandwidth-efficient for real-time video conferencing? [CO1/BL1]

a) Simplex

b) Half-duplex

c) Full-duplex

d) Broadcast

8. The main disadvantage of simplex mode is:

a) Low cost

b) Low utilization of channel

c) Complex hardware

d) High speed

II) Short Answer Questions

1. Name the three types of data floW.
2. Define simplex data flow with an example.
3. Differentiate between half duplex and full duplex.

III) Descriptive Questions

1. Explain simplex, half duplex, and full duplex with real-time examples and diagrams. (BL1)
2. Compare the advantages and disadvantages of each data flow type. (BL4)

1.2 Physical Structures

I) Objective Questions

1.2.1 Networks

1. A network is a collection of interconnected:
   a) Cables
   b) Devices
   c) Hubs
   d) Routers

2.The primary goal of a computer network is to:
a) Increase CPU speed
b) Share resources and data
c) Reduce electricity cost
d) Improve graphics performance

3, The Internet is an example of:
a) LAN
b) WAN
c) MAN
d) VPN

4.Which of the following connects different types of networks together?
a) Switch
b) Router
c) Hub
d) Repeater

5.Which network type is most suitable for connecting offices in different countries?
a) LAN
b) MAN
c) WAN
d) PAN

II) Short Answer Questions

1. Define a computer network.
2. Give two examples of network applications.
3. What is the difference between a host and a node?

III) Descriptive Questions

1. Explain the concept of networks and their importance in communication. (BL1)

2. Analyze the impact of network design on data communication performance. (BL4)

1.2.2 Type of Connection

I) Objective Questions

1. In a point-to-point connection, the communication link is between:
   a) Many devices
   b) Exactly two devices
   c) Only wireless devices
   d) One device and a server
2. Which is more cost-effective for connecting multiple devices?
   a) Point-to-point
   b) Multipoint
   c) Both are equal
   d) None of these
3. Which of the following is an example of a multipoint connection?
   a) USB
   b) Ethernet bus topology
   c) Fiber optic link
   d) Satellite uplink
4. Which connection type is better for video conferencing between two offices?
   a) Point-to-point
   b) Multipoint
   c) Either works
   d) None
5. Multipoint connections are generally:
   a) Faster than point-to-point
   b) Slower than point-to-point
   c) Equal in speed
   d) Wireless only

II) Short Answer Questions

1. List types of physical connections in a network.
2. Define point-to-point connection.
3. State an example of a multipoint connection.

III) Descriptive Questions

1. Explain point-to-point and multipoint connections with neat sketches. (BL1)
2. Compare point-to-point and multipoint connections in terms of bandwidth, cost, and reliability. (BL3)

1.2.3 Physical Topology

I) Objective Questions

1. In a star topology, all nodes are connected to a:
   a) Backbone cable
   b) Central hub
   c) Main router
   d) Server

2. In which topology does a break in one cable affect the entire network?
   a) Star
   b) Bus
   c) Mesh
   d) Hybrid

3. Which topology is most fault-tolerant?
   a) Star
   b) Mesh
   c) Bus
   d) Ring

4. Which topology requires the most cabling for the same number of nodes?
   a) Mesh
   b) Star
   c) Bus
   d) Ring

5. Which topology is most commonly used in Ethernet LANs today?
   a) Star
   b) Bus
   c) Ring
   d) Mesh

6. Hybrid topology is a combination of:

a) Two or more topologies
b) LAN and WAN
c) Guided and unguided media
d) Star and mesh only

7. In ring topology, data travels:
   a) Randomly
   b) Only clockwise
   c) In one direction or both depending on design
   d) Broadcasted to all

8. Bus topology requires:
   a) Hub
   b) Terminators
   c) Router
   d) Switch

9. Which topology is cheapest to install?
   a) Bus
   b) Star
   c) Mesh
   d) Ring

II) Short Answer Questions

1. Define network topology.
2. Name any four physical topologies.
3. Which topology requires the most cable?

III) Descriptive Questions

1. Explain different types of physical topologies with diagrams. (BL1)
2. Evaluate the advantages and disadvantages of bus, star, ring, mesh, and hybrid topologies. (BL4)

1.3 Categories of Network

I) Objective Questions

1.3.1 Local Area Network (LAN)

1. LAN typically covers:
   a) A single building or campus

b) An entire country
c) Multiple cities
d) Across continents

2. Which is **not** a typical LAN characteristic?
   a) High data transfer rate
   b) Limited geographic area
   c) Expensive transmission medium
   d) Controlled privately

3. LAN speed generally ranges from:
   a) 1 kbps to 10 kbps
   b) 10 Mbps to 1 Gbps
   c) 1 Gbps to 1 Tbps
   d) 1 bps to 100 bps

4. Which topology is most commonly used in modern LANs?
   a) Bus
   b) Star
   c) Ring
   d) Tree

1.3.2 Wide Area Network (WAN)

1. WANs cover:
   a) A single room
   b) A city
   c) A country or continent
   d) A small office

2. WANs usually use:
   a) Private cabling only
   b) Public transmission systems
   c) Bluetooth communication
   d) Infrared communication

3. Which device is essential for WAN connection?
   a) Hub
   b) Switch
   c) Router
   d) Access point

1.3.3 Metropolitan Area Network (MAN)

1. MAN usually spans:
   a) 1–10 km
   b) 10–50 km
   c) 50–200 km
   d) More than 200 km

2. Which is an example of a MAN?
   a) Cable TV network in a city
   b) Home Wi-Fi
   c) Bluetooth headset connection
   d) Satellite communication

1.3.4 Internetwork

1. An internetwork connects:
   a) Multiple devices in the same LAN
   b) Two or more separate networks
   c) Only WANs together
   d) Only MANs together

2. The device used to connect different networks in an internetwork is:
   a) Repeater
   b) Switch
   c) Router
   d) Hub

II) Short Answer Questions

1. Expand LAN, WAN, and MAN.
2. Which network type covers the largest geographical area?
3. Give two examples of internetworks.

III) Descriptive Questions

1. Explain LAN, WAN, and MAN with suitable diagrams. (BL2)
2. Compare the characteristics of LAN, WAN, and MAN. (BL3)

1.4 Reference Models

I) Objective Questions

1.4.1 OSI Reference Model

1. Which OSI layer handles data compression?
   a) Session
   b) Presentation
   c) Application
   d) Data link

2. The physical addressing is done at which OSI layer?
   a) Data link
   b) Network
   c) Transport
   d) Application
3. Which OSI layer is responsible for **path determination and logical addressing**?
   a) Data Link Layer
   b) Network Layer
   c) Transport Layer
   d) Session Layer
4. In the OSI model, **encryption and decryption** are handled by:
   a) Session Layer
   b) Presentation Layer
   c) Application Layer
   d) Data Link Layer
5. The main function of the **Transport Layer** is:
   a) Routing packets to destination
   b) Ensuring reliable delivery of messages
   c) Encoding data
   d) Managing sessions
6. Which protocol works at the Application Layer?
   a) IP
   b) HTTP
   c) TCP
   d) ARP
7. Which OSI layer adds **frame headers and trailers**?
   a) Transport
   b) Data Link
   c) Network
   d) Session

8. If a router is malfunctioning, which OSI layer is most likely affected?
   a) Transport Layer
   b) Network Layer
   c) Physical Layer
   d) Data Link Layer
9. Which layer ensures that data from the upper layer is converted into a format suitable for transmission?
   a) Session
   b) Transport
   c) Presentation
   d) Network
10. If an application needs to send encrypted data reliably to a remote host, which OSI layers will be involved in encryption and reliability respectively?
   a) Presentation & Transport
   b) Application & Network
   c) Transport & Data Link
   d) Session & Physical

1.4.2 TCP/IP Reference Model

1. The TCP/IP model has:
   a) 4 layers
   b) 5 layers
   c) 6 layers
   d) 7 layers

2. Which layer in TCP/IP corresponds to OSI's Transport layer?
   a) Network access
   b) Internet
   c) Transport
   d) Application

3. Which protocol works at the Internet layer of TCP/IP?
   a) IP
   b) TCP
   c) HTTP
   d) FTP

1.4.3 Comparison of OSI and TCP/IP

1. Which of the following is **true** about OSI and TCP/IP models?
   a) OSI has fewer layers than TCP/IP

b) TCP/IP combines OSI's session, presentation, and application into one
c) TCP/IP has a separate presentation layer
d) TCP/IP is only theoretical

2. Which model is considered more practical?
a) OSI
b) TCP/IP

II) Short Answer Questions

1. How many layers are in the OSI model?
2. Name the three lower layers of the OSI model.
3. Which layer is responsible for encryption?
4. How many layers are in the TCP/IP model?
5. Name the layers of the TCP/IP model.
6. Which layer of TCP/IP corresponds to OSI's transport layer?
7. Which model is more commonly implemented in real networks?
8. State one similarity and one difference between OSI and TCP/IP.

III) Descriptive Questions

1. Explain the OSI model with functions of each layer. (BL2)
2. Analyze the role of the OSI model in standardizing network communication. (BL3)
3. Explain the TCP/IP reference model with layer functions. (BL2)
4. Compare TCP/IP with OSI in terms of structure and practical use. (BL4)
5. Compare the OSI and TCP/IP reference models in detail. (BL4)

1.5 Physical Layer

I) Objective Questions

1.5.1 Guided Media – Introduction

1. Guided media transmits signals through:
a) Free space
b) Physical pathway
c) Optical modulation only

d) Satellite only

1.5.2 Twisted Pair Cable

1. Twisted pair cables reduce:
   a) Attenuation
   b) Crosstalk
   c) Reflection
   d) Propagation

2. UTP stands for:
   a) Ultra Transmission Protocol
   b) Unshielded Twisted Pair
   c) Universal Transmission Pair
   d) Unshielded Transmission Protocol

1.5.3 Coaxial Cable

1. The central conductor in a coaxial cable is typically made of:
   a) Steel
   b) Copper
   c) Aluminum
   d) Plastic

2. Coaxial cables are **less** susceptible to:
   a) Noise
   b) Bandwidth
   c) Propagation delay
   d) Cost

1.5.4 Fiber Optic Cable

1. Fiber optic cables use signals in the form of:
   a) Electrical pulses
   b) Light
   c) Magnetic waves
   d) Sound

2. Which is **not** an advantage of fiber optic over copper?
   a) Higher bandwidth
   b) Immunity to EMI
   c) Greater distance coverage
   d) Lower initial cost

1.5.5 Unguided Media – Introduction

1. Unguided media is also known as:
   a) Wired communication
   b) Wireless communication
   c) Optical cable
   d) Digital pathway

2. Which of these is **not** an unguided medium?
   a) Radio waves
   b) Microwaves
   c) Infrared
   d) Coaxial cable

II) Short Answer Questions

1. Define guided media.
2. Give two examples of guided media.
3. Name two types of twisted pair cables.
4. Which twisted pair is used for high-speed LANs?
5. What are the main parts of a coaxial cable?
6. Give one advantage of coaxial over twisted pair.
7. Name two modes of fiber optic transmission.
8. Which medium offers the highest bandwidth?
9. Define unguided media.

III) Descriptive Questions

1. Explain types of guided media with diagrams. (BL1)
2. Explain UTP and STP with structure, advantages, and disadvantages. (BL2)
3. Describe the structure and uses of coaxial cable. (BL1)
4. Explain the construction, working, and applications of fiber optic cables. (BL2)
5. Discuss different types of unguided media: radio waves, microwaves, and infrared. (BL1)

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Academics Strengthening & Advancement (AS&A)

R – 23

III B.Tech. I Semester

COMPUTER NETWORKS

Learning Material

UNIT II DATA LINK LAYER

Course Objectives

- Understanding of the How the Data Link Layer Perform Flow and Error control methods.
- Basic Knowledge of Elementary Protocols Supported by Data Link Layer

Course Outcomes

Students can able to

- differentiate different framing methods used by data link layer
- apply the error detection and error correction techniques
- understand the different types of noisy channels protocols

Syllabus

2.1 Data Link Layer Design issues:
- **2.1.1** Framing-fixed size framing, variable size framing
- **2.1.2** Error control
- **2.1.3** Error detection

2.2 Error correction,
- **2.2.1** Checksum
- **2.2.2** CRC
- **2.2.3** hamming code.

2.3 Elementary Data Link Layer protocol
- **2.3.1** simplex protocol
- **2.3.2** Simplex Stop - and Wait Protocol,
- **2.3.3** Simplex protocol for Noisy Channel

2.4 Sliding window protocol
- **2.4.1** one bit
- **2.4.2** Go back N
- **2.4.3** selective repeat,

2.5 Data link layer in HDLC.

2.6 Data link layer in PPP.

What is DLL (Data Link Layer)?

The Data Link Layer is the second layer in the OSI model, above the Physical Layer, which ensures that the error free data is transferred between the adjacent nodes in the network. It breaks the datagram passed down by above layers and converts them into frames ready for transfer. This is called **Framing.**
It provides two main functionalities.

- Reliable data transfer service between two peer network layers
- Flow Control mechanism which regulates the flow of frames such that data congestion is not there at slow receivers due to fast senders.

2.1 The Data Link Layer Design Issues

(1) Framing.
(2) Error Control.
(3) Flow Control.

2.2.1 Framing

The frame contains

1. Frame header
2. Pay load field for holding packet
3. Frame trailer

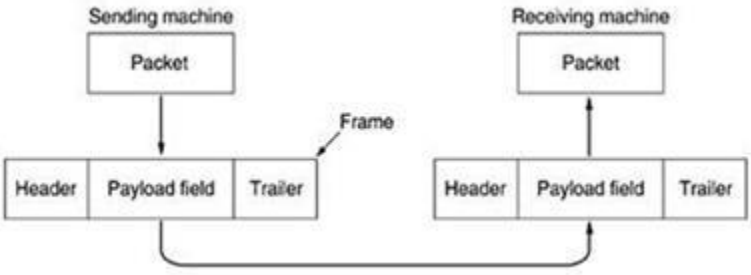


Fig 3.1: Relationship between packets and frames

Services provided to the network layer

- The function of the data link layer is to provide services to the network layer. The principal service is transferring data from the network layer on the source machine to the network layer on the destination machine.
- The data link layer can be designed to offer various services. The actual services offered can vary from system to system. Three reasonable possibilities that are

commonly provided are

1) Unacknowledged Connectionless service
2) Acknowledged Connectionless service
3) Acknowledged Connection-Oriented service

1. **Unacknowledged connection less service**
   - Source machine sends independent frames to destination machine having destination machine acknowledge them
   - No logical connection
   - Used when error rate is very low
   - Good for real-time traffic (voice)

2. **Acknowledged connectionless service**

   - No logical connection
   - Each frame sent is individually acknowledged
   - Useful over unreliable channels (i.e. wireless systems)

3. **Acknowledged connection-oriented service**

   - Source and destination machines establish a connection before any data are transferred
   - Each frame is numbered

   - DLL guarantees that...
   - Each frame is received
   - Each frame is received exactly once
   - Each frame is received in the right order

FRAMING:

- Data transmission in the physical layer means moving bits in the form of a signal from the source to the destination.
- The physical layer provides bit synchronization to ensure that the sender and receiver use the same bit durations and timing.
- The data link layer, on the other hand, needs to pack bits into frames, so that each frame is distinguishable from another. Our postal system practices a type of framing.

Framing in the data link layer separates a message from one source to a destination, or from other messages to other destinations, by adding a sender address and a destination address.

The destination address defines where the packet is to go; the sender address helps the recipient acknowledge the receipt.

There are two types of framing

1. Fixed-Size Framing
2. Variable-Size Framing

Fixed-Size Framing:

- Frames can be of fixed or variable size.
- In fixed-size framing, there is no need for defining the boundaries of the frames; the size itself can be used as a delimiter.
- An example of this type of framing is the ATM wide-area network, which uses frames of fixed size called cells

Variable-Size Framing:

In variable-size framing, we need a way to define the end of the frame and the beginning of the next. Historically, two approaches were used for this purpose:

1. character-oriented approach
2. Bit-oriented approach

Character-Oriented Protocols:

In character - oriented framing, data is transmitted as a sequence of bytes, from an 8-bit coding system like ASCII. The parts of a frame in a character - oriented framing are –

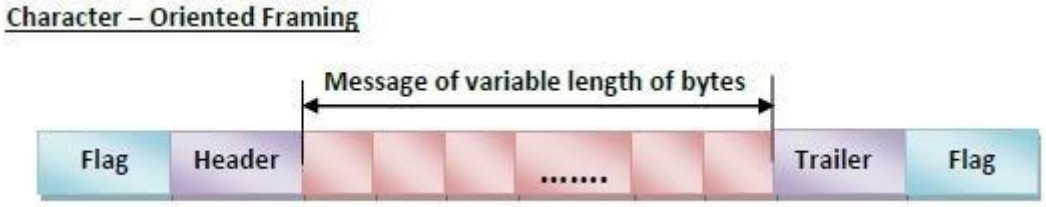


Fig 3.2: Frame format of Character Oriented Protocol

- **Frame Header** – It contains the source and the destination addresses of the frame in form of bytes.
- **Payload field** – It contains the message to be delivered. It is a variable sequence of data bytes.
- **Trailer** – It contains the bytes for error detection and error correction.
- **Flags** – Flags are the frame delimiters signaling the start and end of the frame. It is of 1- byte denoting a protocol - dependent special character.

- Character - oriented protocols are suited for transmission of texts.
- The flag is chosen as a character that is not used for text encoding.
- However, if the protocol is used for transmitting multimedia messages, there

are chances that the pattern of the flag byte is present in the message byte sequence.

- In order that the receiver does not consider the pattern as the end of the frame, **byte stuffing** mechanism is used.

Byte stuffing:

- Here, a special byte called the escape character (ESC) is stuffed before every byte in the message with the same pattern as the flag byte.
- If the ESC sequence is found in the message byte, then another ESC byte is stuffed before it.

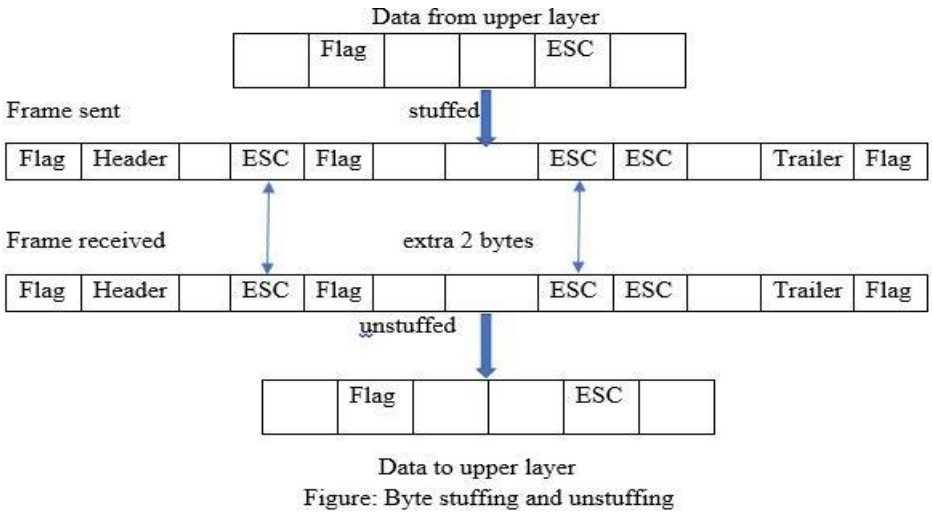


Fig 3.3: Byte Stuffing

Bit-oriented Protocol:

In bit-oriented framing, data is transmitted as a sequence of bits that can be interpreted in the upper layers both as text as well as multimedia data.

The parts of a frame in a character - oriented framing are –

- **Frame Header** – It contains bits denoting the source and the destination addresses of the frame.
- **Payload field** – It contains the message to be delivered. It is a variable sequence of bits.
- **Trailer** – It contains the error detection and error correction bits.
- **Flags** – Flags are a bit pattern that act as the frame delimiters signaling the start and end of the frame. It is generally of 8-bits and comprises of six or more consecutive 1s. Most protocols use the 8-bit pattern 01111110 as flag.

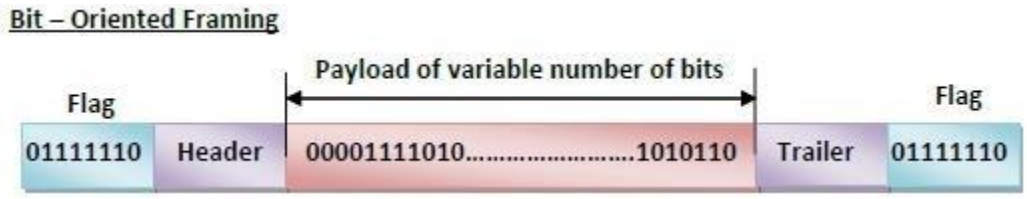


Fig 3.4: Frame format of Bit Oriented Protocol

- Bit - oriented protocols are suited for transmitting any sequence of bits.
- So there are chances that the pattern of the flag bits is present in the message.
- In order that the receiver does not consider this as end of frame, **bit-stuffing** mechanism is used

Bit Stuffing:

Bit stuffing is the process of adding one extra 0 whenever five consecutive 1's appear in the data, so that the receiver does not mistake the pattern 0111110 for a flag.

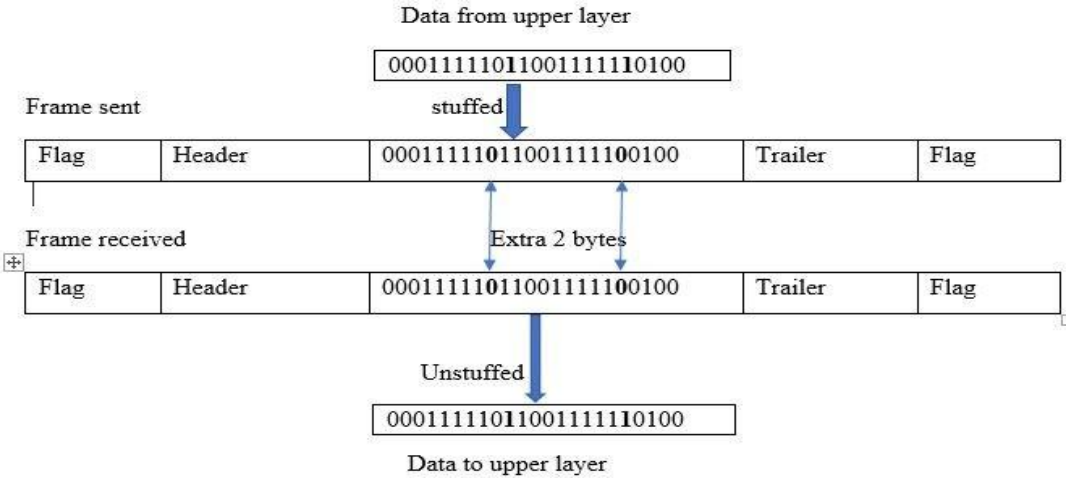


Fig 3.5: Bit Stuffing

2.1.2 Error Control

When data is transmitted over a cable or channel, there is always a chance that some of the bits will be changed (corrupted) due to noise, signal distortion or attenuation. If errors do occur, then some of the bits will either change from 0 to 1 or from 1 to 0.

TYPES OF ERRORS:

- **Single bit error**
- **Burst error**

Single bit error: The term single-bit error means that only one bit of given data unit (such as a byte, character, or data unit) is changed from 1 to 0 or from 0 to 1 as shown in fig

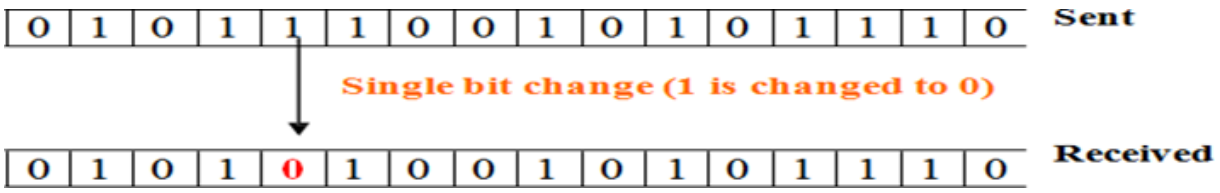


Fig 3.6: Single Bit Error

Single bit errors are least likely type of errors in serial data transmission

Burst error:

- More than one bit gets corrupted very common in serial transmission of data occurs when the duration of noise is longer than the duration of one bit.
- The noise affects data; it affects a set of bits.
- The number of bits affected depends on the data rate and duration of noise

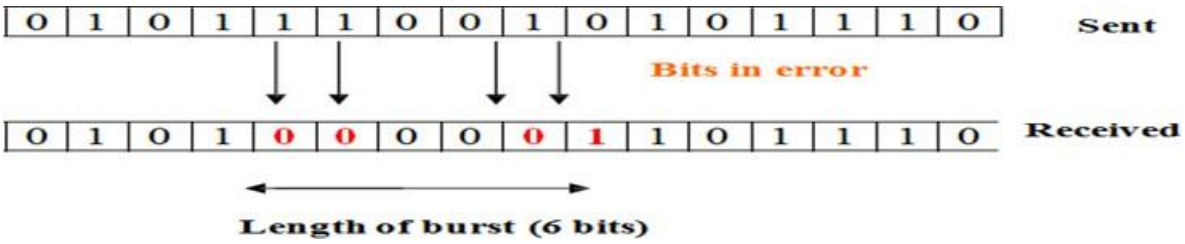


Fig 3.7: Burst Error

2.1.3 Error Detection

Error Control allows the receiver to inform the sender of any frames lost or damaged in transmission and coordinates the retransmission of those frames by the sender. Error control is divided in two main categories:

- Error Detection
- Error Correction

Error Detection It allows a receiver to check whether received data has been corrupted during transmission. It can, for example, request a retransmission.

Error Correction This type of error control allows a receiver to reconstruct the original information when it has been corrupted during transmission.

Error Detection Techniques:

- Checksum
- Cyclic redundancy check

2.2.1 CHECKSUM

- In checksum error detection scheme, the data is divided into k segments each of m bits.
- In the sender's end the segments are added using 1's complement arithmetic to get the sum.
- The sum is complemented to get the checksum. The checksum segment is sent along with the data segments

Example 1:
Sender

10101001	subunit1
00111001	subunit 2
11100010	sum
00011101	Complement of sum

10101001	00111001	00011101
Data		checksum

Reciever:

10101001	subunit1
00111001	subunit2
00011101	Checksum
11111111	sum
00000000	complement

Conclusion = Accept data.

Example 2: K= 10110011, 10101011, 01011111, 11010101

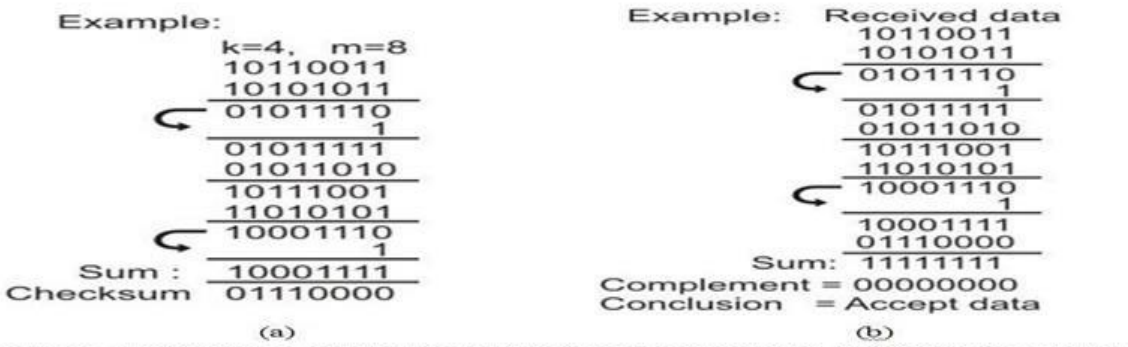


Figure 3.2.5 (a) Sender's end for the calculation of the checksum, (b) Receiving end for checking the checksum

2.2.2 CYCLIC REDUNDANCY CHECK

- At the sender side, the data unit to be transmitted is divided by a predetermined divisor (binary number) in order to obtain the remainder. This remainder is called CRC.
- The CRC has one bit less than the divisor. It means that if CRC is of n bits, divisor is of n+ 1 bit.
- The sender appends this CRC to the end of data unit such that the resulting data unit becomes exactly divisible by predetermined divisor i.e. remainder becomes zero.
- At the destination, the incoming data unit i.e. data + CRC is divided by the

same number (predetermined binary divisor).

- If the remainder after division is zero then there is no error in the data unit & receiver accepts it.
- If remainder after division is not zero, it indicates that the data unit has been damaged in transit and therefore it is rejected.
- This technique is more powerful than the parity check and checksum error detection.

Requirements of CRC :

A CRC will be valid if and only if it satisfies the following requirements:

- It should have exactly one less bit than divisor.
- Appending the CRC to the end of the data unit should result in the bit sequence which is exactly divisible by the divisor.

Cyclic Redundancy Check:

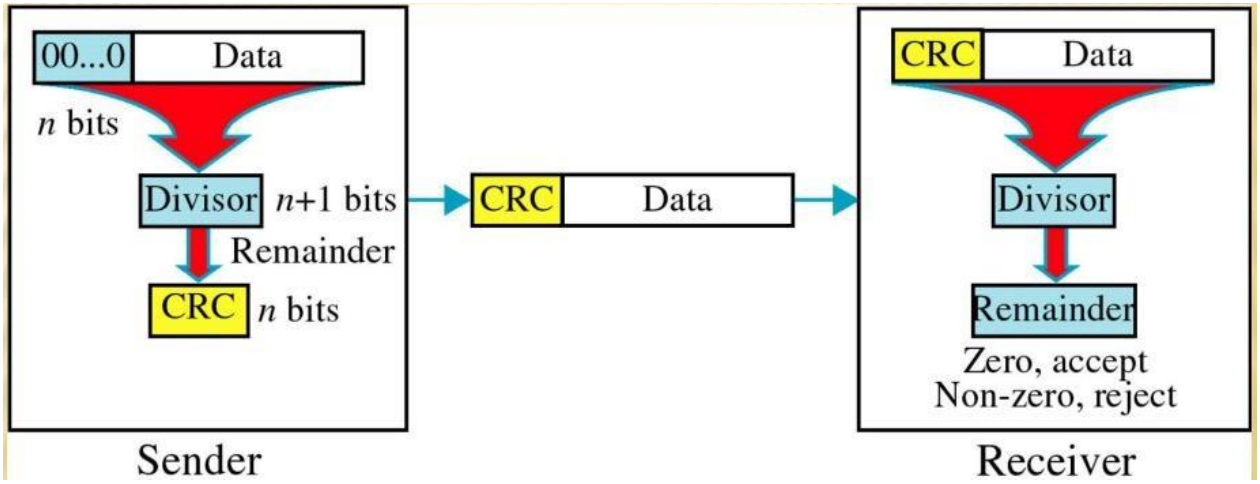
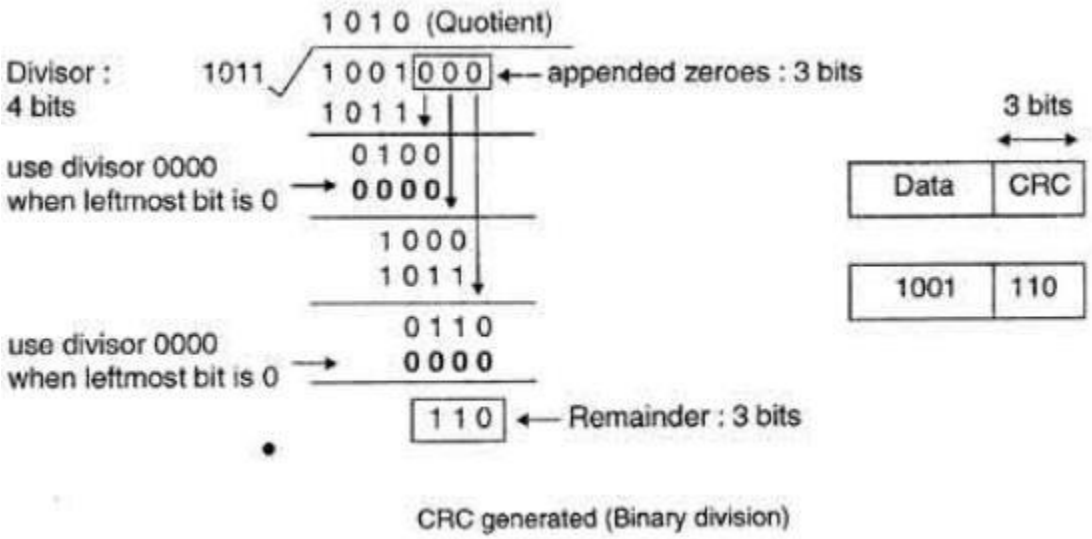


Fig 3.8: Cyclic Redundancy Check

Example1:

1. Data unit 1011000 is divided by 1011.



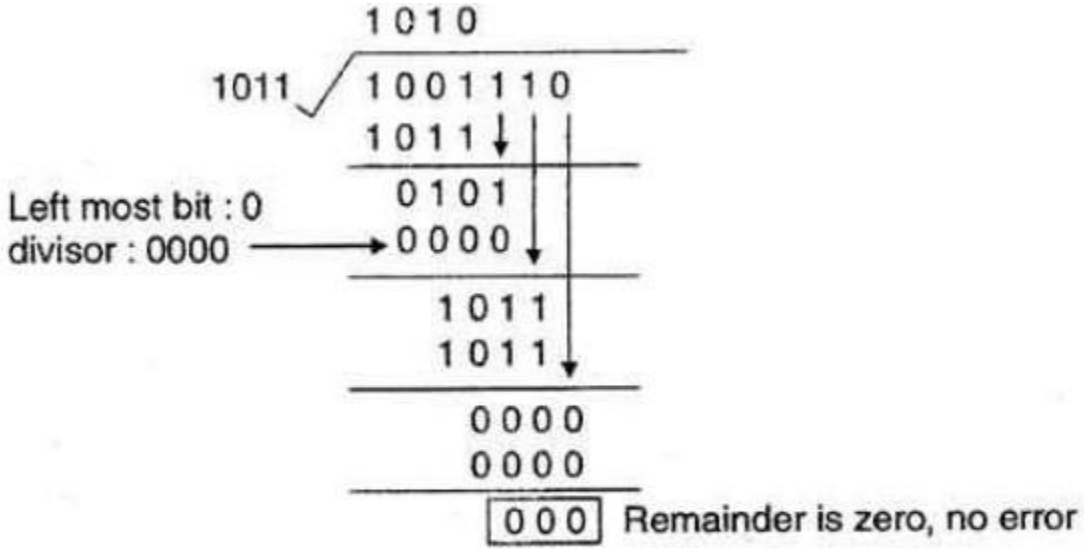
CRC generated (Binary division)

2. During this process of division, whenever the leftmost bit of dividend or remainder is 0, we use a string of Os of same length as divisor. Thus in this case divisor 1011 is replaced by 0000.

3. At the receiver side, data received is 1001110.

4. This data is again divided by a divisor 1011.

5. The remainder obtained is 000; it means there is no error.



CRC decoded (binary division)

CRC can detect all the burst errors that affect an odd number of bits.

Polynomial codes:

- A pattern of Os and 1s can be represented as a polynomial with coefficient of o and 1
- Here, the power of each term shows the position of the bit and the coefficient shows the values of the bit.
- For example, if binary pattern is 100101, its corresponding polynomial representation is x5 + x2 + 1

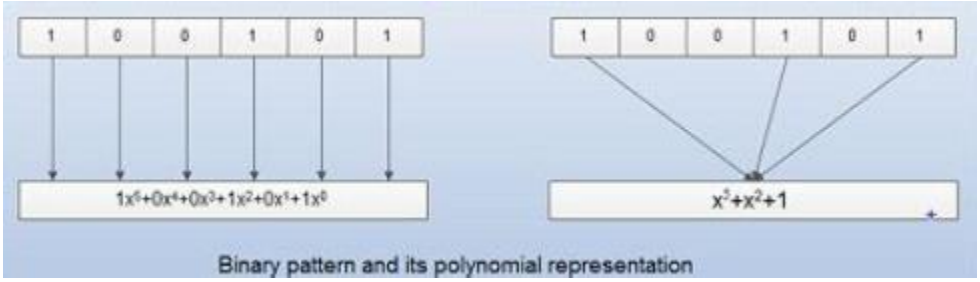


Fig 3.9: Binary Pattern and it's Polynomial representation

- The benefits of using polynomial codes is that it produces short codes. For example here a 6-bit pattern is replaced by 3 terms.
- In polynomial codes, the degree is 1 less than the number of bits in the binary pattern. The degree of polynomial is the highest power in polynomial. For example as shown in fig degree of polynomial x5 +x2 + 1 are 5. The bit pattern in this case is 6.
- Addition of two polynomials is based on modulo-2 method. In such as case, addition and subtraction is same.
- Addition or subtraction is done by combining terms and deleting pairs of identical terms. For example adding x5 + x4 + x2 and x6 + x4 + x2 give x6 + x5. The terms x4 and x2 are deleted.
- In case of multiplication of two polynomials, their powers are added. For example, multiplying x5 + x3 + x2 + x with x2+ x+ 1 yields:

  (X5 + x3 + x2 + x) (x2 + x + 1)

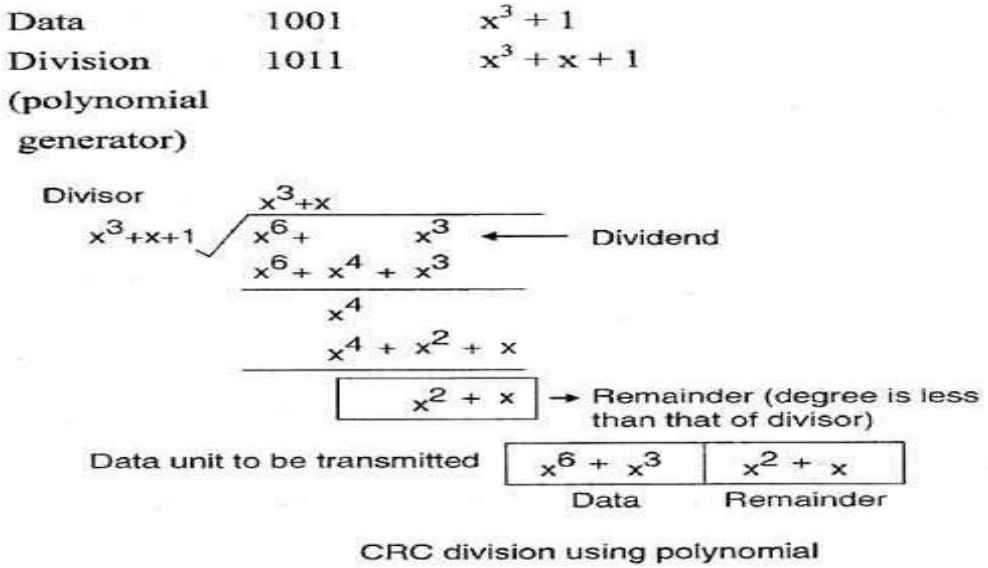
  = x7 + x6+ x5+ x5+ x4+ x3+ x4+ x3+ x2+ x3+ x2+ x

  =X7+x6+x3+X

- In case of division, the two polynomials are divided as per the rules of binary division, until the degree of dividend is less than that of divisor.

CRC generator using polynomials

- If we consider the data unit 1001 and divisor or polynomial generator 1011their polynomial representation is:



CRC division using polynomial

Now string of n 0s (one less than that of divisor) is appended to data. Now data is 1001000 and its corresponding polynomial representation is x6 + x3.

The division of x6+x3 by x3+x+ 1 is shown in fig.

The polynomial generator should have following properties:

1. It should have at least two terms.
2. The coefficient of the term x0 should be 1.
3. It should not be divisible by x.
4. It should be divisible by x+ 1

There are several different standard polynomials used by popular protocols for CRC generation. These are:

Name	Polynomial	Application
CRC-8	$x^8 + x^2 + x + 1$	ATM header
CRC-10	$x^{10} + x^9 + x^5 + x^4 + x^2 + 1$	ATM AAL
CRC-16	$x^{16} + x^{12} + x^5 + 1$	HDLC
CRC-32	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$	LANs

2.2.3 Hamming Code

- Hamming code is technique developed by R.W. Hamming for error correction.
- In Computer Networks, Hamming code is used for the set of error-correction codes which may occur when the data is moved from the sender to the receiver.
- The hamming method corrects the error by finding the state at which the error has occurred.

Redundant Bits

- Redundant bits are extra binary bits that are generated and added to the information-carrying bits of data transfer to ensure that no bits were lost during the data transfer.
- The redundancy bits are placed at certain calculated positions to eliminate the errors and the distance between the two redundancy bits is called "Hamming Distance".

Error Correction Code –

- This is the relationship between data bits and redundancy bits to correct a single-bit error.
- A-frame consists of M data bits and R redundant bits.
- Suppose the total length of the frame be N (N=M+R).
- An N-bit unit containing data and the check bit is often referred to as an N-bit code word.

The following formula is used to find the number of redundant bits.

Number of single-bit errors = M + R

Number of states for no error = 1

So, the number of redundant bits (R) that represent all states (M+R+1) must satisfy –

$$2^R \geq M + R + 1$$

where R = Redundant bit, and M = data bit.

Steps to find the Hamming Code:

- The algorithm of the hamming code in computer networks comprises two major steps which are encoding and decoding a message. The data is encoded at the sender's side and decoded at the receiver's side.

Encode a message: We encode the message at the sender's side and then it is transmitted over the network. The following steps are involved in encoding a message.

- Write the bit position in the binary format.
- Calculate the number of redundant bits for a message.
- Do the positioning of redundant bits.
- Then do the calculations and calculate the value of each redundant bit.
- Append the redundant bits with actual data and transmit.

Decode the message: At the destination end, the receiver receives the encoded message and performs the recalculations to calculate the errors. If the message has an error, then these errors are corrected. To perform recalculations following steps are performed.

- Calculate the number of redundant bits using the same formula.
- Do the positioning of redundant bits.
- The parity checking technique is used to check parity bits.
- Perform error detection and correction.
- The hamming method uses the extra parity bits to allow the identification of a single-bit error.

Hamming Code Example

Let's take a hamming code example of a bit word 1011 is to be transmitted from the sender to the destination machine. Construct the even parity seven-bit hamming code for this data.

The binary format of bits.

	R4	R3	R2	R1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1

R1-->1,3,5,7,9,11
R2-->2,3,6,7,10,1
R3 -->4,5,6,7
R4-->8,9,10,11

Positioning of redundant bits.

The positions which correspond to the power of 2 are filled with redundant bits. It means at positions 1,2, and 4 redundant bits are placed.

As in the above example, Number of data bits = 4

So the number of redundant of bits = 3

total number of bits to be transmitted = 4+3 = 7

D7	D6	D5	R4	D3	R2	R1

So put the bits into the above code word.

1	0	1	R4	1	R2	R1

Decide the Redundant bits

- We calculate R1 bit using all bit positions whose binary representation includes 1 in the least significant position. It means bits 1,3,5,7 are used to calculate the value of the R1 bit.
- We calculate the R2 bit using all the bit positions whose binary representation includes 1 in the second position from the least significant bits. It means bits 2,3,6,7 are used to calculate the value of the R2 bit
- We calculate the R4 bit using all the bit positions whose binary representation includes 1 in the third position from the least significant bits. It means bits 4,5,6,7 are used to calculate the value of the R4 bit.
- We have to construct an even parity bit hamming code, as mentioned in the above example.
- To calculate the redundant bit, use the even parity checking technique. It means the number 1's should be even as shown in the example below.

Decide R1 - R1,D3, D5, D7
as D3 D5 D7 =111 put R1 = 1 in order to have even parity

Decide R2 - R2,D3,D6,D7
as D3 D6 D7 = 101 hence Set R2= 0

Decide R4 - R4,D5,D6,D7
as D5 D6 D7 = 101 so Set R4=0

Now we have calculated the valve of redundant bits to fill the value of redundant bits into the corresponding position. So following is the complete codeword to transmit.

1	0	1	0	1	0	1

Hamming Distance:

Hamming distance is the number of bit positions in which the two bits are different.

The Hamming distance between two strings, a and b is denoted as d(a,b).

Calculation of Hamming Distance

In order to calculate the Hamming distance between two strings, and , we perform their XOR operation, (a⊕ b), and then count the total number of 1s in the resultant string.

Example

Suppose there are two strings 1101 1001 and 1001 1101.

11011001 ⊕ 10011101 = 01000100. Since, this contains two 1s, the Hamming distance, d(11011001, 10011101) = 2.

Minimum Hamming Distance

In a set of strings of equal lengths, the minimum Hamming distance is the smallest Hamming distance between all possible pairs of strings in that set.

Example

Suppose there are four strings 010, 011, 101 and 111.

010 ⊕ 011 = 001, d(010, 011) = 1.

010 ⊕ 101 = 111, d(010, 101) = 3.

010 ⊕ 111 = 101, d(010, 111) = 2.

011 ⊕ 101 = 110, d(011, 101) = 2.

011 ⊕ 111 = 100, d(011, 111) = 1.

101 ⊕ 111 = 010, d(011, 111) = 1.

Hence, the Minimum Hamming Distance, *dmin* = 1.

Flow and Error Control:

The most important responsibilities of the data link layer are flow control and error Control. Collectively, these functions are known as data link control.

Flow control: It refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment.

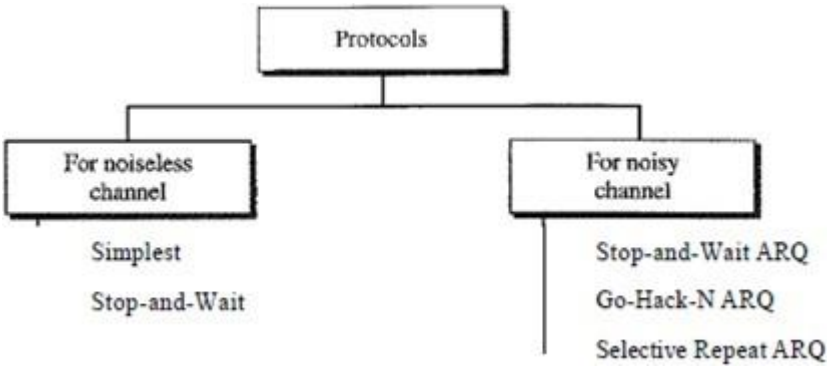
Error Control:

It is based on Automatic Repeat Request, which is the retransmission of data.

2.3 Elementary DLL protocols

The data link layer can combine framing, flow control, and error control to achieve the delivery of data from one node to another.

The protocols are normally implemented in software by using one of the common programming languages.



Protocols are classified into various types depending on different channels.

They are:

1) noiseless channel

2) noisy channel

Noiseless channel:

Protocols for noiseless channel are further classified into two

i) simplest protocol
ii) stop-and-wait protocol

2.3.1 Simplest Protocol

- It is a unidirectional protocol in which data frames are traveling in only one direction-from the sender to receiver. We assume that the receiver can immediately handle any frame it receives with a processing time that is small enough to be negligible.
- The data link layer of the receiver immediately removes the header from the frame and hands the data packet to its network layer, which can also accept the packet immediately.
- In other words, the receiver can never be overwhelmed with incoming frames.

The design of the simplest protocol with no flow or error control

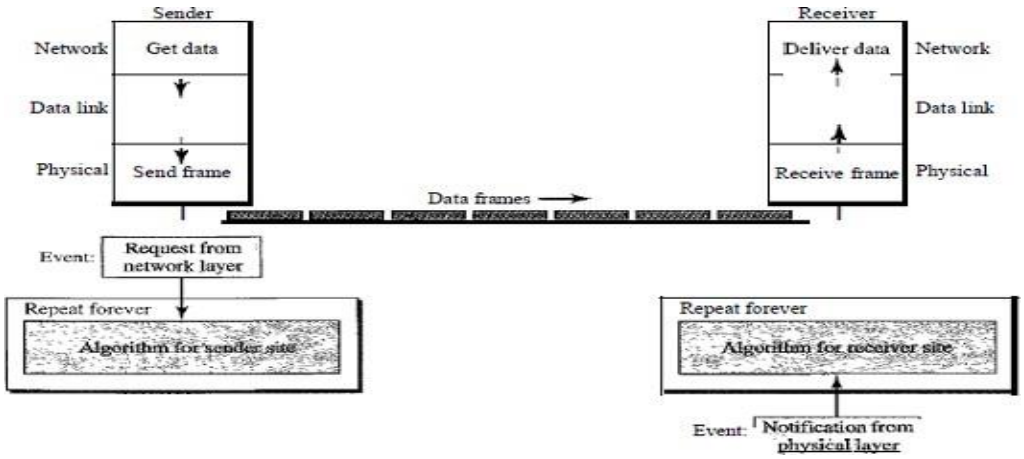


Fig 3.10: Design of the simplest protocol with no flow or error control

- The sender site cannot send a frame until its network layer has a data packet to send. The receiver site cannot deliver a data packet to its network layer until a frame arrives.
- If the protocol is implemented as a procedure, we need to introduce the idea of events in the protocol. The procedure at the sender site is constantly running; there is no action until there is a request from the network layer.
- The procedure at the receiver site is also constantly running, but there is no action until notification from the physical layer arrives.
- Both procedures are constantly running because they do not know when the corresponding events will occur.

Flow diagram for Simplex protocol:

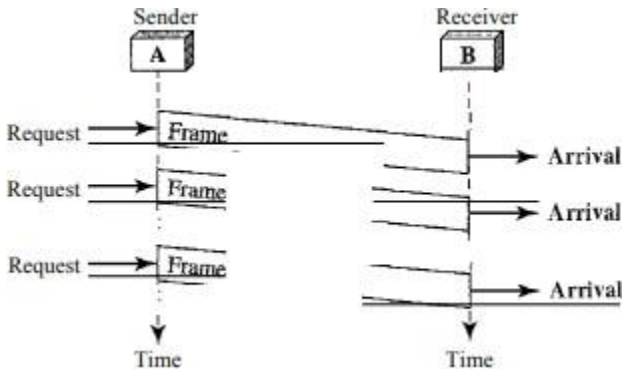


Fig 3.11: Flow diagram for Simplex protocol

It is very simple. The sender sends a sequence of frames without even thinking about the receiver. To send three frames, three events occur at the sender site and three events at the receiver site. Note that the data frames are shown by tilted boxes; the height of the box defines the transmission time difference between the first bit and the last bit in the frame

2.3.2 Stop-and-Wait Protocol

- If data frames arrive at the receiver site faster than they can be processed, the frames must be stored until their use.
- Normally, the receiver does not have enough storage space, especially if it is receiving data from many sources.
- This may result in either the discarding of frames or denial of service. To prevent the receiver from becoming overwhelmed with frames, we somehow need to tell the sender to slow down. There must be feedback from the receiver to the sender
- The protocol we discuss now is called the Stop-and-Wait Protocol because the sender sends one frame, stops until it receives confirmation from the receiver (okay to go ahead), and then sends the next frame.
- We still have unidirectional communication for data frames, but auxiliary ACK frames (simple tokens of acknowledgment) travel from the other direction.

Flow diagram for stop and wait protocol:

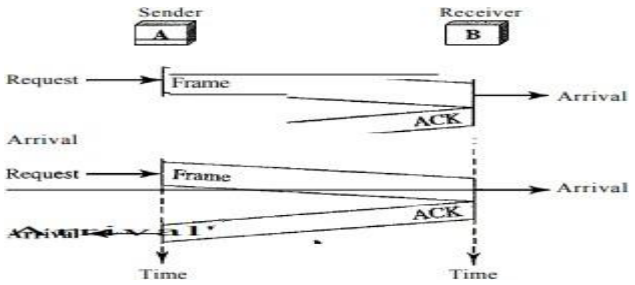


Fig 3.12: Flow diagram for Stop and Wait Protocol

It is still very simple. The sender sends one frame and waits for feedback from the receiver. When the ACK arrives, the sender sends the next frame. Note that sending two frames in the protocol involves the sender in four events and the receiver in two events.

2.3.3 SIMPLEX PROTOCOL FOR NOISY CHANNELS

Noiseless protocols specify how to control the flow of data, whereas noisy protocols specified how to control the error. Noisy protocols are classified as follows:

i) stop and wait automatic Repeat Request (ARQ)

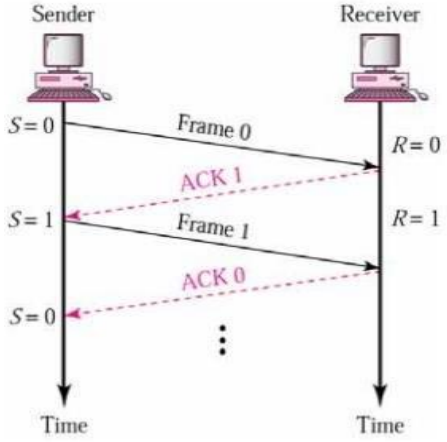
ii) sliding window protocol using Go-back-n ARQ

iii) sliding window protocol using selective Repeat ARQ

2.4 Sliding Window Protocol

2.4.1 Stop-and-Wait Automatic Repeat Request:

- Sender keeps a copy of the last frame until it receives an acknowledgement
- For identification, both data frames and acknowledgements (ACK) frames are numbered alternatively 0 and 1.
- Sender has a control variable (S) that holds the number of the recently sent frame. (0 or 1)
- Receiver has a control variable that holds the number of the next frame expected (0 or 1).
- Sender starts a timer when it sends a frame. If an ACK is not received within a allocated time period, the sender assumes that the frame was lost or damaged and resends it
- Receiver send only positive ACK if the frame is intact.
- ACK number always defines the number of the next expected frame



Stop-and-Wait ARQ,

Fig 3.13: Flow diagram for Stop and Wait ARQ

Damaged frame:

- When a receiver receives a damaged frame, it discards.
- After the timer at the sender expires, another copy of frame 1 is sent.

Lost Frame:

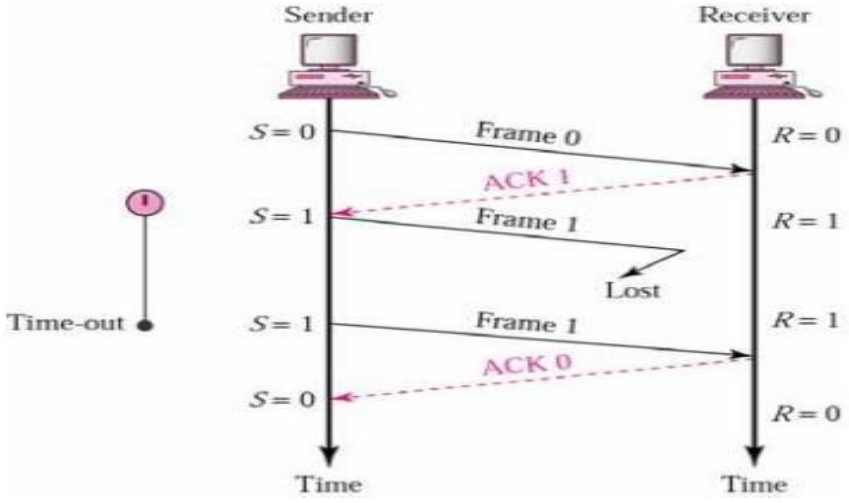


Fig 3.14: Flow diagram for Stop and Wait ARQ lost frame

Stop-and-Wait, lost ACK frame:

- If the sender receives a damaged ACK, it discards it.
- When the timer of the sender expires, the sender retransmits frame 1.
- Receiver has already received frame 1 and expecting to receive frame 0 (R=0). Therefore it discards the second copy of frame 1.

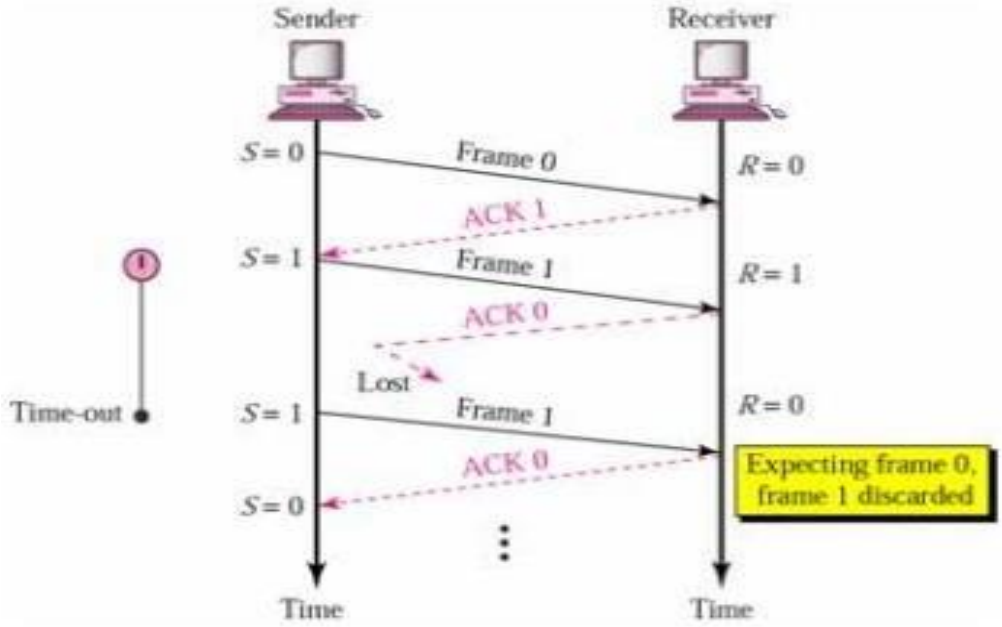


Fig 3.15: Flow diagram for Stop and Wait, Lost ACK

Stop-and-Wait, delayed ACK frame:

- The ACK can be delayed at the receiver or due to some problem

- It is received after the timer for frame 0 has expired.
- Sender retransmitted a copy of frame 0. However, R =1 means receiver expects to see frame 1. Receiver discards the duplicate frame 0.
- Sender receives 2 ACKs, it discards the second ACK

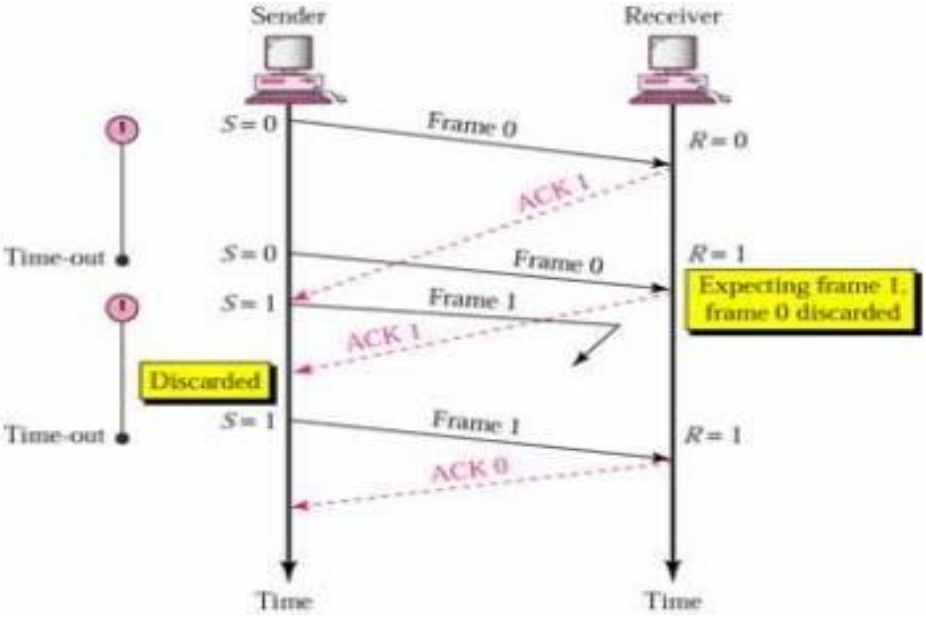


Fig 3.16: Flow diagram for Stop and Wait, Delayed ACK frame

Piggybacking:

- A method to combine a data frame with ACK.
- Station A and B both have data to send.
- Instead of sending separately, station A sends a data frame that includes an ACK.
- Station B does the same thing.
- Piggybacking saves bandwidth.

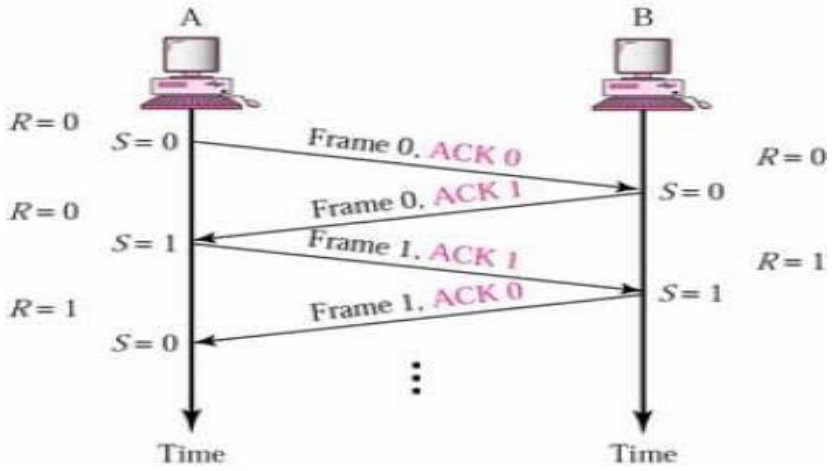


Fig 3.17: Flow diagram for Stop and Wait with Piggy backing

Disadvantage of Stop-and-Wait:

- In stop-and-wait, at any point in time, there is only one frame that is sent and waiting to be acknowledged.
- This is not a good use of transmission medium.
- To improve efficiency, multiple frames should be in transition while waiting for ACK.
- Two protocol use the above concept,
  – Go-Back-N ARQ
  – Selective Repeat ARQ

2.4.2 Go-Back-N ARQ

- We can send up to W frames before worrying about ACKs.
- We keep a copy of these frames until the ACKs arrive.
- This procedure requires additional features to be added to Stop-and-Wait ARQ.

Sequence Numbers:

- Frames from a sender are numbered sequentially.
- We need to set a limit since we need to include the sequence number of each frame in the header.
- If the header of the frame allows m bits for sequence number, the sequence numbers range from 0 to 2m – 1. for m = 3, sequence numbers are: 1, 2, 3, 4, 5, 6, 7.
- We can repeat the sequence number.
  Sequence numbers are:
- 0, 1, 2, 3, 4, 5, 6, 7, 0, 1, 2, 3, 4, 5, 6, 7, 0, 1, ...

Sender Sliding Window

- At the sending site, to hold the outstanding frames until they are acknowledged, we use the concept of a window.
- The size of the window is at most 2m -1 where m is the number of bits for the sequence number.
- Size of the window can be variable, e.g. TCP.
- The window slides to include new unsent frames when the correct ACKs are Received.

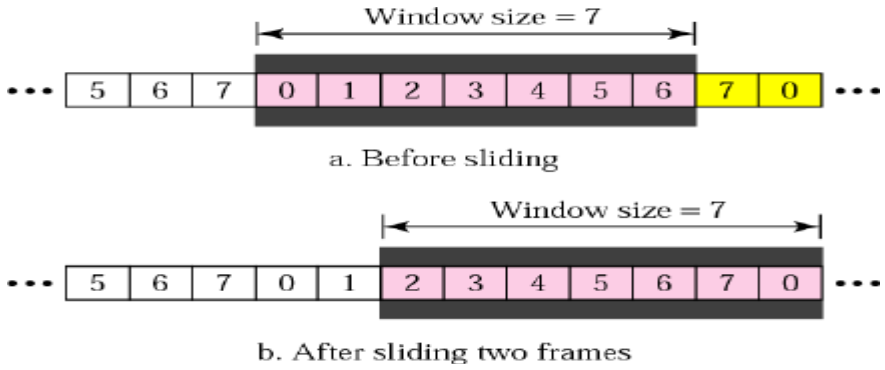


Fig 3.18: Sender Sliding Window

Receiver Sliding Window

- Size of the window at the receiving site is always 1 in this protocol.
- Receiver is always looking for a specific frame to arrive in a specific order.
- Any frame arriving out of order is discarded and needs to be resent.
- Receiver window slides as shown in fig. Receiver is waiting for frame 0 in

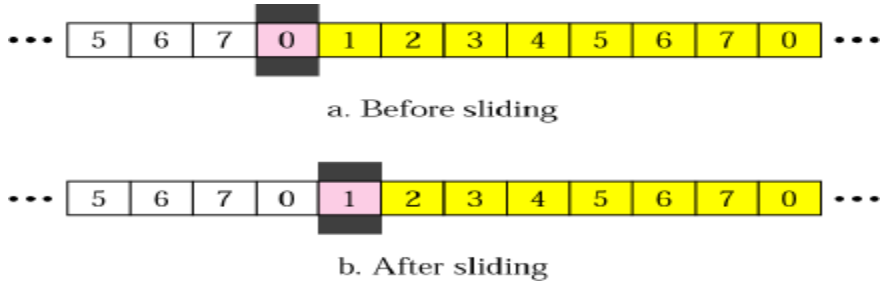


Fig 3.19: Receiver Sliding Window

Control Variables

- Sender has 3 variables: S, SF, and SL
- S holds the sequence number of recently sent frame
- SF holds the sequence number of the first frame
- SL holds the sequence number of the last frame
- Receiver only has the one variable, R, that holds the sequence number of theframe it expects to receive. If the seq. no. is the same as the value of R, theframe is accepted, otherwise rejected.

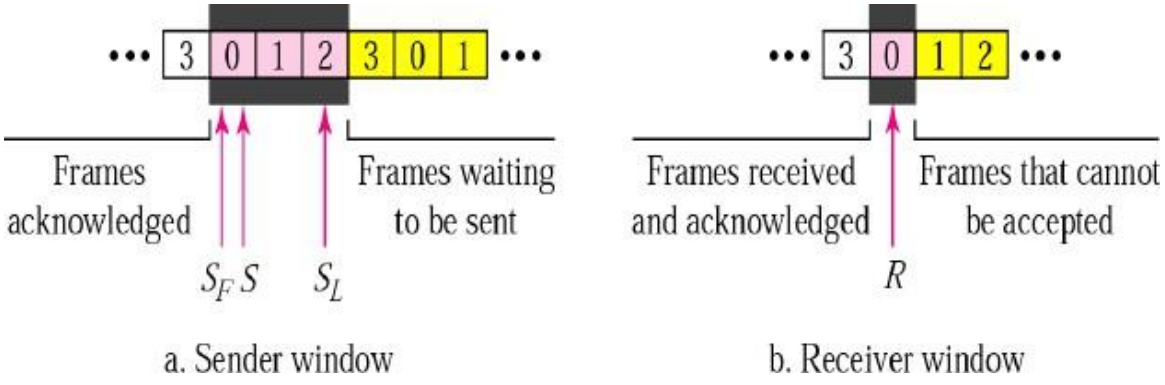


Fig 3.20: Control Variables of Sender window and Receiver window

- Receiver sends positive ACK if a frame arrived safe and in order.
- If the frames are damaged/out of order, receiver is silent and discards all subsequent frames until it receives the one it is expecting.
- The silence of the receiver causes the timer of the unacknowledged frame to expire.
- Then the sender resends all frames, beginning with the one with the expired timer.
- For example, suppose the sender has sent frame 6, but the timer for frame 3 expires (i.e. frame 3 has not been acknowledged), then the sender goes back and sends frames 3, 4, 5, 6 again. Thus it is called Go-Back-N-ARQ
- The receiver does not have to acknowledge each frame received, it can send one cumulative ACK for several frames.

Go-Back-N ARQ, normal operation:

- The sender keeps track of the outstanding frames and updates the variables and windows as the ACKs arrive.

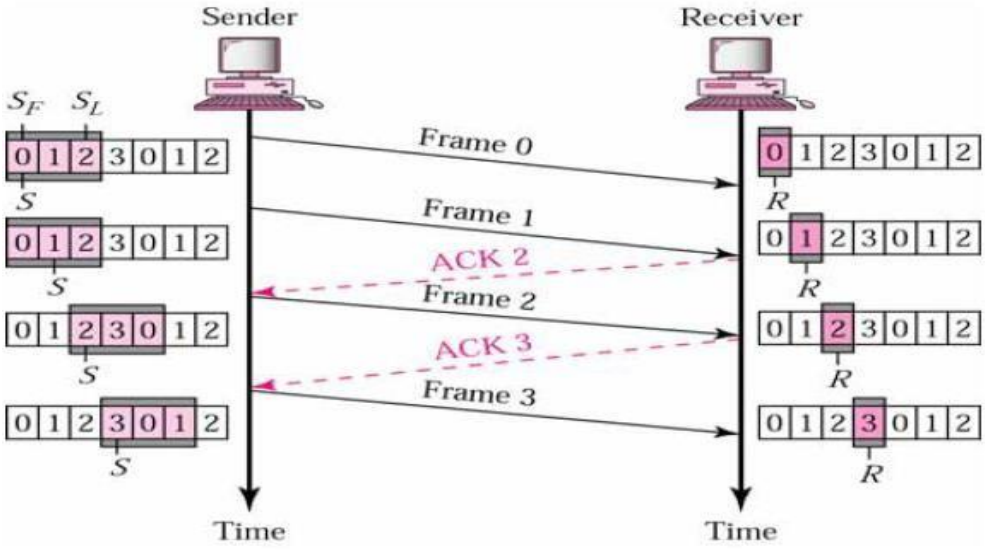


Fig 3.21: Go-Back-N ARQ, normal operation

Go-Back-N ARQ, lost frame *(Example)*

- Frame 2 is lost
- When the receiver receives frame 3, it discards frame 3 as it is expecting frame 2 (according to window).
- After the timer for frame 2 expires at the sender site, the sender sends frame 2 and 3. (go back to 2)

Go-Back-N ARQ, damaged/lost/delayed ACK

- If an ACK is damaged/lost, we can have two situations:
- If the next ACK arrives before the expiration of any timer, there is no need
- for retransmission of frames because ACKs are cumulative in this protocol.
- If ACK1, ACK2, and ACk3 are lost, ACK4 covers them if it arrives before the timer expires.
- If ACK4 arrives after time-out, the last frame and all the frames after that are resent.
- Receiver never resends an ACK.
    - A delayed ACK also triggers the resending of frames.

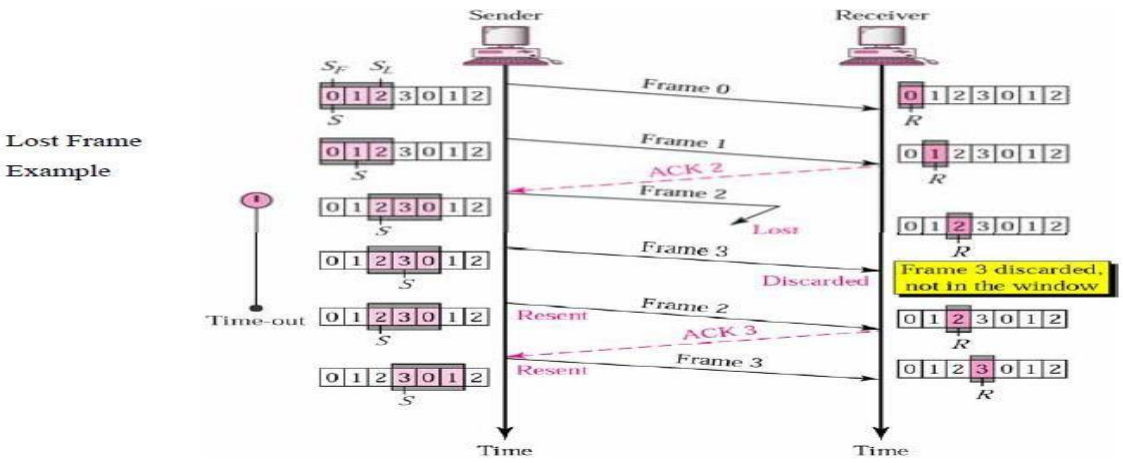


Fig 3.22: Go-Back-N ARQ, damaged/lost/delayed ACK

2.4.3 Selective Repeat ARQ, sender and receiver windows

- Go-Back-N ARQ simplifies the process at the receiver site. Receiver only keeps track of only one variable, and there is no need to buffer out-of-order frames, they are simply discarded.
- However, Go-Back-N ARQ protocol is inefficient for noisy link. It bandwidth inefficient and slows down the transmission.
- In Selective Repeat ARQ, only the damaged frame is resent. More bandwidth
- efficient but more complex processing at receiver.
- It defines a negative ACK (NAK) to report the sequence number of a
- damaged frame before the timer expires.

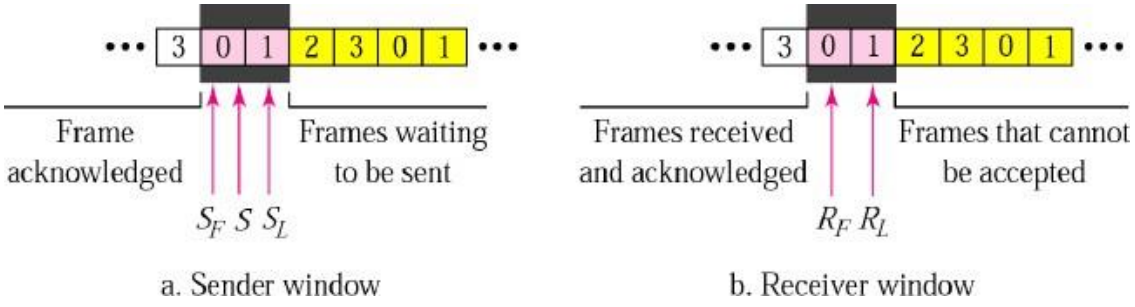


Fig 3.23: Control Variables of Sender window and Receiver window

Selective Repeat ARQ, lost frame:

- Frames 0 and 1 are accepted when received because they are in the range specified by the receiver window. Same for frame 3.
- Receiver sends a NAK2 to show that frame 2 has not been received and then
- sender resends only frame 2 and it is accepted as it is in the range of the window.

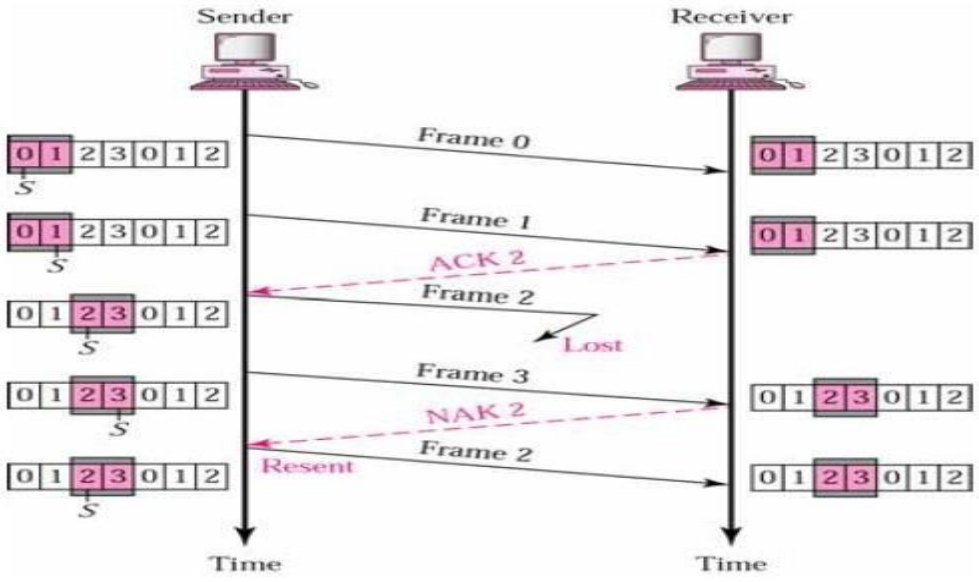


Fig 3.24: Selective Repeat ARQ, lost frame

2.5 HDLC (High Level Data Link Control)

- HDLC is a bit-oriented protocol
- It was developed by the International Organization for Standardization (ISO).
- It falls under the ISO standards ISO 3309 and ISO 4335.
- it supports both half-duplex and full-duplex communication lines, point-to-point (peer to peer) and multi-point networks, and switched or non-switched channels.
- HDLC supports several modes of operation, including a simple sliding-window mode for reliable delivery.

HDLC Stations and Configurations:

- HDLC specifies the following three types of stations for data link control:
  - Primary Station
  - Secondary Station
  - Combined Station

Primary Station:

- primary station, it is used as the controlling station on the link.
- It has the responsibility of controlling all other stations on the link (usually secondary stations).
- A primary issues commands and secondary issues responses.

- the primary station is also responsible for the organization of data flow on the link.
- It also takes care of error recovery at the data link level (layer 2 of the OSI model).

Secondary Station:

- If the data link protocol being used is HDLC, and a primary station is present, a secondary station must also be present on the data link
- The secondary station is under the control of the primary station.
- It has no ability, or direct responsibility for controlling the link.
- It is only activated when requested by the primary station.
- It only responds to the primary station
- The secondary station's frames are called responses.
- It can only send response frames when requested by the primary station.
- A primary station maintains a separate logical link with each secondary station.

Combined Station:

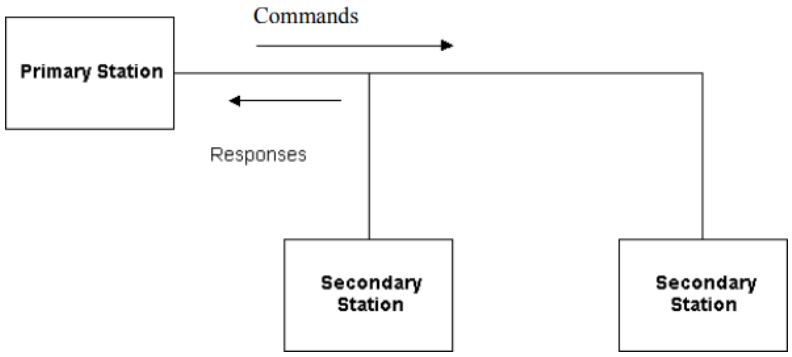
- A combined station is a combination of a primary and secondary station.
- On the link, all combined stations are able to send and receive commands and responses without any permission from any other stations on the link.
- Each combined station is in full control of itself, and does not rely on any other stations on the link
- No other stations can control any combined station.
- May issue both commands and responses.

HDLC Configurations:

- The word configuration refers to the relationship between the hardware devices on a link.
- HDLC also defines three types of configurations for the three types of stations.
- Following are the three configurations defined by HDLC:
  - Unbalanced Configuration
  - Balanced Configuration
  - Symmetrical Configuration

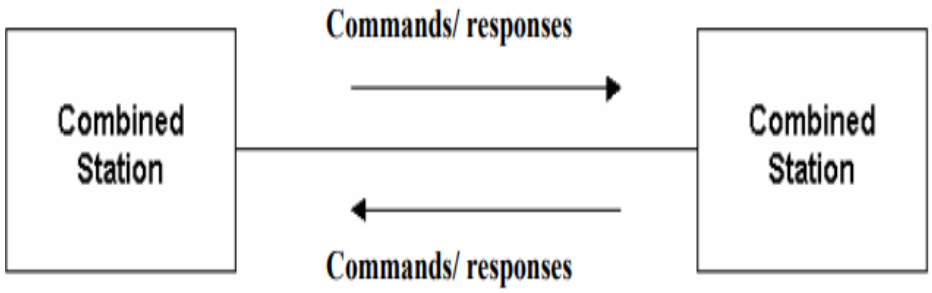
Unbalanced Configuration:

- The unbalanced configuration in an HDLC link consists of a primary station and one or more secondary stations
- The unbalanced condition arises because one station controls the other stations.
- In an unbalanced configuration, any of the following can be used:
  - Full-Duplex or Half-Duplex operation
  - Point to Point or Multi-point networks



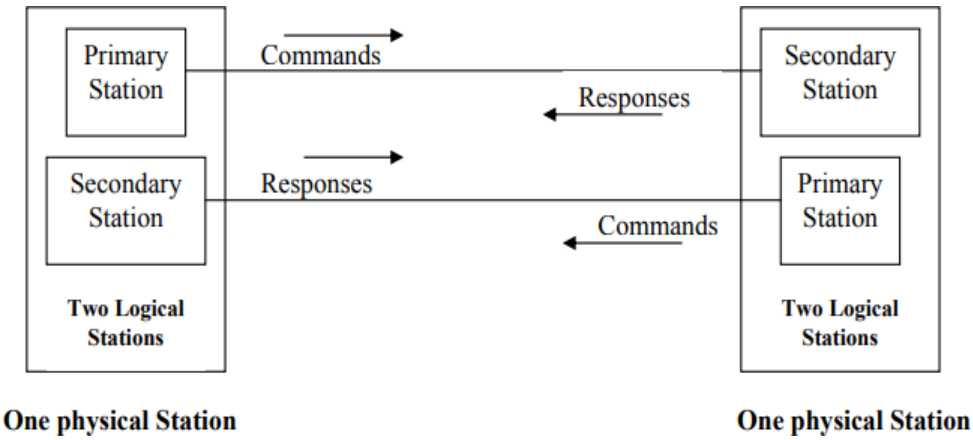
Balanced Configuration:

- The balanced configuration in an HDLC link consists of two or more combined stations.
- Each of the stations has equal and complimentary responsibility compared to each other.
- Full - Duplex or Half - Duplex operation
- Point to Point networks
- HDLC does not support balanced multipoint



Symmetrical Configuration:

- This third type of configuration is not widely in use today.
- It consists of two independent point-to-point, unbalanced station configurations.
- Each station is logically considered as two stations.



HDLC Operational Modes:

- A mode in HDLC is the relationship between two devices involved in an exchange.
- the mode describes who controls the link
- Exchanges over unbalanced configurations are always conducted in normal response mode.
- Exchanges over symmetric or balanced configurations can be set to specific mode using a frame design to deliver the command.

1. Normal Response Mode (NRM)
2. Asynchronous Response Mode (ARM)
3. Asynchronous Balanced Mode (ABM)

Normal Response Mode:

- This is the mode in which the primary station initiates transfers to the secondary station.
- The secondary station can only transmit a response when, and only when, it is instructed to do so by the primary station.

- After receiving permission from the primary station, the secondary station initiates its transmission
- This transmission from the secondary station to the primary station may be much more than just an acknowledgment of a frame.

Normal Response Mode is only used within an unbalanced configuration

Asynchronous Response Mode:

- A secondary may initiate a transmission without permission from the primary whenever channel is idle.
- ARM does not alter the primary-Secondary relation ship in any other way.

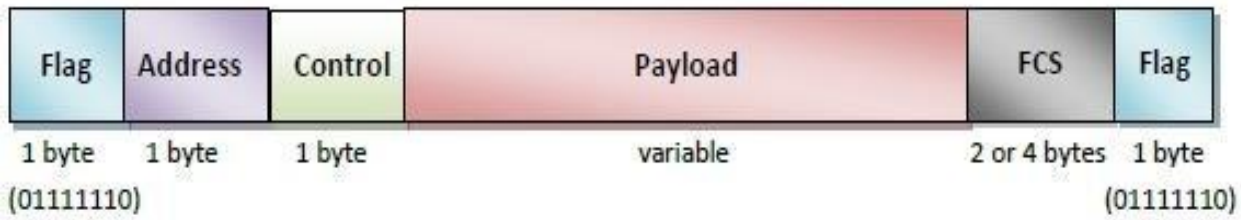
Asynchronous Balanced Mode:

- This mode is used in case of combined stations.
- There is no need for permission on the part of any station in this mode.
- This is because combined stations do not require any sort of instruction
- Normal Response Mode is used most frequently in multi-point lines to perform any task on the link.
- Asynchronous Response Mode is better for point-to point lines

HDLC Modes:

	NRM	ARM	ABM
Station Type	Primary & Secondary	Primary & Secondary	Combined
Initiator	Primary	Either	Any

Frame Format of HDLC:



***Flag* – It is an 8-bit sequence that marks the beginning and the end of the frame. The bit pattern of the flag is 01111110.**

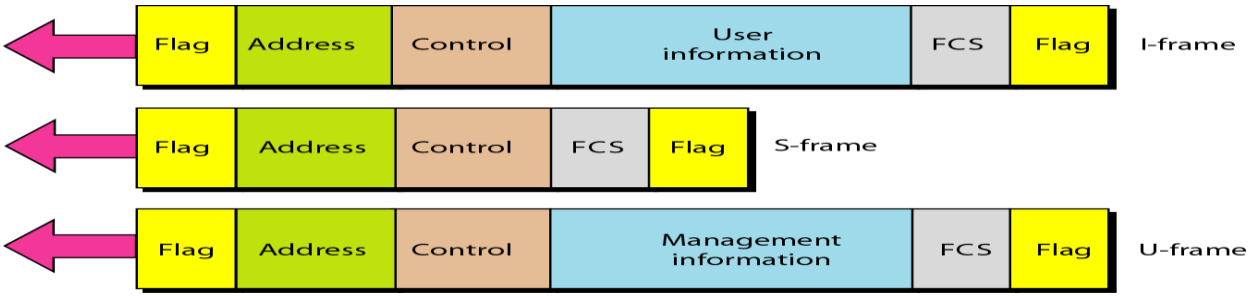
Address: It contains the address of the receiver.If the frame is sent by the primary station,it contains the addresses of the secondary stations.If it is sent by secondary station ,it contains the address of the primary station .The address fiels may be 1 byte

to several bytes.

- **Control –** It is 1 or 2 bytes containing flow and error control information**.**
- **Payload –** This carries the data from the network layer. Its length may vary from one network to another.
- **FCS –** It is a 2 byte or 4 bytes frame check sequence for error detection. The standard code used is CRC (cyclic redundancy code)

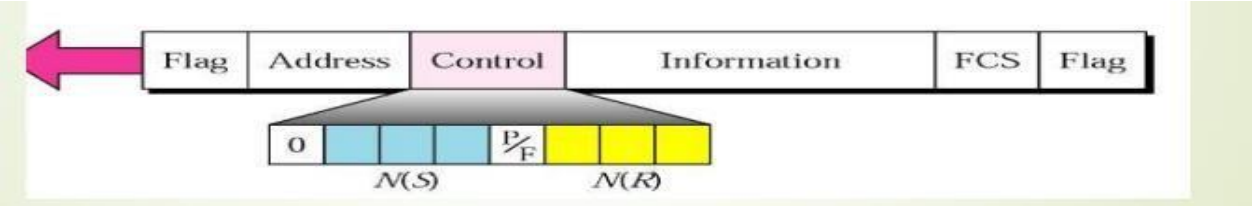
Types of HDLC Frames

- There are three types of HDLC frames. The type of frame is determined by the control field of the frame
- I-Frame
- S-Frame
- U-Frame



Information Frames:

- I-frames carry user's data and control information about user's data.
- I-frame carries user data in the information field.
- The I-frame format is shown in diagram



Control Field:

- Control field is a one or two byte segment of the frame used for flow management.
- Control fields differ depending on frame type
- If the first bit of the control filed is 0,the frame is an I frame.

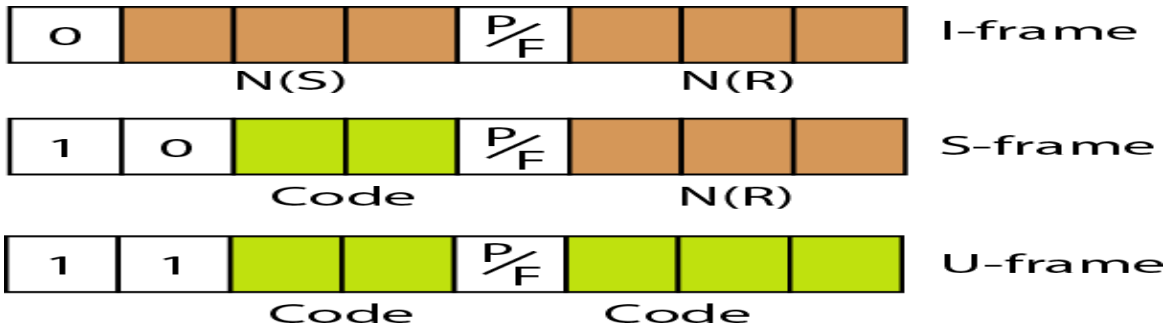
- Control fields of all three types of frames contain a bit called the POLL /FINAL(p/F) bit.
- N(s) → Number of frame being sent
- N(R)→ Number of the frame expected in return Acknowledgement field
- If the last frame receive is error free ,the N(R) number will be that of the next frame in the sequence.
- If the last frame receive was not received Correctly ,the N(R) number will be the number of the damaged frame.

S frame:

- S-frames or Supervisory frames do not contain information field.
- They are used for flow and error control when piggybacking is not required.
- The first two bits of control field of S-frame is 10.



Control field format for the different frame types



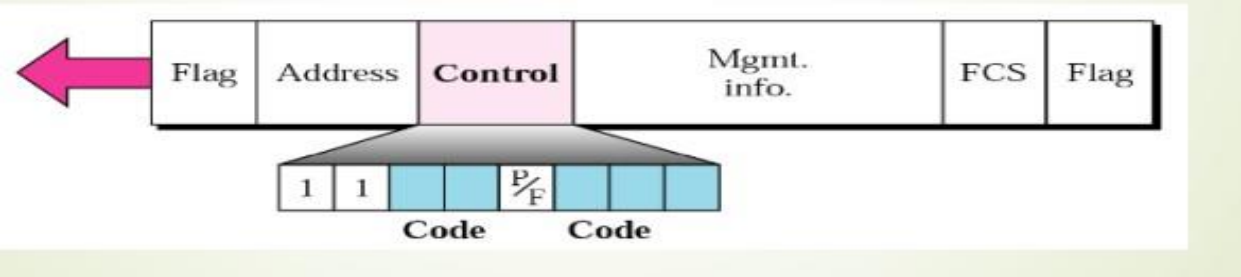
Different supervisory frames:

- SS=00, Receiver Ready (RR), and N(R) ACKs all frames received up to and including the one with sequence number N(R)
- SS=10, Receiver Not Ready (RNR), and N(R) has the same meaning as above
- SS=01, Reject; all frames with sequence number N(R) or higher are rejected, which in turns ACKs frames with sequence number N(R) -1 or lower.
- SS=11, Selective Reject; the receive rejects the frame with sequence number N(R)

U-frame:

- U-frames or Un-numbered frames are used for miscellaneous functions, like link management.
- It may contain an information field, if required.
- The first two bits of control field of U-frame is 11.
- U-frames are reserved for system management and information carried by them is used for managing the link
- U-frames are used to exchange session management and control information between the two connected devices.

Information field in U-frame does not carry user information rather, it carries system management information



U-frame control command and response

Code	*Command*	*Response*	*Meaning*
00 001	SNRM		Set normal response mode
11 011	SNRME		Set normal response mode, extended
11 100	SABM	DM	Set asynchronous balanced mode or disconnect mode
11 110	SABME		Set asynchronous balanced mode, extended
00 000	UI	UI	Unnumbered information
00 110		UA	Unnumbered acknowledgment
00 010	DISC	RD	Disconnect or request disconnect
10 000	SIM	RIM	Set initialization mode or request information mode
00 100	UP		Unnumbered poll
11 001	RSET		Reset
11 101	XID	XID	Exchange ID
10 001	FRMR	FRMR	Frame reject

2.6 POINT-TO-POINT PROTOCOL (PPP)

- One of the most common protocols for point-to-point access is the Point-to-Point Protocol (PPP).

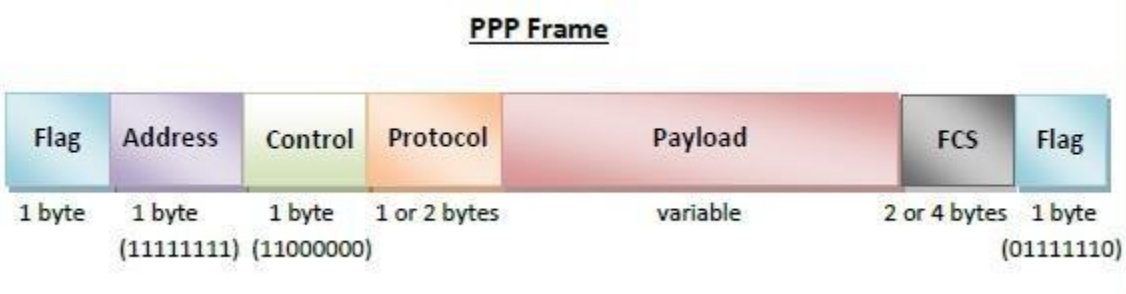
Services: Services Provided by PPP

PPP defines the format of the frame to be exchanged between devices. It also defines how two devices can negotiate the establishment of the link and the exchange of data.

The new version of PPP, called Multilink PPP, provides connections over multiple links.

- **Services Not Provided by PPP**
  - PPP does not provide flow control.

PPP Frame Format



- **Flag:A**PPP frame starts and ends with a 1-byte flag with the bit pattern 01111110.
- **Address:**The address field in this protocol is a constant value and set to 11111111 (broadcast address). Control:This field is set to the constant value 00000011.
- **Protocol:**The protocol field defines what is being carried in the data field: either user data or other information. This field is by default 2 bytes long, but the two parties can agree to use only 1 byte.
- **Payload field**: This field carries either the user data or other information.
- **FCS**: The frame check sequence (FCS) is simply a 2-byte or 4-byte standard CRC.

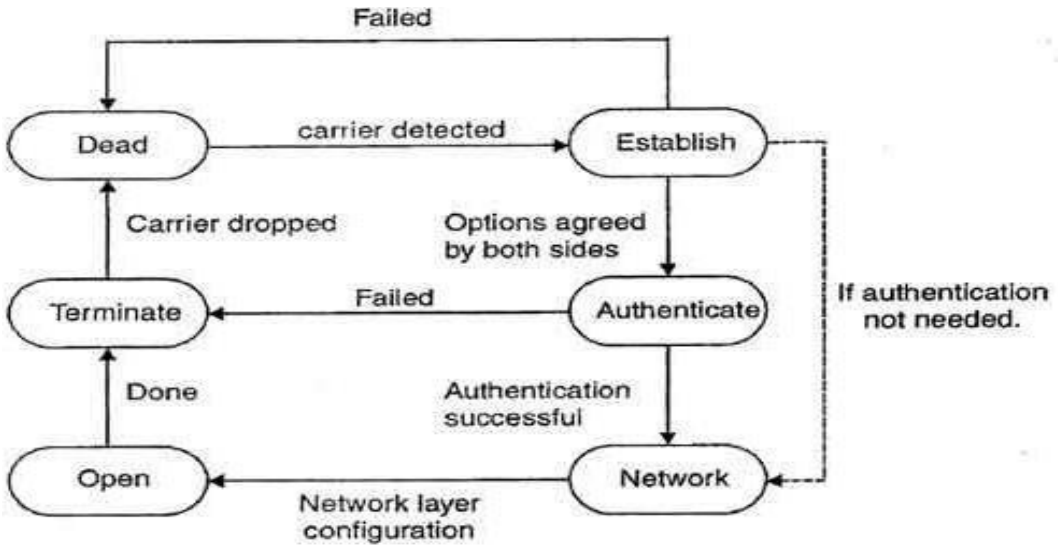
Byte Stuffing:

- Since PPP is a byte-oriented protocol, the flag in PPP is a byte that needs to be escaped whenever it appears in the data section of the frame.
- The escape byte is 01111101, which means that every time the flag like pattern appears in the data, this extra byte is stuffed to tell the receiver that the next byte is not a flag.

Transition Phases:

The PPP connection goes through different states as shown in fig.

1. Dead: In dead phase the link is not used. There is no active carrier and the line is quiet.



Transition phases

2. **Establish**: Connection goes into this phase when one of the nodes start communication. In this phase, two parties negotiate the options. If negotiation is successful, the system goes into authentication phase or directly to networking phase. LCP packets are used for this purpose.

3. **Authenticate**: This phase is optional. The two nodes may decide during the establishment phase, not to skip this phase. However, if they decide to proceed with authentication, they send several authentication packets. If the result is successful, the connection goes to the networking phase; otherwise, it goes to the termination phase.

4. **Network**: In network phase, negotiation for the network layer protocols takes place. PPP specifies that two nodes establish a network layer agreement before data at the network layer can be exchanged. This is because PPP supports several protocols at network layer. If a node is running multiple protocols simultaneously at the network layer, the receiving node needs to know which protocol will receive the data.

5. **Open**: In this phase, data transfer takes place. The connection remains in this phase until one of the endpoints wants to end the connection.

6. **Terminate**: In this phase connection is terminated.

Point-to-point protocol Stack:

PPP uses several other protocols to establish link, authenticate users and to carry the network layer data.

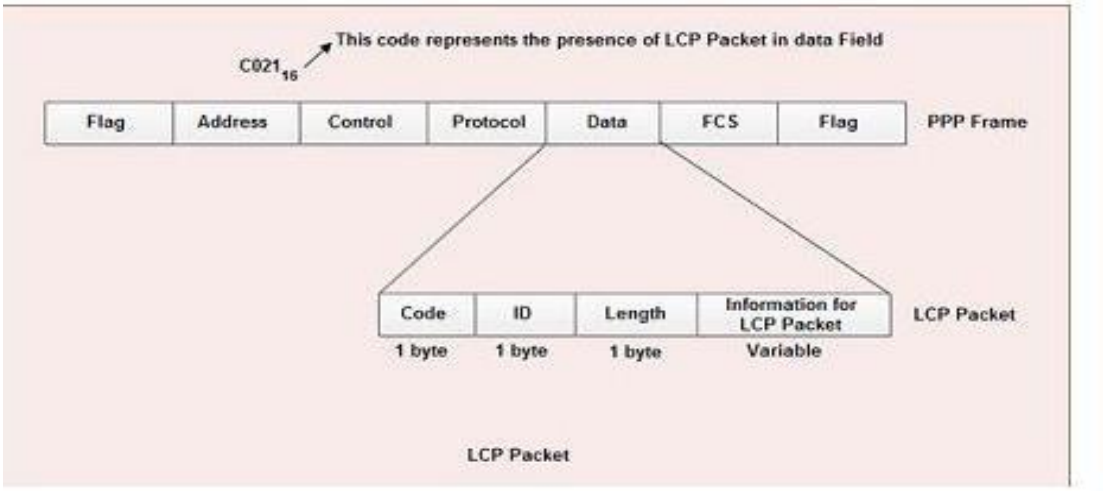
The various protocols used are:

1. Link Control Protocol
2. Authentication Protocol
3. Network Control Protocol

1. Link Control Protocol

It is responsible for establishing, maintaining, configuring and terminating the link.

It provides negotiation mechanism to set options between two endpoints.



• All LCP packets are carried in the data field of the PPP frame.

• The presence of a value $C021_{16}$ in the protocol field of PPP frame indicates that LCP packet is present in the data field.

• The various fields present in LCP packet are:

1. **Code**: 1 byte-specifies the type of LCP packet.

2. I**D**: 1 byte-holds a value used to match a request with the reply.

3. **Length**: 2 byte-specifies the length of entire LCP packet.

4. **Information**: Contains extra information required for some LCP packet.

• There are eleven different type of LCP packets. These are categorized in three groups:

1. **Configuration packet**: These are used to negotiate options between the two ends. For example: configure-request, configure-ack, configure-nak, configure-reject are some configuration packets.

2. **Link termination packets**: These are used to disconnect the link between two end points. For example: terminate-request, terminate-ack, are some link termination packets.

3. **Link monitoring and debugging packets**: These are used to monitor and debug the links. For example: code-reject, protocol-reject, echo-request, echo-reply and discard-request are some link monitoring and debugging packets.

2. Authentication Protocol

Authentication protocols help to validate the identity of a user who needs to access the resources.

There are two authentication protocols:

1. Password Authentication Protocols (PAP)

2. Challenge Handshake Authentication Protocol (CHAP)

1. PAP (Password Authentication Protocol)

This protocol provides two step authentication procedures:

Step 1: User name and password is provided by the user who wants to access a system.

Step 2: The system checks the validity of user name and password and either accepts or denies the connection.

• PAP packets are also carried in the data field of PPP frames.

• The presence of PAP packet is identified by the value $C023_{16}$ in the protocol field of PPP frame.

• There are three PAP packets.

1. **Authenticate-request**: used to send user name & password.

2. **Authenticate-ack**: used by system to allow the access.

3. **Authenticate-ack**: used by system to deny the access.

2. CHAP (Challenge Handshake Authentication Protocol)

• It provides more security than PAP.

• In this method, password is kept secret, it is never sent on-line.

• It is a three-way handshaking authentication protocol:

1. System sends. a challenge packet to the user. This packet contains a value, usually a few bytes.

2. Using a predefined function, a user combines this challenge value with the user password and sends the resultant packet back to the system.

3. System then applies the same function to the password of the user and challenge value and creates a result. If result is same as the result sent in the response packet, access is granted, otherwise, it is denied.

• **There are 4 types of CHAP packets:**

1. Challenge-used by system to send challenge value.

2. Response-used by the user to return the result of the calculation.

3. Success-used by system to allow access to the system.

4. Failure-used by the system to deny access to the system.

3. Network Control Protocol (NCP)

• After establishing the link and authenticating the user, PPP connects to the network layer. This connection is established by NCP.

• Therefore NCP is a set of control protocols that allow the encapsulation of the data coming from network layer.

• After the network layer configuration is done by one of the NCP protocols, the users can exchange data from the network layer.

• PPP can carry a network layer data packet from protocols defined by the Internet, DECNET, Apple Talk, Novell, OSI, Xerox and so on.

• None of the NCP packets carry networks layer data. They just configure the link at the network layer for the incoming data.

CN UNIT-II

Assignment-Cum-Tutorial Questions

PART-A

Objective Questions (1M)

1. The.....layer provides a well-defined service interface to the network layer, determining how the bits of the physical layer are grouped into frames. [CO 2/BL 1]

   a. Data Link b. Physical
   c. Network d. Session

2. In block coding, we divide our message into blocks, each of k bits, called ..... [CO 2/BL 1]

   a. Data word b. Generator c. Code word d. Checker

3.....in the data link layer separates a message from one source to a destination, or from other messages to other destinations, by adding a sender address and a destination address.

   a. Transforming b. Framing c. Separating d. Messaging [CO 2/BL 1]

4. In ....., there is no need for defining the boundaries of the frames; the size itself can be used a delimiter. [CO 2/BL 1]

   a. Standard Size Framing b. Fixed Size Framing
   c. Variable Size Framing d. Constant Size Framing

5. In.....the source machine sends independent frames to the destination machine without having the destination machine acknowledge them. [CO 2/BL 1]

a. Unacknowledged connectionless service b. Acknowledged connectionless service
c. Acknowledged connection oriented service d. Unacknowledged connection oriented service

6.....is the most sophisticated service provided by the data link layer to the network layer. The source and destination machines establish a connection before any data transfer takes place.

a. Unacknowledged connectionless service b. Acknowledged connectionless service
c. Acknowledged connection oriented service d. Unacknowledged connection oriented service [CO 2/BL 1]

7.....is prevalent in LANs, we need a way to define the end of the frame and the beginning of the next. [CO 2/BL 1]

a. Standard Size Framing b. Fixed Size Framing
c. Variable Size Framing d. Constant Size Framing

8. When 2 or more bits in a data unit has been changed during the transmission, the error is called…….. [CO 2/BL1]

a. random error b. burst error

c. inverted error d. none of the mentioned

9. In ……., we need to know the exact number of bits that are corrected and more importantly, their location in the message. [CO 2/BL 1]

a. error searching b. error detection

c. error correction d. error transmission

10. Which of the following is/are the methods used for carrying out framing.

[CO 2/BL 1]

a. Character count b. Starting and ending characters, with character stuffing.

c. Starting and ending flags with bit stuffing. d. All of the above

11. CRC stands for …….. [CO 2/BL 1]

a. cyclic redundancy check b. code repeat check

c. code redundancy check d. cyclic repeat check

12. The checksum of 1111 and 1111 is _____.

a. 0000 b. 1111 c. 1110 d. 0111

13. In cyclic redundancy checking, the divisor is _____the CRC.

a. one bit less than b. one bit more than c. the same size as d . none of the above

14. The _____between two words is the number of differences between corresponding bits.

a. hamming rule b. hamming code c. hamming distance d. none of the above

15. The _____of a polynomial is the highest power in the polynomial.

a. range b. power c. degree d. none of the above

16. In modulo-2 arithmetic, _____give the same results.

a. addition and subtraction b. addition and multiplication

c. addition and division d. none of the above

17. In cyclic redundancy checking, what is the CRC?

a. The quotient b The dividend c. The divisor d. The remainder

18. Which error detection method consists of just one redundant bit per data unit?

a. CRC b. Checksum c. Simple parity check d. Two-dimensional parity check

19. An error-detecting code inserted as a field in a block of data to be transmitted is known as

a. Frame check sequence b. Error detecting code
c. Checksum d. flow control

20. Which one of the following task is not done by data link layer?
A. framing B. error control C. flow control D. channel coding

21. The different types of services provided by data link layer is/are ... [CO 2/BL 1]

a. Unacknowledged connectionless service
b. Acknowledged connectionless service
c. Acknowledged connection oriented service
d. All of the above.

22.....is used to indicate to the network layer that an event has happened, for example, establishment or release of a connection. [CO 2/BL 1]

a. Request b. Indication
c. Response d. Confirm

23.....is the process in which the receiver tries to guess the message by using redundant bits.

[CO 2/BL 1]

a. Forward error correction b. Backward error correction

c. Transmission d. Retransmission

24. In....., the sender sends one frame, stops until it receives confirmation from the receiver, and then sends the next frame. [CO 2/BL 1]

a. Stop and wait protocol b. Simplest protocol
c. Sliding window protocol d. high level data link control protocol(HDLC)

25. In....., the sliding window is an abstract concept that defines the range of sequence numbers that is the concern of the sender and receiver.

a. stop and wait protocol b. simplest protocol
c. sliding window protocol d. high level data link control protocol(HDLC)

PART-B

Short Answer Questions (2M)

1. Define framing and the reason for its need. [CO 2/BL 2]
2. List out different topologies used in computer network. [CO 2/BL 2]
3. Define simplex, half duplex and full duplex communications in digital systems. [CO 2/BL 2]
4. Draw the structure of ISO reference model and TCP/TP reference model. [CO 2/BL 2]
5. How the total internal reflections are done in fiber optical cable. [CO 2/BL 2]
6. What is meant by bit stuffing and byte stuffing. [CO 2/BL 2]
7. List rules in Hamming code generation process. [CO 2/BL 2]

PART-C

Descriptive Questions (5M)

1. Compare and contrast byte-stuffing and bit-stuffing. Which technique is used in byte-oriented protocols? which technique is used in bit-oriented protocols? [CO 3/BL 2]
2. Briefly describe the services provided by the data link layer [CO 2/BL 2]
3. What is the Hamming distance? What is the minimum Hamming distance? [CO 2/BL 1]
4. Define CRC. Describe how CRC works for error detection with an example. [CO 2/BL 1]
5. Explain the working of checksum. [CO 2/BL 2]
6. Explain the working of simple parity checksum with encoder and decoder. [CO 2/BL 1]
7. Find the minimum Hamming distance of the coding scheme in the table [CO 2/BL 3]

Datawords	*Codewords*
00	000
01	011
10	101
11	110

8. Find the minimum Hamming distance of the coding scheme in the table [CO 2/BL 3]

Dataword	*Codeword*
00	00000
01	01011
10	10101
11	11110

9.A code scheme has a Hamming distance dmin == 4. What is the error detection and correction capability of this scheme? [CO 2/BL 3]

10.We need a data word of at least 7 bits. Calculate values of k and n that satisfy this requirement. [CO 2/BL 2]

11.Which of the following g(x) values guarantees that a single-bit error is caught? For each case, what is the error that cannot be caught? [CO 2/BL 3]

a. x + 1

b. x^3

c. 1

12.Find the status of the following generators related to two isolated, single-bit errors. [CO 2/BL 3]

a. x+ 1

b. x^4 + I

c. $x^7 + x^6 + 1$

d. $x^{15}+x^{I4}+1$

13.Find the suitability of the following generators in relation to burst errors of different lengths. [CO 2/BL 3]

a. $x^6 + 1$

b. $x^{18} + x^7 + x + 1$

c. $x^{32}+x^3+x^7+1$

14.How can we represent the number 21 in one's complement arithmetic using only four bits? [CO 2/BL 2]

15.How can we represent the number -6 in one's complement arithmetic using only four bits? [CO 2/BL 2]

16.Apply the following operations on the corresponding polynomials: [CO 2/BL 3]

a. $(x^3 + x^2 + X + 1) + (x^4 + x^2 + x + 1)$

b. $(x^3 + x^2 + x + 1) - (x^4 + x^2 + x + 1)$

c. $(x^3 + x^2)$ X $(x^4 + x^2 + x + 1)$

d. $(x^3 + x^2 + x + 1) / (x^2 + 1)$

17.Given the data word 1010011110 and the divisor 10111, [CO 2/BL 3]

18.Show the generation of the code word at the sender site (using binary division). [CO 3/BL 3]

19. Show the checking of the code word at the receiver site (assume no error). [CO 2/BL 3]

20. Describe the two protocols in noiseless channels with their algorithms. [CO 2/BL 2]

PART-D
Competitive Examination Questions

1. A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 01111100101, then the input bit-string is .....[GATE-2014]

2. The maximum window size for data transmission using the selective reject protocol with n-bit frame sequence numbers is..... [GATE-2005]

3. Suppose we are transmitting frames between two nodes using Stop-and-Wait protocol. The frame size is 3000 bits. The transmission rate of the channel is 2000 bps (bits/ second) and the propagation delay between the two nodes is 100 milliseconds. Assume that the processing times at the source and destination are negligible. Also, assume that the size of the acknowledgement packet is negligible. The following most accurately gives the channel utilization for the above scenario in percentage is ..... [GATE-2025]

4. Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial X3 + X + 1. Suppose the message m4m3m2m1m0 = 11000 is to be transmitted. Check bits c2c1c0 are appended at the end of the message by the transmitter using the above CRC scheme. The transmitted bit string is denoted by m4m3m2m1m0c2c1c0. The value of the check bit sequence c2c1c0 is ..... [GATE-2021]

5. Assume that a 12-bit Hamming code word consisting of 8-bit data and 4 check bits is d8d7d6d5c8d4d3d2c4d1c2c1, where the data bits and the check bits are given in the following tables:

Data bits							
d_8	d_7	d_6	d_5	d_4	d_3	d_2	d_1
1	1	0	x	0	1	0	1

C_8	C_4	C_2	C_1
Y	0	1	0

The correct values of x and y are..... [GATE-2021]

6. A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps (1Kbps = 1000 bits/second). Size of an acknowledgment is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 milliseconds. Assuming no frame is lost, the sender throughput is _____ bytes/second. [GATE-2016]

7. Consider a network connected two systems located 8000 kilometers apart. The bandwidth of the network is 500 × 106 bits per second. The propagation speed of the media is 4 × 106 meters per second. It is needed to design a Go-Back-N sliding window protocol for this network. The average packet size is 107 bits. The network is to be used to its full capacity. Assume that processing delays at nodes are negligible. Then the minimum size in bits of the sequence number field has to be.....[GATE-2015]

8. Suppose that the stop-and-wait protocol is used on a link with a bit rate of 64 kilobits per second and 20 milliseconds propagation delay. Assume that the transmission time for the acknowledgment and the processing time at nodes are negligible. Then the minimum frame size in bytes to achieve a link utilization of at least 50% is _____.
[GATE-2015]

9. Consider a selective repeat sliding window protocol that uses a frame size of 1 KB to send data on a 1.5 Mbps link with a one-way latency of 50 msec. To achieve a link utilization of 60%, the minimum number of bits required to represent the sequence number field is.....[GATE-2014]

10. The message 11001001 is to be transmitted using the CRC polynomial x3 + 1 to protect it from errors. The message that should be transmitted is… [GATE-2007]

SESHADRI RAO GUDLAVALLERU ENGINEERING COLLEGE

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Academics Strengthening & Advancement (AS&A)

Departments of CSE, CSE (AI&ML) and IT

R – 23

III B.Tech. I Semester

COMPUTER NETWORKS

Learning Material

Prepared by: Faculty of CSE

UNIT-3

MEDIA ACCESS CONTROL

Course Objectives

- To familiar with the basics of media access protocols and Wired LAN'S.

Course Outcomes

Students can able to

- Apply different Random-access protocols in data communications
- understand the different types of control access protocols
- understand the different types of Channelization protocols
- understand the Wired LAN'S

Syllabus

3.1 Media access control-Random access

3.1.1 ALOHA

3.1.2 CSMA

3.1.3 CSMA/CD

3.1.4 CSMA/CA

3.2 Controlled Access

3.2.1 Reservation

3.2.2 Polling

3.2.3 Token passing

3.3 Channelization

3.3.1 FDMA

3.3.2 TDMA

3.3.3 CDMA

3.4 Wired LAN'S

3.4.1 Ethernet Protocol

3.4.2 Standard Ethernet

3.4.3 Fast Ethernet

3.4.4 Gigabit Ethernet

3.4.5 10 Gigabit Ethernet

3.1 Multiple Access Protocols

- We can consider the data link layer as two sub layers.
- The upper sub layer is responsible for data link control
- The lower sub layer is responsible for resolving access to the shared media.
- If the channel is dedicated, we do not need the lower sub layer.

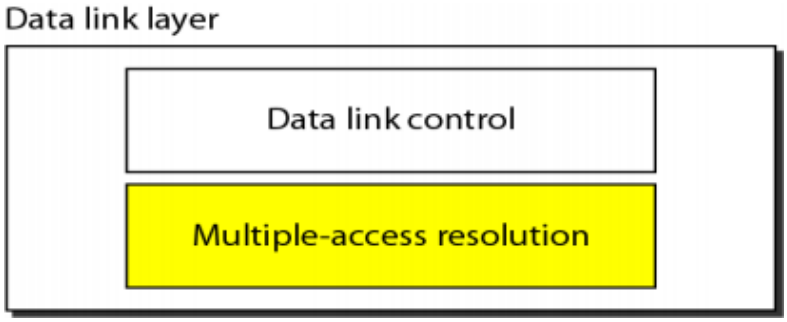


Figure 3.1: Sub layers of Data link layer

- When nodes or stations are connected and use a common link, called a multipoint or broadcast link, we need a multiple-access protocol to coordinate access to the link.
- The problem of controlling the access to the medium is similar to the rules of speaking in an assembly.
- Many algorithms for allocating a multiple access channel are known.
- In the following sections we will study a small sample of the more interesting ones and give some examples of their use.

3.1.1 ALOHA:

- In the 1970s, Norman Abramson and his colleagues at the University of Hawaii devised a new and elegant method to solve the channel allocation problem.
- Their work has been extended by many researchers since then (Abramson, 1985).
- Although Abramson's work, called the ALOHA system, designed for ground-based radio broadcasting or wireless LAN, it can also be used on any shared medium in which uncoordinated users are competing for the use of a single shared channel.

- We will discuss two versions of ALOHA here:

    1. Pure and
    2. Slotted.

  They differ with respect to whether time is divided into discrete slots into which all frames must fit.

- Pure ALOHA does not require global time synchronization; slotted ALOHA does.

Pure ALOHA

- The basic idea of an ALOHA system is simple: let users transmit whenever they have data to be sent.
- There will be collisions, of course, and the colliding frames will be damaged. However, due to the feedback property of broadcasting, a sender can always find out whether its frame was destroyed by listening to the channel, the same way other users do.
- With a LAN, the feedback is immediate; with a satellite, there is a delay of 270 msec before the sender knows if the transmission was successful.
- If listening while transmitting is not possible for some reason, acknowledgements are needed.
- If the frame was destroyed, the sender just waits a random amount of time and sends it again.
- The waiting time must be random or the same frames will collide over and over, in lockstep.
- Systems in which multiple users share a common channel in a way that can lead to conflicts are widely known as contention systems.
- A sketch of frame generation in an ALOHA system is given in Fig. 3.2.

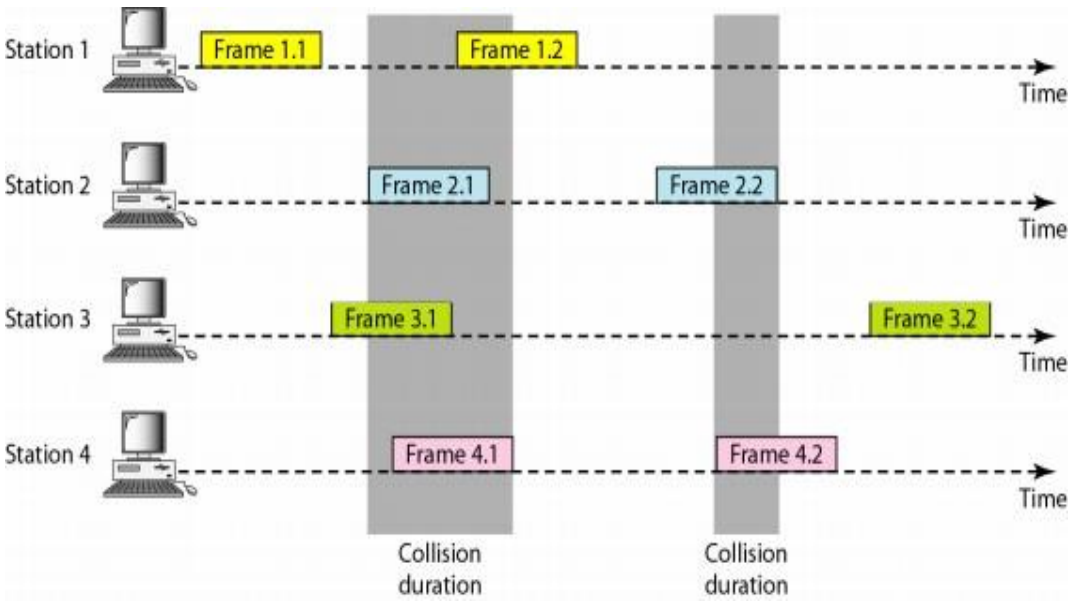


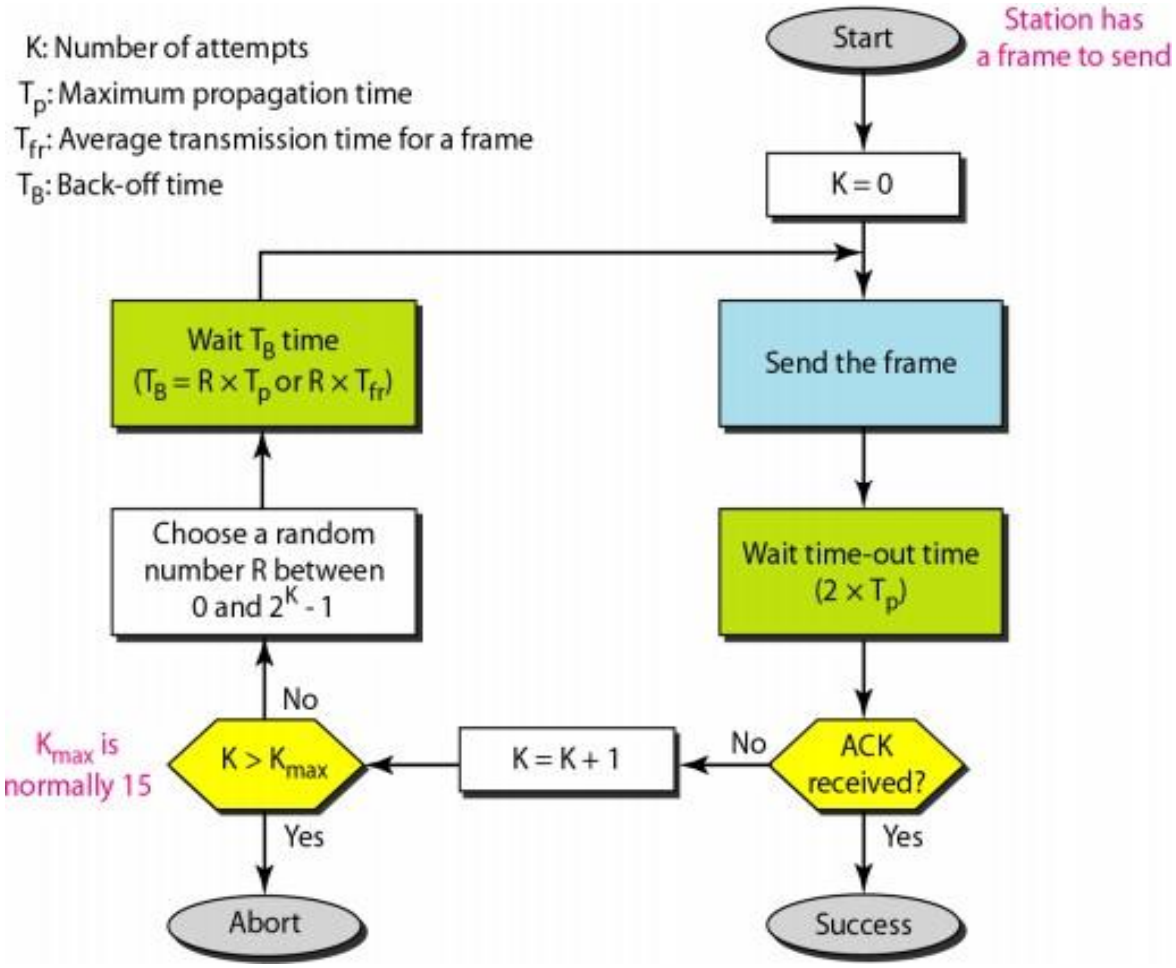
Figure 3.2: In pure ALOHA, frames are transmitted at completely arbitrary times.

- We have made the frames all the same length because the throughput of ALOHA systems is maximized by having a uniform frame size rather than by allowing variable length frames.
- Whenever two frames try to occupy the channel at the same time, there will be a collision and both will be garbled.
- If the first bit of a new frame overlaps with just the last bit of a frame almost finished, both frames will be totally destroyed and both will have to be retransmitted later.
- It is obvious that we need to resend the frames that have been destroyed during transmission.
- The pure ALOHA protocol relies on acknowledgments from the receiver.
- When a station sends a frame, it expects the receiver to send an acknowledgment.
- If the acknowledgment does not arrive after a time-out period, the station assumes that the frame (or the acknowledgment) has been destroyed and resends the frame.
- A collision involves two or more stations.
- If all these stations try to resend their frames after the time-out, the frames will collide again.
- Pure ALOHA dictates that when the time-out period passes, each station waits a random amount of time before resending its frame.

- The randomness will help avoid more collisions.
- We call this time the back-off time TB.

How the repetition interval is estimated in Aloha:

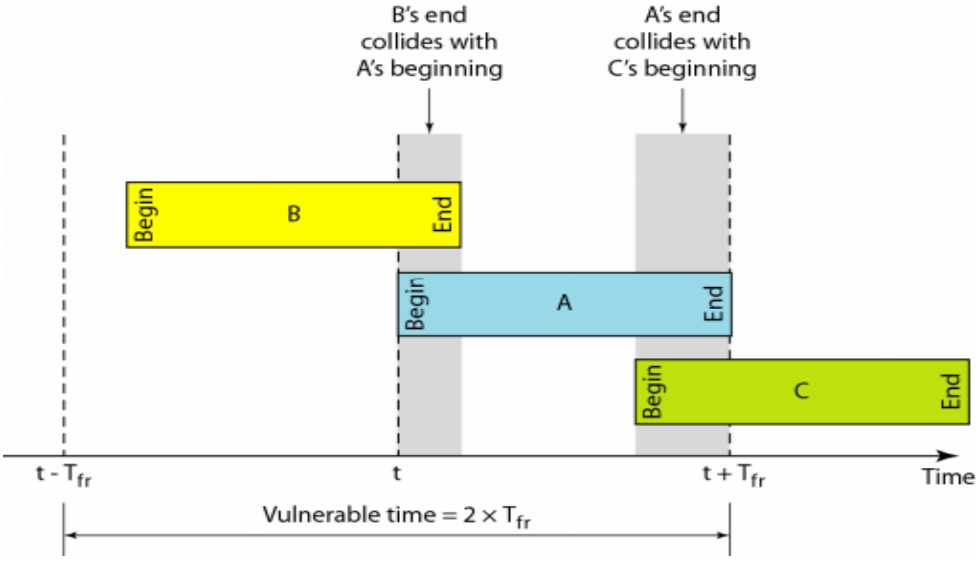
- Pure ALOHA has a second method to prevent congesting the channel with retransmitted frames.
- After a maximum number of retransmission attempts Kmax, a station must give up and try later.
- The time-out period is equal to the maximum possible round-trip propagation delay, which is twice the amount of time required to send a frame between the two most widely separated stations (2 x Tp).
- The back-off time TB is a random value that normally depends on K (the number of attempted unsuccessful transmissions).



- The formula for TB depends on the implementation.
- One common formula is the binary exponential back-off.
- In this method, for each retransmission, a multiplier in the range 0 to 2K - 1 is randomly chosen and multiplied by Tp (maximum propagation time) or Tfr (the average time required to send out a frame) to find TB.
- Note that in this procedure, the range of the random numbers increases after each collision.
- The value of Kmax is usually chosen as 15.

Vulnerable time in Pure Aloha:

- Let us find the length of time, the vulnerable time, in which there is a possibility of collision.
- We assume that the stations send fixed-length frames with each frame taking Tfr S to send.
- Figure shows the vulnerable time for station A.



- Station A sends a frame at time t.
- Now imagine station B has already sent a frame between t - Tfr and t.
- This leads to a collision between the frames from station A and station B.
- The end of B's frame collides with the beginning of A's frame.

- On the other hand, suppose that station C sends a frame between t and t + Tfr .
- Here, there is a collision between frames from station A and station C.
- The beginning of C's frame collides with the end of A's frame.
- We see that the vulnerable time, during which a collision may occur in pure ALOHA, is 2 times the frame transmission time.
- Pure ALOHA vulnerable time = 2 x Tfr

Throughput in Pure Aloha:

- Let us call G the average number of frames generated by the system during one frame transmission time.
- Then it can be proved that the average number of successful transmissions for pure ALOHA is $S = G \times e^{-2G}$.
- The maximum throughput Smax is 0.184, for G = 1/2.
- In other words, if one-half a frame is generated during one frame transmission time (in other words, one frame during two frame transmission times), then 18.4 percent of these frames reach their destination successfully.
- This is an expected result because the vulnerable time is 2 times the frame transmission time.

Slotted ALOHA:

- In 1972, Roberts published a method for doubling the capacity of an ALOHA system (Roberts, 1972).
- His proposal was to divide time into discrete intervals, each interval corresponding to one frame.
- This approach requires the users to agree on slot boundaries.
- One way to achieve synchronization would be to have one special station emit a pip at the start of each interval, like a clock.
- In Roberts' method, which has come to be known as slotted ALOHA, in contrast to Abramson's pure ALOHA, a computer is not permitted to send whenever a carriage return is typed. Instead, it is required to wait for the beginning of the next slot.
- Thus, the continuous pure ALOHA is turned into a discrete one.

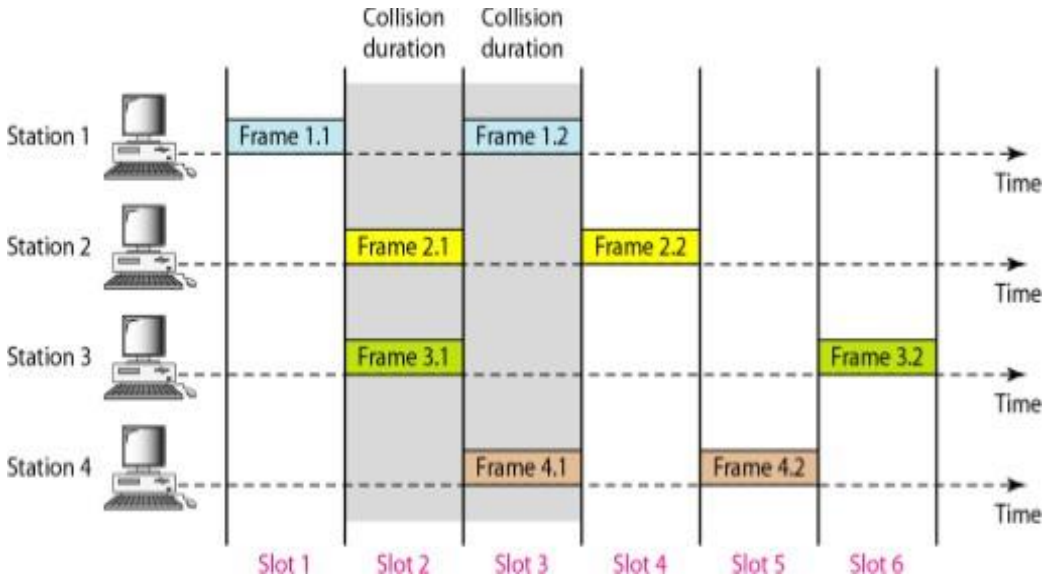


Figure 3.3: Frame transmission in Slotted Aloha

- Because a station is allowed to send only at the beginning of the synchronized time slot, if a station misses this moment, it must wait until the beginning of the next time slot.
- This means that the station which started at the beginning of this slot has already finished sending its frame.
- Of course, there is still the possibility of collision if two stations try to send at the beginning of the same time slot.
- However, the vulnerable time is now reduced to one half, equal to Tfr.

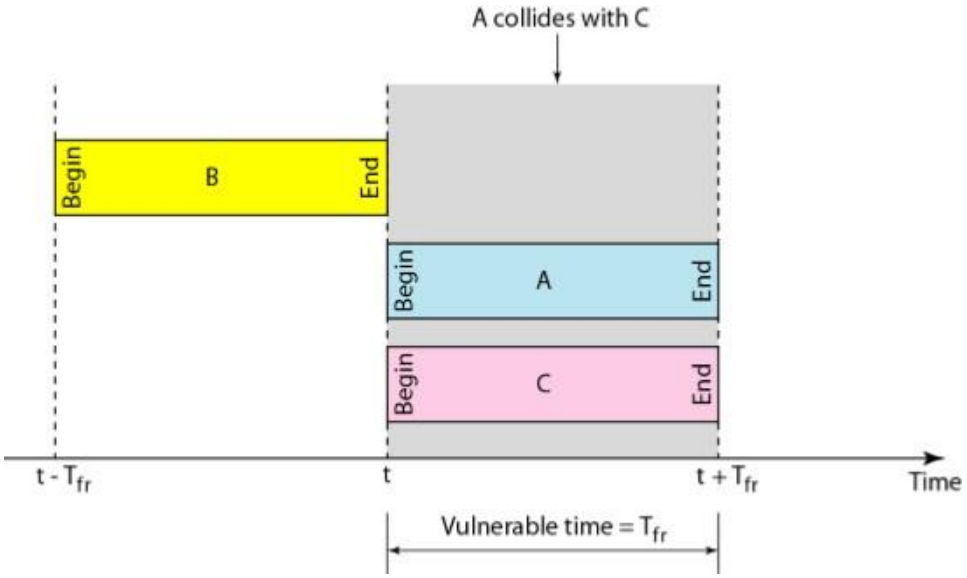


Figure 3.4: Vulnerable time in Slotted Aloha

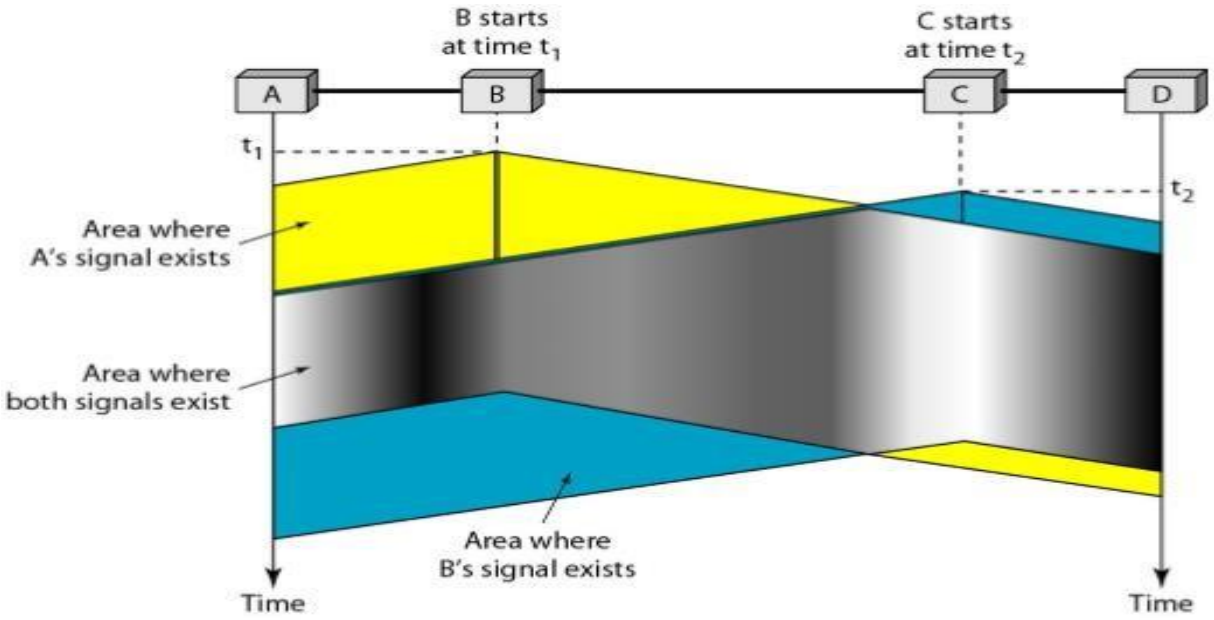
- Since the vulnerable period is now halved, the probability of no other traffic during the same slot as our test frame is e-G which leads to the average number of successful transmissions for slotted ALOHA is $S = G \times e^{-G}$.
- The maximum throughput S max is 0.368, when G = 1, which is twice that of pure ALOHA.
- In other words, if a frame is generated during one frame transmission time, then 36.8 percent of these frames reach their destination successfully.
- A comparison between pure and slotted ALOHA is given below.

S. No	Pure ALOHA	Slotted ALOHA
1	Any station can transmit the data at any time.	Any station can transmit the data at the beginning of any time slot
2	The time is continuous and not globally synchronized	The time is discrete and globally synchronized
3	Vulnerable time in which collision may occur = 2x Tfr	Vulnerable time in which collision may occur = Tfr
4	Probability of successful transmission of data packet = $G \times e^{-2G}$	Probability of successful transmission of data packet = $G \times e^{-G}$
5	Maximum efficiency = 18.4 % (Occurs at G = ½)	Maximum efficiency = 36.8 % (Occurs at G = 1)
6	Main advantage: Simplicity in implementation	Main advantage: It reduces the number of collisions to half and doubles the efficiency of pure Aloha.

3.1.2 CSMA: (Carrier Sense Multiple Access)

- To minimize the chance of collision and, therefore, increase the performance, the CSMA method was developed.
- The chance of collision can be reduced if a station senses the medium before trying to use it.

- Carrier sense multiple access (CSMA) requires that each station first listen to the medium (or check the state of the medium) before sending.
- In other words, CSMA is based on the principle "sense before transmit" or "listen before talk
- "CSMA can reduce the possibility of collision, but it cannot eliminate it.
- Stations are connected to a shared channel.
- The possibility of collision still exists because of propagation delay.
- When a station sends a frame, it still takes time (although very short) for the first bit to reach every station and for every station to sense it.
- In other words, a station may sense the medium and find it idle, only because the first bit sent by another station has not yet been received.
- At time t1, station B senses the medium and finds it idle, so it sends a frame.
- At time t2 (t2> t1), station C senses the medium and finds it idle because, at this time, the first bits from station B have not reached station C.
- Station C also sends a frame. The two signals collide and both frames are destroyed.

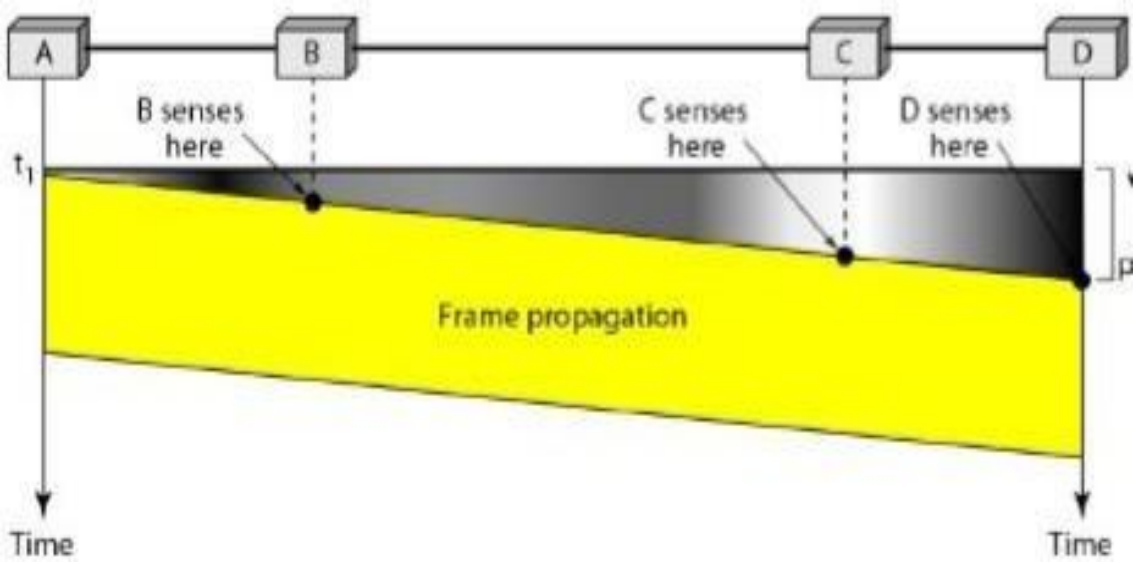


- The vulnerable time for CSMA is the propagation time Tp .
- This is the time needed for a signal to propagate from one end of the medium to the other.
- When a station sends a frame, and any other station tries to send a frame during this time, a collision will result.

- But if the first bit of the frame reaches the end of the medium, every station will already have heard the bit and will refrain from sending.

The worst case scenario:

- The leftmost station A sends a frame at time t1, which reaches the rightmost station D at time t1 + Tp . The gray area shows the vulnerable area in time and space.



Persistence Methods:

- What should a station do if the channel is busy?
- What should a station do if the channel is idle?
- Three methods have been devised to answer these questions:

1-persistent method

non persistent method

p-persistent method.

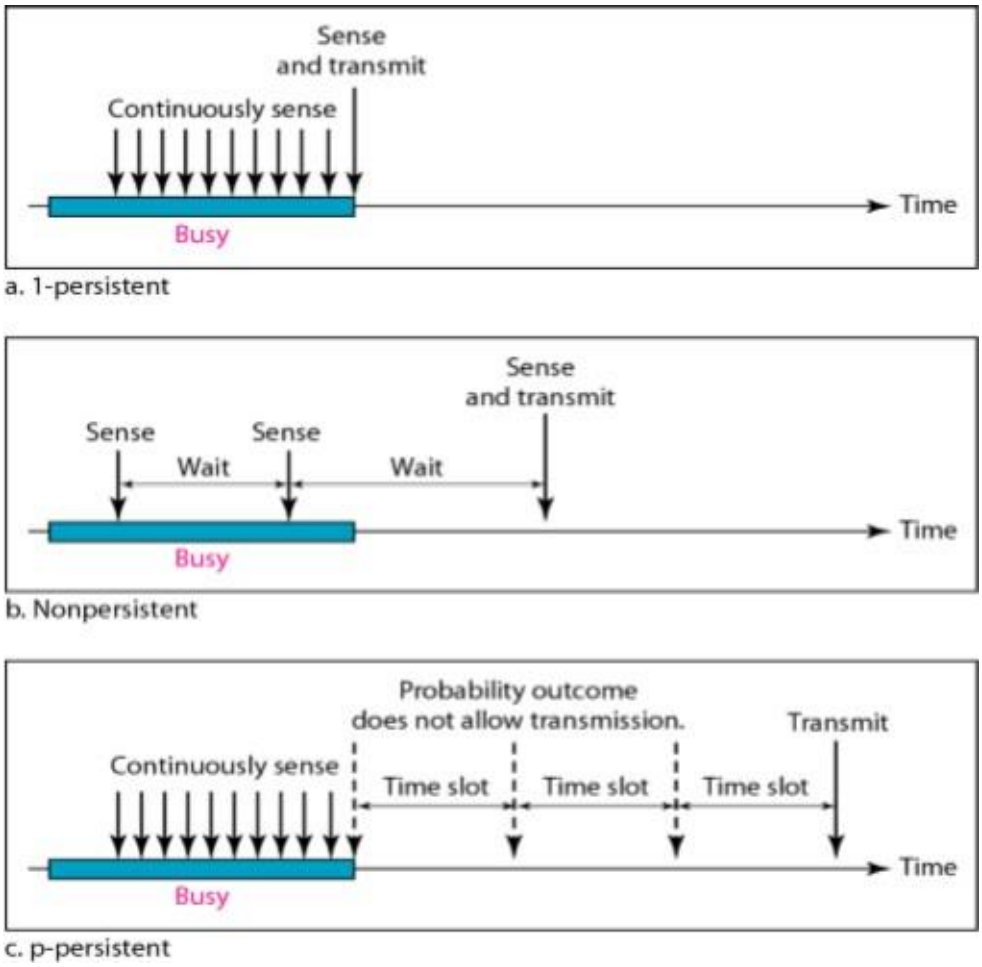


Figure 3.5: Different Persistent Methods

1-persistent method:

- The 1-persistent method is simple and straightforward.
- In this method, after the station finds the line idle, it sends its frame immediately (with probability 1).
- This method has the highest chance of collision because two or more stations may find the line idle and send their frames immediately.

Non persistent Method:

- In the non persistent method, a station that has a frame to send senses the line.
- If the line is idle, it sends immediately.
- If the line is not idle, it waits a random amount of time and then senses the line again.
- The non persistent approach reduces the chance of collision because it is unlikely that two or more stations will wait the same amount of time and retry to send simultaneously.

- However, this method reduces the efficiency of the network because the medium remains idle when there may be stations with frames to send.

P-Persistent :

- The p-persistent method is used if the channel has time slots with a slot duration equal to or greater than the maximum propagation time.
- The p-persistent approach combines the advantages of the other two strategies.
- It reduces the chance of collision and improves efficiency.
- In this method, after the station finds the line idle it follows these steps:
- 1. With probability p, the station sends its frame.
- 2. With probability q = 1 - p, the station waits for the beginning of the next time slot and checks the line again.
- a. If the line is idle, it goes to step 1.
- b. If the line is busy, it acts as though a collision has occurred and uses the back off procedure.

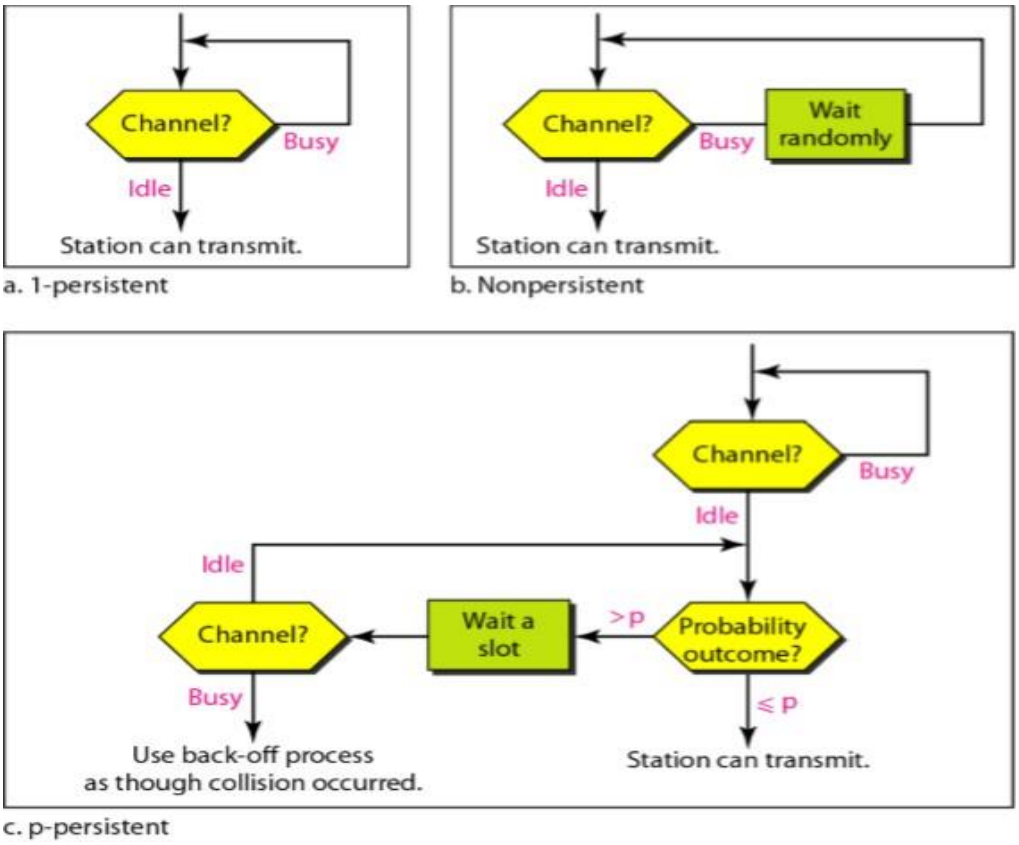


Figure 3.6: Functional behavior of various persistent methods

3.1.3 Carrier Sense Multiple Access with Collision Detection (CSMA/CD)

- The CSMA method does not specify the procedure following a collision.
- Carriers sense multiple access with collision detection (CSMA/CD) augments the algorithm to handle the collision.
- In this method, a station monitors the medium after it sends a frame to see if the transmission was successful.
- If so, the station is finished. If, however, there is a collision, the frame is sent again.
- To better understand CSMA/CD, let us look at the first bits transmitted by the two stations involved in the collision.
- Although each station continues to send bits in the frame until it detects the collision, we show what happens as the first bits collide.
- In figure, stations A and C are involved in the collision.

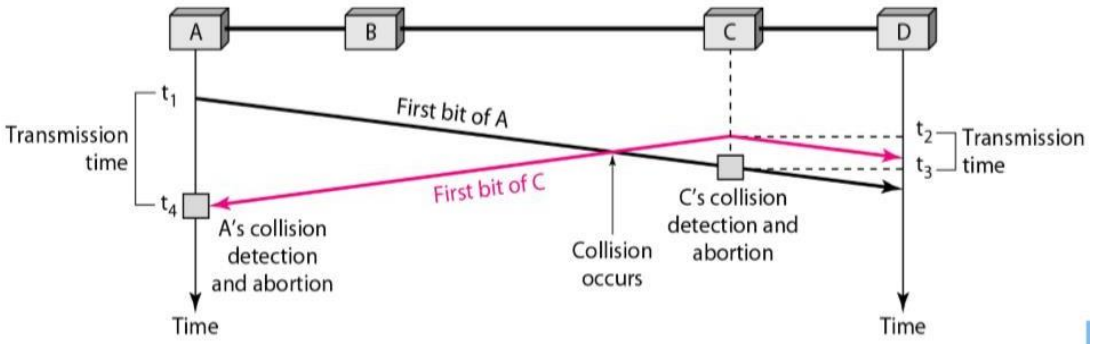


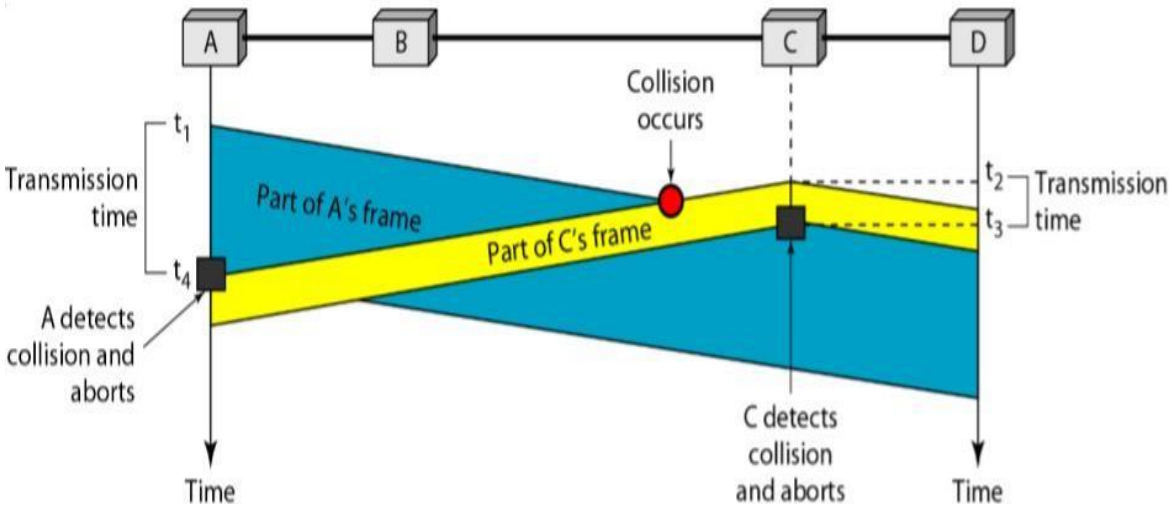
Figure 3.7: Station A and C sending bits and leads to collision

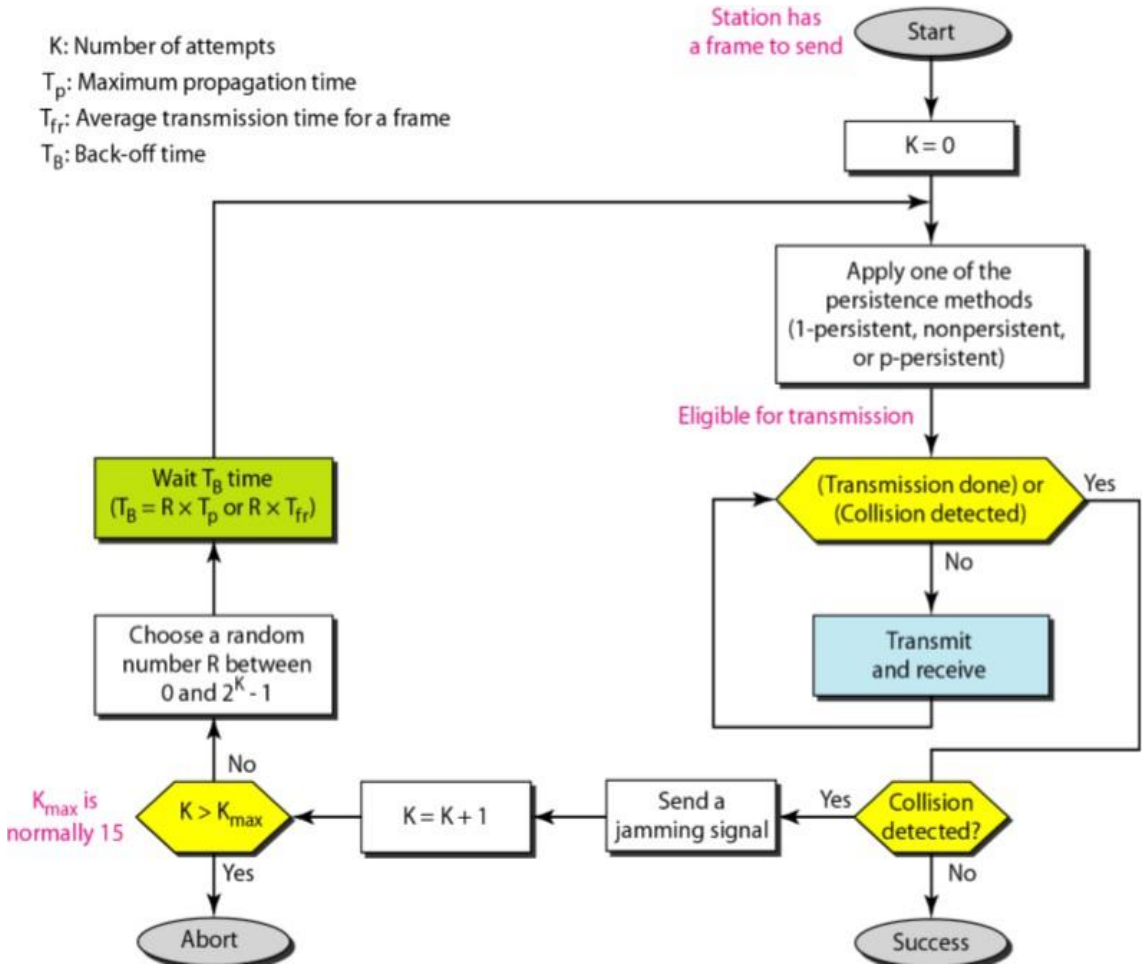
- Looking at the figure, we see that A transmits for the duration t4 – t1;C transmits for the duration t3 - t2'.
- Later we show that, for the protocol to work, the length of any frame divided by the bit rate in this protocol must be more than either of these durations.

Minimum Frame Size:

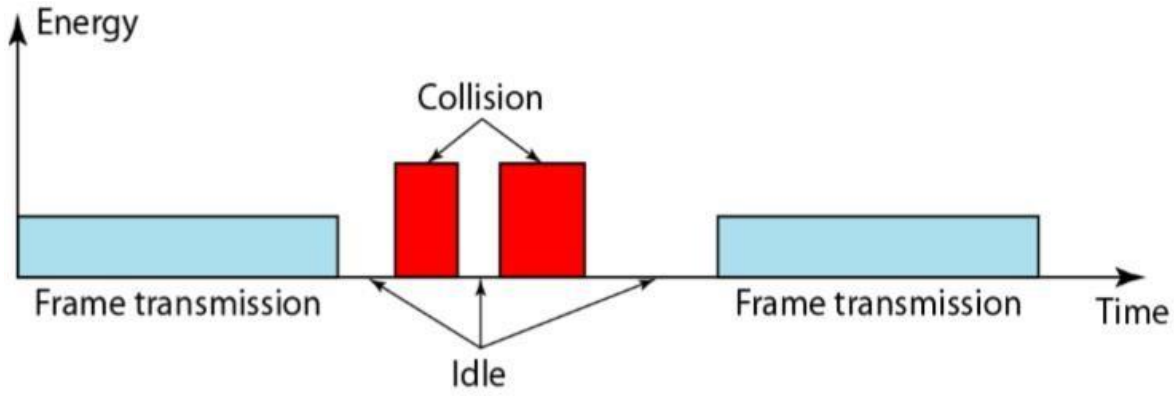
- For CSMA/CD to work, we need a restriction on the frame size.
- Before sending the last bit of the frame, the sending station must detect a collision, if any, and abort the transmission.
- This is so because the station, once the entire frame is sent, does not keep a copy of the frame and does not monitor the line for collision detection.

- Therefore, the frame transmission time Tfr must be at least two times the maximum propagation time Tp.
- If the two stations involved in a collision are the maximum distance apart, the signal from the first takes time Tp to reach the second, and the effect of the collision takes another time Tp to reach the first. So the requirement is that the first station must still be transmitting after 2Tp





- We can say that the level of energy in a channel can have three values:
    - Zero, normal, and abnormal.
- At the zero level, the channel is idle.
- At the normal level, a station has successfully captured the channel and is sending its frame.
- At the abnormal level, there is a collision and the level of the energy is twice the normal level.
- A station that has a frame to send or is sending a frame needs to monitor the energy level to determine if the channel is idle, busy, or in collision mode.



Throughput :

- The throughput of CSMA/CD is greater than that of pure or slotted ALOHA.
- The maximum throughput occurs at a different value of G and is based on the persistence method and the value of p in the p-persistent approach.
- For 1-persistent method the maximum throughput is around 50 percent when G =1.
- For non-persistent method, the maximum throughput can go up to 90 percent when G is between 3 and 8.

3.1.4 CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

CSMA/CA is a media access control protocol, especially for wireless networks, where devices listen to the channel before transmitting data to avoid collisions. It operates by having devices sense the medium for traffic and, if busy, wait a random bakeoff period before trying again. For added protection against collisions, a Request-to-Send/Clear-to-Send (RTS/CTS) handshake can be used, where a device requests permission to send and the receiver grants it, ensuring only one device transmits at a time.

How CSMA/CA Works

1. **Carrier Sensing:**

   Before transmitting, a device monitors the shared wireless medium to detect if any other device is currently sending data.

2. **Backoff Period:**

   If the medium is busy, the device waits for a random, predetermined amount of time before attempting to sense the channel again. This random waiting helps prevent multiple devices from trying to send at the same time after the channel becomes free.

3. **Collision Avoidance (RTS/CTS Handshake):**
    - If the medium appears free after sensing and waiting, the device sends a short Request-to-Send (RTS) frame.
    - This RTS frame signals to all other devices and the access point that a transmission is about to occur.
    - The receiving device (or access point) responds with a Clear-to-Send (CTS) frame, indicating it's ready to receive the data.
    - Only after receiving the CTS frame does the original device send its data.

Why It's Used

1. Wireless Environments:

CSMA/CA is crucial for wireless networks like Wi-Fi because collision detection, the core of CSMA/CD (used in wired networks), is impractical with wireless signals.

2. The Hidden Terminal Problem:

It helps address the "hidden terminal" problem, where two devices are out of each other's direct range but can both communicate with the same access point, leading to collisions if they transmit simultaneously. The RTS/CTS mechanism allows devices to coordinate even when they cannot hear each other.

3. Collision Prevention:

By having devices "listen" and coordinate transmissions, CSMA/CA aims to prevent collisions from happening in the first place, rather than trying to detect and recover from them after they've occurred.

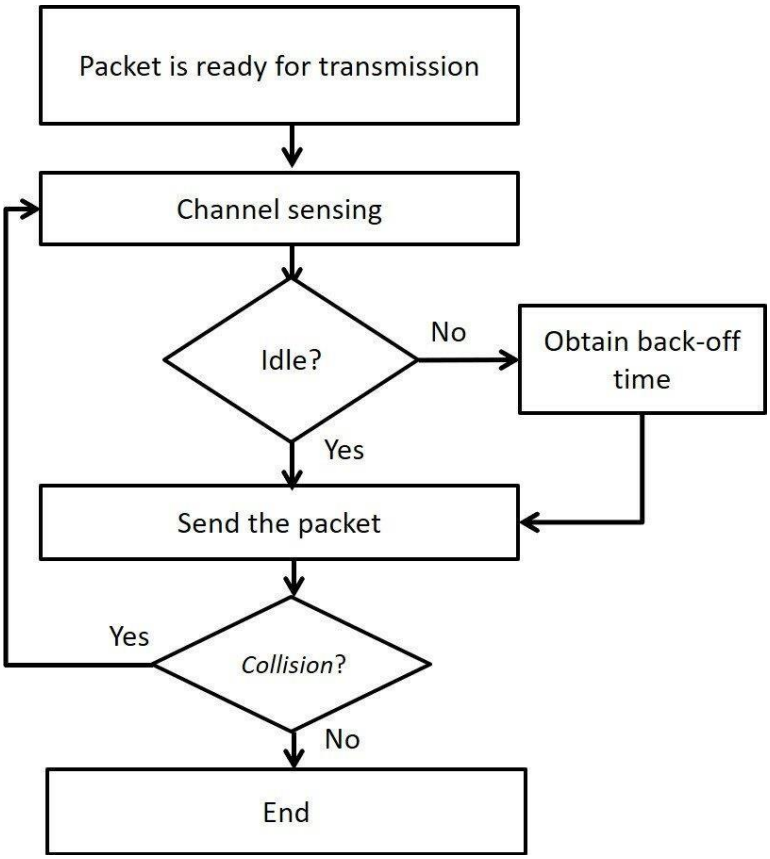


Figure 3.8: Carrier sense multiple access protocol with Collision avoidance

3.2 Controlled access protocols:

A controlled access protocol (CAP) regulates access to a shared network medium, ensuring that only one node transmits data at a time to prevent collisions and interference. This is achieved through coordinated procedures, including reservation, polling, and token passing.

How Controlled Access Works:

1. Consultation:

Before sending, stations consult with each other to determine which station has the right to transmit.

2. Authorization:

A station cannot send data unless it is authorized by the network or other stations.

3. Collision Prevention:

By allowing only one device to send at a time, controlled access protocols eliminate the possibility of data collisions on shared channels.

Three Types of Controlled Access Protocols

1. **Reservation Method:**

Mechanism: Stations divide time into intervals and send a reservation frame, which includes slots for each station, before sending data frames.

Process: A station makes a reservation in its designated mini-slot within the reservation frame. Only stations that have made a reservation can send data in the subsequent data frame interval.

2. **Polling Method:**

Mechanism: One central device, the primary station, is responsible for initiating and controlling the entire process.

Process: The primary station periodically "polls" or queries each secondary station to check if it has data to send. Only the station that is currently being polled is allowed to use the communication channel.

3. **Token Passing Method:**

Mechanism: A special frame, the "token," circulates among the stations arranged in a ring or similar topology.

Process: A station can only send data when it holds the token. After transmitting its data, the station passes the token to the next station in the sequence, ensuring that only one station transmits at a time.

3.2.1 Reservation:

In a reservation-based controlled access protocol, stations must reserve a transmission slot before sending data to prevent collisions. Time is divided into intervals, with a special reservation frame preceding data frames in each interval. This frame contains mini-slots, one for each station, where a station places a bit to indicate its need to transmit. Only stations that have made a reservation can send their data frames during the data transmission period. This method provides predictable access and improves channel utilization by reducing collisions, as the path to the destination can be pre-determined.

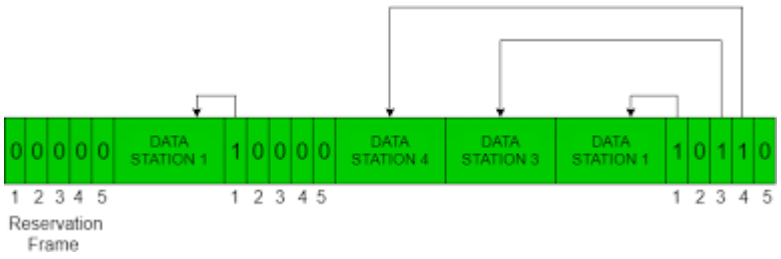


Figure 3.9: Reservation controlled access protocol

How it works:

1. Time Division:

Time is segmented into intervals, each containing a reservation frame followed by a data transmission period.

2. Reservation Frame:

Each reservation frame has a set number of mini-slots, with each mini-slot dedicated to a specific station.

3. Making a Reservation:

If a station needs to send data, it transmits a bit (e.g., a "1") in its corresponding mini-slot within the reservation frame.

4. Data Transmission:

After the reservation frame, stations that have made a reservation in their mini-slots are allowed to send their data frames.

5. Collision Avoidance:

By reserving slots in advance, the system ensures that only stations with a confirmed intention to send data will attempt to transmit, thus avoiding collisions.

Advantages:

1. High Efficiency:

Reduces collisions, leading to better channel utilization and higher throughput.

2. Predictable Access:

Allocates specific time slots, making network access predictable for nodes.

Example:

If there are five stations (1, 2, 3, 4, 5) in the system, the reservation frame will have five mini-slots.

1. Interval 1: Stations 2, 3, and 5 make reservations in their respective slots, while stations 1 and 4 do not.

2. Data Transmission 1: Stations 2, 3, and 5 can then transmit their data frames in the subsequent data transmission period.

3. Interval 2: Only station 2 makes a reservation.

4. Data Transmission 2: Only station 2 transmits its data.

3.2.2 Polling:

In controlled access protocols, polling is a technique where a primary station (or master) controls access to a shared medium by cyclically querying (polling) secondary stations to determine if they have data to send. The primary station sends poll frames, and if a secondary station has data, it responds with a data frame; otherwise, it sends a negative acknowledgment (NAK). This method ensures orderly access and prevents data collisions by assigning transmission turns to specific devices.

How Polling Works:

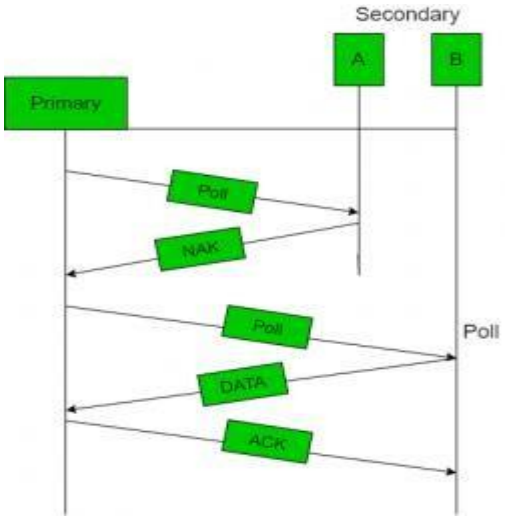


Figure 3.10: Polling controlled access protocol

1. Primary Station Initiates:

The primary station acts as the controller of the network link.

2. Select Function (Sending Data):

If the primary station has data to send, it sends a Select frame to the intended secondary station, waits for an acknowledgment of its readiness, and then transmits the data.

3. Poll Function (Receiving Data):

When the primary station is ready to receive data, it sends a poll frame to each secondary station in turn.

4. Secondary Station Responses:

If a secondary station has data to send, it responds with a data frame containing the data.

If a secondary station has no data to send, it responds with a negative acknowledgment (NAK).

5. Cyclic Access:

The primary station continues polling the secondary stations until it finds one with data to send. This process ensures that each device gets an opportunity to transmit its data without collisions.

Key Characteristics

1. Centralized Control:

The primary station maintains control over the shared medium and grants access to secondary stations.

2. No Collisions:

By assigning turns to each device, polling effectively eliminates data collisions on the network.

3. Predictable Access:

Polling provides a structured and predictable way for devices to access the network.

Drawbacks

1. High Dependence on Primary Station:

The reliability and performance of the entire network depend heavily on the primary station.

2. Overhead:

The constant polling messages from the primary station add overhead to the network traffic.

3. Reduced Throughput (Under Low Loads):

When network traffic is light, the primary station still sends polling messages, which can reduce overall throughput.

3.2.3 Token passing:

Token passing is a controlled access method where stations are arranged in a logical ring, and a special packet called a "token" circulates among them. Only the station holding the token has the right to transmit data. After a station sends its data, it passes the token to the next station in the ring. If a station receives the token but has no data to send, it simply forwards the token to the next station. This method prevents data collisions by ensuring only one station transmits at a time, though potential issues include token loss, duplication, and the latency introduced by waiting for the token.

How Token Passing Works:

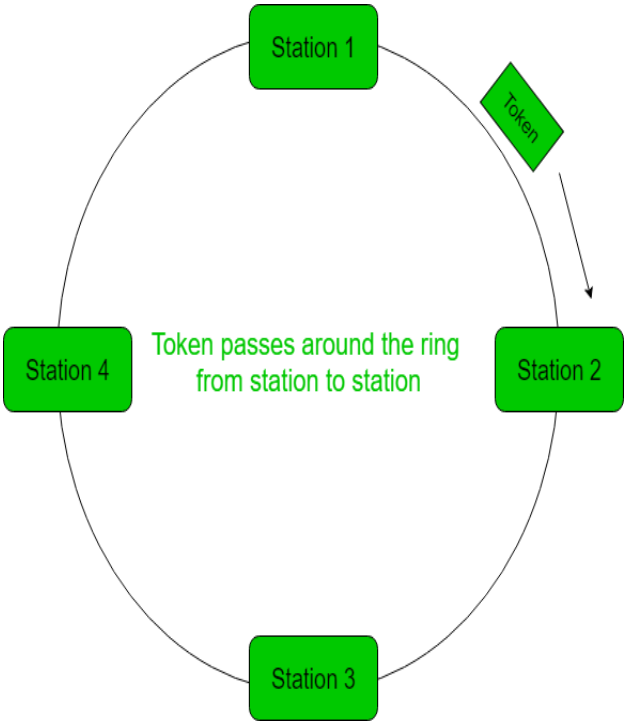


Figure 3.11: Token passing controlled access protocol

1. Logical Ring:

Stations are organized in a logical ring, meaning each station has a predecessor and a successor in a predefined order.

2. Token Circulation:

A special, small message called a "token" circulates through this logical ring.

3. Permission to Send:

The station that possesses the token is granted the exclusive right to transmit data on the network channel.

4. Data Transmission:

When a station receives the token and has data to send, it holds the token, transmits its data, and then passes the token to the next station in the ring.

5. Token Forwarding:

If a station receives the token but has no data to send, it immediately passes the token to its successor without holding it.

6. Collision Prevention:

By allowing only one station to transmit at a time (the one holding the token), token passing completely eliminates the possibility of data collisions on the network.

Challenges and Considerations.

3.3 Channelization:

In computer networks, channelization is a multiple-access method that divides a shared communication channels available bandwidth into smaller sub-channels using time, frequency, or code to allow multiple devices to transmit simultaneously without interference. The most common protocols that implement channelization are FDMA (Frequency Division Multiple Access), which splits the channel by frequency; TDMA (Time Division Multiple Access), which divides the channel into time slots; and CDMA (Code Division Multiple Access), which uses unique codes to distinguish user data.

Types of Channelization Protocols:

FDMA (Frequency Division Multiple Access):

The total available bandwidth is divided into a set of non-overlapping frequency bands. Each device is assigned a specific frequency band, and they transmit in their allocated band without interfering with others.

TDMA (Time Division Multiple Access):

The channel's time is divided into small time slots. Each device gets its own time slot to transmit data, and they take turns sending data over the channel.

CDMA (Code Division Multiple Access):

Instead of dividing time or frequency, CDMA assigns a unique digital code to each station. All devices transmit simultaneously on the same frequency, but each device's data is separated from others' by its unique code.

Benefits of Channelization

- **Improved Efficiency:**

  Allows multiple data streams to share a single physical channel, maximizing bandwidth utilization.

- **Increased Capacity:**

  More users can be supported within a given bandwidth compared to single-access methods.

- **Reduced Interference:**

  By assigning exclusive access (in time, frequency, or by code), channelization minimizes data collisions.

- **Enhanced Security:**

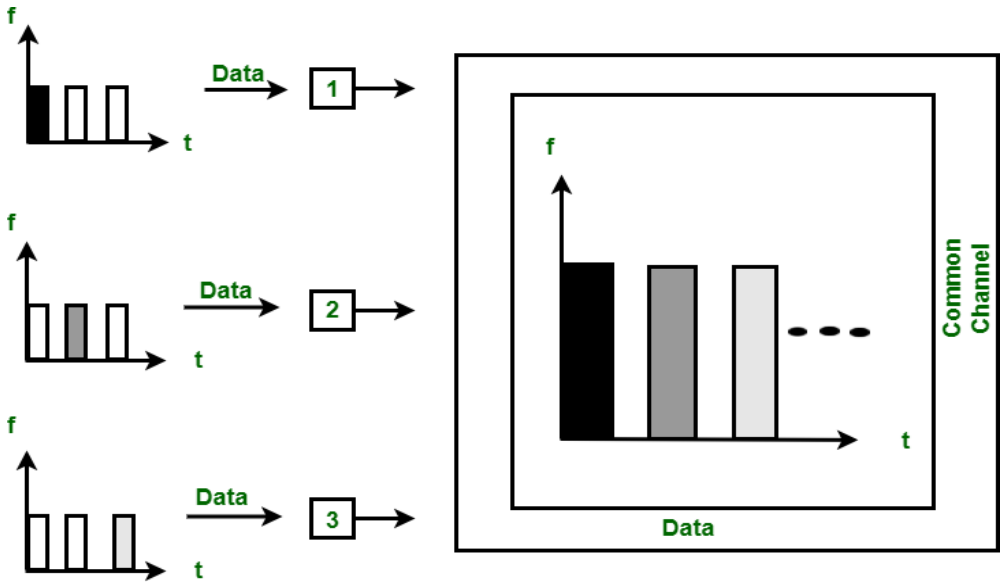
  CDMA, in particular, offers improved security as data streams are harder to decode without the proper code.

3.3.1 FDMA:

In computer networks, FDMA (Frequency Division Multiple Access) is a multiple access method that divides a shared communication channel's bandwidth into smaller, non-overlapping frequency bands, each dedicated to a specific user or application for continuous, independent transmission and reception. While simple and effective for continuous data streams, it requires expensive filters, can lead to wasted bandwidth if users aren't actively

transmitting, and is susceptible to interference between adjacent channels if not properly managed with guard bands.

How FDMA Works:



1. Bandwidth Division:

The total available frequency spectrum is split into multiple individual sub-channels.

2. User Assignment:

Each user is assigned a unique, dedicated frequency band for their communication.

3. Simultaneous Access:

Multiple users can transmit and receive data simultaneously, each on their own assigned frequency band, preventing interference with others.

4. Guard Bands:

To prevent interference between adjacent frequency bands, small, unused frequency spaces called guard bands are inserted between them.

Advantages of FDMA:

- **Simplicity**: It is a straightforward and proven method for multiple access.
- **No Timing Issues**: Unlike **Time Division Multiple Access** (TDMA), it does not require synchronization or complex timing management.

- **Suitable for Continuous Data**: It is well-suited for applications that require a constant, continuous stream of data, such as traditional voice communication.

Disadvantages of FDMA:

1. **Wasted Bandwidth**:

A user's assigned frequency band remains idle and wasted if they are not actively transmitting, making it inefficient for "bursty" (intermittent) traffic.

2. **Expensive Equipment**:

Requires costly bandpass filters and strict linearity requirements for the transmission equipment to minimize adjacent channel interference.

3. **Interference**:

Although guard bands help, there is still a risk of call interference or intermodulation frequencies between adjacent channels due to non-linear effects in amplifiers.

Typical Applications

FDMA is a foundational technique used in various wireless communication systems, including early cellular systems and some satellite communication networks, to allow multiple users to share a single communication medium.

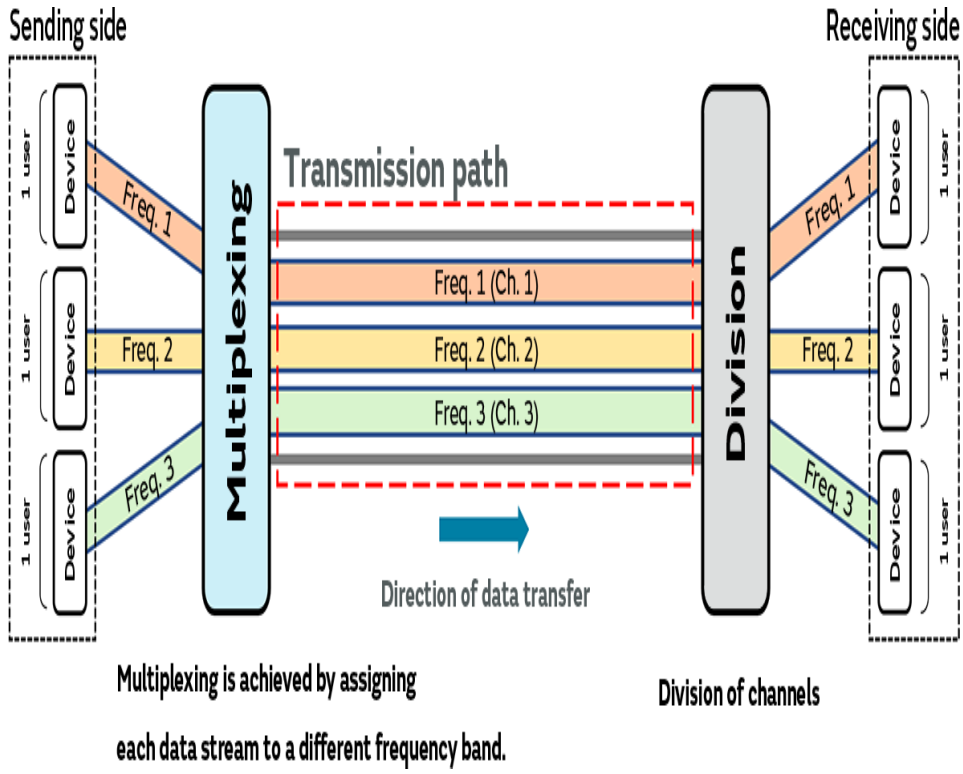


Figure 3.12: Block diagram of a Frequency division multiple access

3.3.2 TDMA

In computer networks, TDMA (Time Division Multiple Access) is a channel access method that allows multiple users to share a single frequency channel by assigning each user a dedicated, repeating time slot for data transmission. This system divides the total time into frames, with each frame partitioned into several time slots, ensuring that only one user transmits at a time to prevent data collisions and channel overlap.

How TDMA Works:

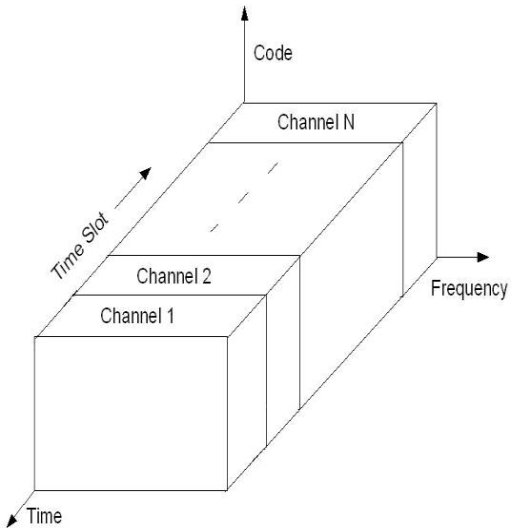


Figure 3.13: Time division multiple access

1. Channel Division: A single frequency channel is divided into specific time slots.

2. Frame Structure: The time slots are organized into frames, which repeat cyclically.

3. Slot Assignment: Each user is allocated a unique time slot within the frame to transmit or receive data.

4. Synchronized Transmission: All users must synchronize their clocks to transmit their data in their assigned time slices, occurring one after the other.

5. Guard Times: A short interval called a "guard time" is added between adjacent TDMA slots to prevent interference and overlapping transmissions.

Advantages of TDMA

- **Efficient Channel Use:** Multiple users can share a single channel, increasing overall network capacity.
- **Cost-Effective:** It is a cost-effective way to convert analog systems to digital.
- **Predictable Delivery:** The synchronized and time-slotted approach allows for predictable message delivery.

Disadvantages of TDMA

- **Guard Time Inefficiency:**

  The necessary guard times consume some of the channel's bandwidth, reducing overall efficiency.

- **Limited Flexibility:**

  Compared to other methods like CDMA, TDMA offers only moderate system flexibility and medium data rates.

- **Potential for Inactivity:**
  - If a user's assigned time slot is not used (i.e., they have no data to send), the channel bandwidth is wasted.

3.3.3 CDMA

In computer networks, CDMA (Code-Division Multiple Access) is a spread-spectrum channel access method that allows multiple users to share the same frequency band simultaneously by assigning each user a unique pseudorandom code to spread and encode their signal. The receiver uses the same unique code to "despread" and recover the signal, separating it from other users' signals that are perceived as noise. This technique enhances network capacity and is used in cellular systems, satellite communications, and military applications.

How CDMA Works:

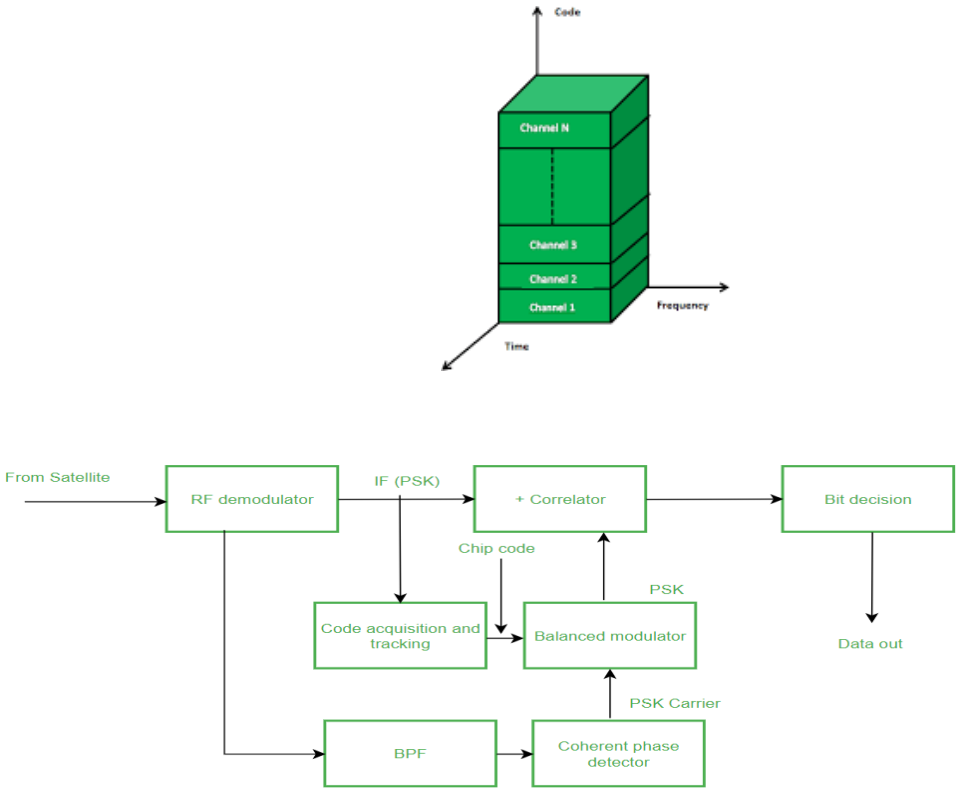


Figure 3.14: Block diagram of a Code division multiple access

1. Spreading Code:

Each user is assigned a unique, high-**speed pseudorandom code.**

2. Signal Spreading:

This code is used to multiply the user's data signal, spreading it over a much wider bandwidth than necessary.

3. Simultaneous Transmission:

Multiple users, each with a different spreading code, transmit their spread signals over the same frequency band at the same time.

4. **Despreading**:

A receiver with the same unique spreading code as the transmitting user can "despread" the signal, isolating it from the other signals on the same band.

5. Signal Recovery:

The receiver effectively filters out all other signals, which are seen as noise because they lack the specific spreading code.

Applications

1. Cellular Networks: Widely used in 2G and 3G cellular systems, with standards including IS-95 and cdma2000.

2. Military Communications: Historically used for anti-jamming and secure communication.

3. Satellite Communications: Used to increase efficiency in satellite-based communication systems.

3.4 Wired LAN'S

ETHERNET:

Ethernet is a wired networking technology for local and wide area networks (LANs/WANs) that connects devices like computers, printers, and servers using physical cables to transmit data. Invented at Xerox PARC in the 1970s, it defines standards for data communication using MAC addresses and protocols like CSMA/CD to manage data flow. Ethernet is known for its high speed, security, and reliability, with evolving standards like Gigabit Ethernet and 10-Gigabit Ethernet supporting increasingly faster speeds.

What is Ethernet?

Wired Technology:

Ethernet uses physical cables, such as twisted-pair or fiber optic cables, to connect network devices.

Standardized Protocol:

It is a standard protocol that defines the rules for how devices communicate and exchange data.

Data Framing:

Data is broken down into small packets called frames, which include header information with source and destination MAC addresses.

Network Connections:

It connects various devices in LANs (Local Area Networks) within buildings or homes and can also extend to WANs (Wide Area Networks) and MANs (Metropolitan Area Networks).

How Ethernet Works

Data Encapsulation: When a device sends data, the data is encapsulated into frames.

MAC Addressing: Each device on the network has a unique MAC address, which is included in the frame's header for destination and source identification.

CSMA/CD Protocol: The Carrier Sense Multiple Access with Collision Detection (CSMA/CD) protocol is used to prevent multiple devices from sending data simultaneously, which would cause a "collision" and data corruption.

Data Transmission: The data is then transmitted over the physical network medium (the cable).

Error Detection: A frame check sequence (FCS) is included to ensure data integrity, and devices check for errors upon receiving data.

Key Benefits

High Speed:

Ethernet offers fast data transfer rates, with modern standards supporting speeds up to 100 Gbps and beyond.

Security:

Wired Ethernet connections are generally more secure and less susceptible to interference than wireless networks.

Reliability:

Ethernet is a reliable technology for consistent data exchange, making it ideal for critical applications like file sharing and printing.

Scalability:

Ethernet technology continues to evolve, with new standards offering higher performance and compatibility with older networks.

3.4.1 Ethernet protocol

Ethernet protocols define rules for data communication on networks, operating at the Physical and Data Link layers of the OSI model. Key aspects include:

MAC Addressing: Uses unique MAC addresses in the Ethernet frame for device identification and routing.

CSMA/CD: Carrier Sense Multiple Access with Collision Detection is a media access method where devices "listen" before transmitting to avoid sending data simultaneously.

Ethernet Frames: Data is packaged into frames with a header (MAC addresses, type), data payload (containing higher-level protocol data), padding, and a Frame Check Sequence (FCS) for error detection.

Standards (IEEE 802.3): Ethernet is governed by the IEEE 802.3 standard, which specifies hardware, cabling, and frame formats.

Types: Ethernet has evolved from slower speeds to Fast Ethernet (100 Mbps), Gigabit Ethernet (1 Gbps), and faster versions, while maintaining backward compatibility.

IEEE standards :

- In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers.

- Project 802 is a way of specifying functions of the physical layer and the data link layer of major LAN protocols.
- The standard was adopted by the American National Standards Institute (ANSI).
- In 1987, the International Organization for Standardization (ISO) also approved it as an international standard under the designation ISO 8802.
- The IEEE has subdivided the data link layer into two sub layers: logical link control (LLC) and media access control (MAC).
- IEEE has also created several physical layer standards for different LAN protocols.

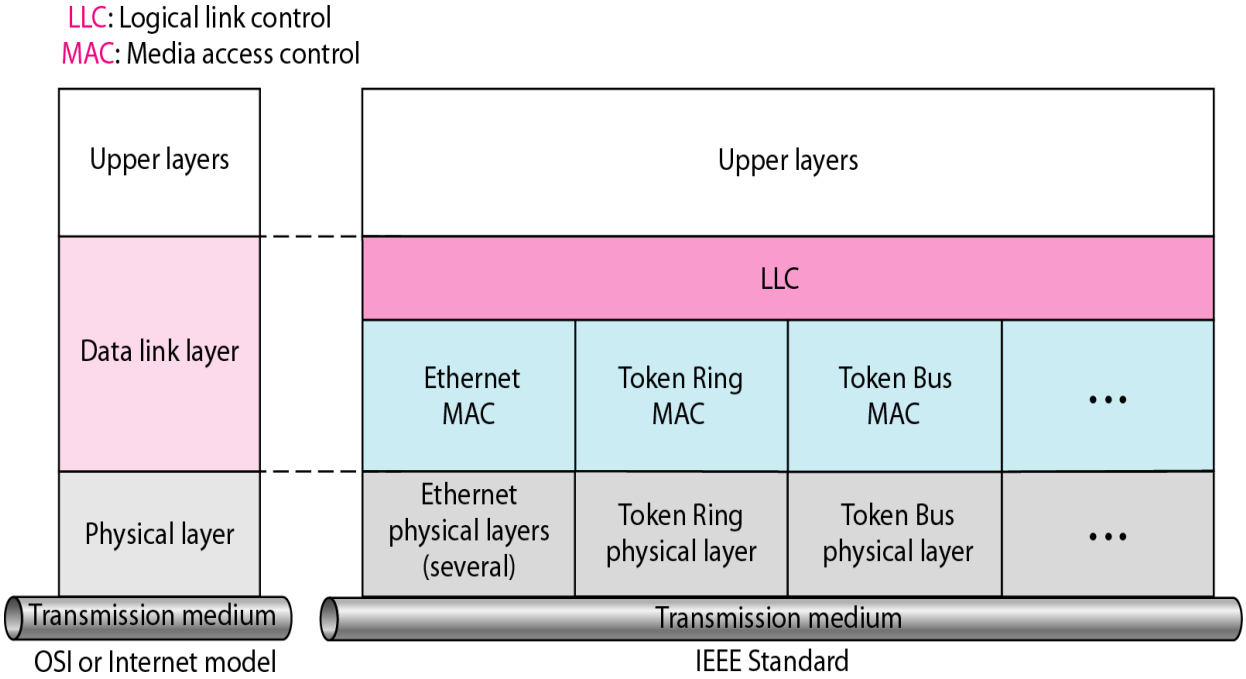


Figure 3.15: The relationship of the 802 Standard to the traditional OSI model

3.4.2 Standard Ethernet :

- The original Ethernet was created in 1976 at Xerox's Palo Alto Research Center (PARC). Since then, it has gone through four generations. We briefly discuss the Standard (or traditional) Ethernet in this section.

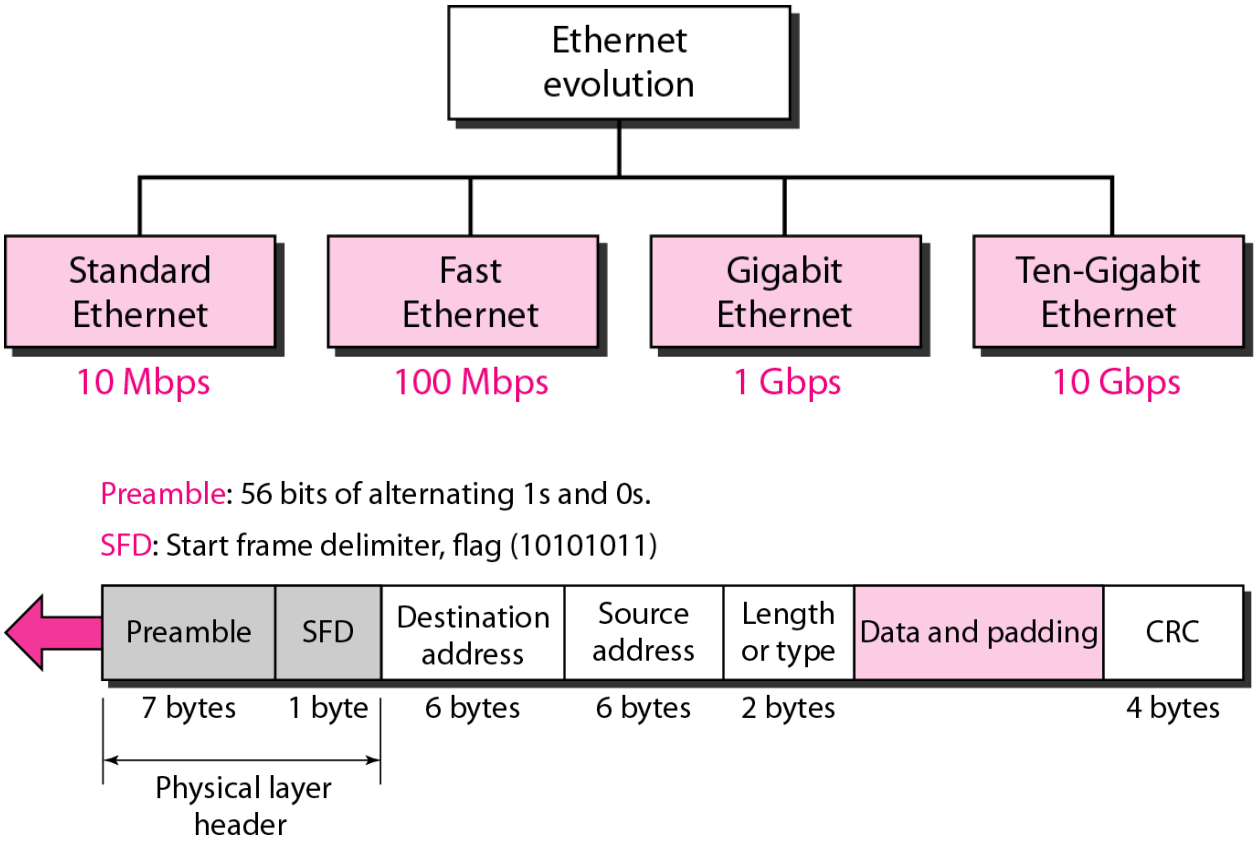


Figure 3.16: MAC SUB LAYER: *802.3 MAC frame*

802.3 MAC frame :

- Ethernet does not provide any mechanism for acknowledging received frames, making it what is known as an unreliable medium
- The Ethernet frame contains seven fields:
  - Preamble
  - SFD
  - DA
  - SA
  - length or type of protocol data unit (PDU)
  - upper-layer data
  - CRC.

Preamble:

- The first field of the 802.3 frame contains 7 bytes (56 bits) of alternating 0s and 1s that alerts the receiving system to the coming frame and enables it to synchronize its input timing.
- The pattern provides only an alert and a timing pulse.

- The preamble is actually added at the physical layer and is not (formally) part of the frame.
-

Start frame delimiter (SFD):

- The second field (l byte: 10101011) signals the beginning of the frame.
- The SFD warns the station or stations that this is the last chance for synchronization.
- The last 2 bits is 11 and alerts the receiver that the next field is the destination address.

Destination address (DA):

- The DA field is 6 bytes and contains the physical address of the destination station or stations to receive the packet.

Source address (SA):

- The SA field is also 6 bytes and contains the physical address of the sender of the packet.

Length or type:

- This field is defined as a type field or length field.
- The original Ethernet used this field as the type field to define the upper-layer protocol using the MAC frame.
- The IEEE standard used it as the length field to define the number of bytes in the data field. Both uses are common today.

Data:

- This field carries data encapsulated from the upper-layer protocols. It is a minimum of 46 and a maximum of 1500 bytes.

CRC:

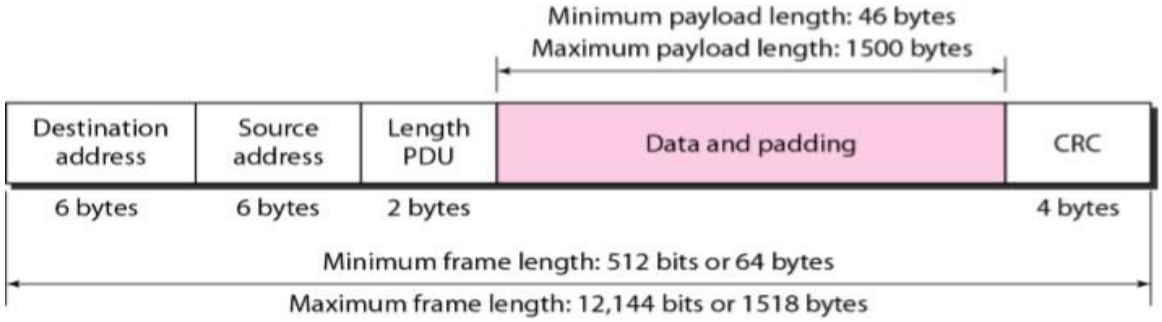
- The last field contains error detection information, in this case a CRC-32

Frame Length :

- Ethernet has imposed restrictions on both the minimum and maximum lengths of a frame,
- The minimum length restriction is required for the correct operation of CSMA/CD .
- An Ethernet frame needs to have a minimum length of 512 bits or 64 bytes. Part of this length is the header and the trailer.
- If we count 18 bytes of header and trailer

- (6 bytes of source address, 6 bytes of destination address, 2 bytes of length or type, and 4 bytes of CRC), then the minimum length of data from the upper layer is 64 - 18 = 46 bytes.

- If the upper-layer packet is less than 46 bytes, padding is added to make up the difference.



3.4.3 Fast Ethernet:

Fast Ethernet is an early standard in computer networks, defined by the IEEE 802.3u standard, that provides a data transmission speed of 100 megabits per second (Mbps). It's a significant improvement over the original 10 Mbps Ethernet, maintaining the same frame format and 48-bit addressing while supporting higher data rates. Fast Ethernet is backward-compatible with older Ethernet standards and can use various cable types, such as Category 5 (CAT5) twisted-pair cables and fiber optic cables, making it a foundational technology for modern networks before the widespread adoption of Gigabit Ethernet.

Key Characteristics of Fast Ethernet

1. Speed:

It transmits data at a maximum rate of 100 Mbps.

2. Standardization:

The IEEE 802.3u standard defines Fast Ethernet, which was a major advancement in the mid-1990s.

3. Compatibility:

It is backward-compatible with standard Ethernet, ensuring that older devices can still connect to the network.

4. Cable Types:

Common variations include:

100BASE-TX: Uses two pairs of Category 5 UTP (unshielded twisted-pair) cable.

100BASE-FX: Uses fiber optic cable for longer distances and better signal integrity.

100BASE-T4: Uses four pairs of Category 3 (Cat3) UTP cable.

5. Frame Format:

It uses the same frame format as original Ethernet, ensuring seamless integration with existing systems.

Applications:

While largely superseded by Gigabit Ethernet, Fast Ethernet is still found in older systems, industrial control environments, and retail point-of-sale (POS) systems where its speed is sufficient.

How it works

Fast Ethernet increased the data rate from 10 Mbps to 100 Mbps by making improvements to the physical layer of the network, allowing for faster data transmission over the same medium or equivalent. Devices use auto-negotiation to determine the highest possible speed and mode of operation. The use of more advanced cabling like Cat5e and switches, which reduce collisions and increase throughput, is also a key component of Fast Ethernet.

3.4.4 Giga-bit Ethernet:

Gigabit Ethernet is a computer networking technology providing data rates of 1 gigabit per second (1 Gbps or 1000 Mbps) and is a standard in modern networks for high-speed connectivity. It operates over either copper (like Cat5e/Cat6 cables) or fiber optic media, uses the IEEE 802.3 standard (802.3ab for copper, 802.3z for fiber), and supports full-duplex operation for increased efficiency. By being compatible with existing Ethernet standards, it allows for a scalable and high-performance network infrastructure that supports demanding applications like 4K streaming and large data transfers.

Key Characteristics and Features

1. High Speed:

Delivers 1 Gbps (1000 Mbps), a tenfold increase over Fast Ethernet.

2. IEEE 802.3 Standard:

Defined by the IEEE 802.3 committee, with specific standards like 802.3z for fiber optics and 802.3ab for copper.

3. Media Support:

Compatible with both copper cabling (Category 5e or higher) and fiber optic cables, offering flexibility for different network needs.

4. Full-Duplex Operation:

Supports simultaneous two-way data transmission, significantly increasing bandwidth and reducing congestion compared to half-duplex.

5. Compatibility:

Designed to be compatible with older Fast Ethernet and Standard Ethernet networks, allowing for smooth upgrades to high-speed infrastructure.

6. Enhanced Performance:

Reduces network bottlenecks, improves performance for data-intensive tasks, and enables seamless streaming of high-resolution content.

Common Gigabit Ethernet Standards (Varieties)

1000BASE-T:

A standard that operates over twisted-pair copper cables, using existing cabling infrastructure like Cat5e and Cat6.

1000BASE-SX:

Uses short-wavelength fiber optics for shorter distances (e.g., 220-550 meters).

1000BASE-LX:

A standard for single-mode fiber optics that allows for much longer distances, up to 5 km.

1000BASE-CX:

An initial fiber standard that uses shielded twisted-pair cables for short segments (up to 25 meters).

Applications

Gigabit Ethernet is used for:

- **Network Backbone:** Connecting high-speed switches and routers in data centers.
- **Enterprise Networks:** Providing fast connectivity between servers, workstations, and network segments.
- **Digital Signage:** Enabling the smooth transmission of high-resolution video content.

- **CCTV Networks:** Interconnecting camera systems and switchers for efficient data transfer.

3.6.5 10 Giga-bit Ethernet:

10 Gigabit Ethernet (10GbE) is a high-speed networking technology offering data transfer rates of 10 gigabits per second (Gbps), a tenfold increase over standard Gigabit Ethernet. Defined by the IEEE 802.3ae standard, it uses optical fiber or copper cables for full-duplex, point-to-point connections, providing enhanced bandwidth for bandwidth-intensive applications like data centers, cloud storage, and enterprise networks. While requiring more expensive hardware, 10GbE is essential for modern networks supporting AI, 5G, and IoT, and can serve as the backbone for metropolitan and wide area networks.

Key Characteristics and Technology

1. Speed:

Achieves 10 Gbps, making it a significant upgrade for high-performance networking.

2. Standards:

Specified by the IEEE 802.3ae-2002 standard.

3. Communication Mode:

Supports only full-duplex, point-to-point links, meaning data flows in both directions simultaneously on a single connection, eliminating the need for older shared-medium technologies like CSMA/CD.

4. Media:

Primarily uses fiber optic cables for longer distances and higher bandwidth, such as 10GBase-SR (short-range) and 10GBase-LR (long-range). For shorter distances or specific applications, copper options like 10GBASE-CX4 and 10GBASE-T exist.

5. Data Encoding:

Employs advanced encoding schemes, such as 64B/66B coding, to achieve the high data rates.

Applications

- **Data Centers & Storage:**

  Provides the necessary bandwidth for fast data access in network-attached storage (NAS) and storage area networks (SANs).

- **Enterprise Networks:**

  Ideal for large organizations requiring high-speed infrastructure to support demanding applications.

- **Metropolitan and Wide Area Networks (MAN/WAN):**

  Serves as a high-performance backbone for connecting larger geographic areas.

- **High-Performance Applications:**

  Essential for handling the large data demands of cloud computing, artificial intelligence (AI), 5G mobile networks, and the Internet of Things (IoT).

Advantages:

1. Increased Bandwidth: Dramatically increases network capacity for data-intensive tasks.

2. Reduced Bottlenecks: Helps prevent data transfer delays and bottlenecks in busy networks.

3. Enhanced Performance: Delivers higher network speed, lower latency, and improved overall connectivity.

CN UNIT-III

Assignment-Cum-Tutorial questions

PART-A

Objective Questions (1M)

1. In _____methods, no station is superior to another station and none is assigned the control over another

a) random access b) controlled access
c) channelization d) none of the above

2. In _____, the chance of collision can be reduced if a station senses the medium before trying to use it.

a) MA b) CSMA
c) FDMA d) CDMA

3. _____requires that each station first listen to the medium before sending.

a) MA b) CSMA
c) FDMA d) CDMA

4. _____augments the CSMA algorithm to detect collision

a) CSMA/CA b) CSMA/CD
c) either (a) or (b) d) both (a) and (b)

5. To avoid collisions on wireless networks, _____was invented.

a) CSMA/CA b) CSMA/CD
c) either (a) or (b) d) both (a) and (b)

6. In _____, collisions are avoided through the use of three strategies: the interframe space, the contention window, and acknowledgments.

a) CSMA/CA b) CSMA/CD
c) either (a) or (b) d) both (a) and (b)

7. In _____methods, a station cannot send unless it has been authorized by other stations.

a) random access b) controlled access
c) channelization d) none of the above

8. In the _____ method, time is divided into intervals. In each interval, a reservation frame precedes the data frames sent in that interval.

a) Reservation b) polling
c) Token passing d) none of the above

9. In _____each station sends a frame whenever it has a frame to send.

a) Pure ALOHA b) slotted ALOHA
c) both (a) and (b) d) neither (a) nor (b)

10. In pure ALOHA, the vulnerable time is _____the frame transmission time.

a) two times b) three times

c) the same as d) none of the above

11. The following are the types of Controlled access protocols

a) Polling and Reservation b) Reservation and token passing
b) Polling and Token passing d) Polling, Reservation & Token passing

12. CDMA stands for

a) Code division multiple accurate b) Code division multiple access
c) Code divide multiple accurate d) Code divide multiple access

13. TDMA stands for

a) Time division multiple accurate b) Time divide multiple accurate
c) Time division multiple access d) Time divide multiple access

14. FDMA stands for

a) Frequency division multiple access b) Frequency division multiple accurate
c) Frequency divide multiple accurate d) Frequency divide multiple access

15. Standard Ethernet provides a data rate of

a) 10 gbps b) 10 mbps
c) 100 mbps d) 1 gbps

16. Fast Ethernet provides a data rate of

a) 10 gbps b) 10 mbps
c) 100 mbps d) 1 gbps

17. Giga-bit Ethernet provides a data rate of

a) 10 gbps b) 10 mbps
c) 100 mbps d) 1 gbps

18. 10 Giga-bit Ethernet provides a data rate of

a) 10 gbps b) 10 mbps
c) 100 mbps d) 1 gbps

PART-B

Short Answer Questions (2M)

1	List the two versions of ALOHA?	L2
2	State the basic idea behind pure ALOHA.	L1
3	What is vulnerable time in pure ALOHA?	L2
4	What is the necessity of controlled access protocols?	L2
5	What is the advantage of CDMA over FDMA and TDMA	L1
6	Classify the Ethernets	L1

PART-C

Descriptive Questions (5M)

1	Two hosts are in a CSMA/CD network and the medium has a data transfer capacity of 1Gbps. The minimum frame length is fixed to 1,000 bits and the propagation speed is 2×10^8 m/s.	L2
2	Discuss the working of CSMA/CD protocol.	L2
3	Explain the Token passing controlled access protocol	L2
4	Explain the working of FDMA with block diagram.	L2
5	Explain the working of CDMA channelization protocol.	L2
6	Explain the MAC layer in IEEE 802.3	L3
7	Explain the following a) Giga-bit Ethernet b) 10 giga-bit Ethernet	L2
8	i) What will be the distance between the hosts? ii) If it is an Ethernet network, what is the efficiency when the hosts have a maximum distance between them? If the distance is reduced to 1 m, does it impact the efficiency?	L3
9	Compare all the Ethernets in data communications	L2

UNIT-IV

Objective: The objective of this unit is to enable students to understand the design issues, functions, and protocols of the Network Layer in computer networks. It focuses on packet switching techniques, routing algorithms, congestion control mechanisms, and internetworking concepts including IPv4 and IPv6 addressing and transition mechanisms.

Outcomes:

After successful completion of this unit, students will be able to:

- Understand the design issues and services offered by the network layer to the transport layer.
- Differentiate between connectionless and connection-oriented services and evaluate virtual circuit vs datagram networks**.**
- Explain and compare various routing algorithms such as shortest path, flooding, distance vector, and link-state routing.
- Analyze the causes and effects of network congestion and apply congestion control and prevention techniques such as leaky bucket, token bucket, and load shedding.
- Describe the concept of internetworking**,** including tunneling, fragmentation, and routing across networks.
- Differentiate and illustrate the features of IPv4 and IPv6, their header structures**,** addressing schemes, and transition mechanisms.

Syllabus:

4.1 The Network Layer Design Issues

- 4.1.1 Store and Forward Packet Switching
- 4.1.2 Services provided to the Transport layer
- 4.1.3 Implementation of Connectionless Service
- 4.1.4 Implementation of Connection Oriented Service
- 4.1.5 Comparison of Virtual Circuit and Datagram Networks.

4.2 Routing Algorithms

- 4.2.1 The Optimality principle
- 4.2.2 Shortest path
- 4.2.3 Flooding
- 4.2.4 Distance vector
- 4.2.5 Link state
- 4.2.6 Hierarchical

4.3 Congestion Control algorithms

- 4.3.1 General principles of congestion control
- 4.3.2 Congestion prevention polices
- 4.3.3 Approaches to Congestion Control
- 4.3.4 Traffic Aware Routing
  - 4.3.4.1 Admission Control
- 4.3.5 Traffic Throttling
  - 4.3.5.1 Load Shedding
- 4.3.6 Traffic Control Algorithm
  - 4.3.6.1 Leaky bucket
  - 4.3.6.2 Token bucket.

4.4 Internet Working

- 4.4.1 How networks differ
- 4.4.2 How networks can be connected
- 4.4.3 Tunnelling

4.4.4 Internetwork routing

4.4.5 Fragmentation

4.5 Network layer in the internet

4.5.1 The IP Version 4 Protocol header

4.5.2 IP addresses

4.5.3 Classful Addressing

4.5.4 Subnets

4.5.5 CIDR- Classless Inter Domain Routing

4.5.6 IPV6

4.5.7 Transition from IPv4 to IPv6 Address

4.5.8 Comparison of IPV4 & IPV6

Learning Material

4.1 Network Layer Design Issues

4.1.1 Store-and-Forward Packet Switching

- The major components of the system are the carrier's equipment (routers connected by transmission lines), shown inside the shaded oval, and the customers' equipment, shown outside the oval.

- Host H1 is directly connected to one of the carrier's routers, A, by a leased line. In contrast, H2 is on a LAN with a router, F, owned and operated by the customer. This router also has a leased line to the carrier's equipment.

- We have shown F as being outside the oval because it does not belong to the carrier, but in terms of construction, software, and protocols, it is probably no different from the carrier's routers. Whether it belongs to the subnet is arguable, but for the purposes of this chapter, routers on customer premises are considered part of the subnet because they run the same algorithms as the carrier's routers (and our main concern here is algorithms).

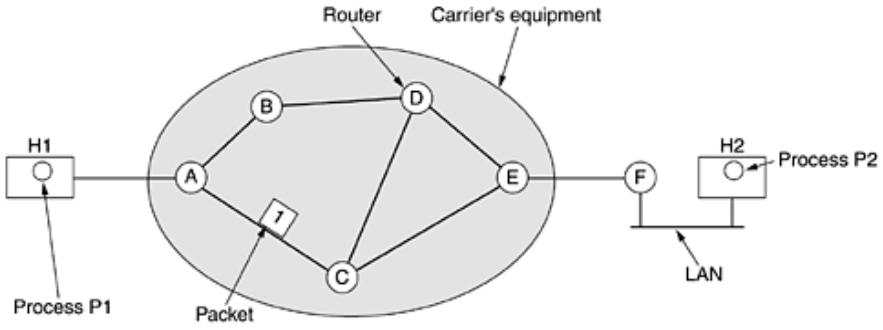


Figure 4.1: The environment of the network layer protocols

- This equipment is used as follows. A host with a packet to send transmits it to the nearest router, either on its own LAN or over a point-to-point link to the carrier. The packet is stored there until it has fully arrived so the checksum can be verified. Then it is forwarded to the next router along the path until it reaches the destination host, where it is delivered.

4.1.2 Services Provided to the Transport Layer

The network layer provides services to the transport layer at the network layer/transport layer interface. An important question is what kind of services the network layer provides to the transport layer. The network layer services have been designed with the following goals in mind.

1. The services should be independent of the router technology.
2. The transport layer should be shielded from the number, type, and topology of the routers present.
3. The network addresses made available to the transport layer should use a uniform numbering plan, even across LANs and WANs.

- Given these goals, the designers of the network layer have a lot of freedom in writing detailed specifications of the services to be offered to the transport layer. This freedom often degenerates into a raging battle between two warring factions. The discussion centers on whether the network layer should provide connection-oriented service or connectionless service.

4. One camp (represented by the Internet community) argues that the routers' job is moving packets around and nothing else. In their view (based on 30 years of actual experience with a real, working computer network), the subnet is inherently unreliable, no matter how it is designed. Therefore, the hosts should accept the fact that the network is unreliable and do error control (i.e., error detection and correction) and flow control themselves.
5. This viewpoint leads quickly to the conclusion that the network service should be connectionless, with primitives SEND PACKET and RECEIVE PACKET and little else. In particular, no packet ordering and flow control should be done, because the hosts are going to do that anyway, and there is usually little to be gained by doing it twice. Furthermore, each packet must carry the full destination address, because each packet sent is carried independently of its predecessors, if any.
6. The other camp (represented by the telephone companies) argues that the subnet should provide a reliable, connection-oriented service. They claim that 100 years of successful experience with the worldwide telephone system is an excellent guide. In this view, quality of service is the dominant factor, and without connections in the subnet, quality of service is very difficult to achieve, especially for real-time traffic such as voice and video.
7. These two camps are best exemplified by the Internet and ATM. The Internet offers connectionless network-layer service; ATM networks offer connection-oriented network-layer service.

4.1.3 Implementation of Connectionless Service

- In connectionless service packets are injected into the subnet individually and routed independently of each other. No advance setup is needed.
- In this context, the packets are frequently called datagrams (in analogy with telegrams) and the subnet is called a datagram subnet.
- If connection-oriented service is used, a path from the source router to the destination router must be established before any data packets can be sent.
- This connection is called a VC (virtual circuit), in analogy with the physical circuits set up by the telephone system, and the subnet is called a virtual-circuit subnet.
- Suppose that the process P1 in figure 4.2 has a long message for P2. It hands the message to the transport layer with instructions to deliver it to process P2 on host H2.

- The transport layer code runs on H1, typically within the operating system. It prepends a transport header to the front of the message and hands the result to the network layer, probably just another procedure within the operating system.

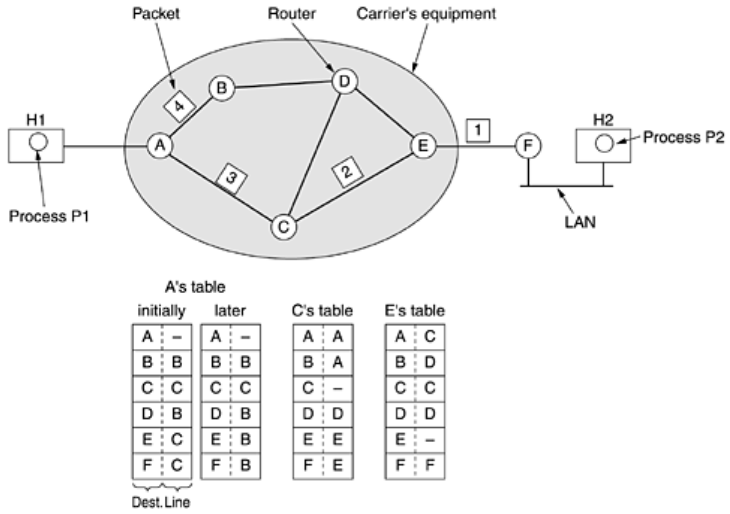


Figure 4.2: Routing within a datagram subnet.

- Let us assume that the message is four times longer than the maximum packet size, so the network layer has to break it into four packets, 1, 2, 3, and 4 and sends each of them in turn to router A using some point-to-point protocol, for example, PPP. At this point the carrier takes over.
- Every router has an internal table telling it where to send packets for each possible destination. Each table entry is a pair consisting of a destination and the outgoing line to use for that destination. Only directly-connected lines can be used.
- For example, in fig 4.2, A has only two outgoing lines—to B and C—so every incoming packet must be sent to one of these routers, even if the ultimate destination is some other router. A's initial routing table is shown in the figure under the label "initially."
- As they arrived at A, packets 1, 2, and 3 were stored briefly (to verify their checksums). Then each was forwarded to C according to A's table. Packet 1 was then forwarded to E and then to F. When it got to F, it was encapsulated in a data link layer frame and sent to H2 over the LAN. Packets 2 and 3 follow the same route.
- However, something different happened to packet 4. When it got to A it was sent to router B, even though it is also destined for F. For some

reason, A decided to send packet 4 via a different route than that of the first three. Perhaps it learned of a traffic jam somewhere along the ACE path and updated its routing table, as shown under the label "later." The algorithm that manages the tables and makes the routing decisions is called the routing algorithm.

4.1.4 Implementation of Connection-Oriented Service

- For connection-oriented service, we need a virtual-circuit subnet.
- The idea behind virtual circuits is to avoid having to choose a new route for every packet sent, as in fig 4.2. Instead, when a connection is established, a route from the source machine to the destination machine is chosen as part of the connection setup and stored in tables inside the routers.
- That route is used for all traffic flowing over the connection, exactly the same way that the telephone system works. When the connection is released, the virtual circuit is also terminated. With connection-oriented service, each packet carries an identifier telling which virtual circuit it belongs to.
- As an example, consider the situation of figure 4.3. Here, host H1 has established connection 1 with host H2. It is remembered as the first entry in each of the routing tables. The first line of A's table says that if a packet bearing connection identifier 1 comes in from H1, it is to be sent to router C and given connection identifier 1. Similarly, the first entry at C routes the packet to E, also with connection identifier 1.

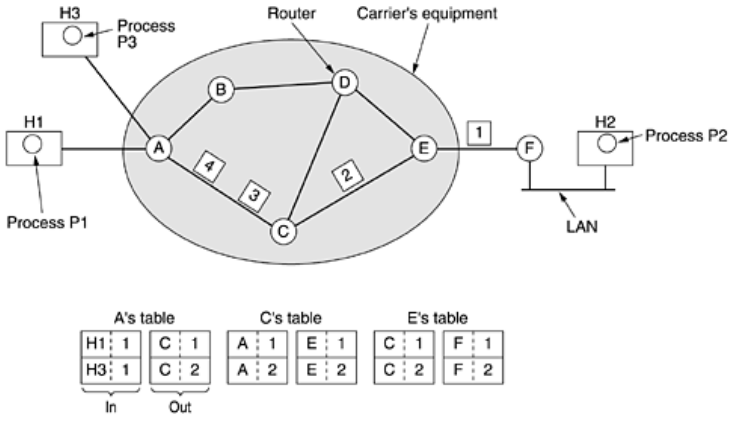


Figure 4.3: Routing within a Virtual-circuit subnet.

4.1.5 Comparison of Virtual-Circuit and Datagram Subnets

Issue	Datagram subnet	Virtual-circuit subnet
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Figure 4.4: Comparison of datagram and virtual-circuit subnets.

4.2 ROUTING ALGORITHMS

- In order to transfer the packets from source to the destination, the network layer must determine the best route through which packets can be transmitted.
- Whether the network layer provides datagram service or virtual circuit service, the main job of the network layer is to provide the best route. The routing protocol provides this job.
- The routing protocol is a routing algorithm that provides the best path from the source to the destination. The best path is the path that has the "least-cost path" from source to the destination.

- Routing is the process of forwarding the packets from source to the destination but the best route to send the packets is determined by the routing algorithm.
- The routing algorithm is that part of the network layer software responsible for deciding which output line an incoming packet should be transmitted on.

Routing algorithms can be grouped into **two** major classes:

1. Non adaptive or Static and
2. Adaptive or Dynamic

Non-Adaptive Routing Algorithms:

Definition:

Non-adaptive routing algorithms use fixed routes that do not change regardless of network conditions.

How it works:

Routes are pre-determined, often based on fixed criteria such as shortest path or predefined paths configured manually.

Characteristics:

- Static and does not respond to network changes.
- Simple to implement.
- Routes remain the same even if there is congestion or link failure.

Advantages:

- Low overhead, no need for route updates or monitoring.

- Simpler and faster in small or stable networks.

Disadvantages:

- Inefficient in large or dynamic networks.
- Can cause packet loss or delays if a chosen path fails or gets congested.

Adaptive Routing Algorithms

- **Definition:**
  Adaptive routing algorithms dynamically adjust the routes based on the current state of the network (like traffic load, failures, or congestion).
- **How it works:**
  They continuously collect and use information such as link status, delay, or congestion to choose the best path in real-time.
- **Characteristics:**
  - Responds to network changes.
  - Can reroute traffic if a path fails or becomes congested.
  - More complex due to constant monitoring and route recalculation.

Examples:

- Distance Vector Routing (e.g., RIP)
- Link State Routing (e.g., OSPF)

Advantages:

- Better at handling dynamic network conditions.
- More reliable and efficient under varying traffic.

Disadvantages:

- Requires more processing power and bandwidth for updates.
- Can suffer from routing loops or instability if not well designed.

Feature	Adaptive Routing	Non-Adaptive Routing
Route changes	Yes, dynamic	No, fixed
Network condition info	Required	Not required
Complexity	Higher	Lower
Efficiency	Better in dynamic networks	Suitable for small/stable nets
Overhead	Higher due to updates	Low
Fault tolerance	Good	Poor

4.2.1 Optimality Principle (Bellman, 1957)

- "An optimal path between two nodes has the property that any intermediate node on the path divides the path into two subpaths, each of which is also optimal."

In simpler terms:

- If you have the best (optimal) path from point A to point C that goes through an intermediate point B,
- Then the path from A to B and the path from B to C must themselves be optimal paths.

Why is it important?

- This principle allows routing algorithms to **break down the routing problem into smaller sub problems**.
- Instead of finding the whole path at once, algorithms can focus on finding optimal routes to intermediate nodes.
- It **simplifies computation** and **enables efficient route calculation** using recursive methods like dynamic programming.
- The set of optimal routes from all sources to a given destination form a tree rooted at the destination. Such a tree is called a sink tree and is illustrated in Fig. 4-5(b), where the distance metric is the number of hops. The goal of all routing algorithms is to **discover** and **use the sink trees** for all routers.

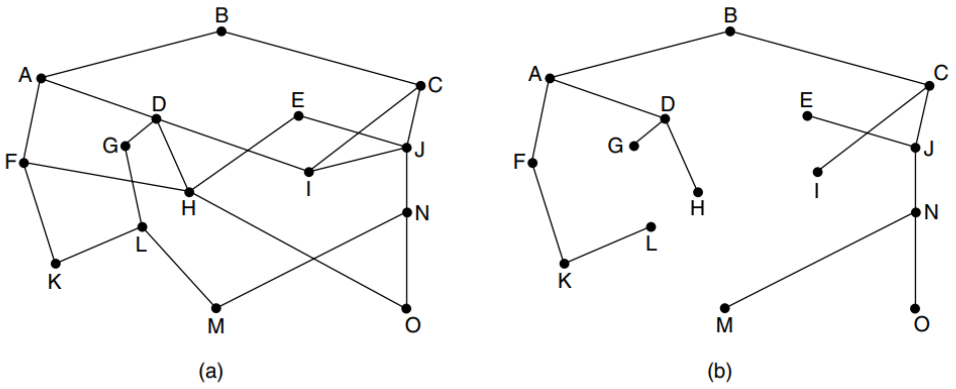


Figure 4.5 (a) A network (b) A sink tree for router B.

4.2.2 Shortest Path Routing

• The idea is to build a graph of the subnet, with each node of the graph representing a router and each arc of the graph representing a communication line (often called a link).

• To choose a route between a given pair of routers, the algorithm just finds the shortest path between them on the graph.

• One way of measuring path length is the number of hops. Another metric is the geographic distance in kilometers. Many other metrics are also possible. For example, each arc could be labeled with the mean queuing and transmission delay for some standard test packet as determined by hourly test runs.

• In the general case, the labels on the arcs could be computed as a function of the distance, bandwidth, average traffic, communication cost, mean queue length, measured delay, and other factors. By changing the weighting function, the algorithm would then compute the "shortest" path measured according to any one of a number of criteria or to a combination of criteria.

The first five steps used in computing the shortest path from A to D. The arrows indicate the working node.

To illustrate how the labelling algorithm works, look at the weighted, undirected graph of Fig), where the weights represent, for example, distance.

- We want to find the shortest path from A to D. We start out by marking node A as permanent, indicated by a filled-in circle.
- Then we examine, in turn, each of the nodes adjacent to A (the working node), relabeling each one with the distance to A.
- Whenever a node is relabelled, we also label it with the node from which the probe was made so that we can reconstruct the final path later.

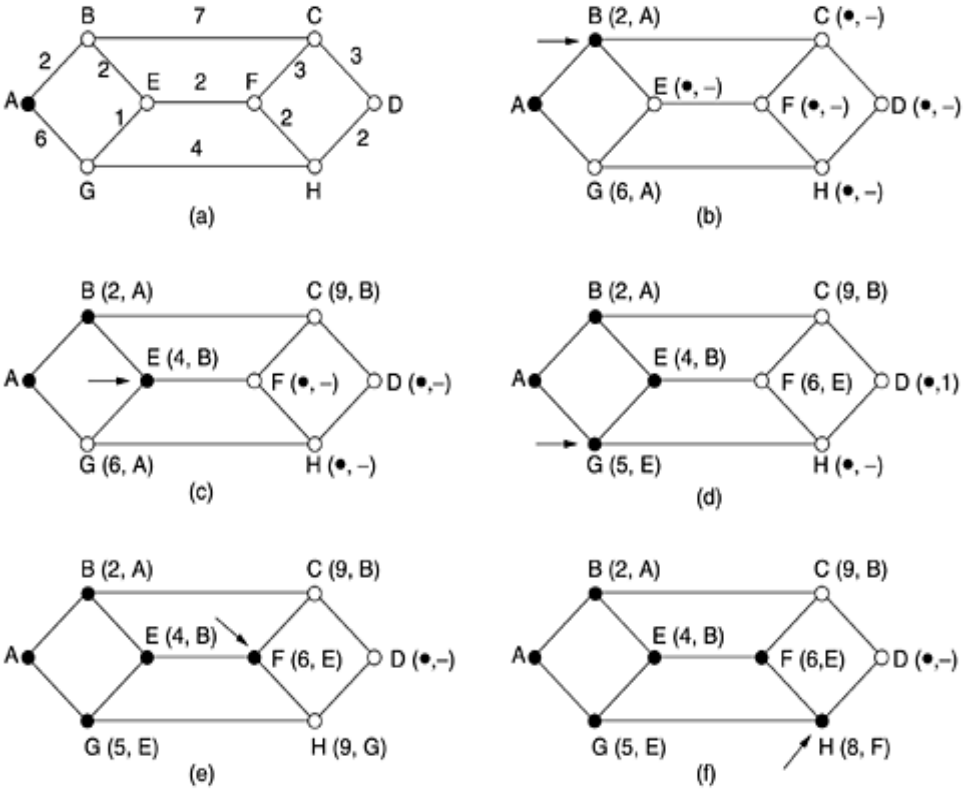


Figure: The first five steps used in computing the shortest path from A to D. The arrows indicate the working node

- Having examined each of the nodes adjacent to A, we examine all the tentatively labelled nodes in the whole graph and make the one with the smallest label permanent, as shown in Fig.
- This one becomes the new working node. We now start at B and examine all nodes adjacent to it. If the sum of the label on B and the distance from B to the node being considered is less than the label on that node, we have a shorter path, so the node is relabelled.
- After all the nodes adjacent to the working node have been inspected and the tentative labels changed if possible, the entire graph is searched for the tentatively-labelled node with the smallest value. This node is made

permanent and becomes the working node for the next round. Figure shows the first five steps of the algorithm.

4.2.3 Flooding

- **Flooding** is a simple routing technique where every incoming packet is sent out on **all outgoing links except the one it arrived on**.

How Flooding Works

- When a node receives a packet, it **broadcasts** the packet to all its neighbors except the sender.
- This process repeats at every node until the packet reaches its destination or a predefined condition stops it.

Characteristics of Flooding

- **No routing tables or path calculations** are needed.
- Ensures **maximum coverage**—the packet reaches every node in the network.
- Can lead to **multiple copies of the same packet** traveling the network.

Problems with Flooding

- **Broadcast Storm:** Excessive packet duplication causes network congestion and wasted bandwidth.
- **Loops:** Packets can circulate indefinitely, causing endless duplication.

Solutions to Flooding Issues

- **Hop Count (TTL):** Limit the packet lifetime by decrementing a hop count field; discard when it reaches zero.
- **Sequence Numbers:** Attach unique IDs to packets so nodes only forward a packet once.

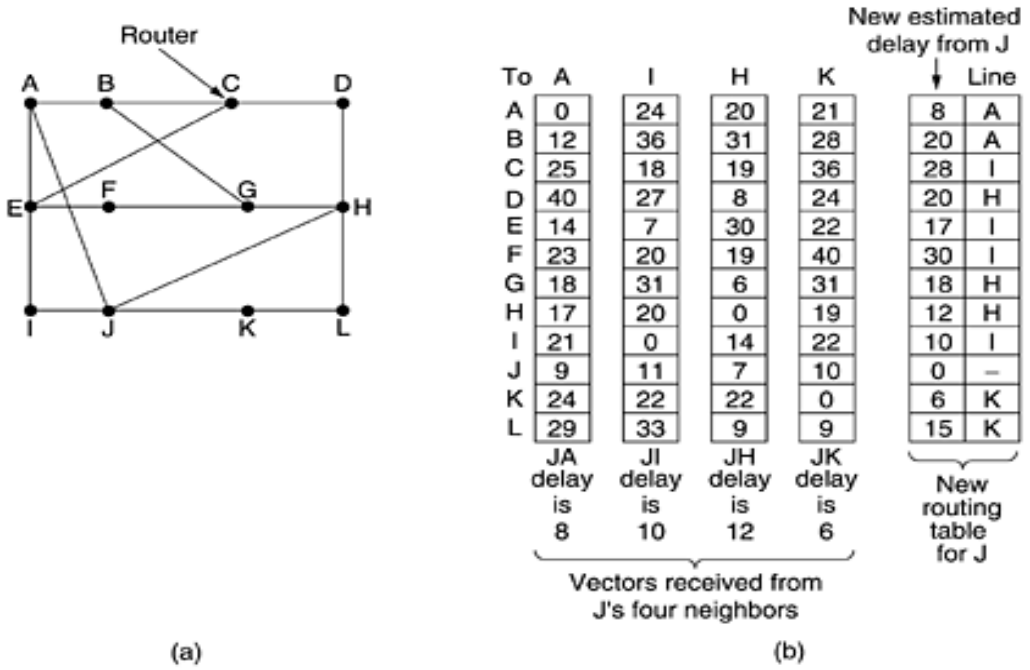
- **Selective Flooding:** Flood only in a certain area or to a subset of neighbors.

When is Flooding Used?

- In **network discovery** or **route discovery** protocols.
- As a **backup or control mechanism** in some routing protocols.
- For **broadcasting control messages** in ad-hoc or sensor networks.

4.2.4 Distance Vector Routing

- Distance vector routing algorithms operate by having each router maintain a table (i.e, a vector) giving the best known distance to each destination and which line to use to get there.
- These tables are updated by exchanging information with the neighbors.
- The distance vector routing algorithm is sometimes called by other names, most commonly the distributed Bellman-Ford routing algorithm and the Ford-Fulkerson algorithm, after the researchers who developed it (Bellman, 1957; and Ford and Fulkerson, 1962).
- It was the original ARPANET routing algorithm and was also used in the Internet under the name RIP.
- Part (a) shows a subnet. The first four columns of part (b) show the delay vectors received from the neighbours of router J.
- A claims to have a 12-msec delay to B, a 25-msec delay to C, a 40-msec delay to D, etc. Suppose that J has measured or estimated its delay to its neighbours, A, I, H, and K as 8, 10, 12, and 6 msec, respectively.



(a) A subnet. (b) Input from A, I, H, K, and the new routing table for J.

- Consider how J computes its new route to router G. It knows that it can get to A in 8 msec, and A claims to be able to get to G in 18 msec, so J knows it can count on a delay of 26 msec to G if it forwards packets bound for G to A. Similarly, it computes the delay to G via I, H, and K as 41 (31 + 10), 18 (6 + 12), and 37 (31 + 6) msec, respectively. The best of these values is 18, so it makes an entry in its routing table that the delay to G is 18 msec and that the route to use is via H. The same calculation is performed for all the other destinations, with the new routing table shown in the last column of the figure.

The Count-to-Infinity Problem

Distance vector routing works in theory but has a serious drawback in practice: although it converges to the correct answer, it may do so slowly. In particular, it reacts rapidly to good news, but leisurely to bad news.

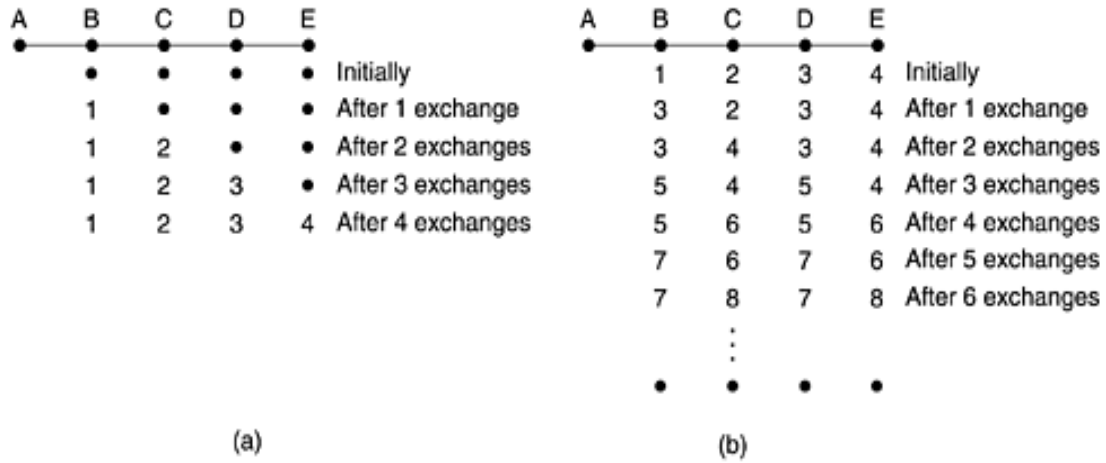


Figure. The count-to-infinity problem.

- Consider the five-node (linear) subnet of Fig. 5-10, where the delay metric is the number of hops. Suppose A is down initially and all the other routers know this. In other words, they have all recorded the delay to A as infinity.
- Now let us consider the situation of Fig. 5-10(b), in which all the lines and routers are initially up. Routers B, C, D, and E have distances to A of 1, 2, 3, and 4, respectively.
- Suddenly A goes down, or alternatively, the line between A and B is cut, which is effectively the same thing from B's point of view.
- At the first packet exchange, B does not hear anything from A. Fortunately, C says: Do not worry; I have a path to A of length 2. Little does B know that C's path runs through B itself. For all B knows, C might have ten lines all with separate paths to A of length 2. As a result, B thinks it can reach A via C, with a path length of 3. D and E do not update their entries for A on the first exchange.
- On the second exchange, C notices that each of its neighbors claims to have a path to A of length 3. It picks one of the them at random and makes its new distance to A 4, as shown in the third row of Fig.

10(b). Subsequent exchanges produce the history shown in the rest of Fig. 10(b).

- From this figure, it should be clear why bad news travels slowly: no router ever has a value more than one higher than the minimum of all its neighbors. Gradually, all routers work their way up to infinity, but the number of exchanges required depends on the numerical value used for infinity. For this reason, it is wise to set infinity to the longest path plus 1. If the metric is time delay, there is no well-defined upper bound, so a high value is needed to prevent a path with a long delay from being treated as down. Not entirely surprisingly, this problem is known as the count-to-infinity problem.

Summary of Count-to-Infinity Problem

- The **Count-to-Infinity Problem** is a routing loop issue where **routers keep increasing the cost to a failed destination** (counting up to infinity), **very slowly**, due to the way distance vector routing works.
- It occurs when a route becomes **invalid** (e.g., a link goes down), but neighboring routers keep telling each other the destination is reachable — with **gradually increasing costs**.
- Since routers only know what their neighbors tell them (not the full topology), they can be misled.

4.2.5 Link State Routing

Distance vector routing was used in the ARPANET until 1979, when it was replaced by link state routing. Two primary problems caused its demise.

First, since the delay metric was queue length, it did not take line bandwidth into account when choosing routes

The second problem also existed, namely, the algorithm often took too long to converge (the count-to-infinity problem).

For these reasons, it was replaced by an entirely new algorithm, now called link state routing. Variants of link state routing are now widely used.

The idea behind link state routing is simple and can be stated as five parts. Each router must do the following:

1. Discover its neighbors and learn their network addresses.

2. Measure the delay or cost to each of its neighbors.

3. Construct a packet telling all it has just learned.

4. Send this packet to all other routers.

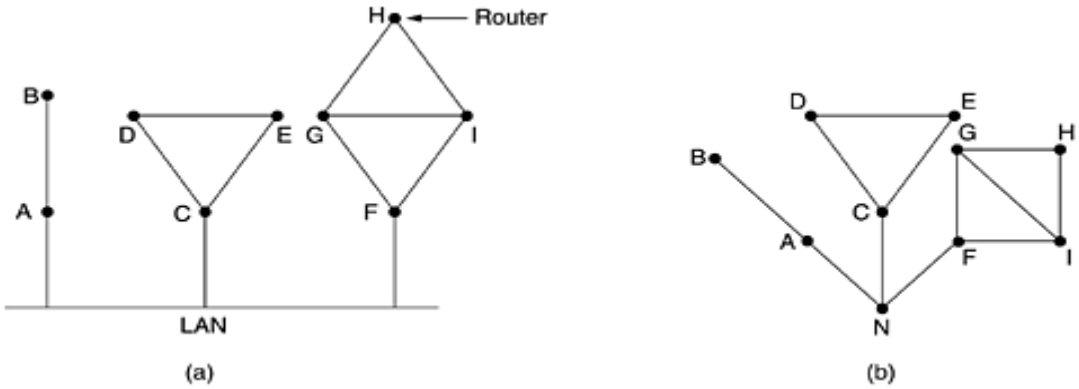
5. Compute the shortest path to every other router.

Learning about the Neighbors

When a router is booted, its first task is to learn who its neighbors are. It accomplishes this goal by sending a special HELLO packet on each point-to-point line. The router on the other end is expected to send back a reply telling who it is.

Fig. 11(a) illustrates a LAN to which three routers, A, C, and F, are directly connected. Each of these routers is connected to one or more additional routers, as shown.

Figure 11. (a) Nine routers and a LAN. (b) A graph model of (a).



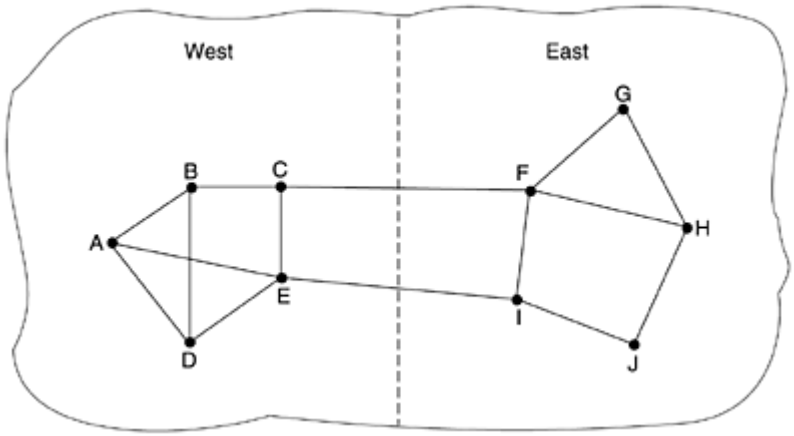
One way to model the LAN is to consider it as a node itself, as shown in Fig. 5-11(b). Here we have introduced a new, artificial node, N, to which A, C, and F are connected. The fact that it is possible to go from A to C on the LAN is represented by the path ANC here.

Measuring Line Cost

The link state routing algorithm requires each router to know, or at least have a reasonable estimate of the delay to each of its neighbors. The most direct way to determine this delay is to send over the line a special ECHO packet that the other side is required to send back immediately. By measuring the round-trip time and dividing it by two, the sending router can get a reasonable estimate of the delay. For even better results, the test can be conducted several times, and the average used. Of course, this method implicitly assumes the delays are symmetric, which may not always be the case.

Unfortunately, there is also an argument against including the load in the delay calculation. Consider the subnet of Fig. 5-12, which is divided into two parts, East and West, connected by two lines, CF and EI.

Figure 12. A subnet in which the East and West parts are connected by two lines.



Suppose that most of the traffic between East and West is using line CF, and as a result, this line is heavily loaded with long delays. Including queueing delay in the shortest path calculation will make EI more attractive. After the new routing tables have been installed, most of the East-West traffic will now go over EI, overloading this line. Consequently, in the next update, CF will appear to be the shortest path. As a result, the routing tables may oscillate wildly, leading to erratic routing and many potential problems

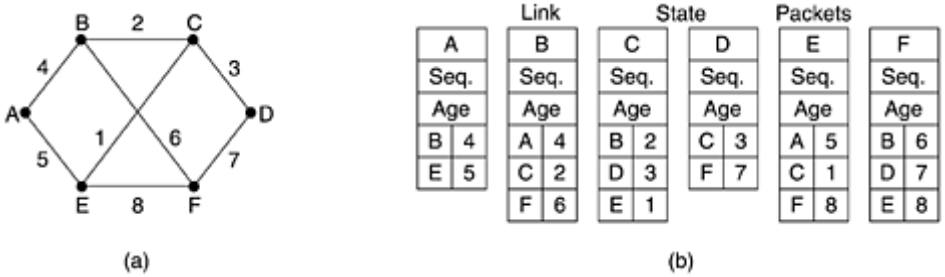
Building Link State Packets

Once the information needed for the exchange has been collected, the next step is for each router to build a packet containing all the data.

The packet starts with the identity of the sender, followed by a sequence number and age, and a list of neighbors.

For each neighbor, the delay to that neighbor is given. An example subnet is given in Fig.13(a) with delays shown as labels on the lines. The corresponding link state packets for all six routers are shown in Fig. 13(b).

Figure 13. (a) A subnet. (b) The link state packets for this subnet.



Building the link state packets is easy. The hard part is determining when to build them. One possibility is to build them periodically, that is, at regular intervals. Another possibility is to build them when some significant event occurs, such as a line or neighbor going down or coming back up again or changing its properties appreciably.

Distributing the Link State Packets

The trickiest part of the algorithm is distributing the link state packets reliably. As the packets are distributed and installed, the routers getting the first ones will change their routes.

Consequently, the different routers may be using different versions of the topology, which can lead to inconsistencies, loops, unreachable machines, and other problems.

First we will describe the basic distribution algorithm. Later we will give some refinements.

The fundamental idea is to use flooding to distribute the link state packets. To keep the flood in check, each packet contains a sequence number that is incremented for each new packet sent. Routers keep track of all the (source router, sequence) pairs they see.

When a new link state packet comes in, it is checked against the list of packets already seen. If it is new, it is forwarded on all lines except the one it arrived on. If it is a duplicate, it is discarded.

If a packet with a sequence number lower than the highest one seen so far ever arrives, it is rejected as being obsolete since the router has more recent data.

This algorithm has a few problems, but they are manageable.

First, if the sequence numbers wrap around, confusion will reign. The solution here is to use a 32-bit sequence number. With one link state packet per second, it would take 137 years to wrap around, so this possibility can be ignored.

Second, if a router ever crashes, it will lose track of its sequence number. If it starts again at 0, the next packet will be rejected as a duplicate.

Third, if a sequence number is ever corrupted and 65,540 is received instead of 4 (a 1-bit error), packets 5 through 65,540 will be rejected as obsolete, since the current sequence number is thought to be 65,540.

Computing the New Routes

Once a router has accumulated a full set of link state packets, it can construct the entire subnet graph because every link is represented. Every

link is, in fact, represented twice, once for each direction. The two values can be averaged or used separately.

Now Dijkstra's algorithm can be run locally to construct the shortest path to all possible destinations. The results of this algorithm can be installed in the routing tables, and normal operation resumed.

4.2.6 Hierarchical Routing

As networks grow in size, the router routing tables grow proportionally. Not only is router memory consumed by ever-increasing tables, but more CPU time and more bandwidth is also increasing.

At a certain point the network may grow to the point where it is no longer feasible for every router to have an entry for every other router, so the routing will have to be done hierarchically, as it is in the telephone network.

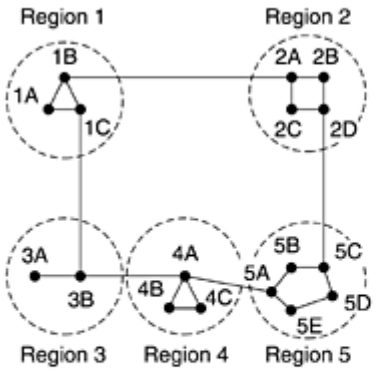
When hierarchical routing is used, the routers are divided into what we will call regions, with each router knowing all the details about how to route packets to destinations within its own region, but knowing nothing about the internal structure of other regions. When different networks are interconnected, it is natural to regard each one as a separate region in order to free the routers in one network from having to know the topological structure of the other ones.

For huge networks, a two-level hierarchy may be insufficient; it may be necessary to group the regions into clusters, the clusters into zones, the zones into groups, and so on.

Figure 15 gives a quantitative example of routing in a two-level hierarchy with five regions. The full routing table for router 1A has 17 entries, as

shown in Fig. 15(b). When routing is done hierarchically, as in Fig. 15(c), there are entries for all the local routers as before, but all other regions have been condensed into a single router, so all traffic for region 2 goes via the 1B -2A line, but the rest of the remote traffic goes via the 1C -3B line. Hierarchical routing has reduced the table from 17 to 7 entries. As the ratio of the number of regions to the number of routers per region grows, the savings in table space increase.

Figure 15. Hierarchical routing.



(a)

Full table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

(c)

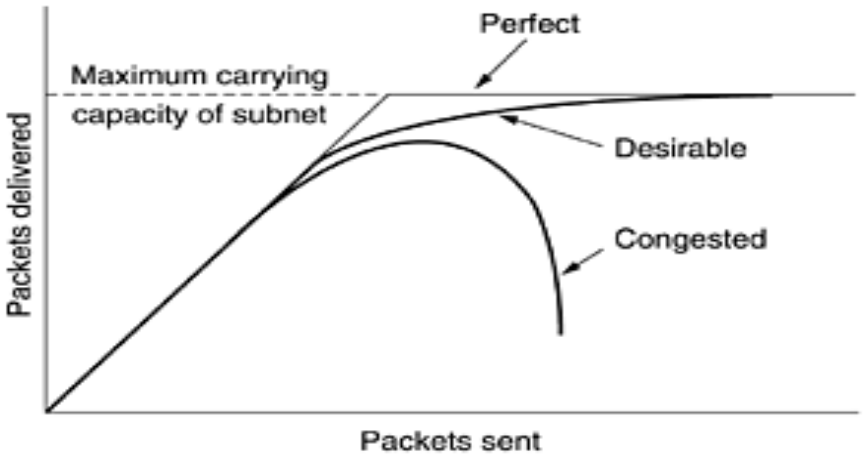
Unfortunately, these gains in space are not free. There is a penalty to be paid, and this penalty is in the form of increased path length. For example, the best route from 1A to 5C is via region 2, but with hierarchical routing all traffic to region 5 goes via region 3, because that is better for most destinations in region 5.

When a single network becomes very large, an interesting question is: How many levels should the hierarchy have? For example, consider a subnet with 720 routers. If there is no hierarchy, each router needs 720 routing table entries. If the subnet is partitioned into 24 regions of 30 routers each, each router needs 30 local entries plus 23 remote entries for a total of 53 entries. If a three-level hierarchy is chosen, with eight clusters, each containing 9 regions of 10 routers, each router needs 10 entries for local routers, 8 entries for routing to other regions within its own cluster, and 7 entries for distant clusters, for a total of 25 entries.

4.3 Congestion Control Algorithms

When too many packets are present in (a part of) the subnet, performance degrades. This situation is called congestion. Figure 25 depicts the symptom. When the number of packets dumped into the subnet by the hosts is within its carrying capacity, they are all delivered (except for a few that are afflicted with transmission errors) and the number delivered is proportional to the number sent. However, as traffic increases too far, the routers are no longer able to cope and they begin losing packets. This tends to make matters worse. At very high trafffic, performance collapses completely and almost no packets are delivered.

Figure: When too much traffic is offered, congestion sets in and performance degrades sharply.



Congestion can be brought on by several factors. If all of a sudden, streams of packets begin arriving on three or four input lines and all need the same output line, a queue will build up. If there is insufficient memory to hold all of them, packets will be lost. Adding more memory may help up to a point, but Nagle (1987) discovered that if routers have an infinite amount of memory, congestion gets worse, not better, because by the time packets get to the front of the queue, they have already timed out (repeatedly) and duplicates have been sent. All these packets will be dutifully forwarded to the next router, increasing the load all the way to the destination.

Slow processors can also cause congestion. If the routers' CPUs are slow at performing the bookkeeping tasks required of them (queueing buffers, updating tables, etc.), queues can build up, even though there is excess line capacity. Similarly, low-bandwidth lines can also cause congestion. Upgrading the lines but not changing the processors, or vice versa, often helps a little, but frequently just shifts the bottleneck. Also, upgrading part, but not all, of the system, often just moves the bottleneck somewhere else. The real problem is frequently a mismatch between parts of the system. This problem will persist until all the components are in balance.

It is worth explicitly pointing out the difference between congestion control and flow control, as the relationship is subtle. Congestion control has to do with making sure the subnet is able to carry the offered traffic. It is a global issue, involving the behavior of all the hosts, all the routers, the store-and-forwarding processing within the routers, and all the other factors that tend to diminish the carrying capacity of the subnet.

Flow control, in contrast, relates to the point-to-point traffic between a given sender and a given receiver. Its job is to make sure that a fast sender cannot continually transmit data faster than the receiver is able to absorb it. Flow control frequently involves some direct feedback from the receiver to the sender to tell the sender how things are doing at the other end.

To see the difference between these two concepts, consider a fiber optic network with a capacity of 1000 gigabits/sec on which a supercomputer is trying to transfer a file to a personal computer at 1 Gbps. Although there is no congestion (the network itself is not in trouble), flow control is needed to force the supercomputer to stop frequently to give the personal computer a chance to breathe.

At the other extreme, consider a store-and-forward network with 1-Mbps lines and 1000 large computers, half of which are trying to transfer files at 100 kbps to the other half. Here the problem is not that of fast senders overpowering slow receivers, but that the total offered traffic exceeds what the network can handle.

The reason congestion control and flow control are often confused is that some congestion control algorithms operate by sending messages back to the various sources telling them to slow down when the network gets into trouble. Thus, a host can get a ''slow down'' message either because the

receiver cannot handle the load or because the network cannot handle it. We will come back to this point later.

We will start our study of congestion control by looking at a general model for dealing with it. Then we will look at broad approaches to preventing it in the first place.

4.3.1 General Principles of Congestion Control

This approach leads to dividing all solutions into two groups:

1. open loop
2. closed loop.

- Open loop solutions attempt to solve the problem by good design, in essence, to make sure it does not occur in the first place. Once the system is up and running, midcourse corrections are not made.
- Tools for doing open-loop control include deciding when to accept new traffic, deciding when to discard packets and which ones, and making scheduling decisions at various points in the network. All of these have in common the fact that they make decisions without regard to the current state of the network.

In contrast, closed loop solutions are based on the concept of a feedback loop. This approach has three parts when applied to congestion control:

1. Monitor the system to detect when and where congestion occurs.
2. Pass this information to places where action can be taken.
3. Adjust system operation to correct the problem.

- A variety of metrics can be used to monitor the subnet for congestion. Chief among these are the percentage of all packets discarded for lack of buffer space, the average queue lengths, the number of packets that time out and are retransmitted, the average packet delay, and the standard deviation of packet delay. In all cases, rising numbers indicate growing congestion.

- The second step in the feedback loop is to transfer the information about the congestion from the point where it is detected to the point where something can be done about it. The obvious way is for the router detecting the congestion to send a packet to the traffic source or sources, announcing the problem. Of course, these extra packets increase the load at precisely the moment that more loads is not needed, namely, when the subnet is congested.
- However, other possibilities also exist. For example, a bit or field can be reserved in every packet for routers to fill in whenever congestion gets above some threshold level. When a router detects this congested state, it fills in the field in all outgoing packets, to warn the neighbors.
- Still another approach is to have hosts or routers periodically send probe packets out to explicitly ask about congestion. This information can then be used to route traffic around problem areas. Some radio stations have helicopters flying around their cities to report on road congestion to make it possible for their mobile listeners to route their packets (cars) around hot spots.
- They further divide the open loop algorithms into ones that act at the source versus ones that act at the destination. The closed loop algorithms are also divided into two subcategories: explicit feedback versus implicit feedback. In explicit feedback algorithms, packets are sent back from the point of congestion to warn the source. In implicit algorithms, the source deduces the existence of congestion by making local observations, such as the time needed for acknowledgements to come back.
- The presence of congestion means that the load is (temporarily) greater than the resources (in part of the system) can handle. Two solutions come to mind: increase the resources or decrease the load.
- However, sometimes it is not possible to increase the capacity, or it has already been increased to the limit. The only way then to beat back the congestion is to decrease the load. Several ways exist to reduce the load, including denying service to some users, degrading service to some or all users, and having users schedule their demands in a more predictable way.

4.3.2 Congestion Prevention Policies

Let us begin our study of methods to control congestion by looking at open loop systems. These systems are designed to minimize congestion in the first place, rather than letting it happen and reacting after the fact. They try to achieve their goal by using appropriate policies at various levels. In Fig. 5-26 we see different data link, network, and transport policies that can affect congestion (Jain, 1990).

Figure 5-26. Policies that affect congestion.

Layer	Policies
Transport	• Retransmission policy • Out-of-order caching policy • Acknowledgement policy • Flow control policy • Timeout determination
Network	• Virtual circuits versus datagram inside the subnet • Packet queueing and service policy • Packet discard policy • Routing algorithm • Packet lifetime management
Data link	• Retransmission policy • Out-of-order caching policy • Acknowledgement policy • Flow control policy

Data link layer:

- The retransmission policy is concerned with how fast a sender times out and what it transmits upon timeout. A jumpy sender that times out quickly and retransmits all outstanding packets using go back n will put a heavier load on the system than will a leisurely sender that uses selective repeat.
- Closely related to this is the buffering policy. If receivers routinely discard all out-of-order packets, these packets will have to be transmitted again later, creating extra load. With respect to congestion control, selective repeat is clearly better than go back n.
- Acknowledgement policy also affects congestion. If each packet is acknowledged immediately, the acknowledgement packets generate extra traffic. However, if acknowledgements are saved up to piggyback

onto reverse traffic, extra timeouts and retransmissions may result. A tight flow control scheme (e.g., a small window) reduces the data rate and thus helps fight congestion.

Network layer:

- The choice between using virtual circuits and using datagrams affects congestion since many congestion control algorithms work only with virtual-circuit subnets.
- Packet queueing and service policy relates to whether routers have one queue per input line, one queue per output line, or both. It also relates to the order in which packets are processed (e.g., round robin or priority based).
- Discard policy is the rule telling which packet is dropped when there is no space. A good policy can help alleviate congestion and a bad one can make it worse.
- A good routing algorithm can help avoid congestion by spreading the traffic over all the lines, whereas a bad one can send too much traffic over already congested lines.
- Finally, packet lifetime management deals with how long a packet may live before being discarded. If it is too long, lost packets may clog up the works for a long time, but if it is too short, packets may sometimes time out before reaching their destination, thus inducing retransmissions.

Transport layer

The same issues occur as in the data link layer, but in addition, determining the timeout interval is harder because the transit time across the network is less predictable than the transit time over a wire between two routers. If the timeout interval is too short, extra packets will be sent unnecessarily. If it is too long, congestion will be reduced but the response time will suffer whenever a packet is lost.

4.3.3 Approaches to congestion control

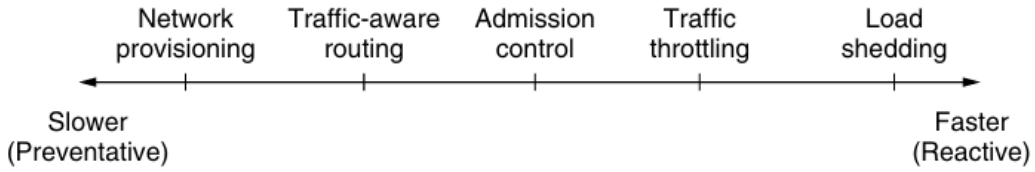


Figure 5-22. Timescales of approaches to congestion control.

The presence of congestion means that the load is (temporarily) greater than the resources (in a part of the network) can handle. Two solutions come to mind: increase the resources or decrease the load. As shown in Fig. 5-22, these solutions are usually applied on different time scales to either prevent congestion or react to it once it has occurred.

- The most basic way to avoid congestion is to build a network that is well matched to the traffic that it carries. If there is a low-bandwidth link on the path along which most traffic is directed, congestion is likely. Sometimes resources can be added dynamically when there is serious congestion, for example, turning on spare routers or enabling lines that are normally used only as backups (to make the system fault tolerant) or purchasing bandwidth on the open market. More often, links and routers that are regularly heavily utilized are upgraded at the earliest opportunity. This is called **provisioning** and happens on a time scale of months, driven by long-term traffic trends.
- To make the most of the existing network capacity, routes can be tailored to traffic patterns that change during the day as network users' wake and sleep in different time zones. For example, routes may be changed to shift traffic away from heavily used paths by changing the shortest path weights. Some local radio stations have helicopters flying around their cities to report on road congestion to make it possible for their mobile listeners to route their packets (cars) around hotspots. This is called **traffic-aware routing**. Splitting traffic across multiple paths is also helpful.
- However, sometimes it is not possible to increase capacity. The only way then to beat back the congestion is to decrease the load. In a

virtual-circuit net work, new connections can be refused if they would cause the network to become congested. This is called **admission control.**

- At a finer granularity, when congestion is imminent the network can deliver feedback to the sources whose traffic flows are responsible for the problem. The network can request these sources to throttle their traffic, or it can slow down the traffic itself.
- Two difficulties with this approach are how to identify the onset of congestion, and how to inform the source that needs to slow down.
- To tackle the first issue, routers can monitor the average load, queueing delay, or packet loss. In all cases, rising numbers indicate growing congestion.
- To tackle the second issue, routers must participate in a feedback loop with the sources. For a scheme to work correctly, the time scale must be adjusted carefully. If every time two packets arrive in a row, a router yells STOP and every time a router is idle for 20 μsec, it yells GO, the system will oscillate wildly and never converge. On the other hand, if it waits 30 minutes to make sure before saying anything, the congestion-control mechanism will react too sluggishly to be of any use.
- Delivering timely feedback is a nontrivial matter. An added concern is having routers send more messages when the network is already congested.
- Finally, when all else fails, the network is forced to discard packets that it cannot deliver. The general name for this is **load shedding**. A good policy for choosing which packets to discard can help to prevent congestion collapse.

4.3.4 Traffic-Aware Routing

These schemes adapted to changes in topology, but not to changes in load. The goal in taking load into account when computing routes is to shift traffic away from hotspots that will be the first places in the network to experience congestion.

- The most direct way to do this is to set the link weight to be a function of the (fixed) link bandwidth and propagation delay plus the (variable) measured load or average queuing delay. Least-weight paths will then favor paths that are more lightly loaded, all else being equal.
- Consider the network of Fig. 5-23, which is divided into two parts, East and West, connected by two links, CF and EI. Suppose that most of the traffic between East and West is using link CF, and, as a result, this link is heavily loaded with long delays. Including queue ing delay in the weight used for the shortest path calculation will make EI more attractive. After the new routing tables have been installed, most of the East-West traffic will now go over EI, loading this link. Consequently, in the next update, CF will appear to be the shortest path. As a result, the routing tables may oscil late wildly, leading to erratic routing and many potential problems.

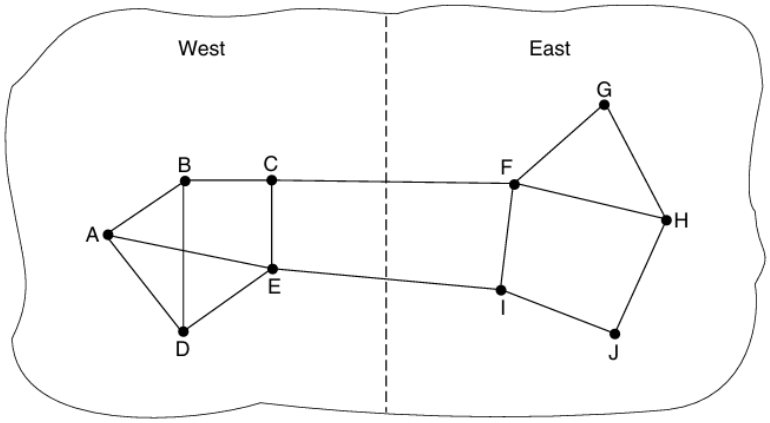


Figure 5-23. A network in which the East and West parts are connected by two links.

- If load is ignored and only bandwidth and propagation delay are considered, this problem does not occur. Attempts to include load but change weights within a narrow range only slow down routing oscillations. Two techniques can contri bute to a successful solution. The first is multipath routing, in which there can be multiple paths from a source to a destination. In our example this means that the traffic can be spread across both of the East to West links. The second one is for the routing scheme to shift traffic across routes slowly enough that it is able to converge, as in the scheme of Gallagher (1977).

- Given these difficulties, in the Internet routing protocols do not generally ad just their routes depending on the load. Instead, adjustments are made outside the routing protocol by slowly changing its inputs. This is called **traffic engineering**.

4.3.4.1 Admission Control

- One technique that is widely used in virtual-circuit networks to keep congestion at bay is admission control.
- The idea is simple: do not set up a new virtual circuit unless the network can carry the added traffic without becoming congested. Thus, attempts to set up a virtual circuit may fail.
- This is better than the alter native, as letting more people in when the network is busy just makes matters worse. By analogy, in the telephone system, when a switch gets overloaded it practices admission control by not giving dial tones.
- The task is then to estimate how many circuits will fit within the carrying capacity of the network without congestion. Suppose that virtual cir cuits that may blast traffic at rates up to 10 Mbps all pass through the same 100 Mbps physical link.
- How many circuits should be admitted? Clearly, 10 circuits can be admitted without risking congestion, but this is wasteful in the normal case since it may rarely happen that all 10 are transmitting full blast at the same time.
- In real networks, measurements of past behavior that capture the statistics of transmissions can be used to estimate the number of circuits to admit, to trade bet ter performance for acceptable risk.
- Admission control can also be combined with traffic-aware routing by considering routes around traffic hotspots as part of the setup procedure. For example, consider the network illustrated in Fig. 5-24(a), in which two routers are congested, as indicated.

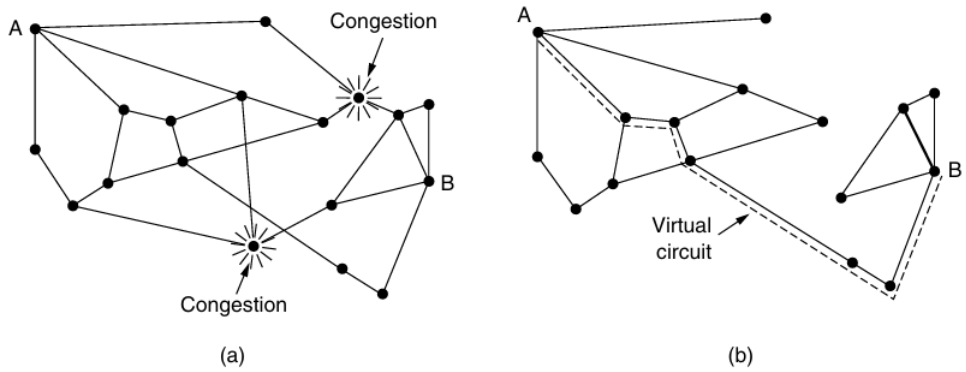


Figure 5-24. (a) A congested network. (b) The portion of the network that is not congested. A virtual circuit from *A* to *B* is also shown.

- Suppose that a host attached to router A wants to set up a connection to a host attached to router B. Normally, this connection would pass through one of the congested routers. To avoid this situation, we can redraw the network as shown in Fig. 5-24(b), omitting the congested routers and all of their lines. The dashed line shows a possible route for the virtual circuit that avoids the congested routers.

4.3.5 Traffic Throttling

In the Internet and many other computer networks, senders adjust their trans missions to send as much traffic as the network can readily deliver. In this setting, the network aims to operate just before the onset of congestion. When congestion is imminent, it must tell the senders to throttle back their transmissions and slow down. This feedback is business as usual rather than an exceptional situation. The term congestion avoidance is sometimes used to contrast this operating point with the one in which the network has become (overly) congested.

Let us now look at some approaches to throttling traffic that can be used in both datagram networks and virtual-circuit networks. Each approach must solve two problems. First, routers must determine when congestion is approaching, ideally before it has arrived. To do so, each router can continuously monitor the resources it is using. Three possibilities are the utilization of the output links, the buffering of queued packets inside the

router, and the number of packets that are lost due to insufficient buffering. Of these possibilities, the second one is the most useful. Averages of utilization do not directly account for the burstiness of most traffic—a utilization of 50% may be low for smooth traffic and too high for highly variable traffic. Counts of packet losses come too late. Congestion has already set in by the time that packets are lost.

The queueing delay inside routers directly captures any congestion experienced by packets. It should be low most of time, but will jump when there is a burst of traffic that generates a backlog. To maintain a good estimate of the queueing delay, d, a sample of the instantaneous queue length, s, can be made periodically and d updated according to

$$d_{new} = ad_{old} + (1 - a)s$$

where the constant α determines how fast the router forgets recent history. This is called an **EWMA (Exponentially Weighted Moving Average).** It smoothes out fluctuations and is equivalent to a low-pass filter. Whenever d moves above the threshold, the router notes the onset of congestion.

The second problem is that routers must deliver timely feedback to the send ers that are causing the congestion. Different schemes use different feedback mechanisms are:

1. Choke Packets
2. Explicit Congestion Notification
3. Hop-by-Hop Backpressure

Choke Packets

A **choke packet** is a type of control message sent by a router to the sender (source node) to signal that there is **congestion** in the network and it needs to **reduce its transmission rate**.

- When a router detects congestion (e.g., its buffer is almost full), it sends a **choke packet** to the source of the data flow.
- The source, upon receiving the choke packet, **reduces its data transmission rate**.

- This is an example of **end-to-end congestion control**.
- Only the **source** is notified.
- The notification is **explicit**.
- Works well for controlling congestion caused by a specific sender.

Explicit Congestion Notification

Instead of generating additional packets to warn of congestion, a router can tag any packet it forwards (by setting a bit in the packet's header) to signal that it is experiencing congestion. When the network delivers the packet, the destination can note that there is congestion and inform the sender when it sends a reply pack et. The sender can then throttle its transmissions as before.

- Two bits in the IP packet header are used to record whether the packet has experienced congestion. Packets are unmarked when they are sent, as illustrated in Fig. 5-25. If any of the routers they pass through is congested, that router will then mark the packet as having experienced congestion as it is forwarded.
- The destination will then echo any marks back to the sender as an explicit congestion signal in its next reply packet. This is shown with a dashed line in the figure to indicate that it happens above the IP level (e.g., in TCP). The sender must then throttle its transmissions, as in the case of choke packets.

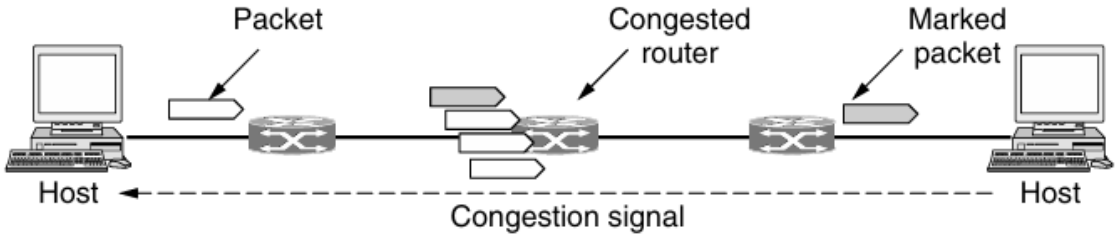


Figure 5-25. Explicit congestion notification

Hop-by-Hop Backpressure

At high speeds or over long distances, many new packets may be transmitted after congestion has been signaled because of the delay before the signal takes effect. Consider, for example, a host in San Francisco (router A in Fig. 5-26) that is sending traffic to a host in New York (router D in Fig. 5-26) at the OC-3 speed of 155 Mbps. If the New York host begins to run out of buffers, it will take about 40 msec for a choke packet to get back to San Francisco to tell it to slow down. An ECN indication will take even longer because it is delivered via the destination. Choke packet propagation is illustrated as the second, third, and fourth steps in Fig. 5-26(a). In those 40 msec, another 6.2 megabits will have been sent. Even if the host in San Francisco completely shuts down immediately, the 6.2 megabits in the pipe will continue to pour in and have to be dealt with. Only in the seventh diagram in Fig. 5-26(a) will the New York router notice a slower flow.

An alternative approach is to have the choke packet take effect at every hop it passes through, as shown in the sequence of Fig. 5-26(b). Here, as soon as the choke packet reaches F, F is required to reduce the flow to D. Doing so will re quire F to devote more buffers to the connection, since the source is still sending away at full blast, but it gives D immediate relief, like a headache remedy in a television commercial. In the next step, the choke packet reaches E, which tells E to reduce the flow to F. This action puts a greater demand on E's buffers but gives F immediate relief. Finally, the choke packet reaches A and the flow genuinely slows down.

The net effect of this hop-by-hop scheme is to provide quick relief at the point of congestion, at the price of using up more buffers upstream. In this

way, congestion can be nipped in the bud without losing any

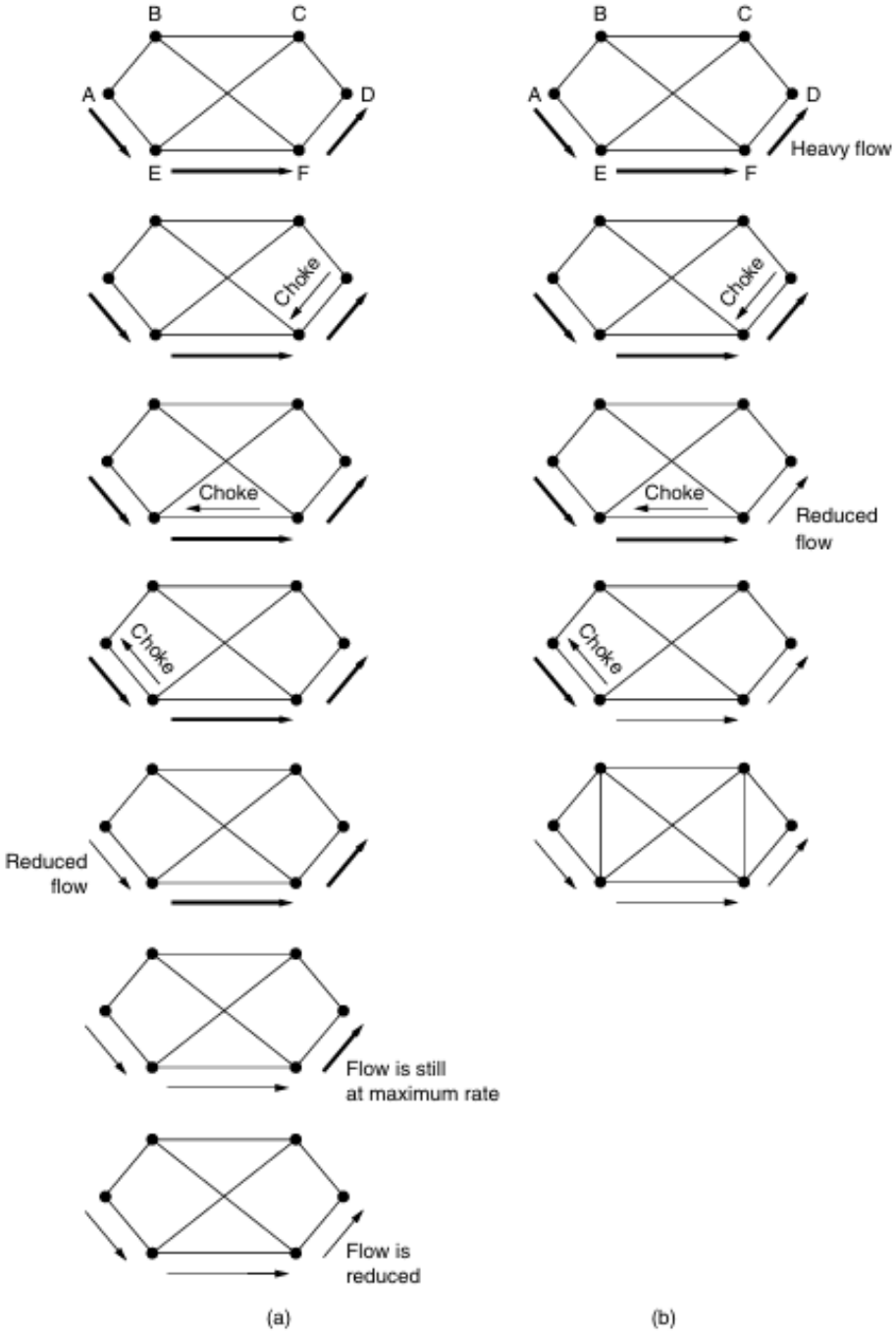


Figure 5-26. (a) A choke packet that affects only the source. (b) A choke packet that affects each hop it passes through.

packets.

Load Shedding

When none of the above methods make the congestion disappear, routers can bring out the heavy artillery: load shedding. Load shedding is a fancy way of saying that when routers are being inundated by packets that they cannot handle, they just throw them away. The term comes from the world of electrical power generation, where it refers to the practice of utilities intentionally blacking out certain areas to save the entire grid from collapsing on hot summer days when the demand for electricity greatly exceeds the supply.

A router drowning in packets can just pick packets at random to drop, but usually it can do better than that. Which packet to discard may depend on the applications running. For file transfer, an old packet is worth more than a new one because dropping packet 6 and keeping packets 7 through 10 will cause a gap at the receiver that may force packets 6 through 10 to be retransmitted (if the receiver routinely discards out-of-order packets). In a 12-packet file, dropping 6 may require 7 through 12 to be retransmitted, whereas dropping 10 may require only 10 through 12 to be retransmitted. In contrast, for multimedia, a new packet is more important than an old one. The former policy (old is better than new) is often called wine and the latter (new is better than old) is often called milk.

A step above this in intelligence requires cooperation from the senders. For many applications, some packets are more important than others. For example, certain algorithms for compressing video periodically transmit an entire frame and then send subsequent frames as differences from the last full frame. In this case, dropping a packet that is part of a difference is preferable to dropping one that is part of a full frame. As another example, consider transmitting a document containing ASCII text and pictures.

Losing a line of pixels in some image is far less damaging than losing a line of readable text.

To implement an intelligent discard policy, applications must mark their packets in priority classes to indicate how important they are. If they do this, then when packets have to be discarded, routers can first drop packets from the lowest class, then the next lowest class, and so on. Of course, unless there is some significant incentive to mark packets as anything other than VERY IMPORTANT— NEVER, EVER DISCARD, nobody will do it.

The incentive might be in the form of money, with the low-priority packets being cheaper to send than the high-priority ones. Alternatively, senders might be allowed to send high-priority packets under conditions of light load, but as the load increased, they would be discarded, thus encouraging the users to stop sending them.

Another option is to allow hosts to exceed the limits specified in the agreement negotiated when the virtual circuit was set up (e.g., use a higher bandwidth than allowed), but subject to the condition that all excess traffic be marked as low priority. Such a strategy is actually not a bad idea, because it makes more efficient use of idle resources, allowing hosts to use them as long as nobody else is interested, but without establishing a right to them when times get tough.

4.3.6 Traffic shaping (Traffic Control Algorithm)

In the network layer, before the network can make Quality of service guarantees, it must know what traffic is being guaranteed. One of the main causes of congestion is that traffic is often bursty.

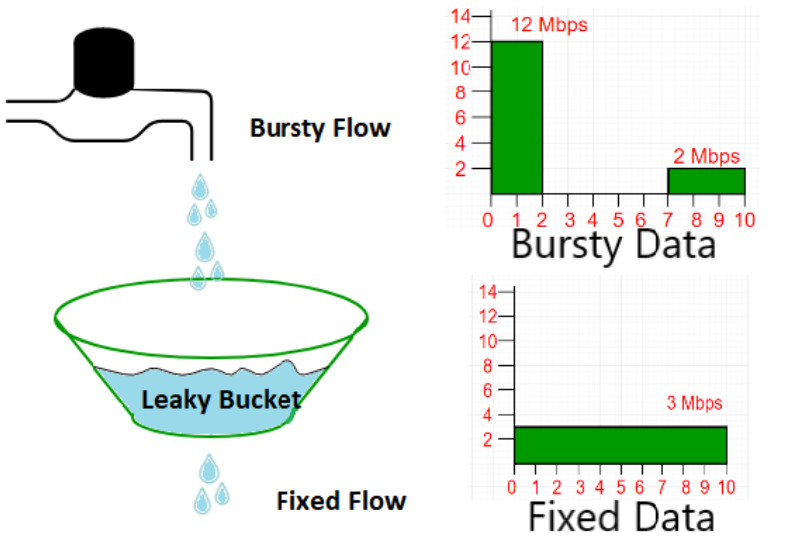
To understand this concept first we have to know little about traffic shaping. Traffic Shaping is a mechanism to control the amount and the rate of the traffic sent to the network. Approach of congestion management is called **Traffic shaping.** Traffic shaping helps to regulate rate of data transmission and reduces congestion.

There are 2 types of traffic shaping algorithms:

1. Leaky Bucket
2. Token Bucket

Suppose we have a bucket in which we are pouring water in a random order but we have to get water in a fixed rate, for this we will make a hole at the bottom of the bucket. It will ensure that water coming out is in a some fixed rate, and also if bucket will full we will stop pouring in it.

The input rate can vary, but the output rate remains constant. Similarly, in networking, a technique called leaky bucket can smooth out bursty traffic. Bursty chunks are stored in the bucket and sent out at an average rate.



In the figure, we assume that the network has committed a bandwidth of 3 Mbps for a host. The use of the leaky bucket shapes the input traffic to make it conform to this commitment. In Figure the host sends a burst of data at a rate of 12 Mbps for 2 s, for a total of 24 Mbits of data. The host is silent for 5 s and then sends data at a rate of 2 Mbps for 3 s, for a total of 6 Mbits of data. In all, the host has sent 30 Mbits of data in 10 s. The leaky bucket smooths the traffic by sending out data at a rate of 3 Mbps during the same 10 s.

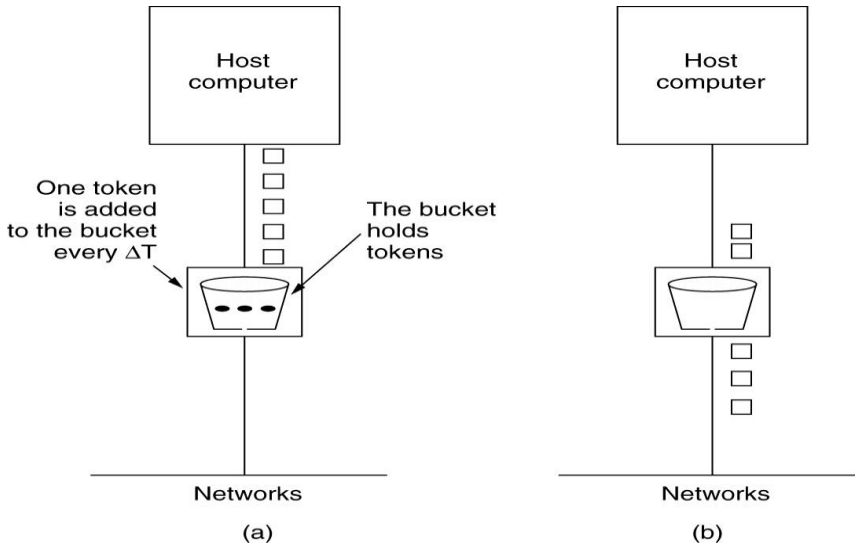
Without the leaky bucket, the beginning burst may have hurt the network by consuming more bandwidth than is set aside for this host. We can also see that the leaky bucket may prevent congestion.

Token Bucket Algorithm

The leaky bucket algorithm enforces output pattern at the average rate, no matter how bursty the traffic is. So in order to deal with the bursty traffic we need a flexible algorithm so that the data is not lost. One such algorithm is token bucket algorithm.

Steps of this algorithm can be described as follows:

1. In regular intervals tokens are thrown into the bucket.
2. The bucket has a maximum capacity.
3. If there is a ready packet, a token is removed from the bucket, and the packet is sent.
4. If there is no token in the bucket, the packet cannot be sent.



Let's understand with an example, In figure (A) we see a bucket holding three tokens, with five packets waiting to be transmitted. For a packet to be transmitted, it must capture and destroy one token. In figure (B) We see that three of the five packets have gotten through, but the other two are stuck waiting for more tokens to be generated.

Aspect	Leaky Bucket	Token Bucket
Visual Analogy	Bucket with a hole, water leaks steadily	Bucket collects tokens, used to send packets
Traffic Behavior	Smooth, steady output regardless of bursty input	Bursty traffic allowed (if tokens are available)
Incoming Packet	Added to a queue if there's room	Sent only if enough tokens exist
Overflow Behavior	Packets are dropped if bucket is full	Packets are delayed/dropped if not enough tokens
Control Parameter	Fixed leak rate (output rate)	Fixed token generation rate
Allows Burst?	✗ No	☑ Yes
Shaping vs Policing	Pure **shaping**	Can be used for **shaping or policing**
Example Scenario	VoIP, live audio streaming (constant rate)	Web browsing, video streaming (bursty traffic)
Output Rate	Constant	Can be variable (up to a max burst rate)

4.4 Internetworking: Until now, we have implicitly assumed that there is a single homogeneous network, with each machine using the same protocol in each layer. Unfortunately, this assumption is wildly optimistic. Many different networks exist, including PANs, LANs, MANs, and WANs. We have described Ethernet, Internet over cable, the fixed and mobile telephone networks, 802.11, 802.16, and more. Numerous protocols are in widespread use across these networks in every layer.

When two or more networks are connected to form an internetwork, or more simply an internet.

The Internet is the prime example of this interconnection. (We will write Internet with a capital "I" to distinguish it from other internets, or connected net works.) The purpose of joining all these networks is to allow users on any of them to communicate with users on all the other ones.

Since networks often differ in important ways, getting packets from one net work to another is not always so easy. We must address problems of heterogeneity, and also problems of scale as the resulting internet grows very large.

In summary, **Internetworking** = Interconnecting multiple **heterogeneous** networks (different protocols, architectures, or technologies) to function as one large logical network.

As an example of how different networks might be connected, consider the example of Fig. Here we see a corporate network with multiple locations tied together by a wide area ATM network. At one of the locations, an FDDI optical backbone is used to connect an Ethernet, an 802.11 wireless LAN, and the corporate data center's SNA mainframe network.

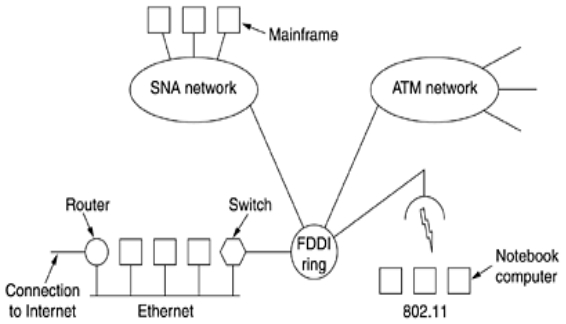


Fig: Collection of interconnected networks.

4.4.1 How Networks Differ

- Networks can differ in many ways. Some of the differences, such as different modulation techniques or frame formats, are internal to the physical and data link layers.
- These differences will not concern us here. Instead, in table, list some of the differences that can be exposed to the network layer. It is papering over these differences that makes internetworking more difficult than operating within a single network.

Item	Some Possibilities
Service offered	Connectionless versus connection oriented
Addressing	Different sizes, flat or hierarchical
Broadcasting	Present or absent (also multicast)
Packet size	Every network has its own maximum
Ordering	Ordered and unordered delivery
Quality of service	Present or absent; many different kinds
Reliability	Different levels of loss
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, packet, byte, or not at all

Fig: Some of the many ways networks can differ.

- When packets sent by a source on one network must transit one or more foreign networks before reaching the destination network, many problems can occur at the interfaces between networks.
- To start with, the source needs to be able to address the destination. What do we do if the source is on an Ethernet network and the destination is on a WiMAX network? Assuming we can even specify a WiMAX destination from an Ethernet network, packets would cross from a connectionless network to a connection-oriented one.
- This may require that a new connection be set up on short notice, which injects a delay, and much overhead if the connection is not used for

many more packets. Many specific differences may have to be accommodated as well.

- How do we multicast a packet to a group with some members on a network that does not support multicast? The differing max packet sizes used by different networks can be a major nuisance, too.
- How do you pass an 8000-byte packet through a net work whose maximum size is 1500 bytes? If packets on a connection-oriented network transit a connectionless network, they may arrive in a different order than they were sent. That is something the sender likely did not expect, and it might come as an (unpleasant) surprise to the receiver as well.
- These kinds of differences can be papered over, with some effort. For ex ample, a gateway joining two networks might generate separate packets for each destination in lieu of better network support for multicasting.
- A large packet might be broken up, sent in pieces, and then joined back together. Receivers might buffer packets and deliver them in order.
- Networks also can differ in large respects that are more difficult to reconcile. The clearest example is quality of service. If one network has strong QoS and the other offers best effort service, it will be impossible to make bandwidth and delay guarantees for real-time traffic end to end.
- In fact, they can likely only be made while the best-effort network is operated at a low utilization, or hardly used, which is unlikely to be the goal of most ISPs.
- Security mechanisms are problematic, but at least encryption for confidentiality and data integrity can be layered on top of networks that do not already include it. Finally, differences in accounting can lead to unwelcome bills when normal usage suddenly becomes ex pensive, as roaming mobile phone users with data plans have discovered.

4.4.2 How Networks can be connected

There are two basic choices for connecting different networks: we can build devices that translate or convert packets from each kind of network into packets for each other network, or, like good computer scientists, we can try to solve the problem by adding a layer of indirection and building a common layer on top of the different networks. In either case, the devices are placed at the boundaries between networks.

We have discussed several different devices that connect networks, including repeaters, hubs, switches, bridges, routers, and gateways. Repeaters and hubs just move bits from one wire to another.

They are mostly analog devices and do not understand anything about higher layer protocols. Bridges and switches operate at the link layer. They can be used to build networks, but only with minor protocol translation in the process, for example, between 10, 100 and 1000 Mbps Ethernet switches.

Let us first explore at a high level how interconnection with a common net work layer can be used to interconnect dissimilar networks. An internet comprised of 802.11, MPLS, and an Ethernet network is shown in Fig. 5-39(a). Suppose that the source machine on the 802.11 network wants to send a packet to the destination machine on the Ethernet network. Since these technologies are different, and they are further separated by another kind of network (MPLS), some added processing is needed at the boundaries between the networks.

Because different networks may, in general, have different forms of ad dressing, the packet carries a network layer address that can identify any host

a cross the three networks. The first boundary the packet reaches is when it transitions from an 802.11 network to an MPLS network. 802.11 provide a connectionless service, but MPLS provides a connection-oriented service.

This means that a virtual circuit must be set up to cross that network. Once the packet has traveled along the virtual circuit, it will reach the Ethernet network. At this boundary, the packet may be too large to be carried, since 802.11 can work with larger frames than Ethernet.

To handle this problem, the packet is divided into fragments, and each fragment is sent separately. When the fragments reach the destination, they are reassembled. Then the packet has completed its journey. The protocol processing for this journey is shown in Fig. 5-39(b).

The source accepts data from the transport layer and generates a packet with the common net work layer header, which is IP in this example. The network header contains the ultimate destination address, which is used to determine that the packet should be sent via the first router. So the packet is encapsulated in an 802.11 frame whose destination is the first router and transmitted. At the router, the packet is removed from the frame's data field and the 802.11 frame header is discarded.

The router now examines the IP address in the packet and looks up this address in its routing table. Based on this address, it decides to send the packet to the second router next. For this part of the path, an MPLS virtual circuit must be established to the second router and the packet must be encapsulated with MPLS headers that travel this circuit. At the far end, the MPLS header is discarded and the network address is again consulted to find the next network layer hop. It is the destination itself. Since the packet is too

long to be sent over Ethernet, it is split into two portions. Each of these portions is put into the data field of an Ethernet frame and sent to the Ethernet address of the destination. At the destination, the Ethernet header is stripped from each of the frames, and the contents are reassembled. The packet has finally reached its destination.

Observe that there is an essential difference between the routed case and the switched (or bridged) case. With a router, the packet is extracted from the frame and the network address in the packet is used for deciding where to send it. With a switch (or bridge), the entire frame is transported on the basis of its MAC ad dress. Switches do not have to understand the network layer protocol being used to switch packets. Routers do.

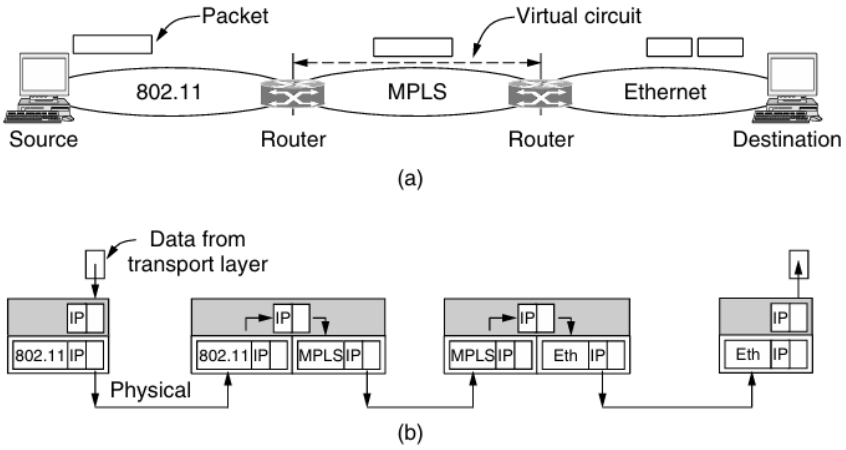


Figure 5-39. (a) A packet crossing different networks. (b) Network and link layer protocol processing.

Unfortunately, internetworking is not as easy as we have made it sound. In fact, when bridges were introduced, it was intended that they would join different types of networks, or at least different types of LANs. They were to do this by translating frames from one LAN into frames from another LAN. However, this did not work well, for the same reason that internetworking is difficult: the differences in the features of LANs, such as different maximum

packet sizes and LANs with and without priority classes, are hard to mask. Today, bridges are predominantly used to connect the same kind of network at the link layer, and routers connect different networks at the network layer.

Internetworking has been very successful at building large networks, but it only works when there is a common network layer. There have, in fact, been many network protocols over time. Getting everybody to agree on a single format is difficult when companies perceive it to their commercial advantage to have a proprietary format that they control. Examples besides IP, which is now the near-universal network protocol, were IPX, SNA, and AppleTalk. None of these protocols are still in widespread use, but there will always be other protocols.

The most relevant example now is probably IPv4 and IPv6. While these are both versions of IP, they are not compatible (or it would not have been necessary to create IPv6).

A router that can handle multiple network protocols is called a multiprotocol router. It must either translate the protocols, or leave connection for a higher protocol layer. Neither approach is entirely satisfactory. Connection at a higher layer, say, by using TCP, requires that all the networks implement TCP (which may not be the case). Then, it limits usage across the networks to applications that use TCP (which does not include many real-time applications).

4.4.3 Tunneling

A technique of inter-networking called **Tunneling** is used when source and destination networks of the same type are to be connected through a network

of different types. Tunneling uses a layered protocol model such as those of the OSI or TCP/IP protocol suite.

As an example, think of an international bank with an IPv6 network in Paris, an IPv6 network in London and connectivity between the offices via the IPv4 Internet. This situation is shown in Fig. 5-40.

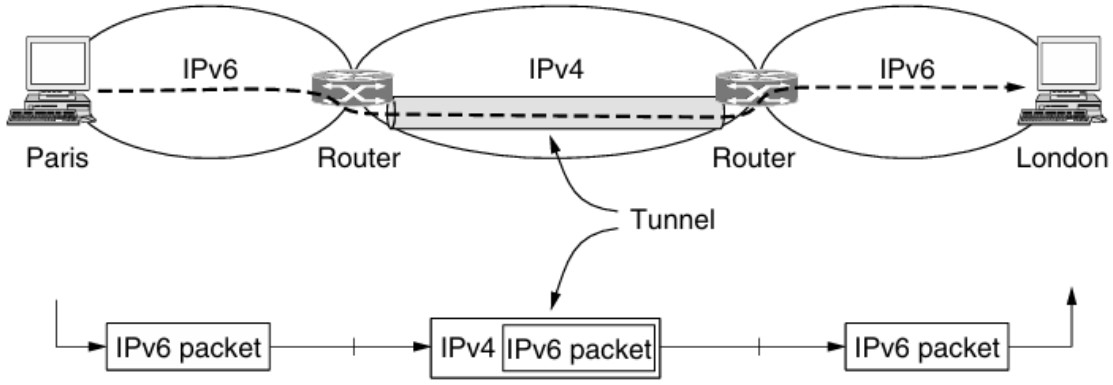


Figure 5-40. Tunneling a packet from Paris to London.

The solution to this problem is a technique called tunneling. To send an IP packet to a host in the London office, a host in the Paris office constructs the packet containing an IPv6 address in London, and sends it to the multiprotocol router that connects the Paris IPv6 network to the IPv4 Internet.

When this router gets the IPv6 packet, it encapsulates the packet with an IPv4 header addressed to the IPv4 side of the multiprotocol router that connects to the London IPv6 net work. That is, the router puts a (IPv6) packet inside a (IPv4) packet. When this wrapped packet arrives, the London router removes the original IPv6 packet and sends it onward to the destination host. The path through the IPv4 Internet can be seen as a big tunnel extending from one multiprotocol router to the other.

The IPv6 packet just travels from one end of the tunnel to the other, snug in its nice box. It does not have to worry about dealing with IPv4 at all. Neither do

the hosts in Paris or London. Only the multi protocol routers have to understand both IPv4 and IPv6 packets. In effect, the en tire trip from one multiprotocol router to the other is like a hop over a single link.

4.4.4 Internetwork Routing

Internetwork routing is the process of **finding a path** across multiple interconnected networks to deliver data from a source to a destination — possibly in **different networks or domains**.

Two major types of internetwork routing protocols:

1. **Interior Gateway Protocols (IGPs)** – Used **within** a single Autonomous System (AS)
2. **Exterior Gateway Protocols (EGPs)** – Used **between** different Autonomous Systems.

An **Autonomous System (AS)** is: A collection of IP networks and routers under the control of a **single organization** that presents a **common routing policy** to the Internet.

Each AS has:

- A unique **AS number (ASN)** assigned by IANA or regional internet registries.
- Complete internal control over routing within the system.
- May connect to multiple other ASes.

Examples:

- An Internet Service Provider (ISP)
- A large enterprise network (e.g., Google has its own AS)

Interior Gateway Protocol (IGP)

IGPs are used to exchange routing information **within a single AS**.

Protocol	Type	Description
RIP (Routing Information Protocol)	Distance Vector	Simple, uses hop count (max 15), slow convergence
OSPF (Open Shortest Path First)	Link State	Fast convergence, uses Dijkstra's algorithm
IS-IS	Link State	Similar to OSPF, more common in ISPs
EIGRP	Advanced Distance Vector	Cisco proprietary, hybrid approach

Exterior Gateway Protocol (EGP)

EGPs are used to exchange routing information **between different ASes** (inter-domain routing).

Main Protocol:

Protocol	Description
BGP (Border Gateway Protocol)	The **only** widely-used EGP today; used to route traffic between ASes on the Internet

BGP Characteristics:

- Path vector protocol (uses AS-path to prevent loops)
- Can implement complex routing policies (e.g., based on business relationships)
- **Scalable** for global Internet
- Routers using BGP are called **edge routers** or **border routers**

4.4.5 Packet Fragmentation:

Each network or link imposes some maximum size on its packets. These limits have various causes, among them

1. Hardware (e.g., the size of an Ethernet frame).
2. Operating system (e.g., all buffers are 512 bytes).
3. Protocols (e.g., the number of bits in the packet length field).
4. Compliance with some (inter)national standard.
5. Desire to reduce error-induced retransmissions to some level.
6. Desire to prevent one packet from occupying the channel too long.

The result of all these factors is that the network designers are not free to choose any old maximum packet size they wish. Maximum payloads for some common technologies are 1500 bytes for Ethernet and 2272 bytes for 802.11. IP is more generous, allows for packets as big as 65,515 bytes. Hosts usually prefer to transmit large packets because this reduces packet overheads such as bandwidth wasted on header bytes.

An obvious internetworking problem appears when a large packet wants to travel through a network whose maximum packet size is too small. This nuisance has been a persistent issue, and solutions to it have evolved along with much experience gained on the Internet.

Fragmentation happens when an IP packet is **too large** for the Maximum Transmission Unit (**MTU**) of a network link along its path.

- **MTU** = Maximum size (in bytes) of a packet that can be transmitted over a link without fragmentation.
- **Example:**
  Ethernet MTU = 1500 bytes → if an IP packet is 2000 bytes, it must be fragmented.

Two types of fragmentation are:

1. Transparent Fragmentation.
2. Non Transparent Fragmentation.

Feature	Transparent Fragmentation	Non-Transparent Fragmentation
Definition	The router fragments a packet and **reassembles** it before sending it to the next network.	The router fragments a packet and forwards **each fragment separately** to the destination.
Reassembly Point	Occurs **at the next router (hop)** before forwarding.	Occurs **only at the final destination host**.
Visibility to Next Network	Next network sees the **original, reassembled packet**.	Next network receives **fragments** of the original packet.
Processing Overhead	High — routers must store and reassemble packets.	Lower — routers just forward fragments.
Buffer Requirement	Requires **large buffer space** at intermediate routers.	Requires **no reassembly buffer** at routers.
Transmission Delay	Higher delay due to reassembly at each hop.	Lower delay, since fragments are forwarded immediately.
Path of Fragments	Only **one path** (since fragments are reassembled before next hop).	Fragments may travel through **different paths**.
Reliability	More reliable at link level, but inefficient.	Less reliable — if one fragment is lost, entire packet is discarded.
Commonly Used In	Special cases (e.g., satellite links, VPN tunnels).	Traditional **IPv4 networks**.
Efficiency	Less efficient due to repeated reassembly.	More efficient for end-to-end transmission.

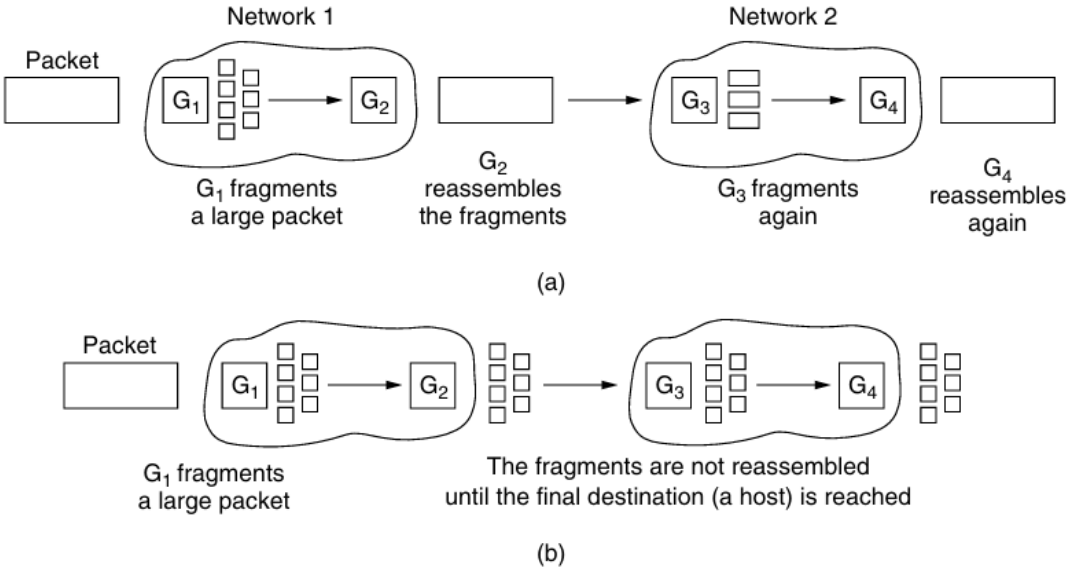


Figure 5-42. (a) Transparent fragmentation. (b) Nontransparent fragmentation.

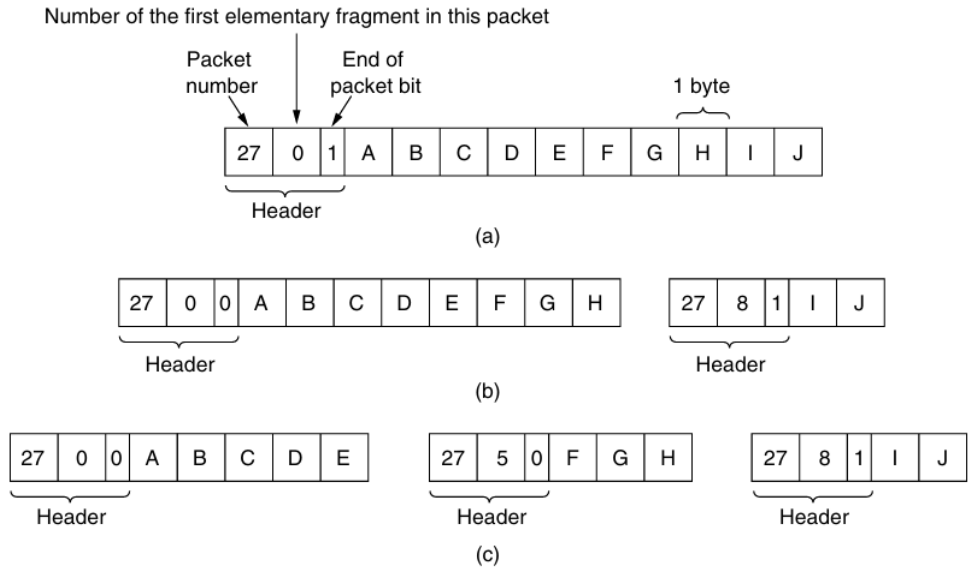


Figure 5-43. Fragmentation when the elementary data size is 1 byte. (a) Original packet, containing 10 data bytes. (b) Fragments after passing through a network with maximum packet size of 8 payload bytes plus header. (c) Fragments after passing through a size 5 gateway.

Path MTU discovery:

Path MTU Discovery is a technique used by IP hosts to determine the **smallest Maximum Transmission Unit (MTU)** along the path between a source and destination, so packets can be sent **without fragmentation**.

How it Works (IPv4)

1. The sender sets the **Don't Fragment (DF)** bit in the IP header.
2. If a router encounters a link with a smaller MTU, it **drops** the packet and sends back an **ICMP "Fragmentation Needed" (Type 3, Code 4)** message.
3. The sender reduces its packet size and retries.
4. Repeat until no ICMP errors occur → discovered Path MTU.

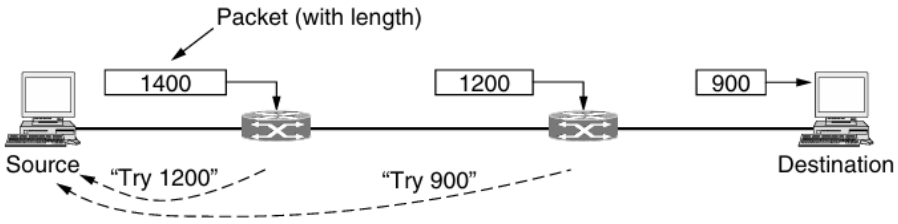


Figure 5-44. Path MTU discovery.

Aspect	Benefit	Drawback
Fragmentation Handling	Avoids fragmentation completely	Fails if ICMP blocked
Efficiency	Improves throughput and lowers router load	Increases initial discovery time
Reliability	Reduces loss due to missing fragments	Can fail silently on ICMP-filtered paths
Suitability	Ideal for IPv6 and modern IPv4 networks	Needs careful firewall configuration

4.5 The Network layer in the Internet

4.5.1 The IP Version 4 Protocol

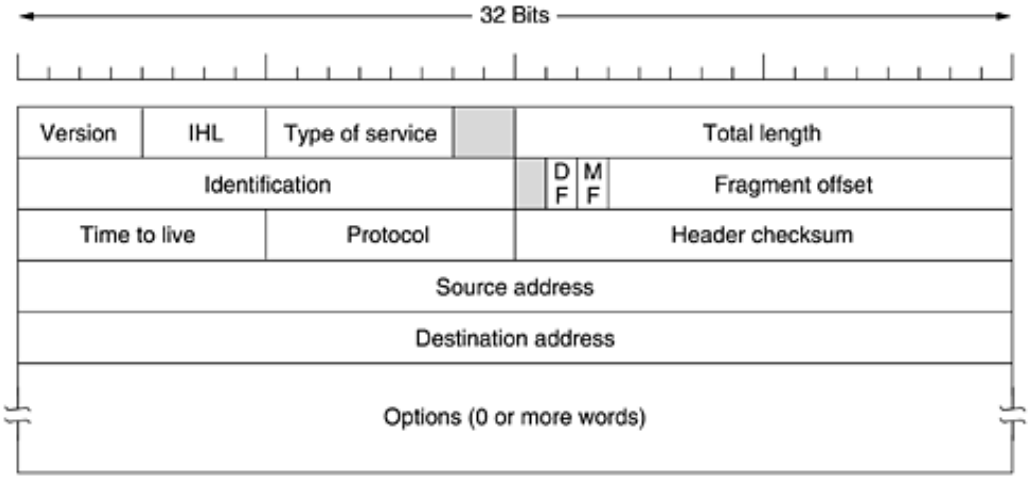


Fig: IPv4 Header format

- The Version field keeps track of which version of the protocol the datagram belongs to.
- Since the header length is not constant, a field in the header, IHL, is provided to tell how long the header is, in 32-bit words. The minimum value is 5, which applies when no options are present. The maximum value of this 4-bit field is 15, which limits the header to 60 bytes, and thus the Options field to 40 bytes. For some options, such as one that records the route a packet has taken, 40 bytes is far too small, making that option useless.

- Type of service field is one of the few fields that has changed its meaning (slightly) over the years. It was and is still intended to distinguish between different classes of service. Various combinations of reliability

and speed are possible. For digitized voice, fast delivery beats accurate delivery. For file transfer, error-free transmission is more important than fast transmission.

- The Total length includes everything in the datagram—both header and data. The maximum length is 65,535 bytes.
- The Identification field is needed to allow the destination host to determine which datagram a newly arrived fragment belongs to.
- DF stands for Don't Fragment. It is an order to the routers not to fragment the datagram because the destination is incapable of putting the pieces back together again. For example, when a computer boots, its ROM might ask for a memory image to be sent to it as a single datagram. By marking the datagram with the DF bit, the sender knows it will arrive in one piece, even if this means that the datagram must avoid a small-packet network on the best path and take a suboptimal route. All machines are required to accept fragments of 576 bytes or less.
- MF stands for More Fragments. All fragments except the last one have this bit set. It is needed to know when all fragments of a datagram have arrived.
- The Fragment offset tells where in the current datagram this fragment belongs. All fragments except the last one in a datagram must be a multiple of 8 bytes, the elementary fragment unit. Since 13 bits are provided, there is a maximum of 8192 fragments per datagram, giving a maximum datagram length of 65,536 bytes, one more than the Total length field.
- The Time to live field is a counter used to limit packet lifetimes. It is supposed to count time in seconds, allowing a maximum lifetime of 255 sec. It must be decremented on each hop and is supposed to be decremented multiple times when queued for a long time in a router. In

practice, it just counts hops. When it hits zero, the packet is discarded and a warning packet is sent back to the source host. This feature prevents datagrams from wandering around forever, something that otherwise might happen if the routing tables ever become corrupted.

- When the network layer has assembled a complete datagram, it needs to know what to do with it. The Protocol field tells it which transport process to give it to. TCP is one possibility, but so are UDP and some others. The numbering of protocols is global across the entire Internet. Protocols and other assigned numbers were formerly listed in RFC 1700, but nowadays they are contained in an on-line data base located at www.iana.org.
- The Header checksum verifies the header only. Such a checksum is useful for detecting errors generated by bad memory words inside a router. The algorithm is to add up all the 16-bit halfwords as they arrive, using one's complement arithmetic and then take the one's complement of the result. For purposes of this algorithm, the Header checksum is assumed to be zero upon arrival. This algorithm is more robust than using a normal add. Note that the Header checksum must be recomputed at each hop because at least one field always changes (the Time to live field), but tricks can be used to speed up the computation.
- The Source address and Destination address indicate the network number and host number.
- The Options field was designed to provide an escape to allow subsequent versions of the protocol to include information not present in the original design, to permit experimenters to try out new ideas, and to avoid allocating header bits to information that is rarely needed. The options are variable length. Each begins with a 1-byte code identifying the option. Some options are followed by a 1-byte option length field, and

then one or more data bytes. The Options field is padded out to a multiple of four bytes.

4.5.2 IP addresses

IPV4

- **IPv4 (Internet Protocol version 4)** is the **fourth version** of the Internet Protocol — the fundamental protocol that defines how data is addressed and routed across networks.
- It was developed in the **early 1980s** and is defined in **RFC 791**.
- IPv4 provides the **logical addressing** and **packet forwarding** mechanism that enables data to travel from a source device to a destination device across interconnected networks (the Internet).

Feature	Description
Address Size	32 bits (4 bytes) — allows around **4.3 billion unique addresses** (2^{32}).
Address Representation	Written in **dotted decimal format**, e.g., 192.168.10.5.
Connectionless Protocol	Each packet is treated independently; no prior connection setup.
Best-Effort Delivery	No guarantee of delivery, order, or error correction — handled by upper layers (like TCP).
Hierarchical Addressing	Divides the address into **network** and **host** portions.
Supports Fragmentation	Large packets can be split into smaller fragments to fit the MTU of a link.

Feature	Description
Header includes Checksum	Used for basic error detection of the header data.
Uses ARP Protocol	Maps an IPv4 address to a MAC (hardware) address.

IPv4 Address Structure

An IPv4 address is **32 bits long**, divided into **4 octets** (each 8 bits):

Example: 192.168.1.10

Binary: 11000000.10101000.00000001.00001010

Parts:

- **Network ID** – identifies the network.
- **Host ID** – identifies a specific device (host) on that network.

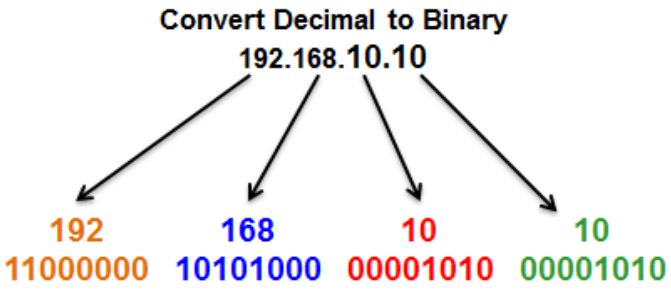
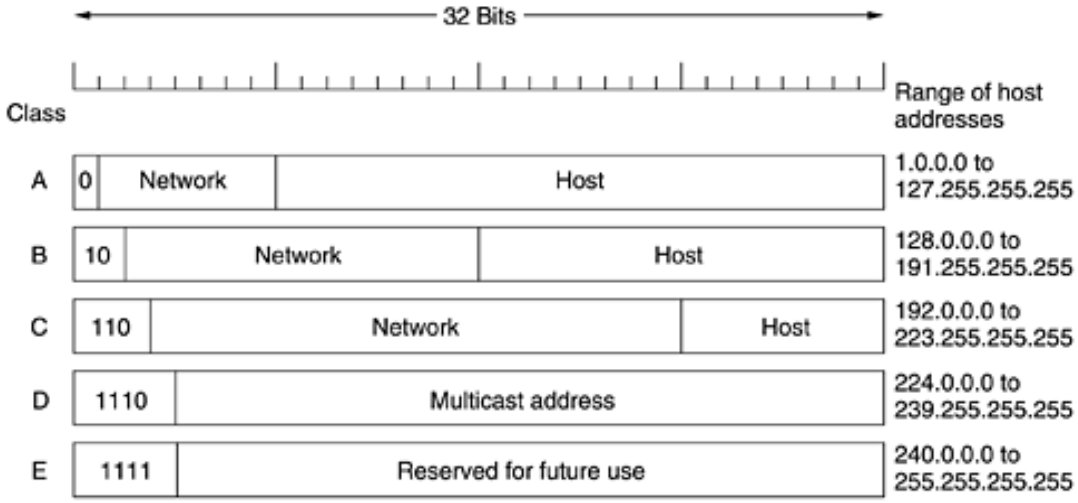


Figure: Converting from Decimal to Binary Conversions

4.5.3 Classful Addressing

- ✓ IPV4 addressing, at its inception, used the concept of classes.
- ✓ In classful addressing, the address space is divided into five classes : A,B,C,D and E.

Figure 5-55. IP address formats.



- ✓ One problem with classful addressing is that each class is divided into a fixed number of blocks with each block having a fixed size as shown in figure

Classless Addressing

- ✓ To overcome address depletion and give more organizations access the internet classless addressing was designed and implemented.
- ✓ In this scheme, there are no classes, but the addresses are still granted in blocks.

Address Blocks

- ✓ In classless addressing, any host wanted to connect internet, it is granted a block(range) of address.
- ✓ The block ovaries based on nature and size.

Restrictions

- ✓ The address in a block must be contiguous, one after another.
- ✓ The no.of address in a block must be a power of 2 (1,2,4,8,….)
- ✓ The first address must be evenly divisible by the number of addresses.
- ✓ Network Address Translation (NAT)

- ✓ To access Internet, one public IP address is needed, but we can use private IP address in our private network.
- ✓ The idea of NAT is to allow multiple devices to access Internet through a single public address.

4.5.4 Subnets

Subnetting is the process of dividing a large IP network into smaller, manageable subnetworks (subnets).

It helps in:

- Efficient IP address utilization
- Improved network performance
- Better security and management

Definition:

Subnetting is a technique that allows you to take a single IP network (like a Class A, B, or C) and divide it into smaller logical networks by borrowing bits from the host portion of the IP address.

Why Subnetting is Needed

Without subnetting:

- All devices are in one large broadcast domain → more traffic, less efficiency.
- Wastage of IP addresses.
- Difficult to manage large networks.

With subnetting:

- You can create **smaller, isolated networks** (e.g., per department, building, or region).
- Reduces **broadcast traffic**.
- Allows **hierarchical routing** and **better security**.

How Subnetting Works

Every IPv4 address has two parts:

IP Address = Network ID + Host ID

Subnetting **borrows some bits** from the Host ID part to create a **Subnet ID**.

So after subnetting:

IP Address = Network ID + Subnet ID + Host ID

Subnet Mask

The **Subnet Mask** tells which part of the IP address is the **network/subnet** and which is the **host**.

Example:

IP Address: 192.168.10.5

Subnet Mask: 255.255.255.0

In binary:

IP: 11000000.10101000.00001010.00000101

Mask: 11111111.11111111.11111111.00000000

→ First 24 bits = Network part

→ Last 8 bits = Host part

Example: Subnetting a Class C Network

Given:
Network = 192.168.10.0
Default Mask = 255.255.255.0 (/24)

If we **borrow 2 bits** from host part:

- New Mask = 255.255.255.192 (/26)

- $2^2 = 4$ subnets
- Each subnet has $2^6 - 2 = 62$ hosts

Result:

Subnet	Range	Broadcast
1	192.168.10.0 – 192.168.10.63	192.168.10.63
2	192.168.10.64 – 192.168.10.127	192.168.10.127
3	192.168.10.128 – 192.168.10.191	192.168.10.191
4	192.168.10.192 – 192.168.10.255	192.168.10.255

4.5.5 CIDR- Classless Inter Domain Routing

Introduced in: 1993 (replaced classful addressing like Class A, B, C)

Purpose:

To overcome:

- Waste of IP addresses due to classful boundaries (A, B, C)
- Rapid exhaustion of IPv4 addresses
- Large routing table sizes in routers

CIDR allows **variable-length subnet masking (VLSM)** — meaning, you can divide or group IP addresses flexibly, without being restricted by class boundaries.

Example:

Old classful: 192.168.0.0 (Class C, fixed /24 mask)

CIDR: 192.168.0.0/22

Here, /22 means **22 bits for network** and **10 bits for host**.

So, /22 combines 4 Class C networks:

192.168.0.0

192.168.1.0

192.168.2.0

192.168.3.0

This represents a single large network block — reducing routing entries.

Route Aggregation (Route Summarization)

Definition:
It's the process of combining **multiple contiguous IP network routes** into a **single summary route** (called a *supernet*).

Example:

Suppose a router has routes:

192.168.0.0/24

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Instead of advertising 4 separate routes, it can **aggregate** them into:

192.168.0.0/22

This reduces routing table size and simplifies routing updates.

Benefits:

- Smaller routing tables (improves efficiency)
- Faster lookup by routers
- Reduces bandwidth for routing updates

Supernetting:

Definition:
Supernetting is **the opposite of subnetting**.
While subnetting divides a large network into smaller parts, **supernetting merges multiple smaller networks into a larger one**.

How it works:

To supernet, you **reduce the number of network bits** in the subnet mask.

Example:

Two Class C networks:

192.168.0.0/24

192.168.1.0/24

Supernet mask → /23

Result → 192.168.0.0/23

Now the address block covers:

192.168.0.0 – 192.168.1.255

CIDR is the overall system that enables **supernetting** and **route aggregation**. All three help in **efficient IP address management** and **reduce Internet routing complexity**.

IPV6:

Figure 5-68. The IPv6 fixed header (required).

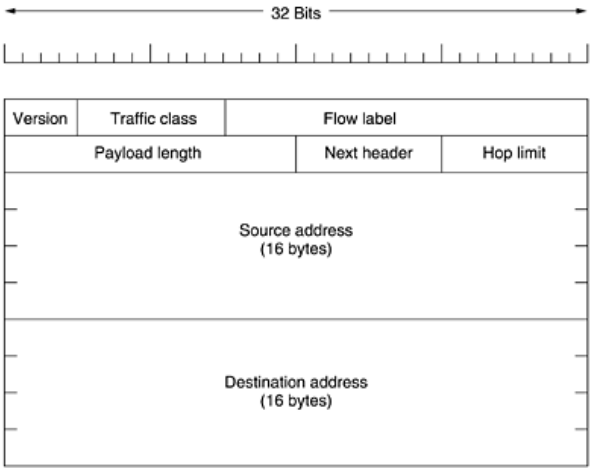


Figure 5-69. IPv6 extension headers.

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

Advantages

- ✓ Larger address space.
- ✓ Better header format.
- ✓ New options.—additional functionalities
- ✓ Provide for extensions.--- suport new technologies
- ✓ Support for resource allocation.--- avoid traffic
- ✓ Support for more security.--confidentiality
- ✓ Structure
- ✓ An IPv6 address consists of
    - 16 bytes (octets); it is 128 bits long.

- ✓ Hexadecimal colon notation:
- ✓ To make addresses more readable, IPv6 specifies hexadecimal colon notation.
- ✓ In this notation, 128 bits is divided into eight sections, each 2 bytes in length.
- ✓ Two bytes in hexadecimal notation requires four hexadecimal digits.
- ✓ Therefore, the address consists of 32 hexadecimal digits, with every four digits separated by a colon.
- ✓ Abbreviation:
- ✓ Although the IP address, even in hexadecimal format, is very long, many of the digits are zeros.
- ✓ In this case, we can abbreviate the address. The leading zeros of a section (four digits between two colons) can be omitted.
- ✓ Only the leading zeros can be dropped, not the trailing zeros.
- ✓ Using this form of abbreviation, 0074 can be written as 74, OOOF as F, and 0000 as 0.
- ✓ Note that 3210 cannot be abbreviated.
- ✓ Address Space
- ✓ IPV6 has a much larger address space; 2^128 addresses are available.
- ✓ The designers of IPv6 divided the address into several categories.
- ✓ A few leftmost bits, called the type prefix, in each address define its category.
- ✓ The type prefix is variable in length.
- ✓ In this way, there is no ambiguity; when an address is given, the type prefix can easily be determined.
- ✓ The third column shows the fraction of each type of address relative to the whole address space.
- ✓ Unicast Addresses:

- ✓ A unicast address defines a single computer. The packet sent to a unicast address must be delivered to that specific computer.
- ✓ IPv6 defines two types of unicast addresses: geographically based and provider-based.
- ✓ We discuss the second type here; the first type is left for future definition.
- ✓ The provider-based address is generally used by a normal host as a unicast address.
- ✓ Fields for the provider-based address are as follows:
- ✓ Type identifier. This 3-bit field defines the address as a provider-based address.
- ✓ Registry identifier. This 5-bit field indicates the agency that has registered the address.
- ✓ Currently three registry centers have been defined.
- ✓ 1. INTERNIC (code11000) is the center for North America;
- ✓ 2.RIPNIC (code 01000) is the center for European registration;
- ✓ 3. APNIC (code 10100) is for Asian and Pacific countries.
- ✓ Provider identifier: This variable-length field identifies the provider for Internet access (such as an ISP). A 16-bit length is recommended for this field.
- ✓ Subscriber identifier: When an organization subscribes to the Internet through a provider, it is assigned a subscriber identification. A 24-bit length is recommended for this field.
- ✓ Subnet identifier: Each subscriber can have many different sub networks, and each sub network can have an identifier. A 32 bit length is recommended for this field.
- ✓ Node identifier: The last field defines the identity of the node connected to a subnet. A length of 48 bits is recommended for this field.
- ✓ 3+8+16+32+48=128
- ✓ Multicast Addresses

- ✓ Multicast addresses are used to define a group of hosts instead of just one. A packet sent to a multicast address must be delivered to each member of the group.
- ✓ The second field is a flag that defines the group address as either permanent or transient.
- ✓ A permanent group address is defined by the Internet authorities and can be accessed at all times.
- ✓ A transient group address, on the other hand, is used only temporarily. Systems engaged in a teleconference, for example, can use a transient group address.
- ✓ The third field defines the scope of the group address. Many different scopes have been defined.
- ✓ Anycast Addresses
- ✓ IPv6 also defines anycast addresses. An anycast address, like a multicast address, also defines a group of nodes.
- ✓ However, a packet destined for an anycast address is delivered to only one of the members of the anycast group.
- ✓ Reserved Addresses
- ✓ Another category in the address space is the reserved address. These addresses start with eight O's (type prefix is 00000000).
- ✓ A few subcategories are defined in this category represent in diagram.
- ✓ An unspecified address is used when a host does not know its own address and sends an inquiry to find its address.
- ✓ A loopback address is used by a host to test itself without going into the network.
- ✓ A compatible address is used during the transition from IPv4 to IPv6.
    - It is used when a computer using IPv6 wants to send a message to another computer using IPv6, but the message

needs to pass through a part of the network that still operates in IPv4.

- ✓ Mapped address is also used during transition. However, it is used when a computer that has migrated to IPv6 wants to send a packet to a computer still using IPv4.
- ✓ Local Addresses
- ✓ These addresses are used when an organization wants to use IPv6 protocol without being connected to the global Internet.
- ✓ In other words, they provide addressing for private networks.
- ✓ A link local address is used in an isolated subnet; a site local address is used in an isolated site with several subnets.

4.5.7 Transition from IPv4 to IPv6 Address

The expansion of users in the Internet and the devices connecting to it, the **Internet Protocol version 4 (IPv4)** having 32-bit address is running out of capacity. To overcome this problem, the **Internet Protocol version 6 (IPv6)** is introduced having 128-bit addresses and therefore allows trillions of unique IPs through which many devices can connect easily.

The transition from IPv4 to IPv6 not only solves the issue of limitation of addresses but also brings improvements in network efficiency, security, and performance. In this article, we will look into what are the different methods through which we can transition or switch from IPv4 to IPv6.

How Transition Happens From IPv4 to IPv6?

Various organizations are currently working with IPv4 technology and in a very short time, we cannot switch directly from IPv4 to IPv6. Instead of only

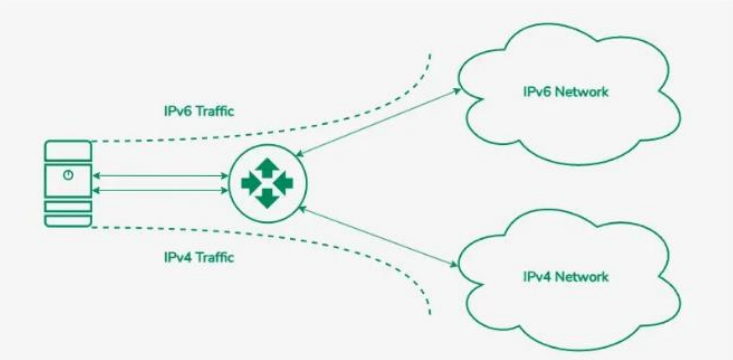
using IPv6, we use a combination of both and transition means not replacing IPv4 but co-existing of both.

When we want to send a request from an IPv4 address to an IPv6 address, it is not possible because IPv4 and IPv6 transition is not compatible. For a solution to this problem, we use some technologies that help in an easy transition from IPv4 to IPv6. These technologies are mentioned below:

1. Dual Stack Routers
2. Tunneling
3. NAT Protocol Translation

Dual-Stack Routers

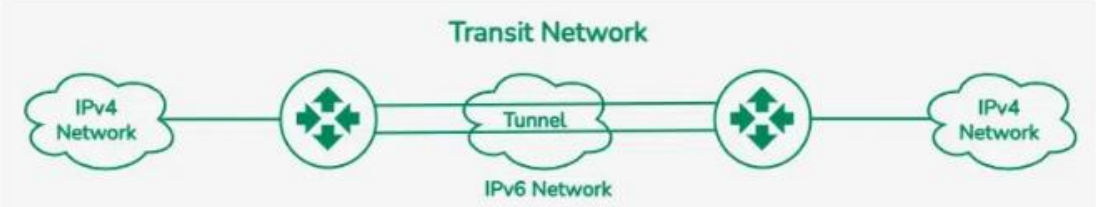
A **dual-stack router** is a network device that can support both IPv4 and IPv6 protocols simultaneously. It allows communication between devices using any of the protocol, making it a key component during the transition from IPv4 to IPv6. In dual-stack router, A router's interface is attached with IPv4 and IPv6 addresses configured are used in order to transition from IPv4 to IPv6.



In this above diagram, A given server with both IPv4 and IPv6 addresses configured can communicate with all hosts of IPv4 and IPv6 via dual-stack router (DSR). The dual stack router (DSR) gives the path for all the hosts to communicate with the server without changing their IP addresses.

Tunneling

Tunneling is a technique used to enable communication between IPv4 and IPv6 networks during the transition from IPv4 to IPv6. **Tunneling** encapsulates IPv6 packets within IPv4 packets (or vice versa). Tunneling is used as a medium to communicate the transit network with the different IP versions.



In this above diagram, the different IP versions such as IPv4 and IPv6 are present. The IPv4 networks can communicate with the transit or intermediate network on IPv6 with the help of the Tunnel. It's also possible that the IPv6 network can also communicate with IPv4 networks with the help of a Tunnel.

NAT Protocol Translation

NAT (Network Address Translation) Protocol Translation (NAT-PT), is a technique used to enable communication between IPv4 and IPv6 networks by translating one protocol to the other. With the help of the NAT Protocol Translation technique, the IPv4 and IPv6 networks can also communicate with each other without understanding the address of different IP version.

Generally, an IP version doesn't understand the address of different IP version, for the solution of this problem we use NAT-PT device which removes the header of first (sender) IP version address and add the second (receiver) IP version address so that the Receiver IP version address understand that the request is sent by the same IP version, and its vice-versa is also possible.



In the above diagram, an IPv4 address communicates with the IPv6 address via a NAT-PT device to communicate easily. In this situation, the IPv6 address understands that the request is sent by the same IP version (IPv6) and it responds.

In conclusion, the transition from IPv4 to IPv6 is important to support the increasing number of devices connected to the internet. Range of IPv4 Address is limited and therefore it creates challenges like full utilization of IP Addresses. This results in **IPv6**, with increase in address range. Ipv6 not only solves the problem of address limitation, IPv6 also bring improvements in network performance, network security, and scalability of networks. With different solutions like dual-stack, tunneling, and NAT, these technologies helps in the transition from IPv4 to IPv6 can be used in parallel until IPv6 is fully adopted.

4.5.8 Comparison of IPV4 & IPV6

Feature	IPv4	IPv6
Address Length	32-bit address	128-bit address
Address Format	Decimal format (e.g., 192.168.0.1)	Hexadecimal format (e.g., 2001:0db8::1)
Configuration	Manual and DHCP configuration	Auto-configuration and renumbering supported

Feature	IPv4	IPv6
Connection Integrity	End-to-end integrity is unachievable	End-to-end integrity is achievable
Security	No built-in security; external tools like IPSec needed	IPSec is built-in for encryption and authentication
Fragmentation	Performed by sender and routers	Performed only by the sender
Flow Identification	Not available	Uses Flow Label field in header for packet flow identification
Checksum Field	Present	Not present
Transmission Scheme	Supports broadcast	Uses multicast and anycast; no broadcast
Header Size	Variable: 20–60 bytes	Fixed: 40 bytes
Conversion	Can be converted to IPv6	Not all IPv6 addresses can be converted to IPv4
Field Structure	4 fields separated by dots (.)	8 fields separated by colons (:)
Address Classes	Has address classes (A, B, C, D, E)	No concept of address classes
VLSM Support	Supports Variable	Does not support VLSM

Feature	IPv4	IPv6
	Length Subnet Mask (VLSM)	
Example	66.94.29.13	2001:0000:3238:DFE1:0063:0000:0000:FEFB

The Internet Protocol (IP) gives every device a unique address so it can send and receive data online.

- **IPv4:** The older version, uses a 32-bit address (about 4 billion addresses). With the rise of the internet and IoT devices, these addresses are nearly used up.
- **IPv6:** The newer version, uses a 128-bit address (virtually unlimited addresses). It was created to overcome the shortage and support the future growth of the internet.

Drawbacks of IPv4

1. **Limited Addresses:** IPv4's 32-bit space cannot meet the growing demand for internet-connected devices.
2. **Complex Setup:** Requires manual configuration or DHCP, which can be error-prone.
3. **Inefficient Routing:** Larger, complex headers slow down data processing.
4. **Weak Security:** No built-in security; extra measures are needed.
5. **Poor QoS Support:** Limited ability to prioritize traffic, affecting video/voice quality.
6. **Packet Fragmentation:** Routers may split packets, causing inefficiency and possible data loss.
7. **Broadcast Overhead:** Uses broadcasting, which increases unnecessary traffic and reduces performance.

Benefits of IPv6 over IPv4

The recent Version of IP IPv6 has a greater advantage over IPv4. Here are some of the mentioned benefits:

1. **Larger Address Space:** IPv6 uses 128-bit addresses (vs. 32-bit in IPv4), allowing many more devices to connect.
2. **Improved Security:** IPv6 has built-in features like data authentication and encryption, making connections safer..
3. **Simpler Header:** IPv6 has a streamlined header, improving speed and reducing processing cost.
4. **Better QoS Support:** IPv6 provides stronger Quality of Service, improving video, audio, and website performance.
5. **Mobile-Friendly:** IPv6 offers better, faster, and more secure support for mobile connections.

SECTION-A
Assignment-Cum-Tutorial Questions
Part- A
Objective Questions

1. Which of the following is not a function of the Network Layer?
   A) Routing
   B) Congestion control
   C) Error detection
   D) Internetworking
   Answer: _____

2. In store-and-forward packet switching, each packet is:
   A) Sent directly to the destination without delay
   B) Stored at intermediate nodes and forwarded later
   C) Delivered only when the full message is received
   D) Used only for connection-oriented communication
   Answer: _____

3. Which service does the Network Layer provide to the Transport Layer?
   A) Reliable delivery
   B) End-to-end process communication
   C) Host-to-host packet delivery
   D) Flow control
   Answer: _____

4. In connectionless service, each packet:
   A) Follows the same route
   B) Is guaranteed to arrive in order
   C) Is independent and routed separately
   D) Requires prior connection setup
   Answer: _____

5. In connection-oriented service, communication takes place through:
   A) Datagram networks
   B) Virtual circuits
   C) Tunneling
   D) IP datagrams

Answer: _____

6. Which of the following correctly compares Virtual Circuit and Datagram Networks?

A) Virtual circuits are connectionless, datagrams are connection-oriented
B) Virtual circuits use routing decisions per packet
C) Datagram networks are faster for long data transfer
D) Virtual circuits establish a path before data transmission

Answer: _____

7. The optimality principle in routing states that:

A) Every router should have the same routing table
B) The best path is the one with the shortest delay
C) Subpaths of an optimal path are also optimal
D) Congestion must be minimized at all times

Answer: _____

8. Which routing algorithm always delivers packets through all possible paths?

A) Flooding
B) Distance vector
C) Link state
D) Hierarchical routing

Answer: _____

9. The distance vector routing algorithm uses which protocol in the Internet?

A) RIP
B) OSPF
C) BGP
D) ARP

Answer: _____

10. In link-state routing, each router must:

A) Share only the next-hop cost
B) Send information about its entire neighborhood
C) Maintain the routing table of all routers
D) Send periodic keep-alive messages

Answer: _____

11. In hierarchical routing, the main advantage is:

A) Faster convergence
B) Reduced routing table size
C) Loop-free paths
D) Use of virtual circuits

Answer: _____

12. Which of the following is not a congestion control policy?

A) Admission control
B) Traffic throttling
C) Load shedding
D) Routing based on shortest path only

Answer: _____

13. The leaky bucket algorithm is mainly used for:

A) Error detection
B) Congestion control
C) Encryption
D) Routing optimization

Answer: _____

14. The token bucket algorithm allows:

A) Fixed rate of packet transmission
B) Bursty traffic at a controlled average rate
C) Only one packet per second
D) Packets to be dropped immediately

Answer: _____

15. In internetworking, the technique of sending packets of one protocol inside another is called:

A) Encapsulation
B) Tunneling
C) Fragmentation
D) Multiplexing

Answer: _____

16. When a packet is too large for a network, it is divided into smaller units called:

A) Segments
B) Frames
C) Fragments
D) Tokens

Answer: _____

17. The IPv4 address consists of:

A) 32 bits
B) 64 bits
C) 128 bits
D) 48 bits

Answer: _____

18. The Classless Inter-Domain Routing (CIDR) technique helps to:

A) Increase routing table entries
B) Reduce address wastage and routing table size
C) Replace IP addresses with MAC addresses
D) Improve congestion control

Answer: _____

19. IPv6 uses addresses of size:

A) 32 bits
B) 48 bits
C) 64 bits
D) 128 bits
Answer: _____

20. Which of the following statements about IPv4 vs IPv6 is true?
A) IPv6 uses smaller header size
B) IPv6 supports NAT by default
C) IPv6 provides improved security and larger address space
D) IPv4 supports auto-configuration

Answer: _____

21. What is the subnet mask for the CIDR notation /26?
a) 255.255.255.0
b) 255.255.255.128
c) 255.255.255.192
d) 255.255.255.224

Answer: _____

22. You are given the IP address 192.168.1.0/24. How many subnets can you create if you borrow 3 bits from the host portion?
a) 6
b) 8

c) 16
d) 32

Answer: _____

23. For the subnet 192.168.1.0/27, what is the range of usable host IP addresses?
a) 192.168.1.1 to 192.168.1.30
b) 192.168.1.1 to 192.168.1.31
c) 192.168.1.2 to 192.168.1.30
d) 192.168.1.2 to 192.168.1.31

Answer: _____

24. You need at least 100 usable host IP addresses per subnet. What is the minimum subnet mask you should use?
a) /24
b) /25
c) /26
d) /27

Answer: _____

25. Given the IP address 172.16.0.0/16, how many subnets can you create by borrowing 4 bits from the host portion?
a) 16
b) 32
c) 64
d) 128

Answer: _____

Part- B

Short Answer Questions (2M)

1. What is the main function of the Network Layer?
2. Define store-and-forward packet switching.
3. What is a virtual circuit?
4. What is a datagram network?
5. Define connectionless service.

6. Define connection-oriented service.
7. Name two services provided by the Network Layer to the Transport Layer.
8. What is the main principle of shortest path routing?
9. Define flooding in routing.
10. Define distance vector routing.
11. Define link-state routing.
12. What is hierarchical routing?
13. Define congestion in networking.
14. What is traffic-aware routing?
15. Define admission control.
16. Define traffic throttling.
17. What is load shedding?
18. Name two traffic control algorithms.
19. Define leaky bucket algorithm.
20. Define token bucket algorithm.
21. What is internetworking?
22. Define tunneling in networking.
23. What is fragmentation?
24. What is the size of an IPv4 address?
25. What is the size of an IPv6 address?
26. Name one feature of IPv6 over IPv4.
27. What is CIDR?
28. Define subnetting.

PART-C
Descriptive Questions (5M)

1. Explain the optimality principle used in routing algorithms. (B.L-2)
2. Compare distance vector and link-state routing algorithms. (B.L-2)
3. Illustrate how the leaky bucket algorithm controls congestion. (B.L-2)
4. Summarize the differences between IPv4 and IPv6**.** (B.L-2)
5. Explain the purpose of fragmentation in packet transmission. (B.L-2)

6. Apply the shortest path algorithm to find the minimum-cost route between network nodes. (B.L-3)
7. Using an example, demonstrate how CIDR (Classless Inter-Domain Routing) reduces address wastage. (B.L-3)
8. Show how token bucket allows controlled bursts in network traffic. (B.L-3)

9. Apply admission control policy to handle congestion in a network scenario. (B.L-3)
10. Construct an example network and show how tunneling helps connect heterogeneous systems. (B.L-3)

11. Analyze the trade-offs between virtual circuit networks and datagram networks in terms of reliability and flexibility. (B.L-4)
12. Examine how traffic-aware routing affects congestion in a large-scale network. (B.L-4)
13. Analyze the impact of fragmentation on data transmission performance. (B.L-4)
14. Compare hierarchical routing with flat routing in terms of scalability and complexity. (B.L-4)
15. Identify factors that cause congestion and explain their interrelationships. (B.L-4)

16. Evaluate the effectiveness of leaky bucket vs token bucket in controlling bursty traffic. (B.L-5)
17. Justify the need for transition mechanisms from IPv4 to IPv6. (B.L-5)
18. Assess how routing algorithms influence overall network performance. (B.L-5)
19. Critique the limitations of distance vector routing in dynamic networks. (B.L-5)
20. Evaluate different congestion prevention policies and recommend the best approach for real-time applications. (B.L-5)

21. Design a hybrid congestion control mechanism combining the best features of leaky bucket and token bucket. (B.L-6)

22. Propose a new routing strategy that adapts dynamically to network load and delay variations. (B.L-6)
23. Develop a simulation model to test congestion control algorithms under different traffic conditions. (B.L-6)
24. Formulate a network design using both IPv4 and IPv6 nodes with tunneling for compatibility. (B.L-6)
25. Create a conceptual framework showing how all network layer services (routing, congestion control, addressing, and fragmentation) interact to ensure efficient data delivery. (B.L-6)

PART-D
Competitive Examination Questions

You have been assigned the IP address 192.168.10.0/24 for a new network segment. Subnet the given IP address into three subnets, ensuring that each subnet has a sufficient number of host addresses.
[NIELIT Scientific Assistant A (January 2025)]

- **Sub-questions:**
    1. Calculate the subnet mask, network address, broadcast address, and valid range of IP addresses for each subnet.
    2. Determine the number of usable host addresses in each subnet. Provide your answers in CIDR notation and dotted-decimal format.

Which of the following is a network layer protocol? [NET 2019]

a) SMTP
b) HTTP
c) IP
d) TCP

In the following pairs of OSI protocol layer/sub-layer and its functionality, the INCORRECT pair is [2014 SET 3]

a) Network layer and Routing
b) Data Link Layer and Bit synchronization
c) Transport layer and End-to-end process communication
d) Medium Access Control sub-layer and Channel sharing

Assume that Source S and Destination D are connected through an intermediate router R. How many times does a packet have to visit the network layer and data link layer during a transmission from S to D? [NET 2018]

a) Network layer – 4 times, Data link layer – 4 times
b) Network layer – 4 times, Data link layer – 6 times
c) Network layer – 2 times, Data link layer – 4 times
d) Network layer – 3 times, Data link layer – 4 times

Which of the following internet services is appropriate to access the computer of your office from home? [NET 2017]

a) WWW b) IRC
c) FTP d) Telnet

UNIT – V: Transport and Application Layers

Objectives:

- To familiarize with the services provided by the Transport and Application Layers.

Outcomes:

At the end of the unit, students will be able to:

- describe the process how the services could be achievedprovided at higher layers.
- exemplify the importance of congestion control
- discuss the architecture of various protocols at application layer.

Syllabus:

5.1 The Transport LayerProtocols

- 5.1.1 Introduction
- 5.1.2 Services
- 5.1.3 Port number

5.2 User data gram protocol (UDP)

- 5.2.1 User datagram
- 5.2.2 UDP services
- 5.2.3 UDP applications

5.3 Transmission control protocol (TCP)

- 5.3.1 TCP services
- 5.3.2 TCP features
- 5.3.3 Segment
- 5.3.4 A TCP connection
- 5.3.5 Windows in TCP
- 5.3.6 Flow control
- 5.3.7 Error control
- 5.3.8 Congestion control in TCP

5.4 Application Layer

- 5.4.1 World Wide Web (WWW)
- 5.4.2 HTTP
- 5.4.3 Electronic Mail-Architecture, web-based mail, Email security
- 5.4.4 TELNET-local versus remote Logging
- 5.4.5 Domain Name System

5.1Transport layer

5.1.1 Introduction

- The transport layer is located between the application layer and the network layer. It provides a process-to-process communication between two application layers, one at the local host and the other at the remote host.
- Communication is provided using a logical connection, which means that the two application layers, which can be located in different parts of the globe, assume that there is an imaginary direct connection through which they can send and receive messages.

5.1.2 Services

- Each protocol provides a different type of service and should be used appropriately.
- UDP is an unreliable connectionless transport-layer protocol used for its simplicity and efficiency in applications where error control can be provided by the application-layer process.
- TCP is a reliable connection-oriented protocol that can be used in any application where reliability is important.
- SCTP is a new transport-layer protocol that combines the features of UDP and TCP.

5.1.3 Port Numbers

- A transport-layer protocol usually has several responsibilities. One is to create a process-to-process communication; these protocols use port numbers to accomplish this.
- Port numbers provide end-to-end addresses at the transport layer and allow multiplexing and demultiplexing at this layer, just as IP addresses do at the network layer. The figure 5.1 shows some common port numbers for all three protocols

Port	*Protocol*	*UDP*	*TCP*	*SCTP*	*Description*
7	Echo	√	√	√	Echoes back a received datagram
9	Discard	√	√	√	Discards any datagram that is received
11	Users	√	√	√	Active users
13	Daytime	√	√	√	Returns the date and the time
17	Quote	√	√	√	Returns a quote of the day
19	Chargen	√	√	√	Returns a string of characters
20	FTP-data		√	√	File Transfer Protocol
21	FTP-21		√	√	File Transfer Protocol
23	TELNET		√	√	Terminal Network
25	SMTP		√	√	Simple Mail Transfer Protocol
53	DNS	√	√	√	Domain Name Service
67	DHCP	√	√	√	Dynamic Host Configuration Protocol
69	TFTP	√	√	√	Trivial File Transfer Protocol
80	HTTP		√	√	HyperText Transfer Protocol
111	RPC	√	√	√	Remote Procedure Call
123	NTP	√	√	√	Network Time Protocol
161	SNMP-server	√			Simple Network Management Protocol
162	SNMP-client	√			Simple Network Management Protocol

Figure 5.1. Some well-known ports used with UDP and TCP

5.2User Datagram Protocol (UDP)

- The User Datagram Protocol (UDP) is called a connectionless, unreliable transport protocol. It does not add anything to the services of IP except to provide process-to-process communication instead of host-to-host communication.
- UDP performs very limited error checking. UDP is a very simple protocol using a minimum of overhead. If a process wants to send a small message and does not care much about reliability, it can use UDP.

5.2.1 User Datagram

- UDP packets, called user datagrams, have a fixed-size header of 8 bytes. The figure 5.2. shows the format of a user datagram. The fields in the header are given below

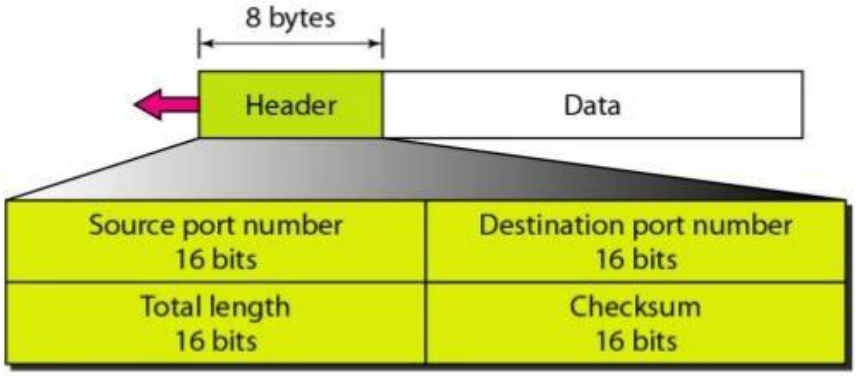


Figure 5.2. UDP Header

- **Source port number:** This is the port number used by the process running on the source host. It is 16 bits long, which means that the port number can range from 0 to 65,535.
- **Destination port number**. This is the port number used by the process running on the destination host. It is also 16 bits long.
- **Length:** This is a 16-bit field that defines the total length of the user datagram, header plus data. The 16 bits can define a total length of 0 to 65,535 bytes. How- ever, the total length needs to be much less because a UDP user datagram is stored in an IP datagram with a total length of 65,535.
  UDP length = IP length - IP header's length
- **Checksum**. This field is used to detect errors over the entire user datagram (header plus data).The UDP checksum calculation is different from the one for IP and ICMP.

5.2.2 UDP Services

Process-to-Process Communication

- UDP provides process-to-process communication using socket addresses, a combination of IP addresses and port numbers.

Connectionless Services

- UDP provides a connectionless service. This means that each user datagram sent by UDP is an independent datagram. There is no relationship between the different user datagrams even if they are coming from the same source process and going to the same destination program.

Flow and Error Control

- UDP is a very simple, unreliable transport protocol. There is no flow control and hence no window mechanism. The receiver may overflow with incoming messages. There is no error control mechanism in UDP except for the checksum. This means that the sender does not know if a message has been lost or duplicated. When the receiver detects an error through the checksum, the user datagram is silently discarded.

Checksum

- UDP checksum includes three sections: a pseudo header, the UDP header, and the data coming from the application layer. The pseudo header is the part of the header of the IP packet in which the user data- gram is to be encapsulated with some fields filled with 0s, which is given in the figure 5.3.

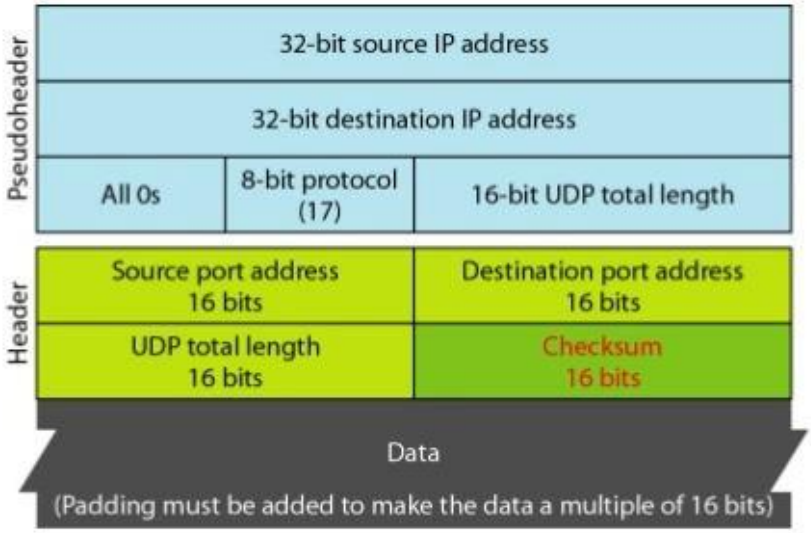


Figure 5.3. Pseudo header and header of UDP

Encapsulation and Decapsulation

- To send a message from one process to another, the UDP protocol encapsulates and decapsulates messages in an IP datagram.

Queuing

- In UDP, queues are associated with ports. At the client site, when a process starts, it requests a port number from the operating system. Some implementations create both an incoming and an outgoing queueassociated with each process. Other implementations create only an incoming queue associated with each process.

Multiplexing and Demultiplexing

- In a host running a TCP/IP protocol suite, there is only one UDP but possibly several processes that may want to use the services of UDP. To handle this situation, UDP multiplexes and demultiplexes.

5.2.3 UDP Applications

The following shows some typical applications that can benefit more from the services of UDP than from those of TCP.

- UDP is suitable for a process that requires simple request-response communication with little concern for flow and error control. It is not usually used for a process such as FTP that needs to send bulk data.
- UDP is suitable for a process with internal flow and error control mechanisms. For example, the Trivial File Transfer Protocol (TFTP) process includes flow and error control. It can easily use UDP.
- UDP is a suitable transport protocol for multicasting. Multicasting capability is embedded in the UDP software but not in the TCP software.
- UDP is used for management processes such as SNMP.
- UDP is used for some route updating protocols such as Routing Information Protocol (RIP).
- UDP is normally used for interactive real-time applications that cannot tolerate uneven delay between sections of a received message

5.3Transmission Control Protocol

- Transmission Control Protocol (TCP)is a connection-oriented, reliable protocol.
- TCP explicitly defines connection establishment, data transfer, and connection teardown phases to provide a connection-oriented service.
- TCP uses a combination of GBN and SR protocols to provide reliability. To achieve this goal, TCP uses checksum (for error detection), retransmission of lost or corrupted packets, cumulative and selective acknowledgments, and timers.

5.3.1 TCP services

Process-to-Process Communication

- As with UDP, TCP provides process-to-process communication using port numbers. and some of the port numbers used by TCP, which are shown infigure 5.1.

Stream Delivery Service

- TCP, unlike UDP, is a stream-oriented protocol. In UDP, a process sends messages with predefined boundaries to UDP for delivery. UDP adds its own header to each of these messages and delivers it to IP for transmission. Each message from the process is called a *user datagram*, and becomes, eventually, one IP datagram.
- TCP, on the other hand, allows the sending process to deliver data as a stream of bytes and allows the receiving process to obtain data as a stream of bytes. TCP creates an environment in which the two processes seem to be connected by an imaginary "tube" that carries their bytes across the Internet. This imaginary environment is depicted in figure 5.4. The sending process produces (writes to) the stream and the receiving process consumes (reads from) it.

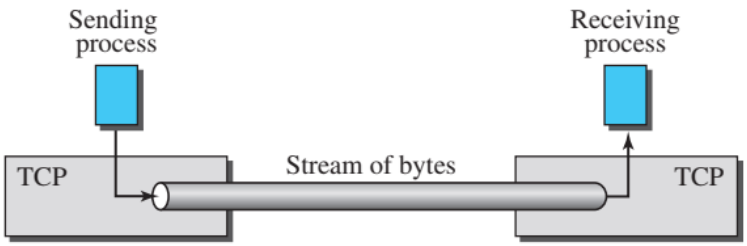


Figure 5.4. Stream delivery

Sending and Receiving Buffers

- Because the sending and the receiving processes may not necessarily write or read data at the same rate, TCP needs buffers for storage. There are two buffers, the sending buffer and the receiving buffer, one for each direction. These buffers are also necessary for flow- and error-control mechanisms used by TCP. One way to implement a buffer is to use a circular array of 1-byte locations as shown in figure 5.5.

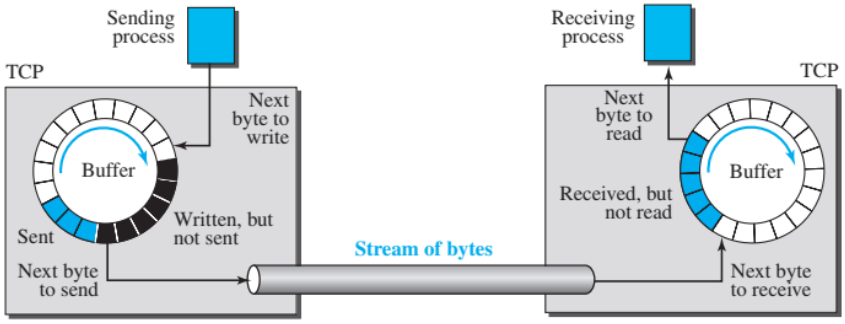


Figure 5.5 *Sending and receiving buffers*

Segments

- Although buffering handles the disparity between the speed of the producing and consuming processes, we need one more step before we can send data. The network layer, as a service provider for TCP, needs to send data in packets, not as a stream of bytes. At the transport layer, TCP groups a number of bytes together into a packet called a *segment*. TCP adds a header to each segment (for control purposes) and delivers the segment to the network layer for transmission. The segments are encapsulated in an IP datagram and transmitted. This entire operation is transparent to the receiving process. Figure 5.6 shows how segments are created from the bytes in the buffers.

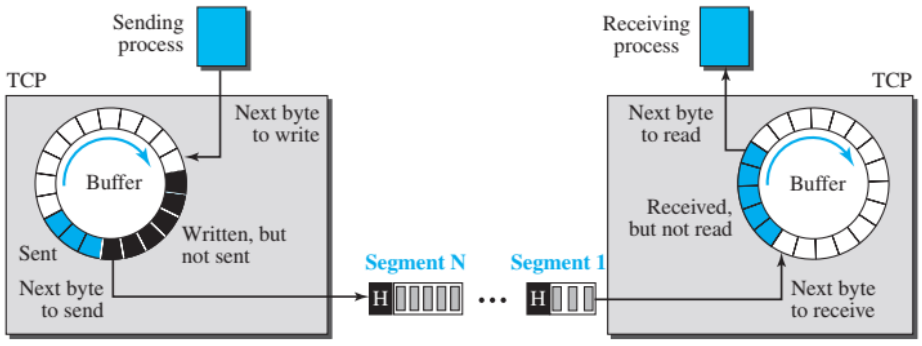


Figure 5.6 TCP segments

Full-Duplex Communication

- TCP offers *full-duplex service,* where data can flow in both directions at the same time. Each TCP endpoint then has its own sending and receiving buffer, and segments move in both directions.

Multiplexing and Demultiplexing : Like UDP, TCP performs multiplexing at the sender and

demultiplexing at the receiver. However, since TCP is a connection-oriented protocol, a connection needs to be established for each pair of processes.

Connection-Oriented Service

- TCP, unlike UDP, is a connection-oriented protocol. When a process at site A wants to send to and receive data from another process at site B, the following three phases occur:
  1. The two TCP's establish a logical connection between them.
  2. Data are exchanged in both directions.
  3. The connection is terminated.
- Note that this is a logical connection, not a physical connection. The TCP segment is encapsulated in an IP datagram and can be sent out of order, or lost or corrupted, and then resent. Each may be routed over a different path to reach the destination. There is no physical connection. TCP creates a stream-oriented environment in which it accepts the responsibility of delivering the bytes in order to the other site.

Reliable Service

- TCP is a reliable transport protocol. It uses an acknowledgment mechanism to check the safe and sound arrival of data. We will discuss this feature further in the section on error control.

5.3.2 TCP Features

TCP has several features that are briefly summarized below.

Numbering System

- Although the TCP software keeps track of the segments being transmitted or received, there is no field for a segment number value in the segment header. Instead, there are two fields, called the *sequence number* and the *acknowledgment number*. These two fields refer to a byte number and not a segment number.

Byte Number

- TCP numbers all data bytes (octets) that are transmitted in a connection. Numbering is independent in each direction. The bytes of data being transferred in each connection are numbered by TCP. The numbering starts with an arbitrarily generated number.

Sequence Number

- After the bytes have been numbered, TCP assigns a sequence number to each segment that is being sent.The sequence number, in each direction, is defined as follows:
  1. The sequence number of the first segment is the ISN (initial sequence number), which is a random number.
  2. The sequence number of any other segment is the sequence number of the previoussegment plus the number of bytes (real or imaginary) carried by the previous segment.

Acknowledgment Number

- The acknowledgment number defines the number of the next byte that the party expects to receive. The acknowledgment number is cumulative.

5.3.3 Segment

A packet in TCP is called a *segment.*

Format

- The format of a segment is shown in figure 5.7. The segment consists of a header of 20 to 60 bytes, followed by data from the application program. The header is 20 bytes if there are no options and up to 60 bytes if it contains options.

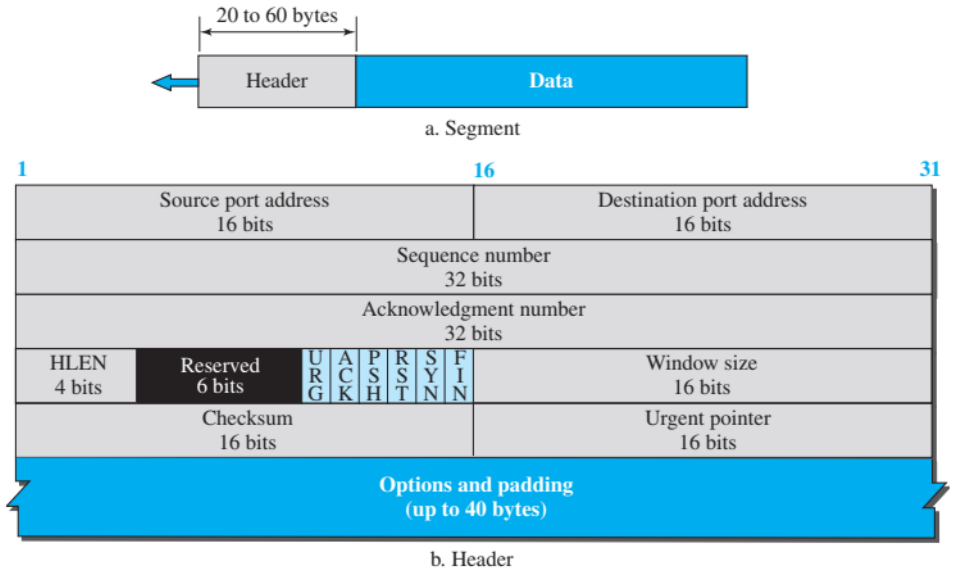


Figure 5.7 TCP segment format

- *Source port address*. This is a 16-bit field that defines the port number of the application program in the host that is sending the segment.
- *Destination port address*. This is a 16-bit field that defines the port number of the application program in the host that is receiving the segment.
- *Sequence number*. This 32-bit field defines the number assigned to the first byte of data contained in this segment. As we said before, TCP is a stream transport protocol. To ensure connectivity, each byte to be transmitted is numbered. The sequence number tells the destination which byte in this sequence is the first byte in the segment.
- *Acknowledgment number*. This 32-bit field defines the byte number that the receiver of the segment is expecting to receive from the other party. If the receiver of the segment has successfully received byte number x from the other party, it returns $x + 1$ as the acknowledgment number. Acknowledgment and data can be piggybacked together.
- *Header length*. This 4-bit field indicates the number of 4-byte words in the TCP header. The length of the header can be between 20 and 60 bytes. Therefore, the value of this field is always between 5 ($5 \times 4 = 20$) and 15 ($15 \times 4 = 60$).
- *Control*. This field defines 6 different control bits or flags, as shown in Figure 5.8.One or more of these bits can be set at a time. These bits enable flow control, connection establishment and termination, connection abortion, and the mode of data transfer in TCP. A brief description of each bit is shown in the figure.

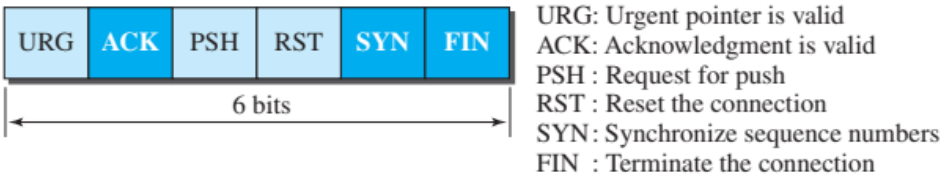


Figure 5.8 *Control field*

- *Window size.* This field defines the window size of the sending TCP in bytes. Note that the length of this field is 16 bits, which means that the maximum size of the window is 65,535 bytes. This value is normally referred to as the receiving window (*rwnd*) and is determined by the receiver.
- *Checksum.* This 16-bit field contains the checksum. The calculation of the checksum for TCP follows the same procedure as the one described for UDP. However, the use of the checksum in the UDP datagram is optional, whereas the use of the checksum for TCP is mandatory.
- *Urgent pointer.* This 16-bit field, which is valid only if the urgent flag is set, is used when the segment contains urgent data. It defines a value that must be added to the sequence number to obtain the number of the last urgent byte in the data section of the segment.
- *Options.* There can be up to 40 bytes of optional information in the TCP header.
- *Encapsulation*A TCP segment encapsulates the data received from the application layer. The TCP segment is encapsulated in an IP datagram, which in turn is encapsulated in a frame at the data-link layer.

5.3.4 A TCP Connection

- TCP is a connection-oriented transport protocol, which establishes a logical path between the source and destination. All of the segments belonging to a message are then sent over this logical path. Using a single logical pathway for the entire message facilitates the acknowledgment process as well as retransmission of damaged or lost frames. The point is that a TCP connection is logical, not physical. TCP operates at a higher level. TCP uses the services of IP to deliver individual segments to the receiver, but it controls the connection itself. If a segment is lost or corrupted, it is retransmitted.
- In TCP, connection-oriented transmission requires three phases: connection establishment, data transfer, and connection termination.

Connection Establishment

- TCP transmits data in full-duplex mode. When two TCPs in two machines are connected, they are able to send segments to each other simultaneously. This implies that each party must initialize communication and get approval from the other party before any data are transferred.

Three-Way Handshaking

- The connection establishment in TCP is called *three-way handshaking.* In our example, an application program, called the *client,* wants to make a connection with another application program, called the *server,* using TCP as the transport-layer protocol.

- The process starts with the server. The server program tells its TCP that it is ready to accept a connection. This request is called a *passive open*. Although the server TCP is ready to accept a connection from any machine in the world, it cannot make the connection itself.
- The client program issues a request for an *active open*. A client that wishes to connect to an open server tells its TCP to connect to a particular server. TCP can now start the three-way handshaking process, as shown in figure 5.10.

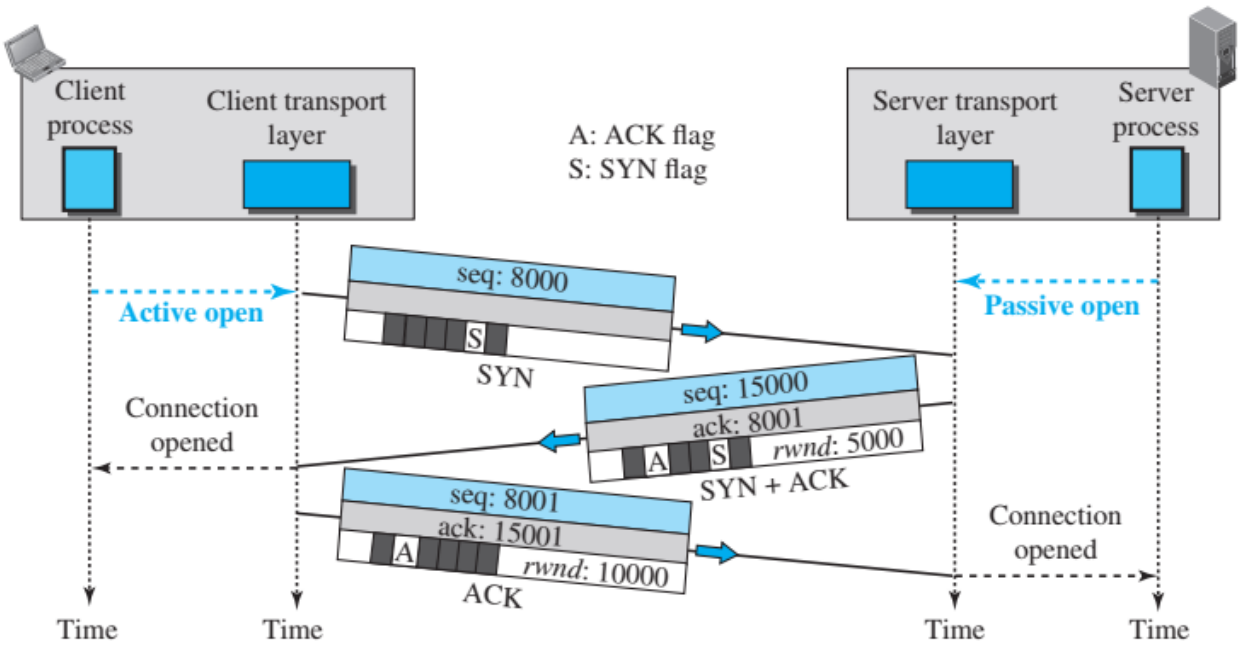


Figure 5.10 Connection establishment using three-way handshaking

1. A SYN segment cannot carry data, but it consumes one sequence number.
2. A SYN segment cannot carry data, but it consumes one sequence number.
3. An ACK segment, if carrying no data, consumes no sequence number.

Data Transfer

- After connection is established, bidirectional data transfer can take place. The client and server can send data and acknowledgments in both directions. Data traveling in the same direction as an acknowledgment are carried on the same segment. The acknowledgment is piggybacked with the data. Figure 5.11 shows an example

Connection Termination

- Either of the two parties involved in exchanging data (client or server) can close the connection, although it is usually initiated by the client. Most implementations today allow two options for connection termination: three-way handshaking and four-way handshaking with a half-close option.

Three-Way Handshaking

- Most implementations today allow three-way handshakingfor connection termination, as shown in figure 5.12.

  1. In this situation, the client TCP, after receiving a close command from the client

process, sends the first segment, a FIN segment in which the FIN flag is set.

2. The server TCP, after receiving the FIN segment, informs its process of the situation and sends the second segment, a FIN + ACK segment, to confirm the receipt of the FIN segment from the client and at the same time to announce the closing of the connection in the other direction.

3. The client TCP sends the last segment, an ACK segment, to confirm the receipt of the FIN segment from the TCP server. This segment contains the acknowledgment number, which is one plus the sequence number received in the FIN segment from the server.

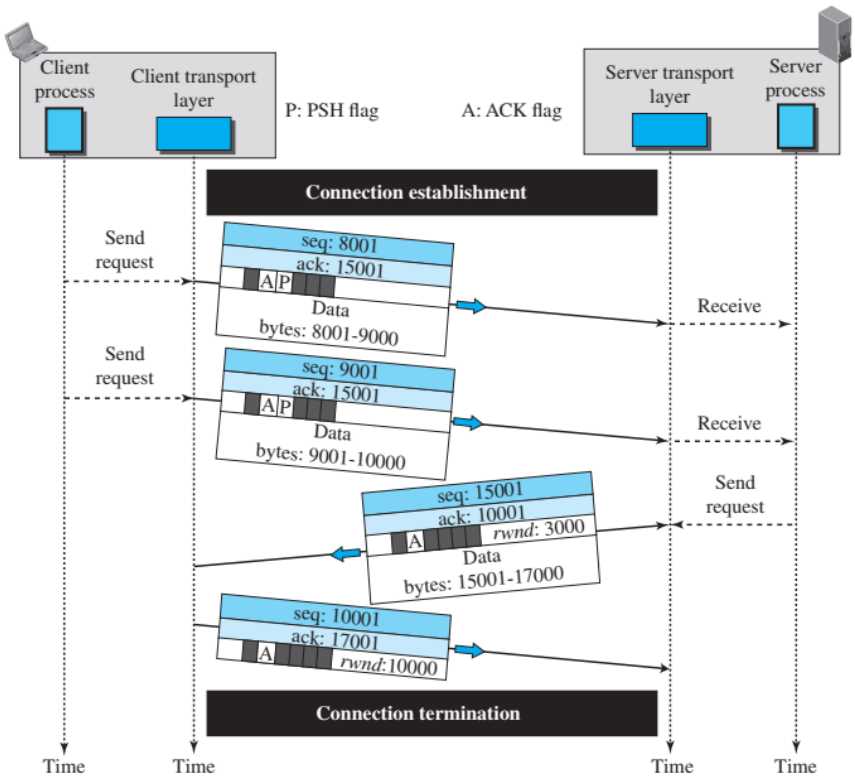


Figure 5.11 *Data transfer*

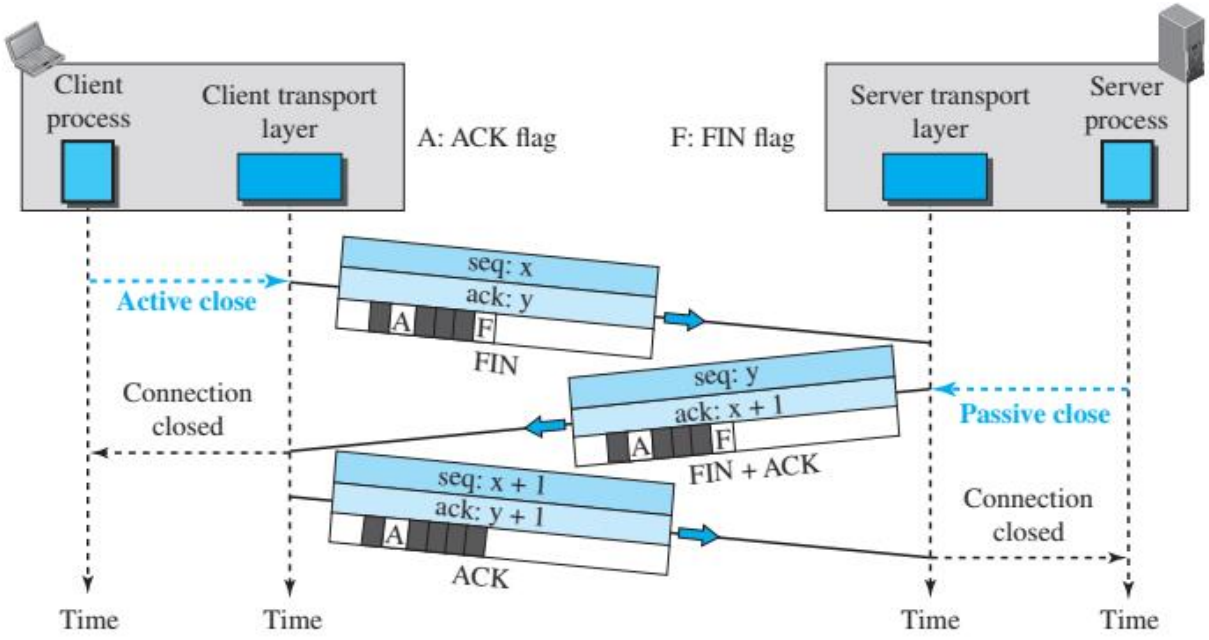


Figure 5.12 Connection termination using three-way handshaking

Half-Close

- In TCP, one end can stop sending data while still receiving data. This is called a *halfclose*. Either the server or the client can issue a half-close request. It can occur when the server needs all the data before processing can begin. Figure 5.13 shows half close.

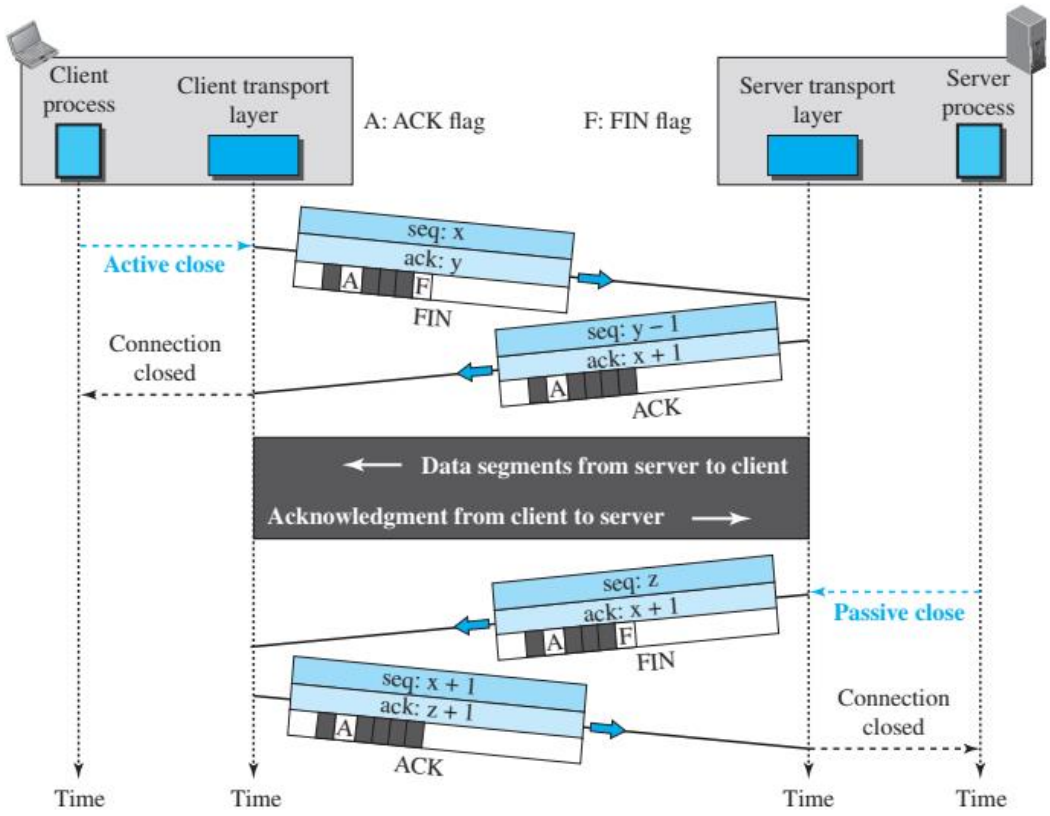


Figure 5.13.*Half-close*

Connection Reset

- TCP at one end may deny a connection request, may abort an existing connection, or may terminate an idle connection. All of these are done with the RST (reset) flag.

5.3.5 Windows in TCP

- TCP uses two windows (send window and receive window) for each direction of data transfer, which means four windows for a bidirectional communication.
- Communication is only unidirectional (say from client to server); the bidirectional communication can be inferred using two unidirectional communications with piggybacking.

Send Window

- Figure 5.14 shows an example of a send window. The window size is 100 bytes, but later we see that the send window size is dictated by the receiver (flow control) and the congestion in the underlying network (congestion control). The figure shows how a send window opens, closes, or shrinks.

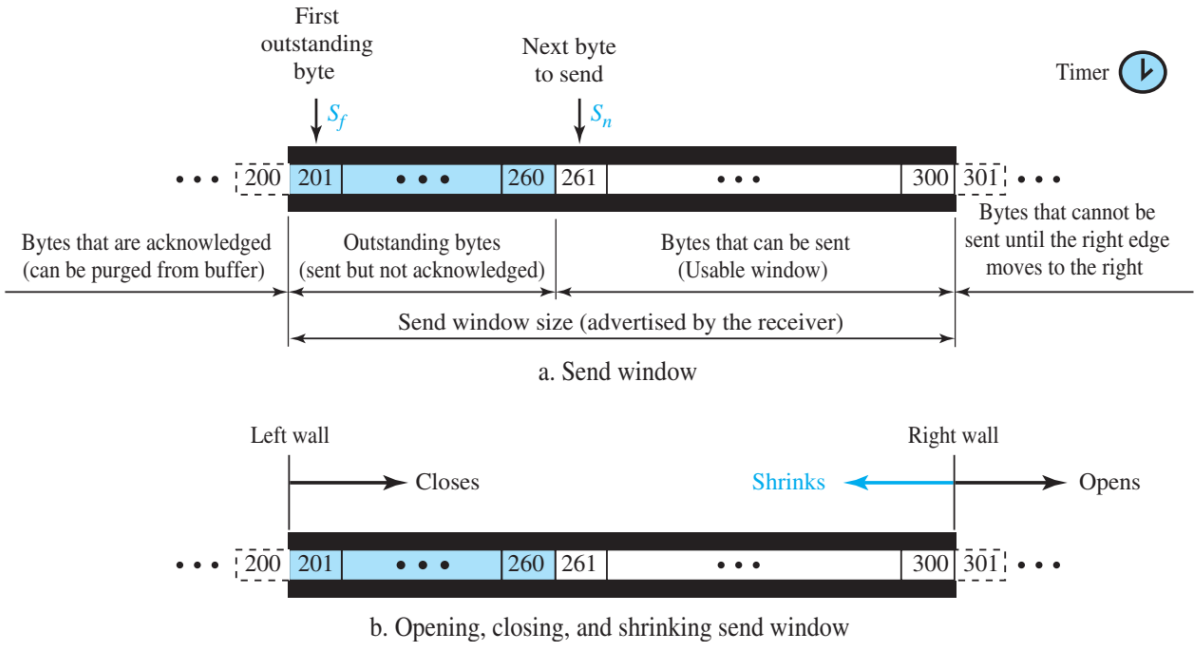


Figure 5.14. Send window in TCP

The send window in TCP is similar to the one used with the Selective-Repeat pro- tocol, but with some differences:

1. One difference is the nature of entities related to the window. The window size in SR is the number of packets, but the window size in TCP is the number of bytes. Although actual transmission in TCP occurs segment by segment, the variables that control the window are expressed in bytes.
2. The second difference is that, in some implementations, TCP can store data received from the process and send them later, but we assume that the sending TCP is capable of sending segments of data as soon as it receives them from its process.
3. Another difference is the number of timers. The theoretical Selective-Repeat protocol may use several timers for each packet sent, but as mentioned before, the TCP protocol uses only one timer.

Receive Window

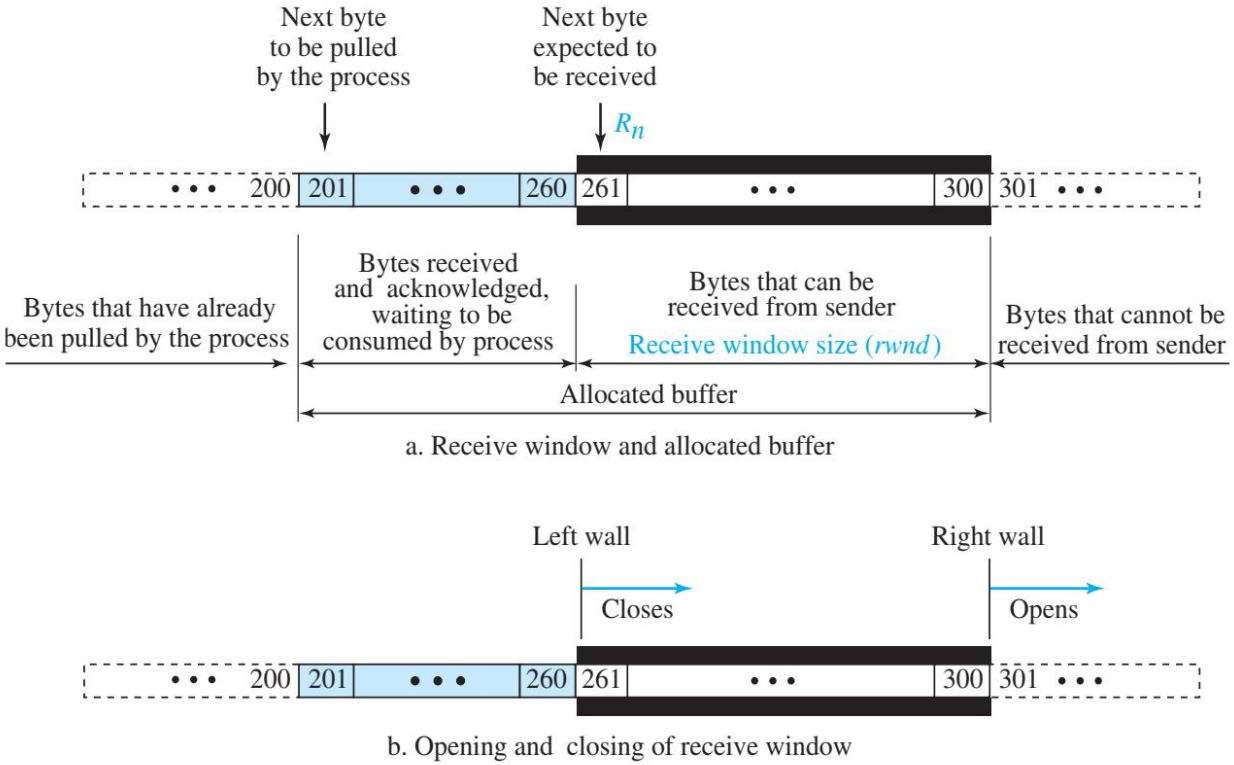
- Figure 5.15 shows an example of a receive window. The window size is 100 bytes. The figure also shows how the receive window opens and closes; in practice, the window should never shrink.

There are two differences between the receive window in TCP and the one we used for SR.

1. The first difference is that TCP allows the receiving process to pull data at its own pace. This means that part of the allocated buffer at the receiver may be occupied by bytes that have been received and acknowledged, but are waiting to be pulled by the receiving process. The receive window size is then always smaller than or equal to the buffer size, as shown in figure 5.15. The receive window size determines the number of bytes that the receive window can accept from the sender before being overwhelmed (flow control). In other words, the receive window size, normally called rwnd, can be determined as:

a. rwnd 5 buffer size 2 number of waiting bytes to be pulled

2. The second difference is the way acknowledgments are used in the TCP protocol. Remember that an acknowledgement in SR is selective, defining the uncorrupted packets that have been received. The major acknowledgment mechanism in TCP is a cumulative acknowledgment announcing the next expected byte to receive (in this way TCP looks



like GBN, discussed earlier). The new version of TCP, how- ever, uses both cumulative and selective acknowledgments; we will discuss these options on the book website.

Figure 5.15. Receive window in TCP

5.3.6 Flow Control

- The flow control balances the rate a producer creates data with the rate a consumer can use the data. TCP separates flow control from error control. In this section we discuss flow control, ignoring error control. We assume that the logical channel between the sending and receiving TCP is error-free.
- Figure 5.16 shows unidirectional data transfer between a sender and a receiver; bidirectional data transfer can be deduced from the unidirectional process.

OpeningandClosingWindows

- Toachieveflowcontrol,TCPforcesthesenderandthereceivertoadjusttheirwindow sizes, although the size of the buffer for both parties is fixed when the connection is established. The receive window closes (moves its left wall to the right) when more bytes arrive from the sender; it opens (moves its right wall to the right) when more bytes are pulled by the process.
- The opening, closing, and shrinking of the send window is controlled by the receiver. The send window closes (moves its left wall to the right) when a new acknowledgmentallowsittodoso.Thesendwindowopens(itsrightwallmovestothe right)whenthereceivewindowsize(*rwnd*)advertisedbythereceiverallowsittodoso

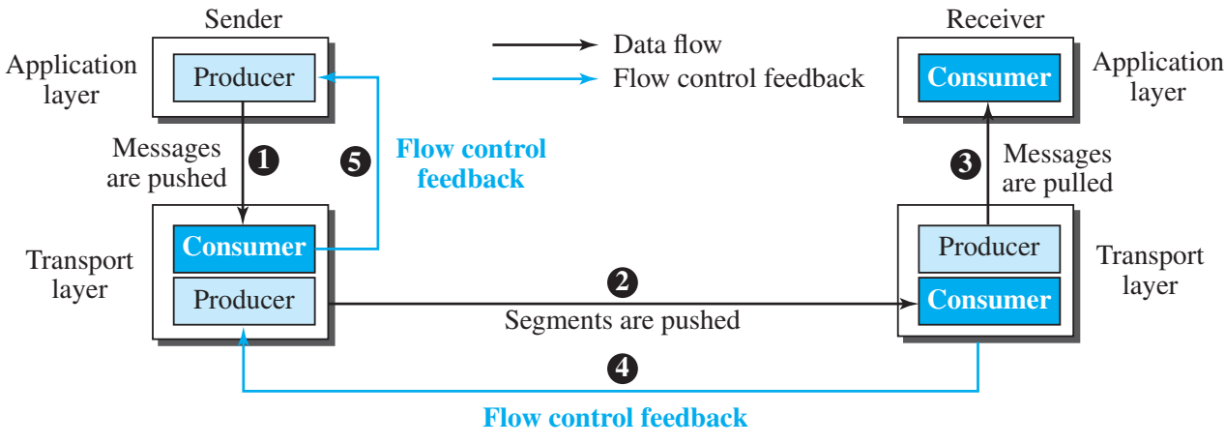


Figure 5.16. Data flow and flow control feedbacks in TCP

- Figure 5.17 shows a simpleexampleofunidirectionaldatatransfer(fromclienttoserver).Forthetimebeing, weignore error control,assumingthat no segment is corrupted, lost, duplicated,or has arrivedoutoforder.Notethatwehaveshownonlytwowindowsforunidirectionaldata transfer.Althoughtheclientdefinesserver'swindowsizeof2000inthethirdsegment, we have not shown that window because the communication is only unidirectional.

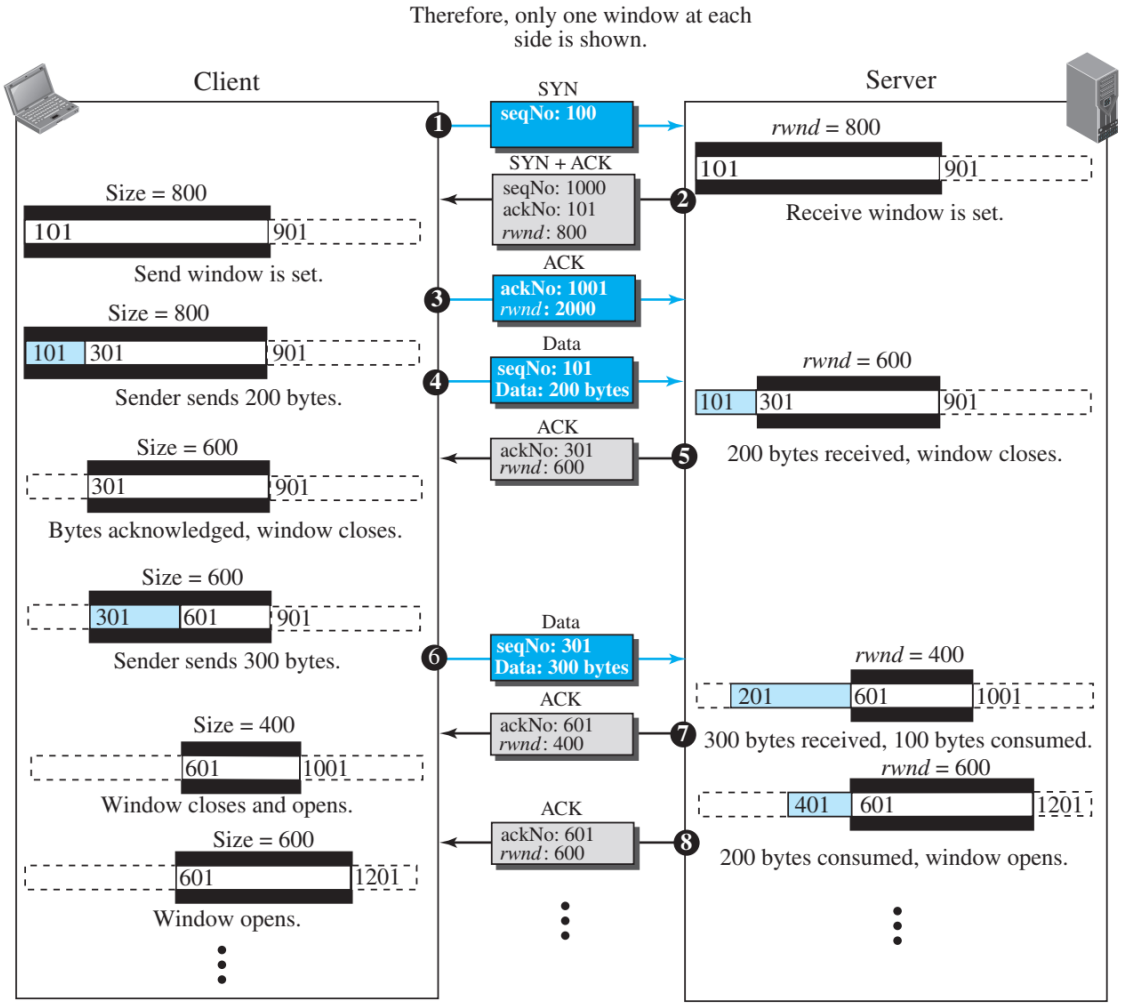


Figure 5.17. An example of flow control

Shrinking of Windows

- The receive window cannot shrink. The send window, on the other hand, can shrink if the receiver defines a value for rwnd that results in shrinking the window. However, some implementations do not allow shrinking of the send window.

Silly Window Syndrome

- A serious problem can arise in the sliding window operation when either the sending application program creates data slowly or the receiving application program consumes data slowly, or both. Any of these situations results in the sending of data in very small segments, which reduces the efficiency of the operation.

Syndrome Created by the Sender

- The sending TCP may create a silly window syndrome if it is serving an application program that creates data slowly, for example, 1 byte at a time. The application pro- gram writes 1 byte at a time into the buffer of the sending TCP. If the sending TCP does not have any specific instructions, it may create segments containing 1 byte of data. The result is a lot of 41-byte segments that are traveling through an internet.
- The solution is to prevent the sending TCP from sending the data byte by byte. The sending TCP must be forced to wait and collect data to send in a larger block. How long should the sending TCP wait? If it waits too long, it may delay the process. If it does not wait long enough, it may end up sending small segments. Nagle found an elegant solution. Nagle's algorithm is simple:

1. The sending TCP sends the first piece of data it receives from the sending application program even if it is only 1 byte.
2. After sending the first segment, the sending TCP accumulates data in the output buffer and waits until either the receiving TCP sends an acknowledgment or until enough data have accumulated to fill a maximum-size segment. At this time, the sending TCP can send the segment.
3. Step 2 is repeated for the rest of the transmission. Segment 3 is sent immediately if an acknowledgment is received for segment 2, or if enough data have accumulated to fill a maximum-size segment.

Syndrome Created by the Receiver

- The receiving TCP may create a silly window syndrome if it is serving an application program that consumes data slowly.
- Two solutions have been proposed to prevent the silly window syndrome created by an application program that consumes data more slowly than they arrive.
- Clark's solution is to send an acknowledgment as soon as the data arrive, but to announce a window size of zero until either there is enough space to accommodate a segment of maximum size or until at least half of the receive buffer is empty.
- The second solution is to delay sending the acknowledgment. This means that when a segment arrives, it is not acknowledged immediately. The receiver waits until there is a decent amount of space in its incoming buffer before acknowledging the arrived segments.

5.3.7 Error Control

- TCP is a reliable transport-layer protocol. This means that an application program that delivers a stream of data to TCP relies on TCP to deliver the entire stream to the application program on the other end in order, without error, and without any part lost or duplicated.
- TCP provides reliability using error control. Error control includes mechanisms for detecting and resending corrupted segments, resending lost segments, storing out-of-order segments until missing segments arrive, and detecting and discarding duplicated segments. Error control in TCP is achieved through the use of three simple tools: checksum, acknowledgment, and time-out.

Checksum

- Each segment includes a checksum field, which is used to check for a corrupted segment. If a segment is corrupted, as detected by an invalid checksum, the segment is discarded by the destination TCP and is considered as lost. TCP uses a 16-bit checksum that is mandatory in every segment.

Acknowledgment

- TCP uses acknowledgments to confirm the receipt of data segments. Control segments that carry no data, but consume a sequence number, are also acknowledged. ACK segments are never acknowledged.
- In the past, TCP used only one type of acknowledgment: cumulative acknowledgment. Today, some TCP implementations also use selective acknowledgment.
- Cumulative Acknowledgment (ACK) TCP was originally designed to acknowledge receipt of segments cumulatively. The receiver advertises the next byte it expects to receive, ignoring all segments received and stored out of order. This is sometimes referred to as positive cumulative acknowledgment, or ACK. The word positive indicates that no feedback is provided for discarded, lost, or duplicate segments. The 32-bit ACK field in the TCP header is used for cumulative acknowledgments, and its value is valid only when the ACK flag bit is set to 1.
- Selective Acknowledgment (SACK) More and more implementations are adding another type of acknowledgment called selective acknowledgment, or SACK. A SACK does not replace an ACK, but reports additional information to the sender. A SACK reports a block of bytes that is out of order, and also a block of bytes that is duplicated, i.e., received more than once. However, since there is no provision in the TCP header for adding this type of information, SACK is implemented as an option at the end of the TCP header.

Generating Acknowledgments

- When does a receiver generate acknowledgments? During the evolution of TCP, several rules have been defined and used by several implementations. The order of a rule does not necessarily define its importance.

1. When end A sends a data segment to end B, it must include (piggyback) an acknowledgment that gives the next sequence number it expects to receive. This rule decreases the number of segments needed and therefore reduces traffic.

2. When the receiver has no data to send and it receives an in-order segment (with expected sequence number) and the previous segment has already been acknowl- edged, the receiver delays sending an ACK segment until another segment arrives or until a period of time (normally 500 ms) has passed. In other words, the receiver needs to delay sending an ACK segment if there is only one out- standing in-order segment. This rule reduces ACK segments.
3. When a segment arrives with a sequence number that is expected by the receiver, and the previous in-order segment has not been acknowledged, the receiver immediately sends an ACK segment. In other words, there should not be more than two in-order unacknowledged segments at any time. This prevents the unnecessary retransmission of segments that may create congestion in the network.
4. When a segment arrives with an out-of-order sequence number that is higher than expected, the receiver immediately sends an ACK segment announcing the sequence number of the next expected segment. This leads to the fast re-transmission of missing segments (discussed later).
5. When a missing segment arrives, the receiver sends an ACK segment to announce the next sequence number expected. This informs the receiver that segments reported missing have been received.
6. If a duplicate segment arrives, the receiver discards the segment, but immediately sends an acknowledgment indicating the next in-order segment expected. This solves some problems when an ACK segment itself is lost.

Retransmission

- The heart of the error control mechanism is the retransmission of segments. When a segment is sent, it is stored in a queue until it is acknowledged. When the retransmission timer expires or when the sender receives three duplicate ACKs for the first segment in the queue, that segment is retransmitted.

Out-of-Order Segments

- TCP implementations today do not discard out-of-order segments. They store them temporarily and flag them as out-of-order segments until the missing segments arrive.

FSMs for Data Transfer in TCP

- Data transfer in TCP is close to the Selective-Repeat protocol with a slight similarity to GBN. Since TCP accepts out-of-order segments, TCP can be thought of as behaving more like the SR protocol, but since the original acknowledgments are cumulative, it looks like GBN. However, if the TCP implementation uses SACKs, then TCP is closest to SR.

Sender-Side FSM

- The simplified FSM for the sender side of the TCP protocol is similar to the SR protocol, but with some changes specific to TCP. The communication is unidirectional and the segments are acknowledged using ACK segments. The selective acknowledgments are ignored and congestion control for the moment. Figure 5.18 shows the simplified FSM for the sender site.

Receiver-Side FSM

- The simplified FSM for the receiver-side TCP protocol is similar to the the SR protocol, but with some changes specific to TCP. The communication is unidirectional and the segments are acknowledged using ACK segments. We also ignore the selective

acknowledgment and congestion control for the moment. Figure 5.19 shows the simplified FSM for the receiver. The silly window syndrome (Clark's solution) and window shutdown are ignored.

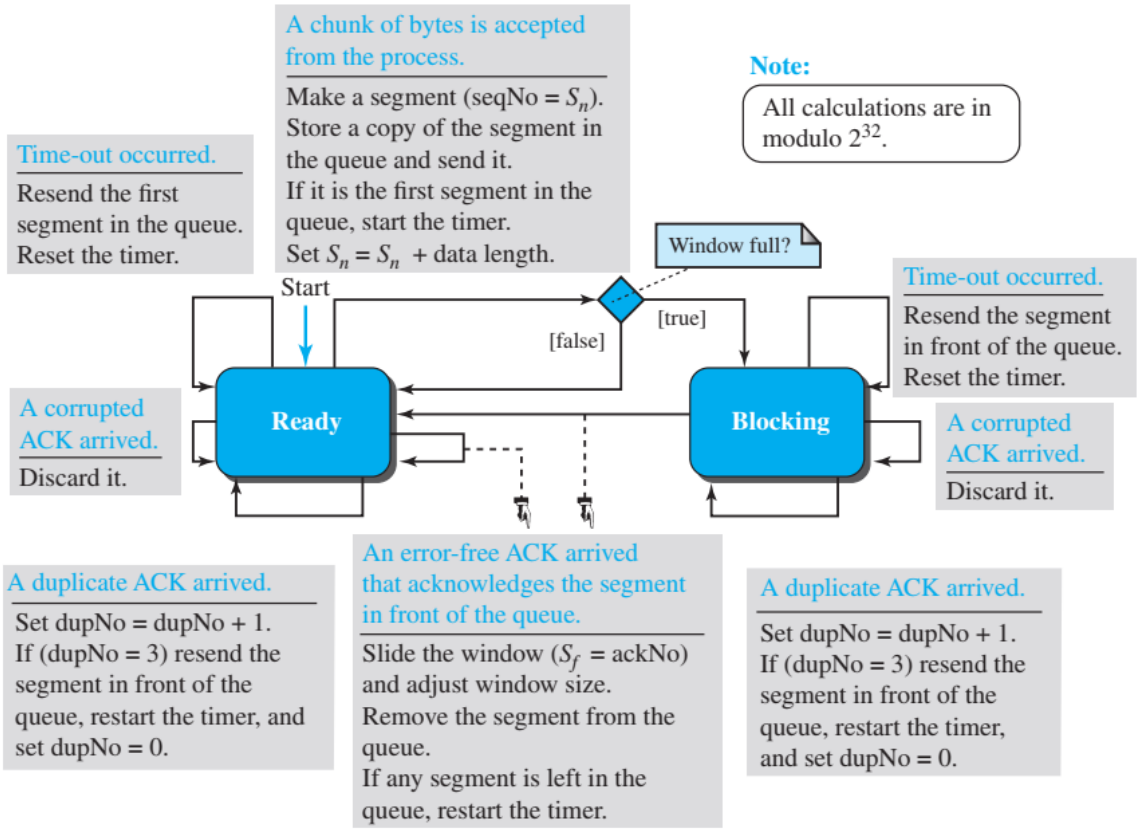


Figure 5.18. Simplified FSM for the TCP sender side

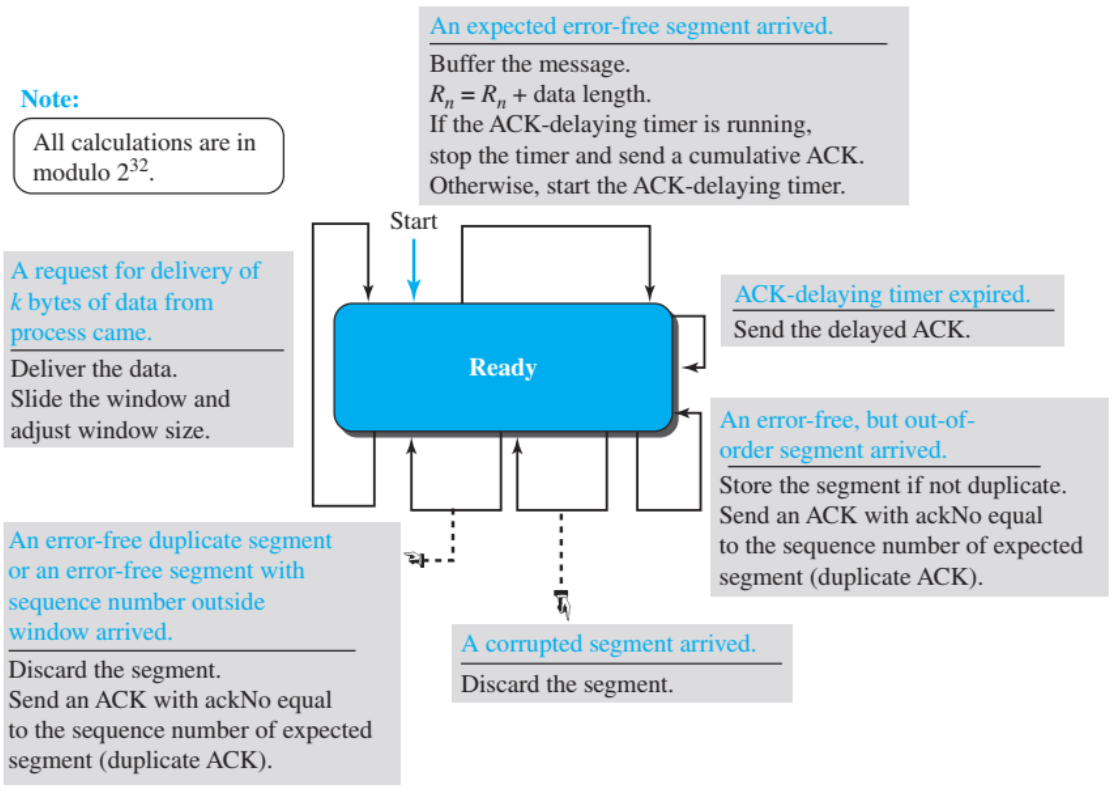


Figure 5.19. Simplified FSM for the TCP receiver side

5.3.8TCP Congestion Control

- TCP uses different policies to handle the congestion in the network.

Congestion Window

- In TCP flow control, the size of the send window is controlled by the receiver using the value of rwnd, which is advertised in each segment traveling in the opposite direction. The use of this strategy guarantees that the receive window is never overflowed with the received bytes (no end congestion). This, however, does not mean that the intermediate buffers, buffers in the routers, do not become congested. A router may receive data from more than one sender. No matter how large the buffers of a router may be, it may be overwhelmed with data, which results in dropping some segments sent by a specific TCP sender.
- TCP is an end-to-end protocol that uses the service of IP. The congestion in the router is in the IP territory and should be taken care of by IP. IP is a simple protocol with no congestion control. TCP, itself, needs to be responsible for this problem.
- TCP cannot ignore the congestion in the network; it cannot aggressively send segments to the network. The result of such aggressiveness would hurt the TCP itself, as we mentioned before. TCP cannot be very conservative, either, sending a small number of segments in each time interval, because this means not utilizing the available band- width of the network. TCP needs to define policies that accelerate the data transmission when there is no congestion and decelerate the transmission when congestion is detected.
- To control the number of segments to transmit, TCP uses another variable called a congestion window, cwnd, whose size is controlled by the congestion situation in the network (as we will explain shortly). The cwnd variable and the rwnd variable together define the size of the send window in TCP. The first is related to the congestion in the middle (network); the second is related to the congestion at the end. The actual size of the window is the minimum of these two, which is represented as follows:

Actual window size 5 minimum (rwnd, cwnd)

Congestion Detection

- TCP sender can detect the possible existence of congestion in the network and then the value of cwnd should be set and changed . The TCP sender uses the occurrence of two events as signs of congestion in the network: time-out and receiving three duplicate ACKs.
- The first is the time-out. If a TCP sender does not receive an ACK for a segment or a group of segments before the time-out occurs, it assumes that the corresponding segment or segments are lost and the loss is due to congestion.
- Another event is the receiving of three duplicate ACKs. However, the congestion in the case of three duplicate ACKs can be less severe than in the case of time-out. When a receiver sends three duplicate ACKs, it means that one segment is missing, but three segments have been received. The network is either slightly congested or has recovered from the congestion.
- The interesting point in TCP congestion is that the TCP sender uses only one feedback

from the other end to detect congestion: ACKs. The lack of regular, timely receipt of ACKs, which results in a time-out, is the sign of a strong congestion; the receiving of three duplicate ACKs is the sign of a weak congestion in the network.

Congestion Policies

TCP's general policy for handling congestion is based on three algorithms: slow start, congestion avoidance, and fast recovery.

1) Slow Start-Exponential Increase:

- The slow-start algorithm is based on the idea that the size of the congestion window (cwnd) starts with one maximum segment size (MSS), but it increases one MSS each time an acknowledgment arrives. The MSS is a value negotiated during the connection establishment, using an option of the same name.
- The name of this algorithm is misleading; the algorithm starts slowly, but grows exponentially, as shown in Figure 5.20. rwnd is much larger than cwnd, so that the sender window size always equals cwnd. Each segment is of the same size and carries MSS bytes. The delayed-ACK policy is ignored and each segment is acknowledged individually.

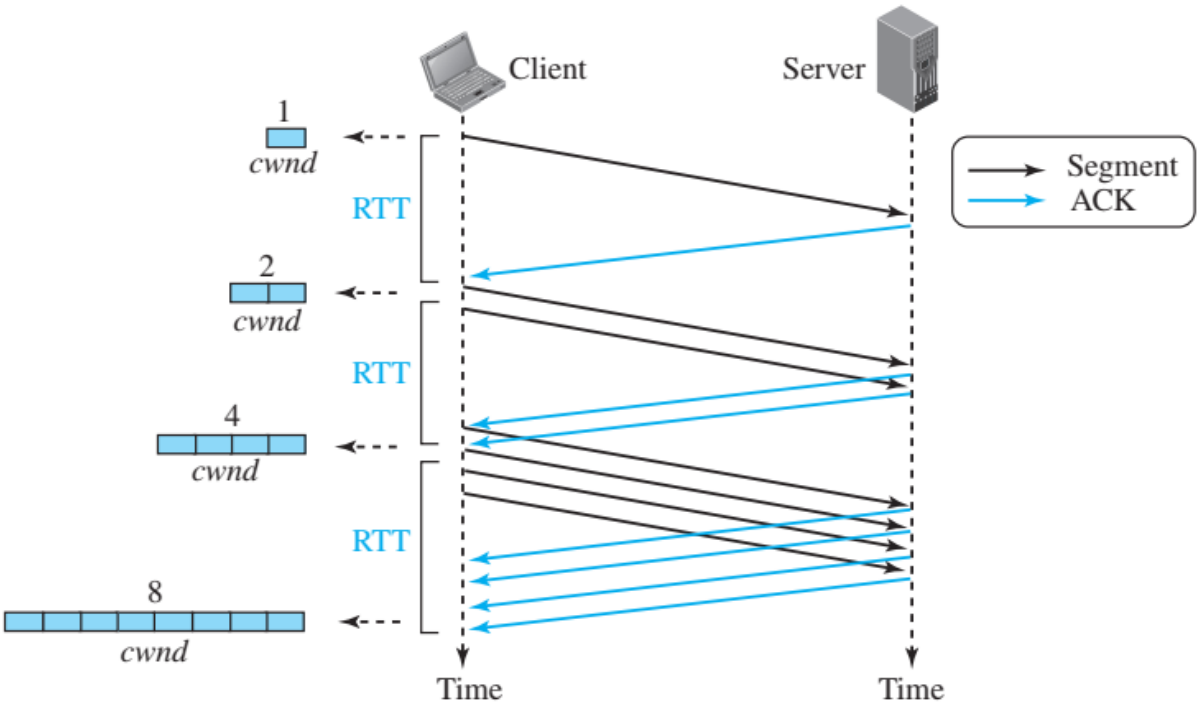


Figure 5.20. Slow start, exponential increase

- The sender starts with cwnd=1. This means that the sender can send only one segment. After the first ACK arrives, the acknowledged segment is purged from the window, which means there is now one empty segment slot in the window. The size of the congestion window is also increased by 1 because the arrival of the acknowledgment is a good sign that there is no congestion in the network. The size of the window is now 2. After sending two segments and receiving two individual acknowledgments for them, the size of the congestion window now becomes 4, and so on.

If an ACK arrives, cwnd =cwnd + 1.

- If we look at the size of the cwnd in terms of round-trip times (RTTs), we find that the growth rate is exponential in terms of each round trip time, which is a very aggressive approach:

Start	→	$cwnd = 1 \rightarrow 2^0$
After 1 RTT	→	$cwnd = cwnd + 1 = 1 + 1 = 2 \rightarrow 2^1$
After 2 RTT	→	$cwnd = cwnd + 2 = 2 + 2 = 4 \rightarrow 2^2$
After 3 RTT	→	$cwnd = cwnd + 4 = 4 + 4 = 8 \rightarrow 2^3$

- A slow start cannot continue indefinitely. There must be a threshold to stop this phase. The sender keeps track of a variable named ssthresh (slow-start threshold). When the size of the window in bytes reaches this threshold, slow start stops and the next phase starts.
- The slow-start strategy is slower in the case of delayed acknowledgments. Remember, for each ACK, the cwnd is increased by only 1. Hence, if two segments are acknowledged cumulatively, the size of the cwnd increases by only 1, not 2. The growth is still exponential, but it is not a power of 2. With one ACK for every two segments, it is a power of 1.5.

2) Congestion Avoidance: Additive Increase

- If the slow-start algorithm is continued, the size of the congestion window increases exponentially. To avoid congestion before it happens, we must slow down this exponential growth. TCP defines another algorithm called congestion avoidance, which increases the cwnd additively instead of exponentially. When the size of the congestion window reaches the slow-start threshold in the case where cwnd=i, the slow-start phase stops and the additive phase begins. In this algorithm, each time the whole "window" of segments is acknowledged, the size of the congestion window is increased by one. A window is the number of segments transmitted during RTT, as shown in figure5.21

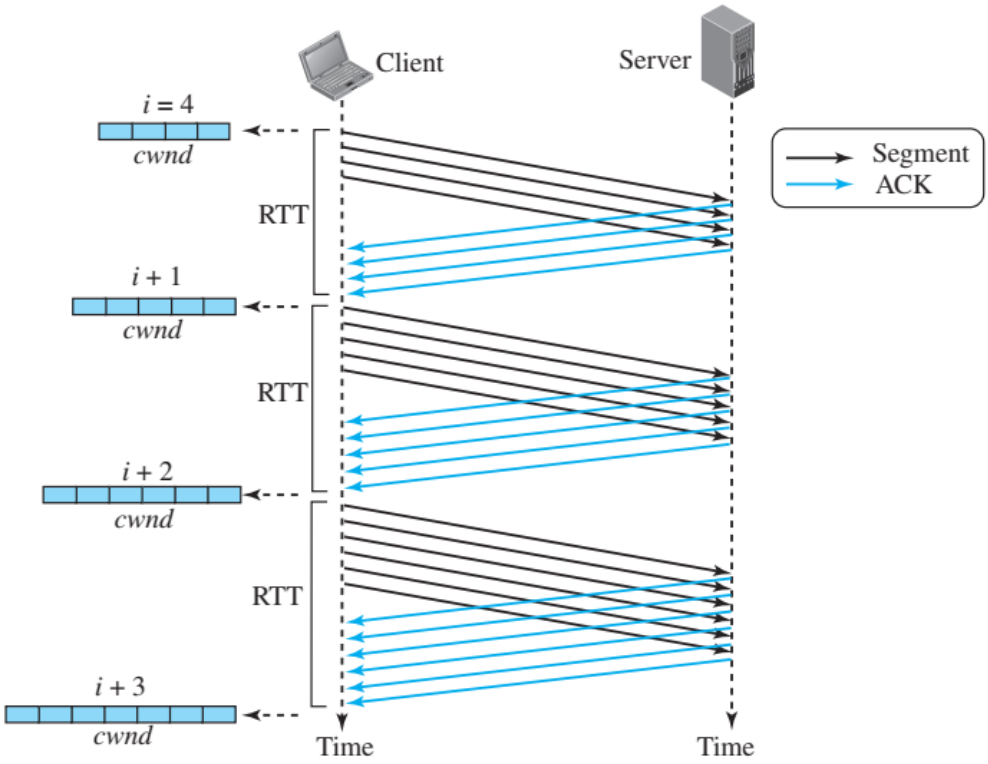


Figure 5.21. Congestion avoidance, additive increase

- The sender starts with cwnd=4. This means that the sender can send only four segments. After four ACKs arrive, the acknowledged segments are purged from the window, which means there is now one extra empty segment slot in the window. The size of the congestion window is also increased by 1. The size of window is now 5. After sending five segments and receiving five acknowledgments for them, the size of the congestion window now becomes 6, and so on. In other words, the size of the congestion window in this algorithm is also a function of the number of ACKs that have arrived and can be determined as follows:

If an ACK arrives, cwnd= cwnd+ (1/cwnd).

- The size of the window increases only 1/cwnd portion of MSS (in bytes).
- If we look at the size of the cwnd in terms of round-trip times (RTTs), we find that the growth rate is linear in terms of each round-trip time, which is much more conservative than the slow-start approach.

Fast Recovery

- The fast-recoveryalgorithm is optional in TCP, which is used in new versions. It starts when three duplicate ACKs arrive, which is interpreted as light congestion in the network. Like congestion avoidance, this algorithm is also an additive increase, but it increases the size of the congestion window when a duplicate ACK arrives and is represented as follows:

If a duplicate ACK arrives, cwnd= cwnd+ (1/cwnd).

5.4 Application Layer

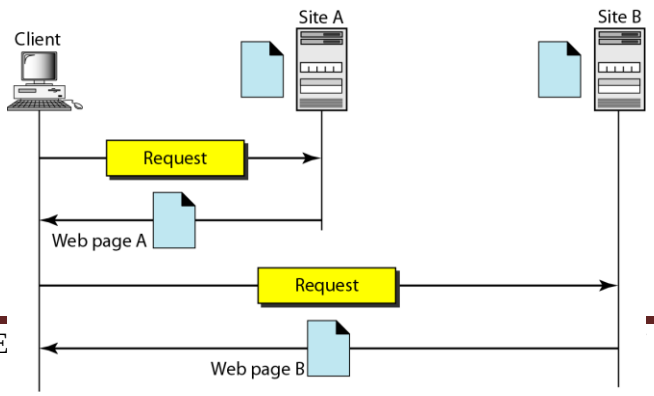
- The application layer provides services to the user.
- Communication is provided using a logical connection, which means that the two application layers assume that there is an imaginary direct connection through which they can send and receive messages.

5.4.1 World Wide Web

- The Web today is a repository of information in which the documents, called web pages,aredistributedallovertheworldandrelateddocumentsarelinkedtogether.

Architecture:

- The WWW today is a distributed client/server service, in which a client using a browser can access a service using a server. However, the service provided is distributed over many locations called sites, as shown in figure 5.22.



- Each site holds one or more documents, referred to as Web pages.
- Each Web page can contain a link to other pages in the same site or at other sites. The pages can be retrieved and viewed by using browsers.
- The client needs to see some information that it knows belongs to site A. It sends a request through its browser, a program that is designed to fetch Web documents. The request, among other information, includes the address of the site and the Web page, called the URL.
- The server at site A finds the document and sends it to the client. When the user views the document, she finds some references to other documents, including a Web page at site B. The reference has the URL for the new site. The user is also interested in seeing this document.
- The client sends another request to the new site, and the new page is retrieved.

Client (Browser)

- A variety of vendors offer commercial browsers that interpret and display a Web document, and all use nearly the same architecture, as shown in figure 5.23. Each browser usually consists of three parts: a controller, client protocol, and interpreters.
- The controller receives input from the keyboard or the mouse and uses the client programs to access the document. After the document has been accessed, the controller uses one of the interpreters to display the document on the screen.
- The client protocol can be one of the protocols described previously such as FTP or HTTP. The interpreter can be HTML, Java, or JavaScript, depending on the type of document.

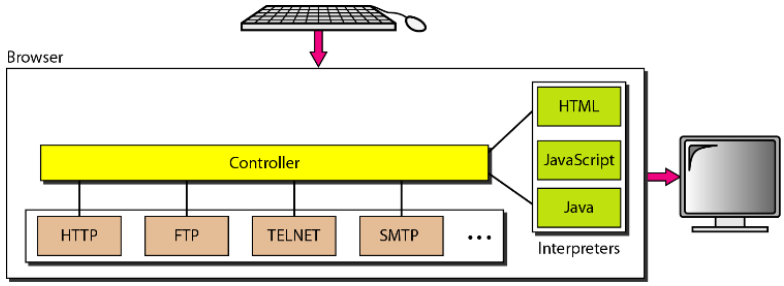


Figure 5.23.Browser

Server

- The Web page is stored at the server. Each time a client request arrives, the corresponding document is sent to the client. To improve efficiency, servers normally store requested files in a cache in memory; memory is faster to access than disk.
- A server can also become more efficient through multithreading or multiprocessing. In this case, a server can answer more than one request at a time.

Uniform Resource Locator (URL)

- A client that wants to access a Web page needs the address. To facilitate the access of documents distributed throughout the world, HTTP uses locators.

- The uniform resource locator (URL) is a standard for specifying any kind of information on the Internet. The URL defines four things: protocol, host computer, port, and path, as shown in figure 5.24.

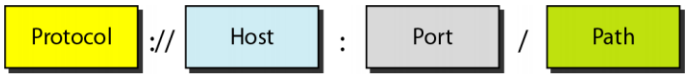


Figure 5.24.uniform resource locator (URL)

- The protocol is the client/server program used to retrieve the document. Many different protocols can retrieve a document; among them are FTP or HTTP. The most common today is HTTP.
- The host is the computer on which the information is located, although the name of the computer can be an alias. Web pages are usually stored in computers, and computers are given alias names that usually begin with the characters "www".
- The URL can optionally contain the port number of the server. If the port is included, it is inserted between the host and the path, and it is separated from the host by a colon.
- Path is the pathname of the file where the information is located.

Web Documents

- The documents in the WWW can be grouped into three broad categories: static, dynamic, and active**.**

Static Documents

- Static documents are fixed-content documents that are created and stored in a server. The client can get a copy of the document only.
- When a client accesses the document, a copy of the document is sent. The user can then use a browser to see the document.
- Static documents are prepared using one of several languages: HyperText Markup Language (HTML), Extensible Markup Language (XML), Extensible Style Language (XSL), and Extensible Hypertext Markup Language (XHTML). We discuss these languages in Appendix C.

Dynamic Documents

- A dynamic document is created by a web server whenever a browser requests the document. When a request arrives, the web server runs an application program or a script that creates the dynamic document.
- The server returns the result of the program or script as a response to the browser that requested the document. Because a fresh document is created for each request, the contents of a dynamic document may vary from one request to another.
- Although the Common Gateway Interface (CGI) was used to retrieve a dynamic document in the past, today's options include one of the scripting languages such as Java Server Pages (JSP), which uses the Java language for scripting, or Active Server Pages (ASP), a Microsoft product that uses Visual Basic language for scripting, or ColdFusion, which embeds queries in a Structured Query Language (SQL) database in the HTML document.

Active Documents

- For many applications, we need a program or a script to be run at the client site. These are called active documents.
- When a browser requests an active document, the server sends a copy of the document or a script. The document is then run at the client (browser) site.
- One way to create an active document is to use Java applets, a program written in Java on the server. It is compiled and ready to be run. The document is in bytecode (binary)

format.

- Another way is to use JavaScripts but download and run the script at the client site.

5.4.2HTTP

- HTTP functions as a request–response protocol in the client–server computing model. A web browser, for example, may be the client and an application running on a computer hosting a website may be the server.
- The client submits an HTTP request message to the server. The server, which provides resources such as HTML files and other content, or performs other functions on behalf of the client, returns a response message to the client. The response contains completion status information about the request and may also contain requested content in its message body.
- HTTP is an application layer protocol designed within the framework of the Internet protocol suite. Its definition presumes an underlying and reliable transport layer protocol,[7] and Transmission Control Protocol (TCP) is commonly used.

HTTP Transaction

- The figure 5.25 illustrates the HTTP transaction between the client and server. AlthoughHTTP uses the services of TCP, HTTP itself is a stateless protocol. The client initializesthe transaction by sending a request message. The server replies by sending a response.

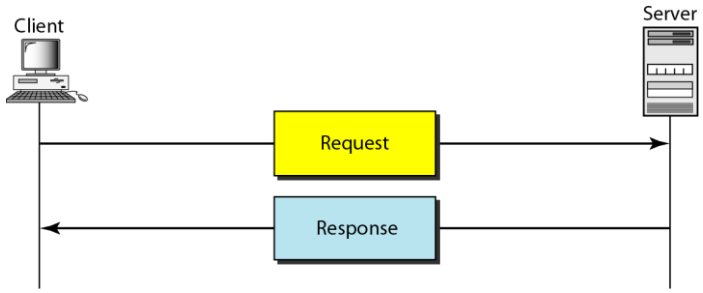


Figure 5.25. HTTP transaction between the client and server.

Messages

- The formats of the request and response messages are similar; both are shown in figure 5.26. A request message consists of a request line, a header, and sometimes a body. A response message consists of a status line, a header, and sometimes a body.

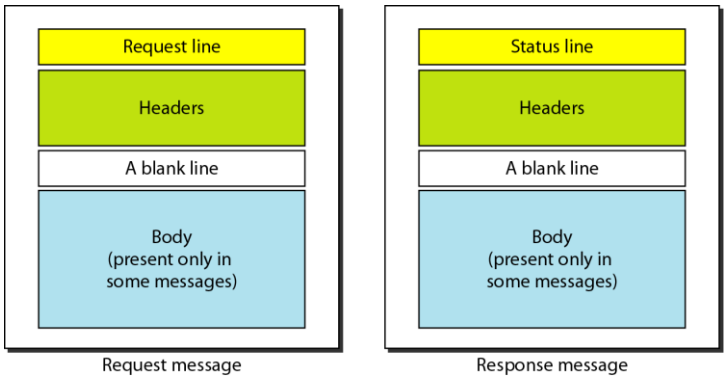


Figure 5.26. Formats of the request and response messages

Request and Status Lines

- The first line in a request message is called a request line; the first line in the response message is called the status line. There is one common field, as shown in figure 5.27.

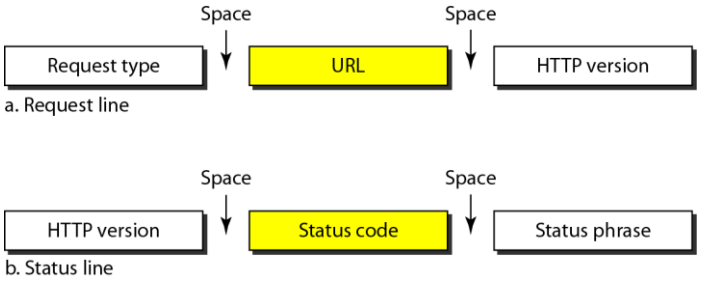


Figure 5.27.Request and Status Lines

Request Type

- This field is used in the request message. In version 1.1 of HTTP, several request types are defined. The request type is categorized into methods as defined in figure 5.28.

Method	*Action*
GET	Requests a document from the server
HEAD	Requests information about a document but not the document itself
POST	Sends some information from the client to the server
PUT	Sends a document from the server to the client
TRACE	Echoes the incoming request
CONNECT	Reserved
OPTION	Inquires about available options

Figure 5.28. Request type categorized into methods

Status code

- Status codefield is used in the response message, which consists of three digits. Whereas the codes in the 100 range are only informational, the codes in the 200 range indicate a successful request, as shown in figure 5.29.

Code	*Phrase*	*Description*
		Informational
100	Continue	The initial part of the request has been received, and the client may continue with its request.
101	Switching	The server is complying with a client request to switch protocols defined in the upgrade header.
		Success
200	OK	The request is successful.
201	Created	A new URL is created.
202	Accepted	The request is accepted, but it is not immediately acted upon.
204	No content	There is no content in the body.

Figure 5.29. Status Codes

Header

- The header exchanges additional information between the client and the server.
- The header can consist of one or more header lines. Each header line has a header name, a

colon, a space, and a header value, as shown in figure 5.30.

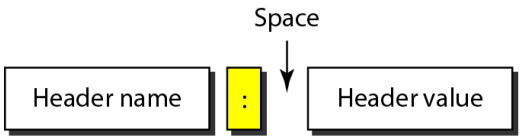


Figure 5.30. Header

- A header line belongs to one of four categories: general header, requestheader, response header, and entity header.
- A request message can contain only general, request, and entity headers. A response message, on the other hand, can contain onlygeneral, response, and entity headers.
- The general header gives general information about the messageand can be present in both a request and a response, as shown in figure 5.31.

Header	*Description*
Cache-control	Specifies information about caching
Connection	Shows whether the connection should be closed or not
Date	Shows the current date
MIME-version	Shows the MIME version used
Upgrade	Specifies the preferred communication protocol

Figure 5.31. General Headers

- The request header can be present only in a request message. Itspecifies the client's configuration and the client's preferred document format, as shown in figure 5.32.

Header	*Description*
Accept	Shows the medium format the client can accept
Accept-charset	Shows the character set the client can handle
Accept-encoding	Shows the encoding scheme the client can handle
Accept-language	Shows the language the client can accept
Authorization	Shows what permissions the client has
From	Shows the e-mail address of the user
Host	Shows the host and port number of the server
If-modified-since	Sends the document if newer than specified date
If-match	Sends the document only if it matches given tag
If-non-match	Sends the document only if it does not match given tag
If-range	Sends only the portion of the document that is missing
If-unmodified-since	Sends the document if not changed since specified date
Referrer	Specifies the URL of the linked document
User-agent	Identifies the client program

Figure 5.32. Request Headers

- The response header can be present only in a response message.It specifies the server's

configuration and special information about the request, as shown in figure 5.33.

Header	*Description*
Accept-range	Shows if server accepts the range requested by client
Age	Shows the age of the document
Public	Shows the supported list of methods
Retry-after	Specifies the date after which the server is available
Server	Shows the server name and version number

Figure 5.33. Response Headers

- The entity header gives information about the body of the document. Although it is mostly present in response messages, some request messages,such as POST or PUT methods, that contain a body also use this type of header, as shown in figure 5.34.

Header	*Description*
Allow	Lists valid methods that can be used with a URL
Content-encoding	Specifies the encoding scheme
Content-language	Specifies the language
Content-length	Shows the length of the document
Content-range	Specifies the range of the document
Content-type	Specifies the medium type
Etag	Gives an entity tag
Expires	Gives the date and time when contents may change
Last-modified	Gives the date and time of the last change
Location	Specifies the location of the created or moved document

Figure 5.34. Entity Headers

- The body can be present in a request or response message. Usually, it contains the document to be sent or received.

5.4.3 Electronic Mail

- Electronic mail (or e-mail) allows users to exchange messages. The nature of this application, however, is different from other applications discussed so far.
- In an application such as HTTP or FTP, the server program is running all the time, waiting for a request from a client.
- When the request arrives, the server provides the service. There is a request and there is a response.
- In the case of electronic mail, the situation is different. This means that the idea of client/server programming should be implemented in another way: using some intermediate computers (servers). The users run only client programs when they want and the intermediate servers apply the client/server paradigm, as we discuss in the next section.

Architecture

- The common architecture of e-mail is shown in figure 5.35. In the common scenario, the

sender and the receiver of the e-mail, Alice and Bob respectively, are connected via a LAN or a WAN to two mail servers. The administrator has created one mailbox for each user where the received messages are stored. A mail- box is part of a server hard drive, a special file with permission restrictions. Only the owner of the mailbox has access to it. The administrator has also created a queue (spool) to store messages waiting to be sent.

- A simple e-mail from Alice to Bob takes nine different steps, as shown in the figure. Alice and Bob use three different agents: a user agent (UA), a message transfer agent (MTA), and a message access agent (MAA). The electronic mail system needs two UAs, two pairs of MTAs (client and server), and a pair of MAAs (client and server).

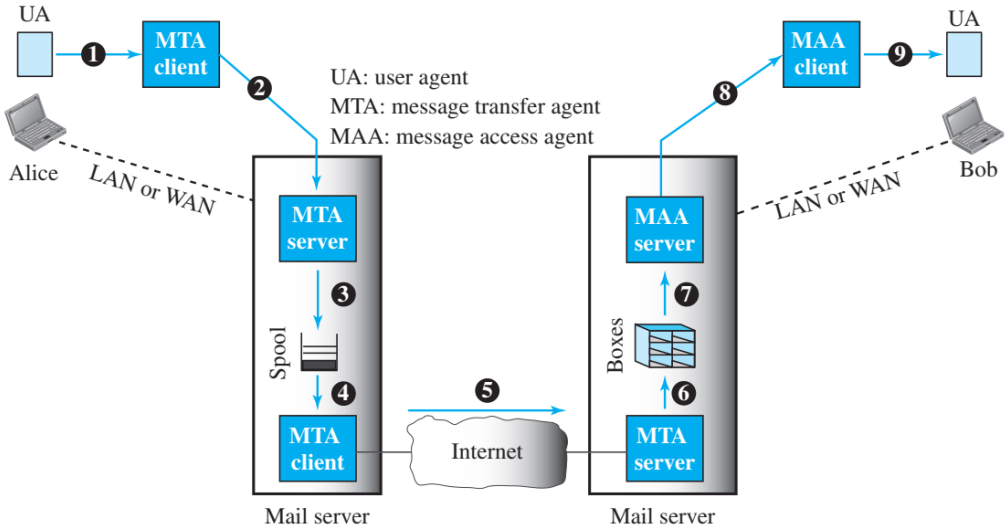


Figure 5.35. Common Architecture of E-mail

User Agent

- The first component of an electronic mail system is the user agent (UA). It provides service to the user to make the process of sending and receiving a message easier. A user agent is a software package (program) that composes, reads, replies to, and for- wards messages. It also handles local mailboxes on the user computers.
- There are two types of user agents: command-driven and GUI-based. Command- driven user agents belong to the early days of electronic mail. They are still present as the underlying user agents. A command-driven user agent normally accepts a one- character command from the keyboard to perform its task.
- Modern user agents are GUI-based. They contain graphical user interface (GUI) components that allow the user to interact with the software by using both the keyboard and the mouse. They have graphical components such as icons, menu bars, and win- dows that make the services easy to access

Sending Mail

- To send mail, the user, through the UA, creates mail that looks very similar to postal mail. It has an envelope and a message, as shown in figure 5.36. The envelope usually contains the sender address, the receiver address, and other information. The message contains the header and the body. The header of the message defines the sender, the receiver, the subject of the message, and some other information. The body of the mes- sage contains the actual information to be read by the recipient.

Receiving Mail

- The user agent is triggered by the user (or a timer). If a user has mail, the UA informs the user with a notice. If the user is ready to read the mail, a list is displayed in which each line contains a summary of the information about a particular message in the mail- box.

The summary usually includes the sender mail address, the subject, and the time the mail was sent or received. The user can select any of the messages and display its contents on the screen.

Addresses

- To deliver mail, a mail handling system must use an addressing system with unique addresses. In the Internet, the address consists of two parts: a local part and a domain name, separated by an @ sign, as shown in figure 5.37.

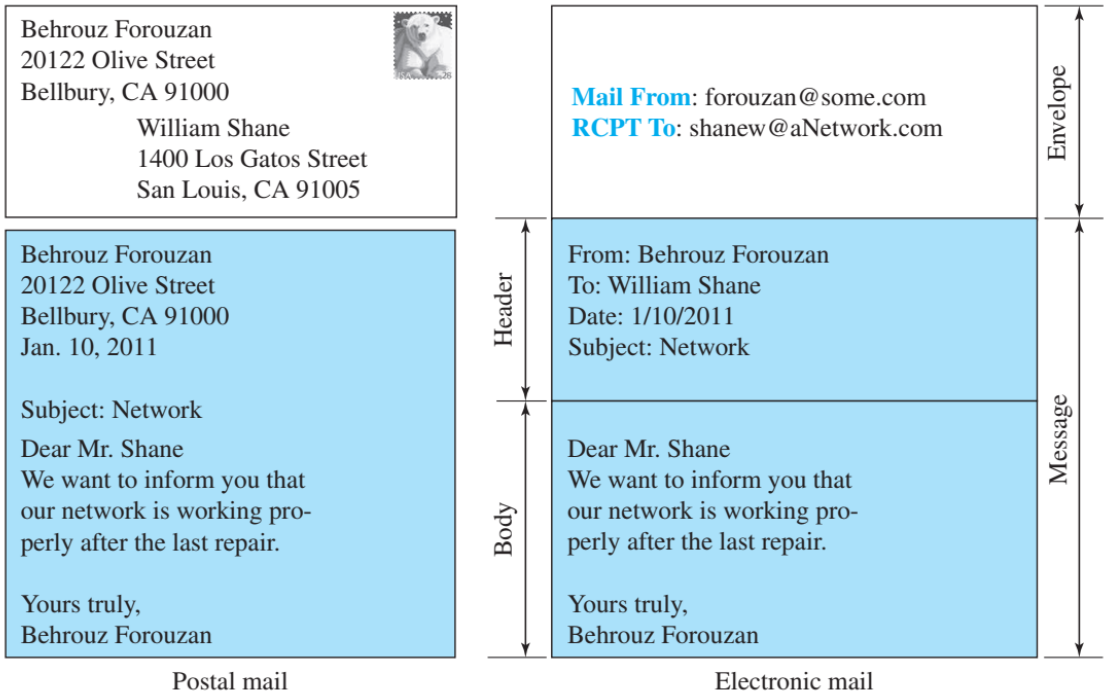


Figure 5.36. Format of an e-mail



Figure 5.37. E-mail address

Message Transfer Agent: SMTP

- The e-mail is one of those applications that needs three uses of client-server paradigms to accomplish its task. It is important that we distinguish these three when we are dealing with e-mail. Figure 5.38 shows these three client-server applications. We refer to the first and the second as Message Transfer Agents (MTAs), the third as Message Access Agent (MAA).
- The formal protocol that defines the MTA client and server in the Internet is called Simple Mail Transfer Protocol (SMTP). SMTP is used two times, between the sender and the sender's mail server and between the two mail servers. As we will see shortly, another protocol is needed between the mail server and the receiver. SMTP simply defines how commands and responses must be sent back and forth.Message Access Agent: POP and IMAP
- The first and second stages of mail delivery use SMTP. However, SMTP is not involved in the third stage because SMTP is a push protocol; it pushes the message from the client to the server. In other words, the direction of the bulk data (messages) is from the client to the server. On the other hand, the third stage needs a pull protocol; the client must pull messages from the server. The direction of the bulk data is from the server to

the client. The third stage uses a message access agent.

- Currently two message access protocols are available: Post Office Protocol, version 3 (POP3) and Internet Mail Access Protocol, version 4 (IMAP4). Figure 26.15 shows the position of these two protocols.

POP3

- Post Office Protocol, version 3 (POP3) is simple but limited in functionality. The client POP3 software is installed on the recipient computer; the server POP3 software is installed on the mail server.
- Mail access starts with the client when the user needs to download its e-mail from the mailbox on the mail server. The client opens a connection to the server on TCP port 110.
- It then sends its user name and password to access the mailbox. The user can then list and retrieve the mail messages, one by one. Figure 5.39 shows an example of downloading using POP3.

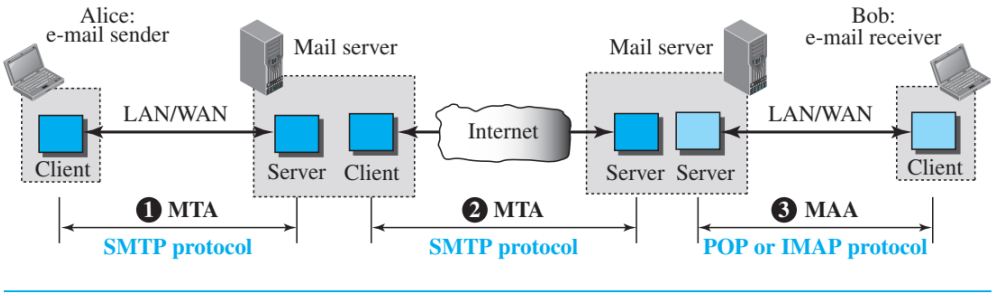


Figure 5.38. Protocols used in electronic mail

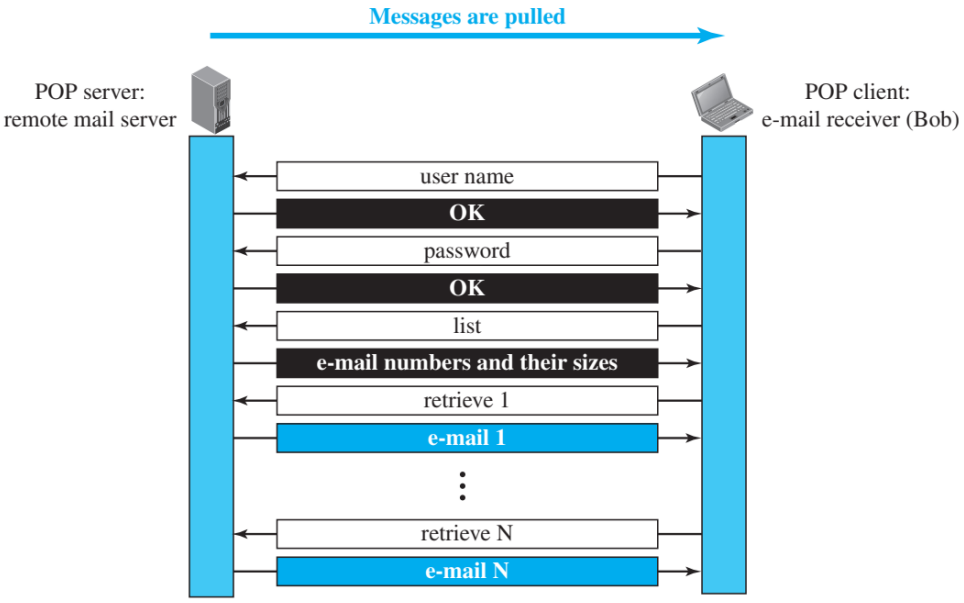


Figure 5.39. POP3

IMAP4

- Another mail access protocol is Internet Mail Access Protocol, version 4 (IMAP4). IMAP4 is similar to POP3, but it has more features; IMAP4 is more powerful and more complex.
- POP3 is deficient in several ways. It does not allow the user to organize her mail on the

server; the user cannot have different folders on the server. In addition, POP3 does not allow the user to partially check the contents of the mail before downloading. IMAP4 provides the following extra functions:

- A user can check the e-mail header prior to downloading.
- A user can search the contents of the e-mail for a specific string of characters prior to downloading.
- A user can partially download e-mail. This is especially useful if bandwidth is limited and the e-mail contains multimedia with high bandwidth requirements.
- A user can create, delete, or rename mailboxes on the mail server.
- A user can create a hierarchy of mailboxes in a folder for e-mail storage.

MIME

- Electronic mail has a simple structure. Its simplicity, however, comes with a price. It can send messages only in NVT 7-bit ASCII format. In other words, it has some limitations. It cannot be used for languages other than English. Also, it cannot be used to send binary files or video or audio data.
- Multipurpose Internet Mail Extensions (MIME) is a supplementary protocol that allows non-ASCII data to be sent through e-mail. MIME transforms non-ASCII data at the sender site to NVT ASCII data and delivers it to the client MTA to be sent through the Internet. The message at the receiving site is transformed back to the original data.
- We can think of MIME as a set of software functions that transforms non-ASCII data to ASCII data and vice versa, as shown in figure 5.40.

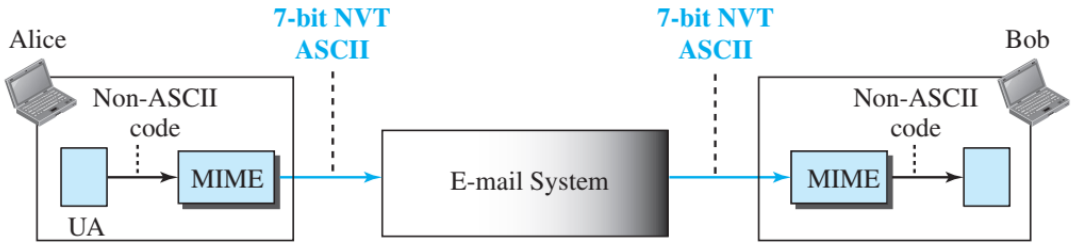
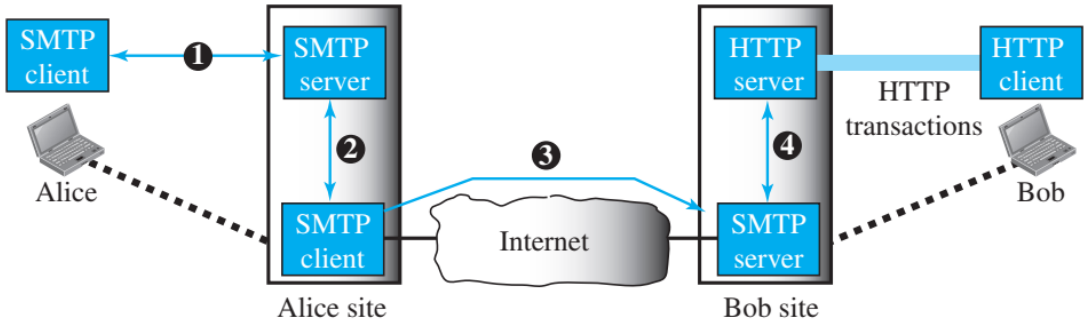


Figure 5.40 MIME

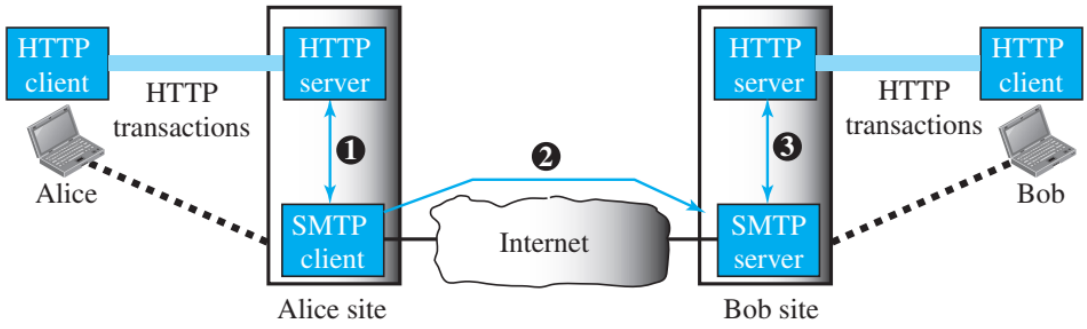
Web-Based Mail

- E-mail is such a common application that some websites today provide this service to anyone who accesses the site. Three common sites are Hotmail, Yahoo, and Google mail. Figure 5.41 shows two cases:
- Case I:
  In the first case, Alice, the sender, uses a traditional mail server; Bob, the receiver, has an account on a web-based server. Mail transfer from Alice's browser to her mail server is done through SMTP. The transfer of the message from the sending mail server to the receiving mail server is still through SMTP. However, the message from the receiving server (the web server) to Bob's browser is done through HTTP. The website sends a form to be filled in by Bob, which includes the log-in name and the password. If the log-in name and password match, the list of e-mails is transferred from the web server to Bob's browser in HTML format. Now Bob can browse through his received e-mails and then, using more HTTP transactions, can get his e-mails one by one.
- Case II:
  In the second case, both Alice and Bob use web servers, but not necessarily the same

server. Alice sends the message to the web server using HTTP transactions. Alice sends an HTTP request message to her web server using the name and address of Bob's mail-box as the URL. The server at the Alice site passes the message to the SMTP client and sends it to the server at the Bob site using SMTP protocol. Bob receives the message using HTTP transactions. However, the message from the server at the Alice site to the server at the Bob site still takes place using SMTP protocol.



Case 1: Only receiver uses HTTP



Case 2: Both sender and receiver use HTTP

Figure 5.41 Web-based e-mail, cases I and II

E-Mail Security

- The e-mail exchanges can be secured using two application-layer securities designed in particular for e-mail systems. Two of these protocols, Pretty Good Privacy (PGP) and Secure/Multipurpose Internet Mail Extensions (S/MIME).

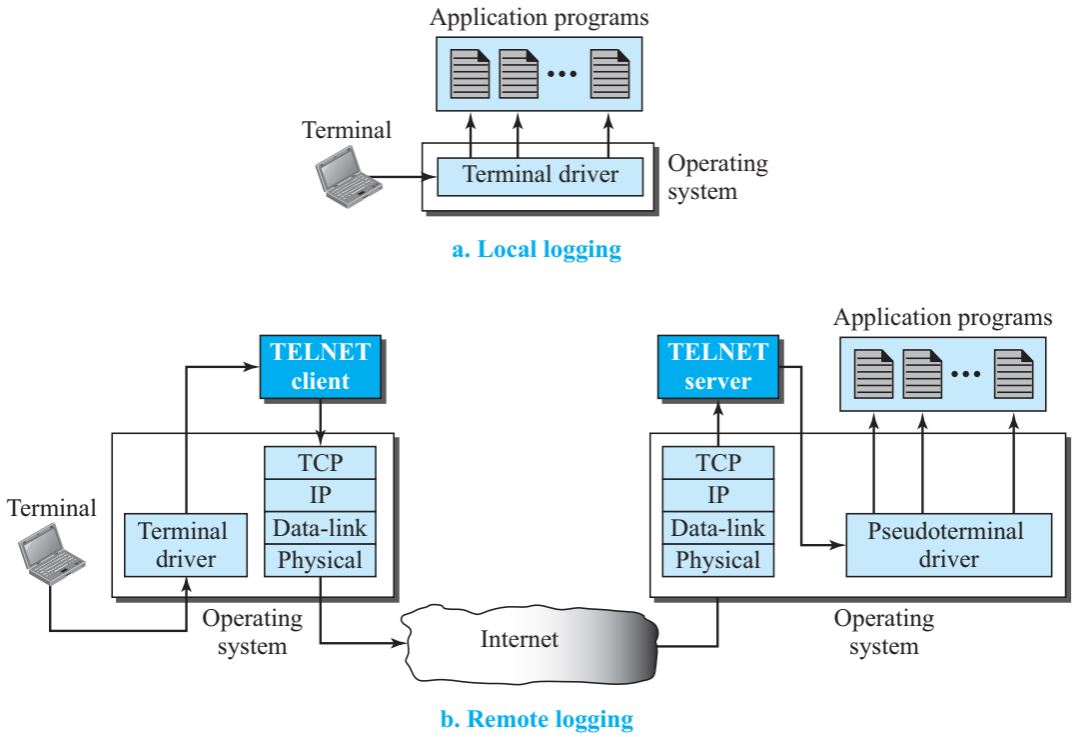
5.4.4 TELNET

- One of the original remote logging protocols is TELNET, which is an abbreviation for TErminaLNETwork. Although TELNET requires a logging name and password, it is vulnerable to hacking because it sends all data including the password in plaintext (not encrypted). A hacker can eavesdrop and obtain the logging name and password. Because of this security issue, the use of TELNET has diminished in favor of another protocol, Secure Shell (SSH), which we describe in the next section. Although TELNET is almost replaced by SSH, we briefly discuss TELNET here for two reasons:
- The simple plaintext architecture of TELNET allows us to explain the issues and challenges related to the concept of remote logging, which is also used in SSH when it serves as a remote logging protocol.
- Network administrators often use TELNET for diagnostic and debugging purposes.

Local versus Remote Logging

- The concept of local and remote logging as shown in figure 5.42.

- When a user logs into a local system, it is called local logging. As a user types at a terminal or at a workstation running a terminal emulator, the keystrokes are accepted by the terminal driver. The terminal driver passes the characters to the operating system.
- The operating system, in turn, interprets the combination of characters and invokes the



desired application program or utility.

Figure 5.42. Local versus remote logging

- However, when a user wants to access an application program or utility located on a remote machine, she performs remote logging. Here the TELNET client and server programs come into use. The user sends the keystrokes to the terminal driver where the local operating system accepts the characters but does not interpret them.
- The characters are sent to the TELNET client, which transforms the characters into a universal character set called Network Virtual Terminal (NVT) characters (discussed below) and delivers them to the local TCP/IP stack.
- The commands or text, in NVT form, travel through the Internet and arrive at the TCP/IP stack at the remote machine. Here the characters are delivered to the operating system and passed to the TELNET server, which changes the characters to the corresponding characters understandable by the remote computer.
- However, the characters cannot be passed directly to the operating system because the remote operating system is not designed to receive characters from a TELNET server; it is designed to receive characters from a terminal driver.
- The solution is to add a piece of software called a pseudoterminal driver, which pretends that the characters are coming from a terminal. The operating system then passes the characters to the appropriate application program.

Network Virtual Terminal (NVT)

- The mechanism to access a remote computer is complex. This is because every computer and its operating system accepts a special combination of characters as tokens.

- We are dealing with heterogeneous systems. If we want to access any remote computer in the world, we must first know what type of computer we will be connected to, and we must also install the specific terminal emulator used by that computer.
- TELNET solves this problem by defining a universal interface called the Network Virtual Terminal (NVT) character set. Via this interface, the client TELNET translates characters (data or commands) that come from the local terminal into NVT form and delivers them to the network. The server TELNET, on the other hand, translates data and commands from NVT form into the form acceptable by the remote computer, which is shown in figure 5.43.

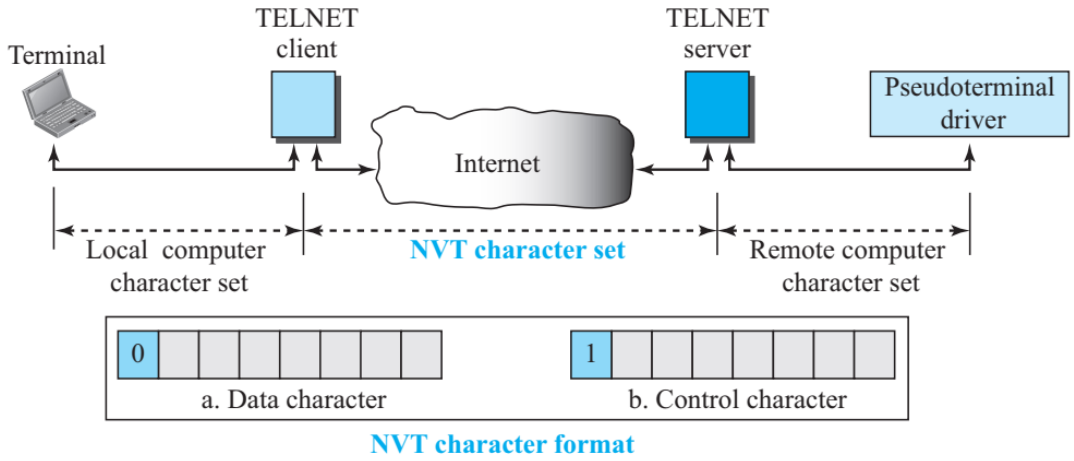


Figure 5.43. Concept of NVT

- NVT uses two sets of characters, one for data and one for control. Both are 8-bit bytes as shown in figure 5.43.
- For data, NVT normally uses what is called NVT ASCII. This is an 8-bit character set in which the seven lowest order bits are the same as US ASCII and the highest order bit is 0. To send control characters between computers (from client to server or vice versa), NVT uses an 8-bit character set in which the highest order bit is set to 1.
- TELNET lets the client and server negotiate options before or during the use of the service. Options are extra features available to a user with a more sophisticated terminal. Users with simpler terminals can use default features.
- The operating system (UNIX, for example) defines an interface with user-friendly commands. An example of such a set of commands can be found in figure 5.44.

Command	*Meaning*	*Command*	*Meaning*
open	Connect to a remote computer	**set**	Set the operating parameters
close	Close the connection	**status**	Display the status information
display	Show the operating parameters	**send**	Send special characters
mode	Change to line or character mode	**quit**	Exit TELNET

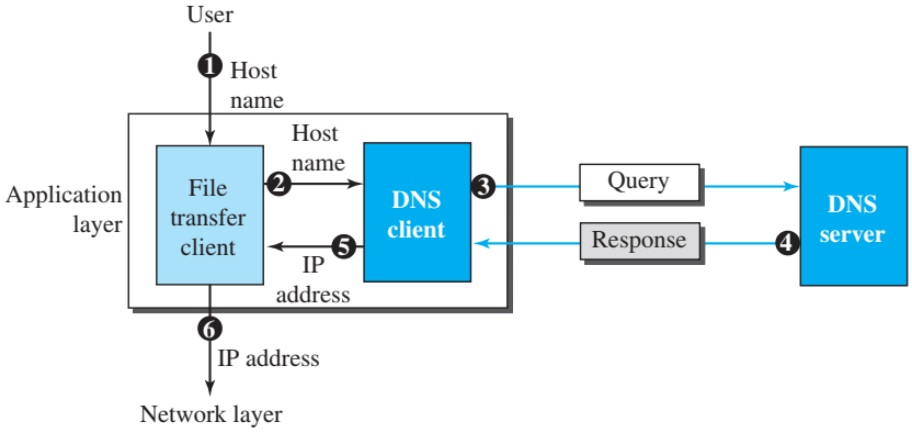
Figure5.44. Examples of interface commands

5.4.5 DOMAIN NAME SYSTEM (DNS)

- Since the Internet is so huge today, a central directory system cannot hold all the mapping. In addition, if the central computer fails, the whole communication network will collapse. A better solution is to distribute the information among many computers in the world. In this method, the host that needs mapping can contact the closest computer

holding the needed information. This method is used by the **Domain Name System (DNS)**.

- Figure 5.45 shows how TCP/IP uses a DNS client and a DNS server to map a name to an address. A user wants to use a file transfer client to access the corresponding file transfer server running on a remote host. The user knows only the file transfer. server name, such as *afilesource.com.* However, the TCP/IP suite needs the IP address of the file transfer server to make the connection.



- The following six steps map the host name to an IP address:
  1. The user passes the host name to the file transfer client.
  2. The file transfer client passes the host name to the DNS client.
  3. Each computer, after being booted, knows the address of one DNS server. TheDNS client sends a message to a DNS server with a query that gives the file transfer server name using the known IP address of the DNS server.
  4. The DNS server responds with the IP address of the desired file transfer server.
  5. The DNS server passes the IP address to the file transfer client.
  6. The file transfer client now uses the received IP address to access the file transferserver.

Name Space

- The names assigned to machines must be carefully selected from a name space with complete control over the binding between the names and IP addresses.
- A name spacethat maps each address to a unique name can be organized in two ways: flat or hierarchical. In a *flat name space*, a name is assigned to an address. A name in this space is a sequence of characters without structure.
- The names may or may not have a common section; if they do, it has no meaning. The main disadvantage of a flat name space is that it cannot be used in a large system such as the Internet because it must be centrally controlled to avoid ambiguity and duplication. In a *hierarchical name space,* each name is made of several parts.
- The first part can define the nature of the organization, the second part can define the name of an organization, the third part can define departments in the organization, and so on. In this case, the authority to assign and control the name spaces can be decentralized.
- A central authority can assign the part of the name that defines the nature of the organization and the name of the organization. The responsibility for the rest of the name can be given to the organization itself.
- The organization can add suffixes (or prefixes) to the name to define its host or resources.

Domain Name Space

- To have a hierarchical name space, a domain name space was designed. In this design the names are defined in an inverted-tree structure with the root at the top. The tree can have

only 128 levels: level 0 (root) to level 127, as shown infigure 5.46.

Label

- Each node in the tree has a label, which is a string with a maximum of 63 characters. The root label is a null string (empty string). DNS requires that children of a node (nodes that branch from the same node) have different labels, which guarantees the uniqueness of the domain names.

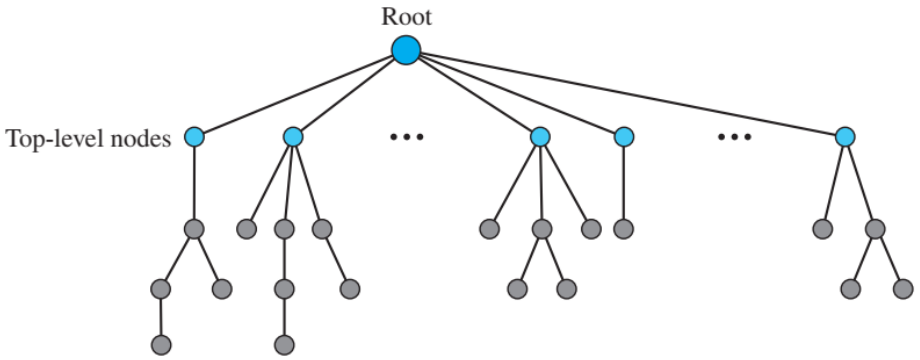


Figure 5.46. Domain name space

Domain Name

- Each node in the tree has a domain name. A full domain name is a sequence of labels separated by dots (.). The domain names are always read from the node up to the root. The last label is the label of the root (null). Figure 5.47 shows some domain names.

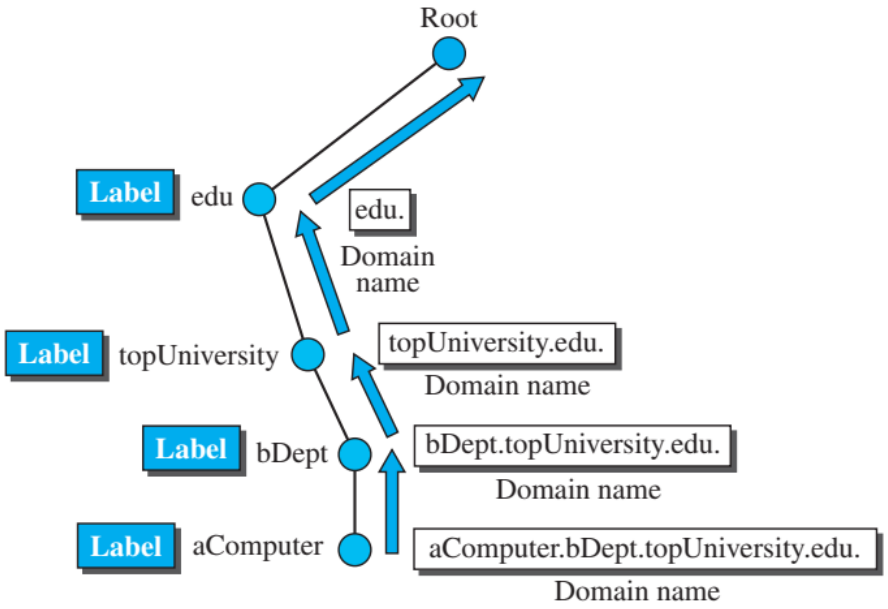


Figure 5.47. Domain names and labels

Domain

- A domain is a subtree of the domain name space. The name of the domain is the name of the node at the top of the subtree. Figure 5.47 shows some domains.

Distribution of Name Space

- The information contained in the domain name space must be stored. However, it is very inefficient and also not reliable to have just one computer store such a huge amount of information.
- It is inefficient because responding to requests from all over the world places a heavy

load on the system. It is not reliable because any failure makes the data inaccessible.

Hierarchy of Name Servers

- The solution to these problems is to distribute the information among many computers called DNS servers. One way to do this is to divide the whole space into many domains based on the first level.
- In other words, we let the root stand alone and create as many domains (subtrees) as there are first-level nodes. Because a domain created this way could be very large, DNS allows domains to be divided further into smaller domains (subdomains).
- Each server can be responsible (authoritative) for either a large or small domain. In other words, we have a hierarchy of servers in the same way that we have a hierarchy of name, as shown in figure 5.48.

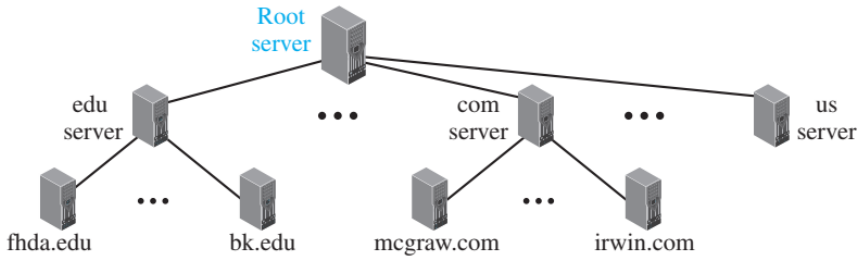


Figure 5.48. Hierarchy of name servers

Zone

- Since the complete domain name hierarchy cannot be stored on a single server, it is divided among many servers.
- What a server is responsible for or has authority over iscalled a zone. We can define a zone as a contiguous part of the entire tree, which is shown in figure 5.49.

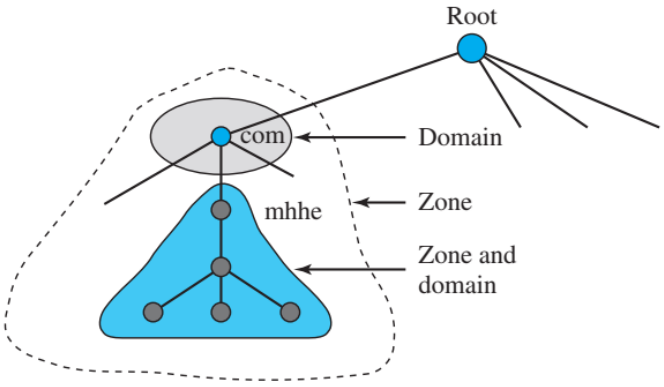


Figure 5.49. Zone

Primary and Secondary Servers

- DNS defines two types of servers: primary and secondary. A primary server is a server that stores a file about the zone for which it is an authority. It is responsible for creating, maintaining, and updating the zone file. It stores the zone file on a local disk.A primary server loads all information from the disk file.
- A secondary server is a server that transfers the complete information about a zone from another server (primary or secondary) and stores the file on its local disk.The secondary server loads all information from the primary server

DNS in the Internet

- DNS is a protocol that can be used in different platforms. In the Internet, the domain name space (tree) was originally divided into three different sections: generic domains,

country domains, and the inverse domains.

- The generic domains define registered hosts according to their generic behavior. Each node in the tree defines a domain, which is an index to the domain name space database, as shown in figure 5.50.
- The country domains section uses two-character country abbreviations. Second labels can be organizational, or they can be more specific national designations, as shown in figure 5.51.

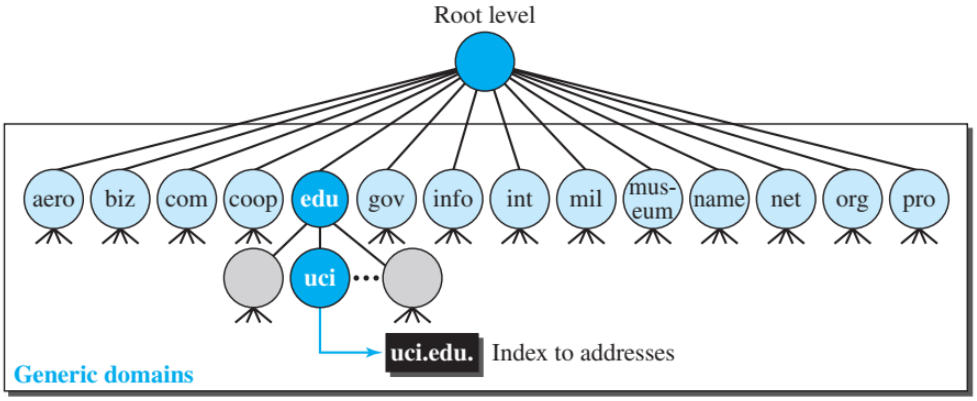


Figure 5.50. Generic domains

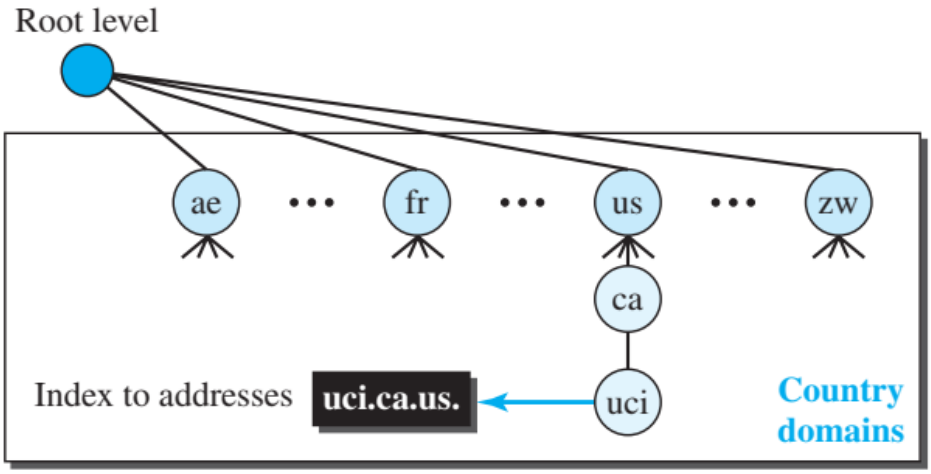


Figure 5.51 Country domains

Resolution

- Mapping a name to an address is called name-address resolution. DNS is designed as a client-server application. A host that needs to map an address to a name or a name to an address calls a DNS client called a resolver. A resolution can be either recursive or iterative.Figure 5.52 shows a simple example of a recursive resolution.

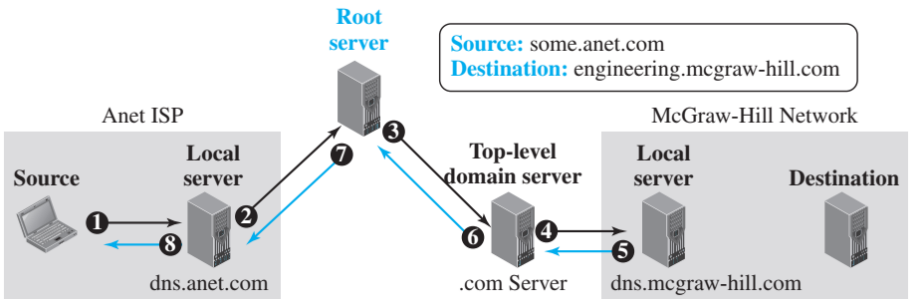
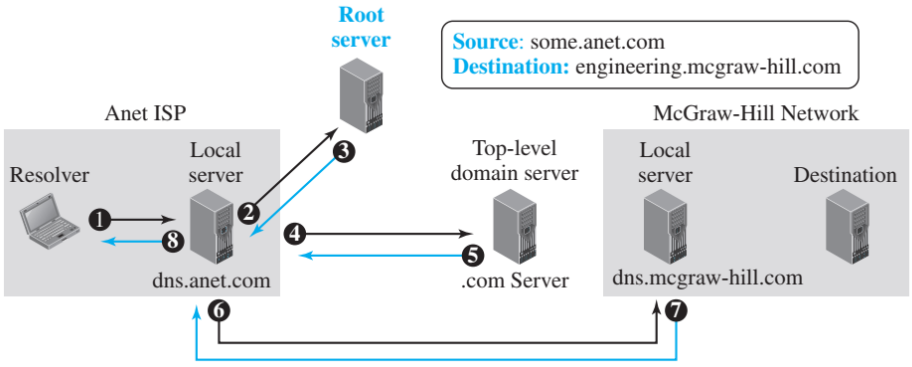


Figure 5.52 Recursive resolution

- In iterative resolution,each server that does not know the mapping sends the IP address of the next server back to the one that requested it. Figure 5.53 shows the flow of information in an iterative resolution in the same scenario as the one depicted in figure 5.52.

Caching



- Each time a server receives a query for a name that is not in its domain, it needs to search its database for a server IP address. Reduction of this search time would increase efficiency. DNS handles this with a mechanism called caching.

Resource Records

- The zone information associated with a server is implemented as a set of *resource records*. In other words, a name server stores a database of resource records. A *resource record* is a 5-tuple structure, as shown below:

(Domain Name, Type, Class, TTL, Value)

- The domain name field is what identifies the resource record. The value defines the information kept about the domain name. The TTL defines the number ofseconds for which the information is valid. The class defines the type of network; we are only interested in the class IN (Internet). The type defines how the value should be interpreted. Figure 5.54 lists the common types and how the value is interpreted for each type.

Type	*Interpretation of value*
A	A 32-bit IPv4 address (see Chapter 18)
NS	Identifies the authoritative servers for a zone
CNAME	Defines an alias for the official name of a host
SOA	Marks the beginning of a zone
MX	Redirects mail to a mail server
AAAA	An IPv6 address (see Chapter 22)

DNS Messages

- To retrieve information about hosts, DNS uses two types of messages: *query* and *response*. Both types have the same format as shown in figure 5.55.

Registrars

- How are new domains added to DNS? This is done through a registrar, a commercial entity accredited by ICANN. A registrar first verifies that the requested domain name is

unique and then enters it into the DNS database. To register, the organization needs to give the name of its server and the IP address of the server.

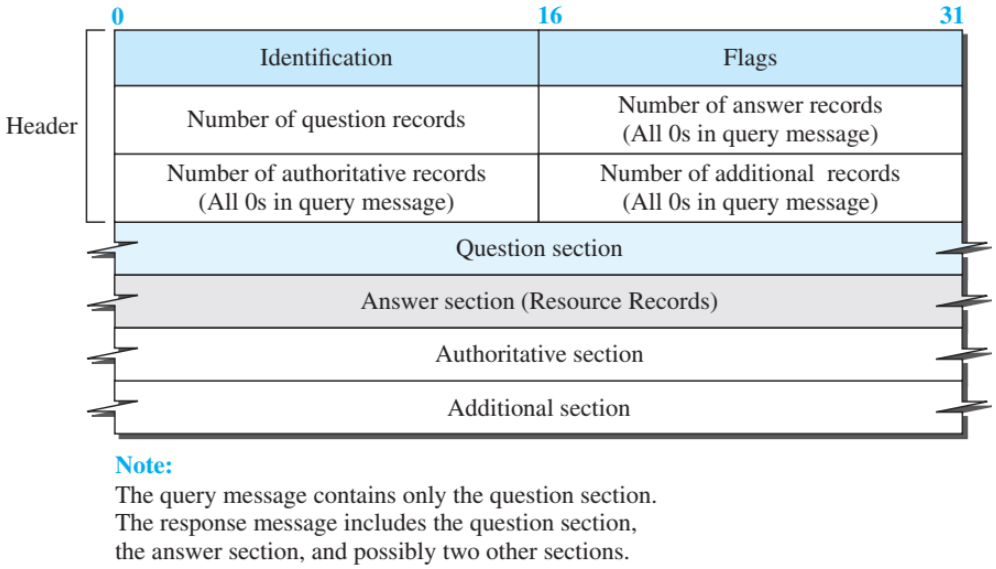


Figure 5.55. DNS message

DDNS

- When the DNS was designed, no one predicted that there would be so many address changes. In DNS, when there is a change, such as adding a new host, removing a host, or changing an IP address, the change must be made to the DNS master file. These types of changes involve a lot of manual updating. The size of today's Internet does not allow for this kind of manual operation.
- The DNS master file must be updated dynamically. The Dynamic Domain Name System (DDNS) therefore was devised to respond to this need. In DDNS, when a binding between a name and an address is determined, the information is sent, usually by DHCP to a primary DNS server.

Security of DNS

- DNS is one of the most important systems in the Internet infrastructure; it provides crucial services to Internet users. Applications such as Web access or e-mail are heavily dependent on the proper operation of DNS. DNS can be attacked in several ways including:
  1. The attacker may read the response of a DNS server to find the nature or names of sitesthe user mostly accesses. This type of information can be used to find the user's profile.To prevent this attack, DNS messages need to be confidential.
  2. The attacker may intercept the response of a DNS server and change it or create atotally new bogus response to direct the user to the site or domain the attackerwishes the user to access. This type of attack can be prevented using message origin authentication and message integrity.
  3. The attacker may flood the DNS server to overwhelm it or eventually crash it. Thistype of attack can be prevented using the provision against denial-of-service attack.
- To protect DNS, IETF has devised a technology named DNS Security (DNSSEC) that provides message origin authentication and message integrity using a security service called digital signature. DNSSECdoes not provide confidentiality for the DNS messages.
- There is no specific protection against the denial-ofservice attack in the specification of

DNSSEC. However, the caching system protectsthe upper-level servers against this attack to some extent.

ASSIGNMENT CUM TUTORIAL QUESTIONS

5.1The Transport Layer Protocols

Objective Questions

1. The transport layer aggregates data from different applications into a single stream before passing it to _____ layer . [ ]
   a) Network b) Application c) Data Link d) Physical
2. An endpoint of an inter-process communication flow across a computer network is called _____. [ ]
   a) Socket b) Pipe c) Port d) Machine

Short Answer Questions

S. No.	Question	BL
1	Give the significance of port numbers in transport layer.	L1

Descriptive Questions

S. No.	Question	BL
1	Explain the services offered by the transport layers to the upper layers.	L2

5.2 User data gram protocol (UDP)

Objective Questions

1. Which of the following services use UDP? [ ]
   (i) DHCP (ii) SMTP (iii) SNMP (iv) FTP (v) HTTP (vi) TFTP
   a) 1, 3 and 6 b) 2 and 4 c) 1, 2 and 4 d) 1, 4 and 5

2. User datagram protocol is called connectionless because [ ]
   a) all UDP packets are treated independently by transport layer.
   b) it sends data as a stream of related packets
   c) it is received in the same order as sent order
   d) it sends data very quickly

3. The main advantage of UDP is: [ ]
   a) High reliability b) Error correction
   c) Low overhead and fast transmission d) Guaranteed delivery

4. Which of the following does UDP NOT provide? [ ]
   a) Flow control b) Error detection c) Multiplexing d) Port addressing

5. UDP is suitable for: [ ]
   a) File transfer b) Email c) Real-time video streaming d) Web browsing

Short Answer Questions

S. No.	Question	BL
1	Why is UDP considered a connectionless protocol?	L2
2	What is the function of the checksum field in UDP?	L1
3	Explain the usage of Encapsulation and Decapsulation process by UDP?	L2

Descriptive Questions

S. No.	Question	BL
1	Describe the UDP Internet Transport Protocol.	L2
2	Draw and explain the structure of UDP Header format.	L2
3	Discuss various applications of UDP.	L2
4	Explain the services offered by UDP.	L2

5.3Transmission Control Protocol

Objective Questions

1. What is the length of port address in TCP/IP? [ ]
a) 16 bits b) 32 bits c) 48 bits d) 64 bits

2. Which of the following ensures reliability in TCP?[ ]
a) Error detection onlyb) Acknowledgment and retransmission
c) Sequencing onlyd) Flow control only

3. The process of establishing a TCP connection is called _____. [ ]
a) Handshakingb) Synchronizationc) Three-way handshaked) Acknowledgment

4. TCP provides:[ ]
a) Error detection onlyb) Error correction and flow control
c) Multiplexing onlyd) flow control

5. In TCP, data is sent in units called _____. [ ]

a) Packets
b) Datagrams
c) Segments
d) Frames

Short Answer Questions

S. No.	Question	BL
1	What mechanism does TCP use to establish a connection?	L1
2	How does TCP terminate a connection?	L2
3	What is the role of the SYN flag in TCP?	L1
4	Why is TCP slower than UDP?	L2
5	What is meant by “reliable byte stream” in TCP?	L1

Descriptive Questions

S. No.	Question	BL
1	Discuss in detail about the connection establishment and release in TCP.	L2
2	Describe the process of connection release in TCP.	L2
3	Explain the services offered by TCP.	L2
4	Discuss various features of TCP.	L2
5	Draw and explain TCP segment format	L2
6	Compare the process of sender and receiver window in TCP.	L4
7	Discuss the concept of Data flow and flow control feedbacks in TCP.	L2
8	Explain the process of Generating Acknowledgments in error control for TCP.	L2
9	Describe various Congestion Policies for TCP.	L2

5.4 Application Layer

Objective Questions

1. Which protocol is primarily used by the World Wide Web?[ ]

a) FTPb) HTTPc) SMTPd) TCP

2. What is a web server?[ ]
a) A computer that provides web pagesb) A website
c) A search engined) A browser plugin

3. Which one of the following allows client to update their DNS entry as their IP address change? [ ]
a) dynamic DNSb) mail transfer agent
c) authoritative name serverd) none of the mentioned

4. In the architecture of an e-mail, we can have how many scenarios?[ ]
a) 2 b) 3 c) 4 d) 6

5. When the sender and the receiver of an e-mail are on the same system, two _____ are needed. [ ]
a) IP b) Domains c) Servers d) User Agents.

6. Which of the following protocol is used by internet mail?[ ]
a) FTP b) UTP c) HTTP d) TCP/IP

7. What does SMTP stands for?[ ]
a) Simple Mail Terminal Protocol b) Simple Mail Transfer Protocol
c) Simple Mail Transport Protocol d) Simple Mail Protocol Terminal

8.The HTTP request message is sent in _____ part of three-way handshake.[ ]
a) First b) Second c) Third d) None

9. The first line of HTTP request message is called_____. [ ]
a). Request line b) Header line c) Status line d) Entity line

10. The values GET, POST, HEAD etc are specified in _____ of HTTP message.[ ]
a) Request line b) Header line c) Status line d) Entity line

11. A DNS client is called _____. [ ]
a) DNS resolverb) DNS updaterc) DNS handlerd) both a and b

12. The type of records used in DNS are _____. [ ]
a) Question Recordb) Answers Recordc) Resource Recordd) Both A and C

Short Answer Questions

S. No.	Question	BL
1	What is the function of a URL?	L1

2	State the parts of the browser.	L1
3	Compare static and dynamic documents.	L4
4	Give the process of HTTP transaction.	L1
5	Differentiate POP3 and IMAP protocols.	L4
6	State the protocols providing E-Mail Security.	L1
7	Compare Local versus Remote Logging.	L4
8	Give the purpose of Domain Name Space.	L1
9	Differentiate the functionality of Primary and Secondary Servers in DNS.	L4

Descriptive Questions

S. No.	Question	BL
1.	Describe the client-server model used in the World Wide Web.	L2
2	Draw and explain the architecture of World Wide Web.	L2
3	Describe in brief about HTTP request and response methods.	L2
4	Draw and explain the common architecture of an E-mail.	L2
5	Discuss the role played by the message transfer agent.	L2
6	Describe the SMTP and POP3 protocols.	L2
7	Describe in brief about sending and receiving e-mail.	L2
8	Explain the importance of DNS in application layer.	L2
9	How DNS service maps domain names to IP addresses.	L2
10	Discuss the different methods of attacking DNS.	L2